Continuous Time Financial Economics

Part 2 of 2

2025

Raman Uppal

EDHEC Business School

Core Course
Continuous Time Financial Economics Continuous Time Financial Economics Continuous Time Financial Economics Continuous Time Financial Economics Continuous Time Financial Economics Continuous Time Financial Economics Continuous Time Financial Economics Continuous Time Financial Economics Continuous Time Financial Economics Continuous Time Financial Economics Continuous Time Financial Economics Continuous Time Financial Economics Continuous Time Financial Economics Continuous Time Financial Economics Continuous Time Financial Economics Continuous Time Financial Economics Continuous Time Financial Economics Continuous Time Financial Economics Continuous Time Financial Economics
Core Course

All Rights Reserved

The text contained in these notes, or any part thereof, may not be reproduced in any form or by any means, electronic or mechanical, including photocopying, recording, storage in an information retrieval system, or otherwise, without the prior written permission of the author.

© Raman Uppal

Brief Table of Contents

0 Syllabus and How This Course Is Organized **21**

. . . . . See Part 1 for Chapters 1–7 . . . . .

8 Valuation of Fixed Income Securities **45**

9 Term Structure of Interest Rates **149**

10 Portfolio Choice: Mean-Variance and Two-Date Models **293**

11 Portfolio Choice Using Martingale Approach **393**

12 Portfolio Choice Using Dynamic Programming **481**

13 Partial- and General-Equilibrium Asset Pricing **553**

14 Course Summary: Key Ideas Covered in Part 2 **613**

. . . . . Other material for the course . . . . .

A Practice Questions for Part 2 **645**

B Homework Questions for Part 2 . . . To be handed in **661**

C Information about the Final Exam **677**

D Sample Final Exam **685**

E Guide to Notation **693**

F List of Ph.D. Elective Courses Offered by EDHEC **701**

Bibliography **709**

Blank page.

Contents

0 Syllabus and How This Course Is Organized **21**
0.1 Instructor for the Course and Contact Information . . . 22
0.2 Course Description . . . 23
0.3 Course Prerequisites . . . 24
0.4 Course Objective . . . 25
0.5 Topics Covered in the Course . . . 27
0.6 Course Method . . . 31
0.7 Course Assessment . . . 32
0.8 Textbooks for the Course . . . 33
0.9 Other Textbooks (listed alphabetically) . . . 34
0.10 Review Articles . . . 35
0.11 My Teaching Style . . . 40
0.12 Your "Style" in Class . . . 41
0.13 The Good News . . . 42
0.14 The Bad News . . . 43

See Part 1 for Chapters 1–7

8 Valuation of Fixed Income Securities **45**
8.1 Plan for the Day . . . 49
8.2 Background . . . 50
8.3 Main Points for Today . . . 51
8.4 Guide to Notation . . . 52
8.5 Overview . . . 53
8.6 Basics: The Yield Curve . . . 54
8.6.1 Yield of a Zero-Coupon Bond . . . 55
8.6.2 Definition and Shapes of the Yield Curve . . . 56
8.6.3 Estimating the Yield Curve . . . 57
8.7 Basics: Yield to Maturity, Duration, Convexity, & Hedging . . . 58
8.7.1 Duration Hedging . . . 61
8.7.2 Continuous Hedging in a Single Factor Model . . . 64
8.8 Basics: LIBOR Interest Rates . . . 70
8.9 Basics: Forward Interest Rates . . . 73
8.10 Swaps . . . 76
8.10.1 Valuation of swaps . . . 79
8.10.2 Setting the Swap Fixed Interest Rate . . . 86
8.11 Fixed-Income Derivatives: Caps and Floors . . . 87
8.11.1 Definition and Interpretation of Caps and Floors . . . 87
8.11.2 Portfolios of Bonds that Pay Spot Rates . . . 91
8.11.3 Forward Price of a Non-Dividend Paying Asset . . . 95
8.11.4 The Market Model for Caplets and Floorlets . . . 96
8.11.5 Put-Call Parity Relation for Caplets and Floorlets . . . 101
8.11.6 Valuing Caplets as Puts on Discount Bonds . . . 103
8.12 Fixed-Income Derivatives: European Swaptions . . . 105
8.13 Inconsistency in Assuming Constant Volatilities for Forward Interest Rates and Forward Swap Rate . . . 111
8.14 Summary . . . 115
8.15 Appendix: Estimating the Yield Curve . . . 123
8.15.1 Bootstrapping Approach . . . 125
8.15.2 Limitations of Bootstrapping . . . 128
8.15.3 Cubic Splines . . . 129
8.15.4 Exponential Splines . . . 136
8.15.5 Implementation Issues with Splines . . . 137
8.15.6 Nelson-Siegel-Svensson Method for Estimating Term Structure . . . 139
8.16 Suggestions for Further Readings . . . 143
8.16.1 Source material for the lecture notes . . . 143
8.16.2 Potential applications and extensions of lecture notes . . . 144

8.16.3 Ph.D. elective courses offered on related topics . . . 148

9 Term Structure of Interest Rates 149
9.1 Plan for the Day . . . 153
9.2 Background . . . 154
9.3 Main Points for Today . . . 155
9.4 Motivation for Dynamic Models of the Term Structure of Interest Rates . . . 156
9.4.1 Differences in Modeling Movements in Equity and Bond Prices . . . 157
9.4.2 State variables and factors . . . 160
9.4.3 An Example of Arbitrage in a Badly Specified Dynamic Model of the Term Structure . . . 161
9.5 Overview and Background of Material To be Covered . . . 166
9.5.1 Plan: Focus on the Vasicek Model . . . 167
9.5.2 List of Topics For Each Model of the Term Structure . . . 168
9.6 The Short Rate and Discount Bond Prices . . . 169
9.7 The Basic Vasicek Model . . . 173
9.7.1 The Short Rate and Its Distribution for the Vasicek Process . . . 174
9.7.2 Bond Prices under the Vasicek Process . . . 187
9.7.3 Hedging in the Vasicek Model . . . 199
9.7.4 Estimating the Vasicek Model . . . 210
9.8 Extensions of Vasicek Model: Hull & White Model and Ho & Lee Model . . . 214
9.8.1 Interest Rates in the Hull-White and Ho-Lee Models . . . 218
9.8.2 Bond Prices in the Extended Vasicek model . . . 220
9.8.3 Bond Returns in the Extended Vasicek Model . . . 223
9.8.4 Fitting Discount Bond Prices and Forward Rates . . . 225
9.8.5 Aside: The Instantaneous Forward Interest Rate . . . 230
9.9 Options on Discount Bonds & Interest Rates (Caps/Floors) . . . 233
9.9.1 Valuation of Options on Discount Bonds . . . 234
9.9.2 Valuation of Caps and Floors . . . 240
9.10 Other Continuous-Time Models of the Term Structure of Interest Rates . . . 244
9.10.1 General Hull and White (1990) Model . . . 246
9.10.2 Cox, Ingersoll, and Ross (1985a) Model . . . 247
9.10.3 Multifactor Cox, Ingersoll, and Ross (1985a) and Longstaff and Schwartz (1992) Models . . . 248
9.10.4 Affine Models of Duffie and Kan (1996) . . . 249
9.10.5 Essentially Affine Model of Duffee (2002) . . . 253
9.10.6 Quadratic Models . . . 254
9.10.7 Heath, Jarrow, and Morton (1992) Model . . . 255
9.11 Summary . . . 257
9.12 Appendix: Discrete-Time Models of Term Structure of Interest Rates . . . 267
9.12.1 The Black, Derman, and Toy (1990) of Dynamics of the Term Structure . . . 269
9.12.2 The Volatility Equation . . . 271
9.12.3 Building the Short-rate Tree for Black, Derman, and Toy (1990) Model . . . 272
9.12.4 Numerical Example from Black, Derman, and Toy (1990) . . . 273
9.13 Suggestions for Further Readings . . . 286
9.13.1 Source material for the lecture notes . . . 286
9.13.2 Potential applications and extensions of lecture notes . . . 287
9.13.3 Ph.D. elective courses offered on related topics . . . 289

10 Portfolio Choice: Mean-Variance and Two-Date Models 293
10.1 Plan for the Day . . . 297
10.2 Main Points for Today . . . 298
10.3 Background . . . 300
10.3.1 Valuation of Redundant Assets . . . 301
10.3.2 Optimal Consumption and Portfolio Policies . . . 302
10.3.3 Plan for Understanding Optimal Portfolio Selection . . . 304
10.4 Understanding Portfolio Choice Via Nobel Pictures . . . 305
10.4.1 A Summary of Main Insights from Modern Portfolio Theory . . . 314
10.5 Mean-Variance Portfolio Choice in a One-Date Model . . . 316
10.5.1 Guide to Notation . . . 318
10.5.2 Finding the Mean-Variance Efficient Portfolio in the Absence of a Riskfree Asset . . . 326
10.5.3 Global Minimum-Variance Portfolio . . . 329
10.5.4 Mean-Variance Efficient Portfolio in Presence of a Risk-Free Asset . . . 331
10.5.5 Characterizing the Tangency Portfolio and the Sharpe Ratio . . . 334

10.5.6 Identifying the Weights of the Tangency Portfolio: Mean-Variance Utility Maximization . . . 338
10.5.7 Relation Between Returns of Individual Assets and Frontier Portfolios: CAPM . . . 341
10.6 Measures of Portfolio Performance . . . 350
10.6.1 Sharpe Ratio . . . 350
10.6.2 The Treynor Ratio . . . 351
10.6.3 Jensen's Alpha . . . 352
10.6.4 Multifactor Alpha . . . 353
10.7 Consumption-Portfolio Choices in a Two-Date Model . . . 357
10.7.1 Understanding the First-Order Condition for Optimal Portfolios . . . 365
10.7.2 Understanding First-Order Condition for Intertemporal Allocation . . . 367
10.7.3 Elasticity of Intertemporal Substitution (EIS) . . . 372
10.7.4 Understanding relation between the SDF and risky asset prices . . . 377
10.7.5 Understanding relation between the SDF and the risk-free rate . . . 378
10.8 Summary . . . 379
10.9 Suggestions for Further Readings . . . 383
10.9.1 Source material for the lecture notes . . . 383
10.9.2 Potential applications and extensions of lecture notes . . . 385
10.9.3 Ph.D. elective courses offered on related topics . . . 392

11 Portfolio Choice Using Martingale Approach **393**
11.1 Plan for the Day . . . 397
11.2 Main Points for Today . . . 398
11.3 Martingale Approach to Consumption and Portfolio Choice in a Multiperiod Discrete-Time Model . . . 399
11.3.1 Steps to Identify Optimal Consumption and Portfolio . . . 402
11.3.2 Summary of Martingale Approach to Consumption-Portfolio Problem . . . 420
11.4 Martingale Approach to Consumption and Portfolio Choice in Continuous Time . . . 423
11.5 Special Case: Single Risky Asset, No Bequest, and Constant Investment Opportunity Set . . . 427
11.5.1 Description of the Setting for this Special Case . . . 427
11.5.2 Expression for the Optimal Portfolio Weight . . . 428
11.5.3 Identifying the Expression $W_M M/W$ . . . 439
11.5.4 Showing that $W(M_t, t) = f(t)\, M_t^{\frac{-1}{\gamma}}$ . . . 441
11.6 General Case: Many risky assets, Bequest, and Stochastic Investment Opportunity Set . . . 446
11.7 Summary . . . 461
11.8 Appendix: Case with Multiple Risky Assets and Power Utility But Constant Investment Opportunity Set . . . 464
11.8.1 Expression for the Optimal Portfolio . . . 465
11.8.2 Showing that $-W_M M/W = 1/\gamma$ . . . 466
11.8.3 Explicit Expression for the Optimal Portfolio . . . 467
11.8.4 Interpreting the Expression for the Optimal Portfolio . . . 468
11.8.5 Showing that $W(M_t, t) = f(t)\, M_t^{\frac{-1}{\gamma}}$ . . . 469
11.9 Suggestions for Further Readings . . . 474
11.9.1 Source material for the lecture notes . . . 474
11.9.2 Potential applications and extensions of lecture notes . . . 475
11.9.3 Ph.D. elective courses offered on related topics . . . 479

12 Portfolio Choice Using Dynamic Programming **481**
12.1 Plan for the Day . . . 485
12.2 Main Points for Today . . . 486
12.3 Background (repeated) . . . 487
12.3.1 Valuation of Redundant Assets . . . 488
12.3.2 Optimal Consumption and Portfolio Policies . . . 489
12.4 General Description of Consumption & Portfolio Choice in Continuous Time . . . 491
12.4.1 Description of the General Setting . . . 491
12.4.2 Assumptions of the General Model . . . 492
12.4.3 Investor's Dynamic Budget Constraint . . . 495
12.4.4 Investor's Objective . . . 498
12.5 Simple Case: Single Risky Asset and Constant Opportunity Set . . . 499
12.6 Special Case with Many Risky Assets and Constant Opportunity Set . . . 514
12.6.1 First-order condition for consumption . . . 516
12.6.2 First-order condition for portfolio choice . . . 517

12.6.3 Implications of expression for optimal portfolio weights . . . 519
12.7 Example of Portfolio Choice with Stochastic Risk-free Rate and Only One Risky Asset . . . 522
12.7.1 Utility Function . . . 522
12.7.2 Securities Market . . . 523
12.7.3 Wealth Process of the Investors . . . 524
12.7.4 State Variables and Bellman Equation . . . 524
12.7.5 First-Order Condition for Consumption and Portfolio . . . 525
12.7.6 Interpretation of the Optimal Portfolio Weight . . . 526
12.8 General Case with Stochastic Opportunity Set and Multiple Assets . . . 527
12.8.1 First-order condition for consumption . . . 529
12.8.2 First-order condition for portfolio choice . . . 530
12.8.3 Understanding the Intertemporal Hedging Portfolio . . . 532
12.8.4 Solving for optimal consumption and portfolio weights . . . 536
12.9 Summary . . . 537
12.10 Appendix: Special Case with Stochastic Investment Opportunities and Log Utility . . . 540
12.11 Suggestions for Further Readings . . . 543
12.11.1 Source material for the lecture notes . . . 543
12.11.2 Potential applications and extensions of lecture notes . . . 549
12.11.3 Ph.D. elective courses offered on related topics . . . 552

13 Partial- and General-Equilibrium Asset Pricing **553**
13.1 Plan for the Day . . . 557
13.2 Main Points for Today . . . 558
13.3 Partial-Equilibrium Asset Returns in Continuous-Time . . . 559
13.3.1 Assumptions of the Merton Model (repeated) . . . 560
13.3.2 Stochastic Discount Factor (repeated) . . . 562
13.3.3 First-Order Condition for Optimal Portfolio Weight . . . 565
13.3.4 Envelope Condition . . . 566
13.3.5 First-Order Condition for Optimal Consumption . . . 567
13.3.6 Definition of Absolute Risk Aversion . . . 569
13.3.7 The Consumption CAPM (CCAPM) . . . 570
13.3.8 The Intertemporal CAPM (ICAPM) . . . 577
13.3.9 The Conditional CAPM . . . 582
13.4 General-Equilibrium Asset Pricing in an Exchange Economy . . . 584
13.4.1 Assumptions of the Lucas Model . . . 586
13.4.2 Asset Prices in the Lucas Model . . . 590
13.4.3 Interpreting the Interest Rate and Market Price of Risk in the Lucas Model . . . 593
13.5 General-Equilibrium Asset Pricing in a Production Economy . . . 595
13.5.1 The Production Process . . . 596
13.5.2 Investor Preferences and Investment Opportunities . . . 599
13.5.3 Optimization Problem and Dynamic Budget Constraint . . . 600
13.5.4 State Variables, Value Function, and Bellman Equation . . . 601
13.5.5 First-order Conditions for Consumption and Portfolio Policy . . . 602
13.5.6 Equilibrium Risk-Free Interest Rate and Its Interpretation . . . 603
13.6 Summary . . . 605
13.7 Suggestions for Further Readings . . . 608
13.7.1 Source material for the lecture notes . . . 608
13.7.2 Potential applications and extensions of lecture notes . . . 609
13.7.3 Ph.D. elective courses offered on related topics . . . 612

14 Course Summary: Key Ideas Covered in Part 2 **613**
14.8 Valuation of Fixed Income Securities . . . 617
14.9 Models of the Term Structure of Interest Rates . . . 621
14.10 Portfolio Choice: Mean-Variance and Two-Date Models . . . 631
14.11 Portfolio Choice Using the Martingale Approach . . . 635
14.12 Portfolio Choice Using Dynamic Programming . . . 638
14.13 Partial- and General-Equilibrium Asset Pricing . . . 641

A Practice Questions for Part 2 **645**

B Homework Questions for Part 2 . . . To be handed in **661**

C Information about the Final Exam **677**
C.1 Information about the exam . . . 679
C.2 Rules for the exam . . . 680
C.3 Suggestions on how to prepare for the exam . . . 681

D Sample Final Exam **685**

E Guide to Notation **693**
E.1 Roman letters . . . 695
E.2 Greek letters . . . 697

F List of Ph.D. Elective Courses Offered by EDHEC **701**
F.1 List of Ph.D. Courses Offered in the Past . . . 703
F.2 List of Ph.D. Courses Scheduled (subject to change) . . . 707

Bibliography **709**

Blank page.

Lecture Notes for 2025

Continuous Time Financial Economics

Chapter 0

Syllabus and How This Course Is Organized

Raman Uppal

EDHEC Business School

0.1 Instructor for the Course and Contact Information

Instructor	Raman UPPAL, Professor of Finance, EDHEC Business School
Office hours	Please send email to set up an appointment
Email	Raman.Uppal@edhec.edu
Subject line	CTFE

0.2 Course Description

- This course builds on material covered in the course on **Financial Economics**, which is set in discrete time.
- This course develops the economic and mathematical foundations of asset pricing and portfolio selection in *continuous time*.
- The course starts by reviewing basic financial economics and introducing the mathematical tools required for continuous time finance.
- The course then applies these tools to the
  - ⋆ valuing of financial derivatives (on equities, currencies, forwards, etc.)
  - ⋆ modeling term structure of interest rates and valuing interest-rate options
  - ⋆ dynamic portfolio choice, and
  - ⋆ dynamic asset pricing in partial and general equilibrium.

0.3 Course Prerequisites

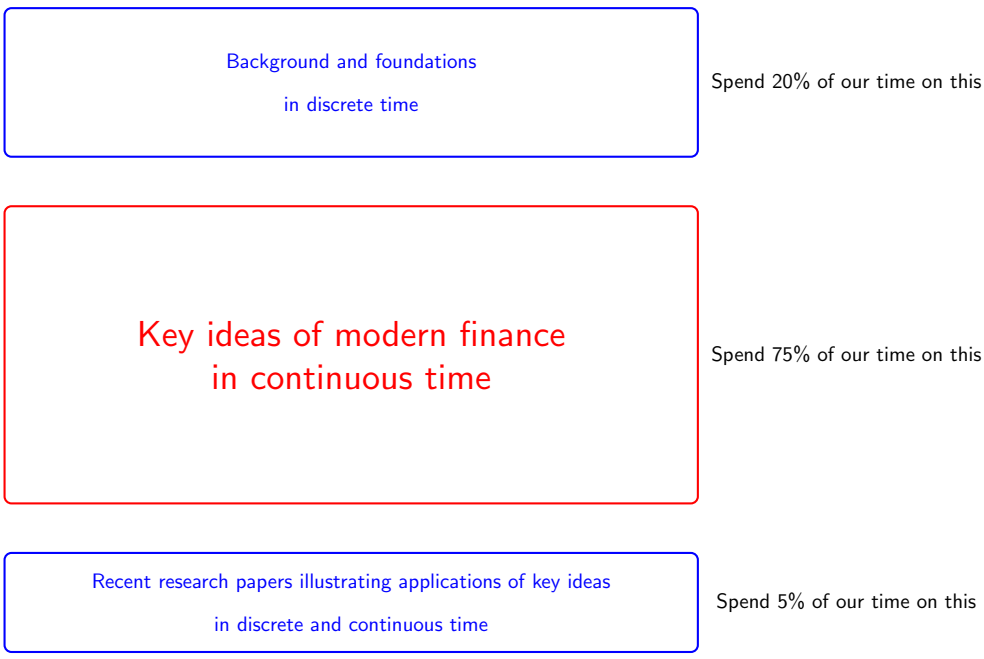
The prerequisites for the course are

- Solid understanding of material covered in course on **Financial Economics**. This means that you should understand very well the material in:
  1. Chapters 1–11 of Back (2017);
  2. Chapters 1–4 of Dumas and Luciano (2017); and
  3. Chapters 1–9 of Campbell (2018).
- Solid understanding of economics, probability, and calculus
- Participate actively and be well-prepared for class
- Do not sit in class without a **name card**
  (make one if you forget to bring your name card)

0.4 Course Objective

The objective of the course is to:

- Give you a **deep** understanding of the **key ideas** of modern finance;
- Provide you with the mathematical tools and theoretical foundations of financial economics that will help you
  - ⋆ understand the academic literature in finance
  - ⋆ solve problems in financial economics, and
  - ⋆ write your dissertation.
- Give you a good sense of the existing literature
  - ⋆ foundational research, which develops the key ideas;
  - ⋆ more recent research, which applies the key ideas to interesting problems.



0.5 Topics Covered in the Course

- **First half of the course**

1. Overview of Asset Pricing & Asset Pricing in Discrete Time
2. Mathematics for Continuous Time Finance
3. Stochastic Discount Factor and Returns
4. Change of Measure/Numeraire and Change of Drift
5. The Black-Scholes Model of Option Pricing
6. Extensions of Black-Scholes Model to
   - Dividends, Non-constant Interest Rates, Non-constant Volatility, and
   - Exchange Options, Options on Forwards & Foreign Currency
7. Summary of Part 1 of the course

- **Second half of the course**

8. Valuation of Fixed-Income Securities
9. Terms Structure of Interest Rates and Interest Rate Derivatives
10. Portfolio Selection: Mean-Variance and Two-Date Models
11. Portfolio Selection Using Martingale Methods
12. Portfolio Selection Using Dynamic Programming
13. Asset Pricing in Representative-Agent General-Equilibrium Economies
14. Summary of Part 2 of the course.

Papers that we study in Part 1 of the course

Paper	Google cite count (as of 2025/02/17)
Chapter 1	
• Cox and Ross (1976) The valuation of options for alternative stochastic processes	4,927
• Cox, Ross, and Rubinstein (1979) Option pricing: A simplified approach	10,942
Chapter 2	
• Merton (1990, Chapter 3) *Continuous time finance*	4,204
Chapter 3	
• Harrison and Kreps (1979) Martingales and arbitrage in multiperiod securities markets	5,779
Chapter 4	
• Geman, El Karoui, and Rochet (1995) Changes of numeraire, changes of probability measure and option pricing	1,027
Chapter 5	
• Black and Scholes (1973) The pricing of options and corporate liabilities	48,776
Chapter 6	
• Merton (1973b) The theory of rational option pricing	17,338
• Black (1976) The pricing of commodity contracts	4,660
• Margrabe (1978) The value of an option to exchange one asset for another	2,636
• Garman and Kohlhagen (1983) Foreign currency option values	1,497

- Note that a Google Scholar cite count of 250 is good and a count of 500 is very good.

Papers that we will study in Part 2 of the course

Paper	Google cite count (as of 2025/04/01)
Chapter 8	
• Brace, Gątarek, and Musiela (1997) The market model of interest rate dynamics	1,632
Chapter 9	
• Vasicek (1977) An equilibrium characterization of the term structure	9,828
Chapter 10	
• Markowitz (1952) Portfolio selection	63,980
• Sharpe (1964) Capital asset prices: A theory of market equilibrium under conditions of risk	35,732
Chapter 11	
• Karatzas, Lehoczky, and Shreve (1987) Optimal portfolio and consumption decisions for a small investor on a finite horizon	1,555
• Cox and Huang (1989) Optimal consumption and portfolio policies when asset prices follow a diffusion process	2,200
Chapter 12	
• Merton (1969) Lifetime portfolio selection under uncertainty: The continuous-time case	7,217
• Merton (1971) Optimum consumption and portfolio rules in a continuous time model	8,527
Chapter 13	
• Merton (1973a) An intertemporal capital asset pricing model	9,994
• Lucas (1978) Asset prices in an exchange economy	7,734
• Cox, Ingersoll, and Ross (1985b) An intertemporal general equilibrium model of asset prices	3,007

- Note that a Google Scholar cite count of 250 is good and a count of 500 is very good.

0.6 Course Method

- Class lectures.
- Practice problems to test your understanding of the material covered in class.
- Homework exercises to extend the material covered in class.
  - ⋆ It is **very important** that you do the homework exercises;
  - ⋆ If you do not do the homeworks, you will learn nothing from this course.
- In addition to doing the assigned homework problems, you should read the two books for the course, Back (2017) and Dumas and Luciano (2017), and do the exercises in these books.
- Readings from the academic literature.

0.7 Course Assessment

- First assignment to be returned on 7 April 2025.
  Second assignment to be returned on 2 June 2025.
- Please email homeworks to `brigitte.bogaerts-chevillotte@edhec.edu`.
- Homeworks should be typed (preferably in LaTeX).
- The assignments will make up 100% (50% each) of the course mark.
- Penalties for late submission of assignments
  - ⋆ Assignments submitted after the deadline will not be accepted, and a grade of 0 will be assigned.
  - ⋆ If neither assignment is submitted, the student will be deemed to have failed the course and will not be allowed to take the comprehensive exam.
- *The final exam for this course is online on 28 June 2025.*

0.8 Textbooks for the Course

The material that we will cover in the course can be found in the two books listed below.

1. Back, K. 2017. *Asset pricing and portfolio choice theory.* Second edition. NY: Oxford University Press.
   - We will cover the material in Chapters 12–18 of this book.
2. Dumas, B., and E. Luciano. 2017. *The economics of continuous-time finance.* Cambridge, MA: MIT Press.
   - We will cover the material in Chapters 5–12 and 16 of this book.

0.9 Other Textbooks (listed alphabetically)

1. Back, K. 2005. *A course in derivative securities: Introduction to theory and computation.* NY: Springer.
2. Campbell, J. Y. 2018. *Finance decisions and markets: A course in asset pricing.* Princeton University Press.
3. Campbell, J. Y., A. W. Lo, and A. C. MacKinlay. 1997. *The econometrics of financial markets.* Princeton, New Jersey: Princeton University Press.
4. Cochrane, J. H. 2005. *Asset pricing.* Revised. Princeton, NJ: Princeton University Press.
5. Cvitanic, J., and F. Zapatero. 2004. *Introduction to the economics and mathematics of financial markets.* First. MIT Press.
6. Duffie, D. 2001. *Dynamic asset pricing theory.* Princeton, NJ: Princeton University Press.
7. Gollier, C. 2001. *The economics of risk and time.* Cambridge, MA: MIT Press.
8. Hull, J. C. 2017. *Options, futures, and other derivatives.* Upper Saddle River, NJ: Prentice Hall.
9. Ingersoll, J. E., Jr. 1987. *Theory of financial decision making.* Maryland: Rowman & Littlefield.
10. LeRoy, S. F., and J. Werner. 2001. *Principles of financial economics.* Cambridge, UK: Cambridge University Press.
11. Merton, R. C. 1990. *Continuous time finance.* Cambridge MA: Basil Blackwell.
12. Munk, C. 2013. *Financial asset pricing theory.* Oxford.
13. Pennacchi, G. 2008. *Theory of asset pricing.* MA: Addison-Wesley.

0.10 Review Articles

1. Sundaresan, S. M. 2000. Continuous-time methods in finance: A review and an assessment. *Journal of Finance* 55 (4): 1569–1622.
2. Campbell, J. Y. 2000. Asset pricing at the millennium. *Journal of Finance* 55 (4): 1515–67.
3. Brunnermeier, M. K., and Y. Sannikov. 2016. Macro, money, and finance: A continuous-time approach. In *Handbook of Macroeconomics,* vol. 2B. 1497-1545: Elsevier.

- There are several books and journals that specialize in publishing review articles. So, when starting your research, you may wish to first look for review papers in your area of interest.
  - ⋆ The books are typically called "*Handbook of . . .* ".
  - ⋆ The journals have names such as:
    - "*Annual Review of Economics*" and "*Annual Review of Financial Economics*"
    - "*Foundations and Trends in Finance*"
- There is also a journal that specializes in publishing critiques of research articles, called the *Critical Finance Review*.

Sample articles from "Foundations and Trends in Finance" (sorted by year)

- Aragon, G. O., and W. E. Ferson. 2007. Portfolio performance evaluation. *Foundations and Trends in Finance* 2 (2): 83–190
- Mehra, R. 2007. The equity premium puzzle: A review. *Foundations and Trends in Finance* 2 (1): 1–81
- Brav, A., W. Jiang, and H. Kim. 2010. Hedge fund activism: A review. *Foundations and Trends in Finance* 4 (3): 185–246
- Gibson, R., F.-S. Lhabitant, and D. Talay. 2010. Modeling the term structure of interest rates: A review of the literature. *Foundations and Trends in Finance* 5 (1–2): 1–156
- Shefrin, H. 2010. Behavioralizing finance. *Foundations and Trends in Finance* 4 (1-2): 1–184
- Ang, A., W. N. Goetzmann, and S. M. Schaefer. 2011. The efficient market theory and evidence: Implications for active investment management. *Foundations and Trends in Finance* 5 (3): 157–242
- Cochrane, J. H. 2012. Continuous-time linear models. *Foundations and Trends in Finance* 6 (3): 165–219
- Agarwal, V., K. A. Mullally, and N. Y. Naik. 2015. The economics and finance of hedge funds: A review of the academic literature. *Foundations and Trends in Finance* 10 (1): 1–111

Sample articles from "Annual Review of Financial Economics" (by year)

- Aït-Sahalia, Y. 2009. Estimating and testing continuous-time models in finance: The role of transition densities. *Annual Review of Financial Economics* 1 (1): 341–359
- Bodie, Z., J. Detemple, and M. Rindisbacher. 2009. Life-cycle finance and the design of pension plans. *Annual Review of Financial Economics* 1 (1): 249–286
- Demirgüç-Kunt, A., and R. Levine. 2009. Finance and inequality: Theory and evidence. *Annual Review of Financial Economics* 1 (1): 287–318
- Jarrow, R. A. 2009. Credit risk models. *Annual Review of Financial Economics* 1 (1): 37–68
- Tufano, P. 2009. Consumer finance. *Annual Review of Financial Economics* 1 (1): 227–247
- Ferson, W. E. 2010. Investment performance evaluation. *Annual Review of Financial Economics* 2 (1): 207–234
- Frydman, C., and D. Jenter. 2010. CEO compensation. *Annual Review of Financial Economics* 2 (1): 75–102
- Jagannathan, R., E. Schaumburg, and G. Zhou. 2010. Cross-sectional asset pricing tests. *Annual Review of Financial Economics* 2 (1): 49–74
- Spatt, C. S. 2010. An informal perspective on the economics and regulation of securities markets. *Annual Review of Financial Economics* 2 (1): 127–143

- Wachter, J. A. 2010. Asset allocation. *Annual Review of Financial Economics* 2 (1): 175–206
- Burnside, C., M. Eichenbaum, and S. Rebelo. 2011. Carry trade and momentum in currency markets. *Annual Review of Financial Economics* 3 (1): 511–535
- Jegadeesh, N., and S. Titman. 2011. Momentum. *Annual Review of Financial Economics* 3 (1): 493–509
- Laeven, L. 2011. Banking crises: A review. *Annual Review of Financial Economics* 3 (1): 17–40
- Musto, D. K. 2011. The economics of mutual funds. *Annual Review of Financial Economics* 3 (1): 159–172
- Wermers, R. 2011. Performance measurement of mutual funds, hedge funds, and institutional accounts. *Annual Review of Financial Economics* 3 (1): 537–574
- Litzenberger, R., J. Castura, and R. Gorelick. 2012. The impacts of automation and high frequency trading on market quality. *Annual Review of Financial Economics* 4 (1): 59–98
- Strahan, P. E. 2013. Too big to fail: causes, consequences, and policy responses. *Annual Review of Financial Economics* 5 (1): 43–61
- Madhavan, A. 2014. Exchange-traded funds: An overview of institutions, trading, and impacts. *Annual Review of Financial Economics* 6 (1): 311–341

- Brav, A., W. Jiang, and H. Kim. 2015. Recent advances in research on hedge fund activism: Value creation and identification. *Annual Review of Financial Economics* 7 (1): 579–595
- Choi, J. J. 2015. Contributions to defined contribution pension plans. *Annual Review of Financial Economics* 7 (1): 161–178
- Hirshleifer, D. 2015. Behavioral finance. *Annual Review of Financial Economics* 7 (1): 133–159
- Bekaert, G., C. R. Harvey, A. Kiguel, and X. Wang. 2016. Globalization and asset returns. *Annual Review of Financial Economics* 8 (1): 221–288
- Ben-David, I., F. Franzoni, and R. Moussawi. 2017. Exchange-traded funds. *Annual Review of Financial Economics* 9 (1): 169–189
- Berk, J. B., and J. H. van Binsbergen. 2017. Mutual funds in equilibrium. *Annual Review of Financial Economics* 9 (1): 147–167
- Zhou, G. 2018. Measuring investor sentiment. *Annual Review of Financial Economics* 10 (1): 239–259
- Allen, F., J. Qian, and M. Qian. 2019. A review of China's institutions. *Annual Review of Financial Economics* 11 (1): 39–64
- Korteweg, A. 2019. Risk adjustment in private equity returns. *Annual Review of Financial Economics* 11 (1): 131–152

0.11 My Teaching Style

- Focus on *applying theory* to problems in financial economics
- Rigorous and technical
- Interactive and informal
  - ⋆ Do not be nervous when I call on you
  - ⋆ Feel free to ask for help from people sitting around you
  - ⋆ Objective of this is **not** to make you uncomfortable

▶ **My job**

- Provide *structure* for thinking about finance theory in a *systematic* manner.
- Provide you with the opportunity to learn this material.
- Office hours: By appointment and by email Raman.Uppal@edhec.edu

0.12 Your "Style" in Class . . .

- **Do not be shy** about asking questions
  - ⋆ if you knew everything you would not be here
  - ⋆ be honest with yourself about what you do/don't know
- You are **strongly encouraged to make mistakes** in responding to questions
- Feel free to interrupt and share your insights/experiences/funny thoughts
  It is important to have a good time in class.
- Be nice to other students—be patient when they ask questions
- Complain—when you disagree with something I say/do;
  Use the *Feedback Form*
- Please do not use electronic devices (phones, laptops, etc.) in class.

0.13 The Good News . . .

- We have two great books:
  - ⋆ Back (2017)
  - ⋆ Dumas and Luciano (2017)

0.14 The Bad News . . .

- YOU have to work through the book.
- Each chapter of Kerry Back's book has a discussion of the existing literature for that topic. You are strongly encouraged to read these papers:
  - ⋆ I will be covering material from many of these papers in class, but will not mention each paper explicitly in the lectures.
  - ⋆ I also do not intend to cover each lecture slide; many of the slides are for your information, but will not be discussed during the lecture.
- What you get out of this course will depend very much on the effort you put into the course.

Lecture Notes for 2025

Continuous Time Financial Economics

Chapter 8

Valuation of Fixed Income Securities

Raman Uppal

EDHEC Business School

Chapter contents

8.1 Plan for the Day . . . 49
8.2 Background . . . 50
8.3 Main Points for Today . . . 51
8.4 Guide to Notation . . . 52
8.5 Overview . . . 53
8.6 Basics: The Yield Curve . . . 54
8.6.1 Yield of a Zero-Coupon Bond . . . 55
8.6.2 Definition and Shapes of the Yield Curve . . . 56
8.6.3 Estimating the Yield Curve . . . 57
8.7 Basics: Yield to Maturity, Duration, Convexity, & Hedging . . . 58
8.7.1 Duration Hedging . . . 61
8.7.2 Continuous Hedging in a Single Factor Model . . . 64
8.8 Basics: LIBOR Interest Rates . . . 70
8.9 Basics: Forward Interest Rates . . . 73
8.10 Swaps . . . 76
8.10.1 Valuation of swaps . . . 79
8.10.2 Setting the Swap Fixed Interest Rate . . . 86
8.11 Fixed-Income Derivatives: Caps and Floors . . . 87
8.11.1 Definition and Interpretation of Caps and Floors . . . 87
8.11.2 Portfolios of Bonds that Pay Spot Rates . . . 91
8.11.3 Forward Price of a Non-Dividend Paying Asset . . . 95
8.11.4 The Market Model for Caplets and Floorlets . . . 96
8.11.5 Put-Call Parity Relation for Caplets and Floorlets . . . 101
8.11.6 Valuing Caplets as Puts on Discount Bonds . . . 103
8.12 Fixed-Income Derivatives: European Swaptions . . . 105
8.13 Inconsistency in Assuming Constant Volatilities for Forward Interest Rates and Forward Swap Rate 111
8.14 Summary . . . 115
8.15 Appendix: Estimating the Yield Curve . . . 123
8.15.1 Bootstrapping Approach . . . 125
8.15.2 Limitations of Bootstrapping . . . 128
8.15.3 Cubic Splines . . . 129
8.15.4 Exponential Splines . . . 136
8.15.5 Implementation Issues with Splines . . . 137
8.15.6 Nelson-Siegel-Svensson Method for Estimating Term Structure . . . 139
8.16 Suggestions for Further Readings . . . 143
8.16.1 Source material for the lecture notes . . . 143
8.16.2 Potential applications and extensions of lecture notes . . . 144
8.16.3 Ph.D. elective courses offered on related topics . . . 148

8.1 Plan for the Day

Last Time

- Extensions of Black and Scholes (1973) to various options on equity and FX.

Today

- **How to value fixed-income securities?**
  - ⋆ Basics: The yield curve, LIBOR, yield-to-maturity, duration, convexity
  - ⋆ Basics: Swaps and forward interest rates
  - ⋆ Market model for caps, floors, and swaptions

For Next Time

- Review the material covered in class.
- Read Back, Chapter 18 and Dumas & Luciano, Chapter 16.
- Do the practice problems.

8.2 Background

Recall, that in this course we wish to answer three questions, which are nested:

1. How to value redundant securities?

   Least general question: we are given prices of fundamental assets and the desired cashflows; need to determine only the replicating portfolio strategy.

2. How to determine optimal consumption and portfolio policies of investors?

   More general than the previous question: we are given prices of the fundamental assets; need to determine both the desired cashflows and the portfolio strategy generating these cashflows.

3. How to determine the equilibrium values in a market with investors who are optimizing their consumption and portfolio policies?

   Most general question: need to determine prices of fundamental assets, desired cashflows, and the portfolio strategy that replicates these cashflows.

8.3 Main Points for Today

1. The yield curve
2. Yield-to-maturity, duration, convexity
3. LIBOR (floating interest rate)
4. Forward interest rates
5. Swaps
6. Caps and floors (which are options on interest rates)
7. Market model for caps, floors, and swaptions

Bottom line: The methods we have developed for equity options apply in a straightforward way to options on claims on interest rates.

8.4 Guide to Notation

What it denotes	Notation
Price of a zero-coupon bond at t maturing at T	$\mathcal{P}^Z(t,T)$
Price of a coupon bond at t maturing at T	$\mathcal{P}^B(t,T)$
Yield on a bond for maturity $\tau = T - t$	$y(\tau)$
Yield-to-maturity on a bond	$\mathbf{y}$
Spot floating LIBOR (simple) interest rate per annum at date i	$\mathcal{R}_i$
Spot fixed (simple) interest rate per annum	$\bar{\mathcal{R}}$
Forward interest rate at t for a loan between t_i and t_{i+1}	$\mathcal{R}_i^{fwd}(t)$
Swap rate at t (spot rate if $t = t_0$; forward rate if $t < t_0$)	$\mathcal{R}^{swap}(t)$

8.5 Overview

- In this section of the course, we will study the pricing and hedging of default-free fixed-income securities and derivatives on fixed-income securities.
- The main challenge is to recognize that:
  - ⋆ Bond prices or interest rates have stochastic volatilities.
  - ⋆ And, volatilities of bonds of different maturities are linked to one other.
- We will consider only securities that have little likelihood of default, and so we will ignore the valuation of default risk.
- We will focus on discount bonds of all maturities, because coupon-paying bonds can be priced simply as a portfolio of discount bonds.
- We start by studying a fundamental construct: the yield curve, which is the yield of **zero-coupon** (discount) bonds of various maturities.

8.6 Basics: The Yield Curve

- The relation between time to maturity and the price of the corresponding discount bond with $\$1$ face value is called the discount function.
- An equivalent concept is the yield curve, which is the relation between time to maturity and the yield of the corresponding discount bond.
- Yields can be quoted according to different compounding conventions (annual; semi-annual, which is the typical case; etc.) but we will often use the convention of continuous compounding.

8.6.1 Yield of a Zero-Coupon Bond

- With the convention of continuous compounding, the yield of a zero-coupon bond with \$1 face value having $\tau = T - t$ years to maturity and price $\mathcal{P}^Z(t,T)$ is defined to be the rate $y(\tau)$ such at:

$$\mathcal{P}^Z(t,T) = \frac{\text{amount received}}{\text{discount factor}} = \frac{1_\tau}{e^{y\,\tau}} = e^{-y\,\tau} \quad \Leftrightarrow \quad y = -\frac{1}{\tau}\log \mathcal{P}^Z(t,T).$$

8.6.2 Definition and Shapes of the Yield Curve

- Denoting the yield y for a bond maturing at date τ by $y(\tau)$, the yield curve is the collection of yields for different maturities.

- The yield curve is usually plotted with maturity τ on the horizontal axis and the yields $y(\tau)$ on the vertical axis.

- Usually, the yield curve is upward sloping ("normal"), implying that rates for longer maturities are higher than rates for shorter maturities.
  - ⋆ But, the yield curve can also be downward sloping ("inverted").
  - ⋆ And, it is possible for the yield curve to slope up for some maturities and slope down for others ("hump shaped").

8.6.3 Estimating the Yield Curve

- In the real world, zero-coupon bonds for all maturities are not traded.
- Thus, the prices of zero-coupon bonds being traded can be used to infer the yield curve or the prices of bonds that are not being currently traded.
- The prices of coupon-paying bonds (or swap rates, to be covered shortly) can also be used to estimate the yield curve.
- The most common way of estimating the yield curve from bond prices is to fit a cubic spline, using the prices of the bonds being traded.
- Details of how to do this are provided in the appendix.

8.7 Basics: Yield to Maturity, Duration, Convexity, & Hedging

- Consider a bond with cashflows $C_1, C_2, \ldots, C_N$ at dates $u_1 < u_2 < \ldots < u_N$ and price P at date $t < u_1$.
- Let $\tau_j = u_j - t$ denote the time remaining until the j-th cash flow is paid.
- Then, the continuously compounded yield to maturity of the bond at date t is defined to be the rate $\mathbf{y}$ such that

$$\mathcal{P}^B = \sum_{j=1}^{N} e^{-\mathbf{y}\tau_j} C_j.$$

Note that $\mathbf{y}$ is different from the yield of a bond, y.

- Viewing the right-hand-side of the above equation as a function of $\mathbf{y}$, we can express the first derivative in differential form as:

$$d\mathcal{P}^B = -\sum_{j=1}^{N} \tau_j \times e^{-\mathbf{y}\tau_j} C_j \, d\mathbf{y}.$$

- Dividing both sides by $\mathcal{P}^B$ gives:

$$\frac{d\mathcal{P}^B}{\mathcal{P}^B} = -\sum_{j=1}^{N} \tau_j \times \frac{e^{-\mathbf{y}\tau_j} C_j}{\mathcal{P}^B} \, d\mathbf{y}.$$

- "Macaulay duration" of the bond, or simply duration is:

$$\sum_{j=1}^{N} \tau_j \times \frac{e^{-\mathbf{y}\tau_j} C_j}{\mathcal{P}^B} \tag{8.1}$$

This is a weighted average of the times-to-maturity τ_j of the cashflows, with the weights being the fraction of bond value that each cashflow constitutes.

- Thus, we have

$$\frac{d\mathcal{P}^B}{\mathcal{P}^B} = -\text{Duration} \times d\mathbf{y}. \tag{8.2}$$

- If the initial yield to maturity is $\mathbf{y}$ and it changes to $\mathbf{y}'$, then the above equation suggests the following approximation for the return on the bond:

$$\frac{\Delta\mathcal{P}^B}{\mathcal{P}^B} \approx -\text{Duration} \times \Delta\mathbf{y},$$

where $\Delta\mathbf{y} = \mathbf{y}' - \mathbf{y}$.

- This is the foundation for "duration hedging."

8.7.1 Duration Hedging

- To duration hedge a liability with a present value of x_L dollars and duration of D_L years, one needs an asset with value x_A and duration D_A such that

$$x_A D_A = x_L D_L.$$

- Note that duration hedging is only approximate; the duration hedge will not be perfect because of "convexity" of the price function.

- In general, changes in the yields on an asset and a liability will not be equal.

- To see this, note that we can write down the price of the bond in two ways:

$$\sum_{j=1}^{N} e^{-\mathbf{y}\tau_j} C_j = \sum_{j=1}^{N} e^{-y(\tau_j)\tau_j} C_j.$$

- Taking differentials of both sides, we get

$$d\mathcal{P}^B = -\mathcal{P}^B \times \text{Duration} \times d\mathbf{y} = -\sum_{j=1}^{N} \tau_j \, e^{-y(\tau_j)\tau_j} C_j \, dy(\tau_j).$$

- If we assume that the changes $dy(\tau_j)$ in the yields of discount bonds are all equal, say to dy, then we have:

$$d\mathcal{P}^B = -\mathcal{P}^B \times \text{Duration} \times d\mathbf{y} = -\mathcal{P}^B \times \text{DurationYields} \times dy(\tau_j),$$

where we define

$$\text{DurationYields} = \sum_{j=1}^{N} \frac{e^{-y(\tau_j)\tau_j} C_j}{\mathcal{P}^B} \tau_j, \tag{8.3}$$

which is the definition of duration in terms of yields, $y(\tau_j)$ as opposed to yield-to-maturity, $\mathbf{y}$.

- Observe that this definition of duration, which is in terms of yields, $y(\tau_j)$ is different from the earlier one, which was in terms of the yield-to-maturity, $\mathbf{y}$.

$$\text{Maculay Duration} = \sum_{j=1}^{N} \tau_j \times \frac{e^{-\mathbf{y}\tau_j} C_j}{\mathcal{P}^B}$$

$$\text{DurationYields} = \sum_{j=1}^{N} \tau_j \times \frac{e^{-y(\tau_j)\tau_j} C_j}{\mathcal{P}^B}.$$

- The two definitions of duration will coincide, and duration hedging will work, only if the changes in all the yields, $dy(\tau_j)$, are equal, which is called a parallel shift in the yield curve.

- But, parallel shifts are not the norm in the real world.

8.7.2 Continuous Hedging in a Single Factor Model

- Another set of circumstances under which one gets perfect hedging is if
  - ⋆ One can rebalance the hedge continuously, and
  - ⋆ There is a single factor (Brownian motion) driving all yields.

- To show this we need to compute the Itô differential of the bond price.

- Denote the bond price at t as a function of the yield-to-maturity at t, $\mathbf{y}(t)$:

$$\mathcal{P}^B(t) = f(t, \mathbf{y}(t)) = \sum_{j=1}^{N} e^{-\mathbf{y}(t)(u_j - t)} C_j.$$

- From Itô's formula applied to $f(t, \mathbf{y}(t))$, we have:

$$d\mathcal{P}^B = \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial \mathbf{y}} d\mathbf{y} + \frac{1}{2} \frac{\partial^2 f}{\partial \mathbf{y}^2} (d\mathbf{y})^2 . \tag{8.4}$$

- We can now see that the convexity of the bond appears in the last term: $\frac{\partial^2 f}{\partial \mathbf{y}^2}$.

- We start by re-writing the last equation on the previous page:

$$d\mathcal{P}^B = \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial \mathbf{y}} d\mathbf{y} + \frac{1}{2} \frac{\partial^2 f}{\partial \mathbf{y}^2} (d\mathbf{y})^2 . \tag{8.4}$$

- Assuming that the process for the yield-to-maturity is given by

$$d\mathbf{y}(t) = \alpha(t) dt + \sigma(t) dB^{\mathbb{P}}(t),$$

where $B^{\mathbb{P}}$ is a Brownian motion under the physical probability measure and α, σ are some functions of time, we have:

$$\frac{d\mathcal{P}^B}{\mathcal{P}^B} = \frac{1}{\mathcal{P}^B} \left(\frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial \mathbf{y}} \alpha(t) + \frac{1}{2} \frac{\partial^2 f}{\partial \mathbf{y}^2} \sigma^2(t) \right) dt + \frac{1}{\mathcal{P}^B} \left(\frac{\partial f}{\partial \mathbf{y}} \sigma(t) \right) dB^{\mathbb{P}}(t). \tag{8.5}$$

- We start by re-writing the last equation on the previous page:

$$\frac{d\mathcal{P}^B}{\mathcal{P}^B} = \frac{1}{\mathcal{P}^B}\left(\frac{\partial f}{\partial t}dt + \frac{\partial f}{\partial \mathbf{y}}\alpha(t) + \frac{1}{2}\frac{\partial^2 f}{\partial \mathbf{y}^2}\sigma^2(t)\right)dt + \frac{1}{\mathcal{P}^B}\left(\frac{\partial f}{\partial \mathbf{y}}\sigma(t)\right)dB^{\mathbb{P}}(t). \tag{8.5}$$

- Re-writing Equation (8.2), which is given on Page 60:

$$\frac{d\mathcal{P}^B}{\mathcal{P}^B} = -\text{Duration} \times d\mathbf{y} \tag{8.2}$$

$$\frac{1}{d\mathbf{y}}\frac{d\mathcal{P}^B}{\mathcal{P}^B} = -\text{Duration} \qquad \ldots \text{multiplying both sides by } \frac{1}{d\mathbf{y}}$$

$$\frac{1}{\mathcal{P}^B}\frac{\partial f}{\partial \mathbf{y}} = -\text{Duration} \qquad \ldots \text{using the definition } \mathcal{P}^B(t) = f(t, \mathbf{y}(t)). \tag{8.6}$$

- Substituting (8.6) into the last term of (8.5) gives:

$$\frac{d\mathcal{P}^B}{\mathcal{P}^B} = \frac{1}{\mathcal{P}^B}\left(\frac{\partial f}{\partial t}dt + \frac{\partial f}{\partial \mathbf{y}}\alpha(t) + \frac{1}{2}\frac{\partial^2 f}{\partial \mathbf{y}^2}\sigma^2(t)\right)dt - \text{Duration}\,\sigma(t)\,dB^{\mathbb{P}}(t). \tag{8.7}$$

- Re-writing the last equation from the previous slide:

$$\frac{d\mathcal{P}^B}{\mathcal{P}^B} = \frac{1}{\mathcal{P}^B}\left(\frac{\partial f}{\partial t}dt + \frac{\partial f}{\partial \mathbf{y}}\alpha(t) + \frac{1}{2}\frac{\partial^2 f}{\partial \mathbf{y}^2}\sigma^2(t)\right)dt - \text{Duration}\,\sigma(t)\,dB^{\mathbb{P}}(t). \tag{8.7}$$

- From the above expression, we can see that:
  - ⋆ if the yields of an asset and liability are driven by the same Brownian motion
  - ⋆ and the duration hedge is adjusted for the relative volatilities of the asset and liability (holding more of the asset if its volatility is lower),
  - ⋆ then a duration hedge will hedge the risky part of the change in the liability value.

- If the hedge is adjusted continuously, then just as in the case of options, it will provide a perfect hedge.

- Two continuous time models that assume a single Brownian motion drives all yields are:
  - ⋆ Vasicek (1977) (discussed in detail in the next chapter), and
  - ⋆ Cox, Ingersoll, and Ross (1985a) (discussed briefly in next chapter).
- Three discrete time models that are single factor models are:
  - ⋆ Ho and Lee (1986) model (discussed briefly in the next chapter),
  - ⋆ Black, Derman, and Toy (1990) (discussed in appendix of next chapter),
  - ⋆ Black and Karasinski (1991) model.
- Empirically, however, the evidence suggests that the yield curve is affected by more than just a single factor. Usually three factors influence the level, slope, and curvature of the yield curve.
  - ⋆ In the next chapter, we will describe briefly some multifactor models of the term structure.

8.8 Basics: LIBOR Interest Rates

- Many fixed-income instruments have cash flows tied to interest rate indices that are quoted as simple (that is, not compounded) interest rates.
- If you
  - ⋆ deposit $\$1$ at date i at an annualized simple interest rate of $\mathcal{R}_i$ for a period of time Δt,
  - ⋆ then at the end of the period you will have: $\$1 + \mathcal{R}_i \Delta t$.

- The most common interest rate index is the London Interbank Offer Rate (LIBOR), which is an average rate in the London interbank market for loans for a specific term.

Example 8.1 (LIBOR)
If $\mathcal{R}_i$ denotes the six-month LIBOR at date i, a $\$1$ million deposit at this rate will grow in six months to

$$\$1\text{million} \times \left[1 + \mathcal{R}_i \, \Delta t\right] = \$1\text{million} \left[1 + \mathcal{R}_i \left(\frac{1}{2}\right)\right].$$

LIBOR Rates Expressed in Terms of Bond Prices

- We will frequently find it convenient to express LIBOR rates in terms of equivalent bond prices.

- As before, let $\mathcal{P}^Z(t,u)$ denote the price at t of a zero-coupon bond with a face value of $\$1$ maturing at date u, having $\Delta t = (u-t)$ years to maturity.

- Then, the relation between the price of this bond and the LIBOR rate (ignoring the risk of default) is:

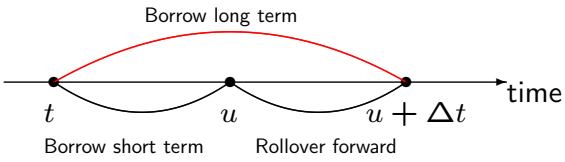
$$\mathcal{P}^Z(t,u) = \frac{1}{1 + \mathcal{R}_i \, \Delta t}, \quad \text{or equivalently:} \quad 1 + \mathcal{R}_i \, \Delta t = \frac{1}{\mathcal{P}^Z(t,u)}.$$

- $\mathcal{R}_i$ is often referred to as the "spot rate" at date t for maturity u.
  This is a "floating rate" because it changes with market conditions.

- Next, we will study "forward" interest rates.

8.9 Basics: Forward Interest Rates

- The forward interest rate allows you to lock in at date t a rate for a loan or deposit that starts at $u > t$ for a period Δt.



- Consider two ways of borrowing $\$1$ from date t to date $u + \Delta t$.
  1. Borrow long term from t to date $u + \Delta t$ at a cost of $1 + \mathcal{R}(t, u + \Delta t)$.
  2. Alternatively, first borrow short term from date t to date u, and then at the forward interest rate from date u to date $u + \Delta t$.
     - ⋆ The cost of the first loan is $[1 + \mathcal{R}(t, u)]$ and
     - ⋆ the cost of the second loan is $[1 + \mathcal{R}^{fwd}(u, u + \Delta t)]$.

- Because the forward rate is set at date t, no-arbitrage requires that the cost of the two ways of doing the transaction be the same:

$$1 + \mathcal{R}(t, u + \Delta t)(u + \Delta t - t) = [1 + \mathcal{R}(t, u)(u - t)] \times [1 + \mathcal{R}^{fwd}(u, u + \Delta t)\Delta t].$$

- Thus, rearranging the terms in the equation above, we see that the forward rate $\mathcal{R}^{fwd}$ is given by:

$$1 + \mathcal{R}^{fwd}(u, u + \Delta t)\Delta t = \frac{1 + \mathcal{R}(t, u + \Delta t)(u + \Delta t - t)}{1 + \mathcal{R}(t, u)(u - t)}$$

$$= \frac{\mathcal{P}^Z(t, u)}{\mathcal{P}^Z(t, u + \Delta t)} \quad \text{. . . expressed in terms of bond prices.}$$

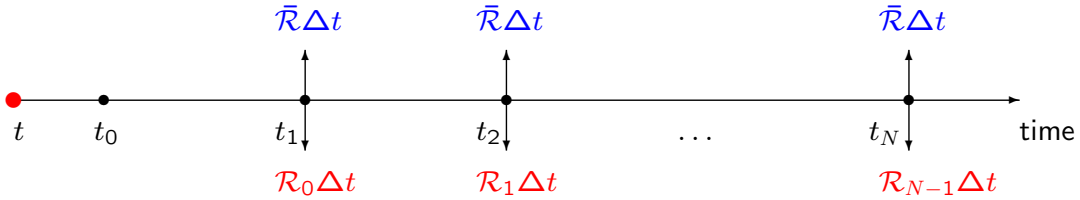
- We will denote forward rates at t for a loan between t_i and t_{i+1} as $\mathcal{R}_i^{fwd}(t)$:

$$1 + \mathcal{R}_i^{fwd}(t)\Delta t = \frac{\mathcal{P}^Z(t, t_i)}{\mathcal{P}^Z(t, t_{i+1})}. \tag{8.8}$$

- Note that for the special case
  - ⋆ when $t = t_i$, $\mathcal{P}^Z(t, t_i) = \mathcal{P}^Z(t_i, t_i) = 1$,
  - ⋆ so $1 + \mathcal{R}_i^{fwd}(t_i)\Delta t = \frac{1}{\mathcal{P}^Z(t, t_{i+1})}$,
  - ⋆ which means that $\mathcal{R}_i^{fwd}(t_i) = \mathcal{R}_i$, the spot interest rate.
- In the next class, we will study instantaneous forward rates, for the case of continuously-compounded interest rather than simple interest.

8.10 Swaps

- A standard (plain vanilla) interest rate swap is the exchange of fixed for floating interest rate payments on a given notional principal.
- We will denote
  - ⋆ the fixed interest rate by $\bar{\mathcal{R}}$; and,
  - ⋆ the floating rate is $\mathcal{R}_i$ (usually LIBOR or some other index rate).



- The swap contract also specifies the payment dates, $t_1, t_2, \ldots, t_N$, when the fixed for floating payments are exchanged, with $t_{i+1} - t_i = \Delta t$.

- These are also called the "reset dates" because the floating interest rate is reset on these dates. Note that the reset dates are $t_0, t_1, \ldots, t_{N-1}$.

  At each reset date t_i, the simple interest rate $\mathcal{R}_i$ for the next period Δt is observed, and this determines the payment at date t_{i+1}.

- One can enter a swap as

  ⋆ a "payer" of the fixed rate, or

  ⋆ a "receiver" of the fixed rate.

- Only the net amount is exchanged, so if $\bar{\mathcal{R}} > \mathcal{R}_i$, then $(\bar{\mathcal{R}} - \mathcal{R}_i)\Delta t$ times the notional principal amount is paid by the payer to the receiver.

- Using the standard convention that a negative cashflow is an outflow,

  ⋆ the cashflow to the receiver is $(\bar{\mathcal{R}} - \mathcal{R}_i)\Delta t$, and

  ⋆ the cashflow to the payer is $(\mathcal{R}_i - \bar{\mathcal{R}})\Delta t$.

- Because the principal is notional, there is no exchange of principals at the initial or terminal dates of the swap.

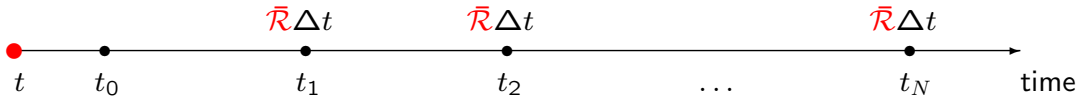
8.10.1 Valuation of swaps

- To find the value of a swap at date $t < t_0$ to a payer of the fixed rate.
  - ⋆ first find the value of the fixed-rate leg;
  - ⋆ then find the value of the floating-rate leg of the swap; and,
  - ⋆ the difference in the value of the two legs then gives the value of the swap.
- We show below that:
  - ⋆ The value of the floating-rate leg is: $\mathcal{P}^Z(t,t_0) - \mathcal{P}^Z(t,t_N)$.
  - ⋆ The value of the fixed-rate leg is: $\bar{\mathcal{R}}\Delta t \sum_{i=1}^{N} \mathcal{P}^Z(t,t_i)$.
- Thus, the value at date t of a fixed-rate payer swap is:

$$[\text{Value of payer swap}]_t = \underbrace{[\mathcal{P}^Z(t,t_0) - \mathcal{P}^Z(t,t_N)]}_{\text{value of floating leg}} - \underbrace{[\bar{\mathcal{R}}\Delta t \sum_{i=1}^{N} \mathcal{P}^Z(t,t_i)]}_{\text{value of fixed leg}} . \tag{8.9}$$

Step 1: Value of the fixed-rate leg of the swap

- The cashflows to the fixed-rate leg of the swap can be illustrated as:



- At each payment date, the fixed-rate leg has cashflows of $\bar{\mathcal{R}}\Delta t$ times the notional principal of $\$1$.
- The present value at t of $\$1$ at t_i is given by $\mathcal{P}^Z(t,t_i)$.
- Thus, the value of the fixed-rate leg of the swap is the sum of the above stream of payments:

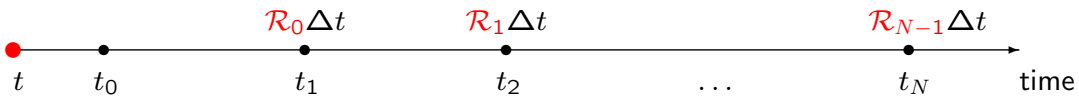
$$[\text{Value of the fixed-rate leg}] = \bar{\mathcal{R}}\Delta t \times \sum_{i=1}^{N} \mathcal{P}^Z(t,t_i). \tag{8.10}$$

Step 2: Value of the floating-rate leg of the swap

- We now explain why the value of the floating-rate leg of the swap is

$$\mathcal{P}^Z(t,t_0) - \mathcal{P}^Z(t,t_N).$$

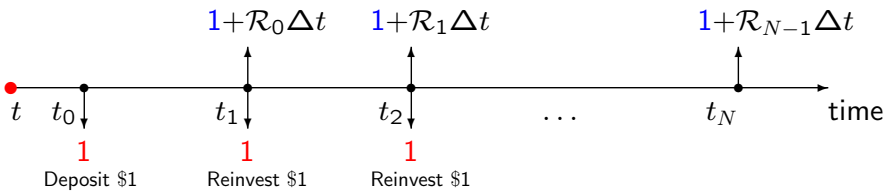
- The cashflows of the floating-rate leg of the swap can be illustrated as:



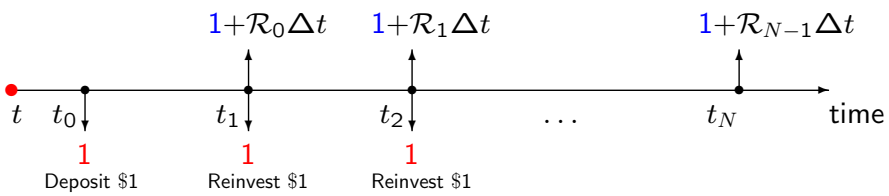
- **Question:**
  - ⋆ How do we replicate/generate this payoff, where $\mathcal{R}_i$ changes at each t_i?
  - ⋆ What is the cost of replicating this payoff?

Let us examine the payoff to the following strategy:

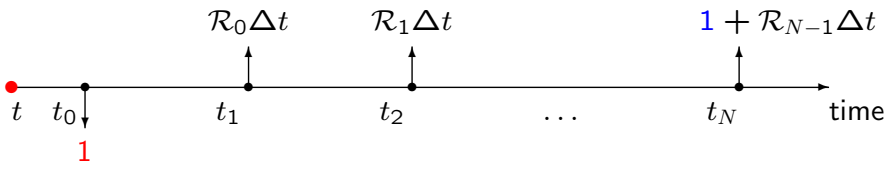
- At t_0:
  - ⋆ Invest (that is, deposit) $\$1$ at the rate $\mathcal{R}_0$
- At t_1:
  - ⋆ Receive the amount $(1 + \mathcal{R}_0\Delta t)$
  - ⋆ Set aside the amount $\mathcal{R}_0\Delta t$, and reinvest $\$1$ at the new rate $\mathcal{R}_1$
- At t_2:
  - ⋆ Receive the amount $(1 + \mathcal{R}_1\Delta t)$
  - ⋆ Set aside the amount $\mathcal{R}_1\Delta t$, and reinvest $\$1$ at the new rate $\mathcal{R}_2$
- ...
- At t_N:
  - ⋆ Receive the amount $(1 + \mathcal{R}_{N-1}\Delta t)$.



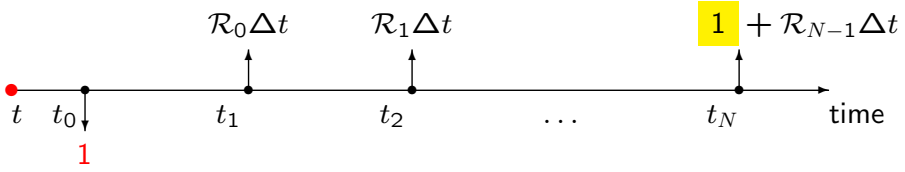
- We start by re-producing the figure from the previous page:



- The net payoff from the above strategy is
(after canceling the receipt and reinvestment of \$1 at $t_1, t_2, \ldots, t_{N-1}$)



- We start by re-producing the figure from the previous page:



- The cost of this at $t < t_0$ is the cost of depositing \$1 at t_0: $\mathcal{P}^Z(t, t_0)$.

- But, this strategy generates a surplus \$1 at t_N that we do not need for replicating the cashflows to the floating rate leg of the swap.

- So, we can sell this \$1 that we will be receiving at t_N, the present value at t of which is: $\mathcal{P}^Z(t, t_N)$.

- Thus, the net cost of replicating the payoff to the floating rate leg gives the value of the floating-rate leg of the swap:

$$\text{Value of floating-rate leg} = \mathcal{P}^Z(t, t_0) - \mathcal{P}^Z(t, t_N). \tag{8.11}$$

8.10.1.1 Summary: Valuation of Interest Rate Swap

- The value of the fixed rate leg is:

$$[\text{Value of fixed-rate leg}]_t = \bar{\mathcal{R}}\Delta t \times \sum_{i=1}^{N} \mathcal{P}^Z(t, t_i). \tag{8.10}$$

- The value of the floating-rate leg of the swap is

$$[\text{Value of floating-rate leg}]_t = \mathcal{P}^Z(t, t_0) - \mathcal{P}^Z(t, t_N). \tag{8.11}$$

- Thus, value of the swap to payer of fixed rate (receiver of floating rate) is:

$$[\text{Value of swap to fixed-rate payer}]_t = \underbrace{[\mathcal{P}^Z(t, t_0) - \mathcal{P}^Z(t, t_N)]}_{\text{value of floating leg}} - \underbrace{[\bar{\mathcal{R}}\Delta t \sum_{i=1}^{N} \mathcal{P}^Z(t, t_i)]}_{\text{value of fixed leg}}. \tag{8.9}$$

8.10.2 Setting the Swap Fixed Interest Rate

- As with the forward price, the swap fixed interest rate is set so that on the initiation date the value of the swap is zero.

- That is, the fixed rate $\bar{\mathcal{R}} = \mathcal{R}^{swap}(t)$ so that

$$\text{Value of swap}_t = 0$$

which implies that $\mathcal{R}^{swap}(t)$ is chosen in such a way that

$$\mathcal{P}^Z(t, t_0) - \mathcal{P}^Z(t, t_N) = \mathcal{R}^{swap}(t)\Delta t \sum_{i=1}^{N} \mathcal{P}^Z(t, t_i). \tag{8.12}$$

- If $t = t_0$, then $\mathcal{R}^{swap}(t)$ is called the spot swap rate.

- If $t < t_0$, then $\mathcal{R}^{swap}(t)$ is called the forward swap rate.

8.11 Fixed-Income Derivatives: Caps and Floors

8.11.1 Definition and Interpretation of Caps and Floors

- Caps and floors are similar to swaps, but
  - ⋆ while swaps are a series of forward contracts,
  - ⋆ caps and floors are like a series of options.
- At each reset date t_i, the simple interest rate $\mathcal{R}_i$ for period Δt is observed, which determines the payment at date t_{i+1}.
- Buyers of caps and floors pay a premium to the seller of the option; all future cashflows are paid by the seller to the buyer.

- For each $\$1$ of notional principal,
  - ⋆ the owner of a cap with cap rate $\bar{\mathcal{R}}$ receives at t_{i+1}

    $$\max(0, \mathcal{R}_i - \bar{\mathcal{R}})\Delta t;$$

  - ⋆ the owner of a floor with floor rate $\bar{\mathcal{R}}$ receives at t_{i+1}

    $$\max(0, \bar{\mathcal{R}} - \mathcal{R}_i)\Delta t.$$

- Caps and floors can be combined to create other positions. For example:
  - ⋆ The combination of a long cap and a short floor at the same $\bar{\mathcal{R}}$ creates the payer side of a swap, in the same way that a long call and a short put creates the receiver side of a forward.
  - ⋆ A long cap at rate $\bar{\mathcal{R}}_c$ and a short floor at rate $\bar{\mathcal{R}}_f$ creates a collar for an underlying floating rate obligation.

- The individual payments on a cap are called "caplets", and so a cap can be viewed as a portfolio of caplets.
- Similarly, individual payments on a floor are called "floorlets."
- A caplet can be viewed
  - ⋆ either as a call option on the spot rate with strike equal to the fixed rate; that is, it is a *bet on interest rates going up*;
  - ⋆ or, equivalently as a bet on *bond prices going down*, and so is a put option on a discount bond.
- Similarly, a floorlet can be viewed as
  - ⋆ either a put option on the spot interest rate,
  - ⋆ or a call option on a discount bond.
- Values of caplets and floorlets are linked by put-call parity.

8.11.2 Portfolios of Bonds that Pay Spot Rates

- One way to value caps and floors is to view them as portfolios of options on rates, which we will do shortly.

- But, to use the formulas we had developed earlier (for options on equities), we need to interpret rates in terms of prices of assets, so that each option can be viewed as an option on the price of an asset.

- From the previous slide, it should be clear how to do this using prices of discount bonds.

- We first look at an example where we wish to find the value, in terms of an asset price, of the rate at $t < t_i$; that is, where t is before the initiation date of the investment, t_i.

- For example, to find the asset price corresponding to the spot rate R_i,
  - ⋆ one needs to invest $\$1$ at date t_i, which at $t < t_i$ is equivalent to buying one unit of the bond maturing at t_i; the cost of this bond is $\mathcal{P}^Z(t, t_i)$;
  - ⋆ investing the $\$1$ at the spot rate at t_i will generate $1 + \mathcal{R}_i \Delta t$ dollars at date t_{i+1}.
  - ⋆ To get rid of the extra dollar, we short one unit of the bond maturing at date t_{i+1}, leaving us with only $\mathcal{R}_i \Delta t$ dollars; the value of shorting this bond is $\mathcal{P}^Z(t, t_{i+1})$.
  - ⋆ Thus, the value at date $t < t_i$ of receiving $\mathcal{R}_i \Delta t$ dollars at date t_{i+1} is:

$$\mathcal{P}^Z(t, t_i) - \mathcal{P}^Z(t, t_{i+1}).$$

- Next, we look at the value of the spot rate $\mathcal{R}_i$ at date t when $t > t_i$; that is, we already know $\mathcal{R}_i$, and we simply have to find the present value of $\mathcal{R}_i \Delta t$.

- So, between the date t_i and t_{i+1}, the value of receiving $\mathcal{R}_i \Delta t$ dollars at date t_{i+1} is the present value of this amount:

$$\frac{\mathcal{R}_i \, \Delta t}{1 + \mathcal{R}_t(t, t_{i+1})(t_{i+1} - t)} = [\mathcal{R}_i \, \Delta t] \, \mathcal{P}^Z(t, t_{i+1}).$$

- Summarizing our discussion so far:
$\mathcal{P}_i(t)$, the value at t of receiving $\mathcal{R}_i \, \Delta t$ dollars at date t_{i+1}, is:

$$\mathcal{P}_i(t) = \begin{cases} \mathcal{P}^Z(t, t_i) - \mathcal{P}^Z(t, t_{i+1}) & \text{if } t < t_i, \\ \mathcal{R}_i \, \Delta t \, \mathcal{P}^Z(t, t_{i+1}) & \text{if } t_i < t < t_{i+1}. \end{cases} \tag{8.13}$$

- To find the value of a caplet, we will view it as an option on a forward contract and then apply the formula from Black (1976) for options on forward.

- To do this, we need to know the forward price of the asset with price $\mathcal{P}_i(t)$ for a contract maturing at date t_{i+1}.

- Let us denote the forward price by $\mathcal{P}_i^{fwd}(t)$.

8.11.3 Forward Price of a Non-Dividend Paying Asset

- We know that forward price of a non-dividend paying asset is its spot price multiplied by the interest factor:

$$
\begin{aligned}
\mathcal{P}_i^{fwd}(t) &= [1 + \mathcal{R}(t, t_{i+1})\Delta t] \times \mathcal{P}_i(t) && \text{... from definition of a forward price} \\
&= \frac{1}{\mathcal{P}^Z(t, t_{i+1})} \mathcal{P}_i(t) && \text{... writing } [1 + \mathcal{R}(t, t_{i+1})\Delta t] \text{ in terms of bond price } 1/\mathcal{P}^Z(t, t_{i+1}) \\
&= \frac{1}{\mathcal{P}^Z(t, t_{i+1})} \times \begin{cases} \mathcal{P}^Z(t, t_i) - \mathcal{P}^Z(t, t_{i+1}) & \text{if } t < t_i, \\ \mathcal{R}_i \, \Delta t \, \mathcal{P}^Z(t, t_{i+1}) & \text{if } t_i < t < t_{i+1}. \end{cases} && \text{... using Equation (8.13)} \\
&= \begin{cases} \frac{\mathcal{P}^Z(t,t_i)}{\mathcal{P}^Z(t,t_{i+1})} - 1 & \text{if } t < t_i, \\ \mathcal{R}_i \, \Delta t & \text{if } t_i < t < t_{i+1}. \end{cases} && \text{... dividing through by } \mathcal{P}^Z(t, t_{i+1}) \\
&= \begin{cases} \mathcal{R}_i^{fwd}(t) \, \Delta t & \text{if } t < t_i, \\ \mathcal{R}_i \, \Delta t & \text{if } t_i < t < t_{i+1}. \end{cases} && \text{... using definition of } \mathcal{R}_i^{fwd}(t) \text{ in Equation (8.8)}
\end{aligned}
\tag{8.14}
$$

8.11.4 The Market Model for Caplets and Floorlets

- To find the value of caps and floors:
  - ⋆ We will use the standard model used in the market to value these options;
  - ⋆ This model applies the formula from Black (1976) to the forward price derived above;
  - ⋆ That is, we interpret the caplet as a call option maturing at date t_i on this forward contract that matures at date t_{i+1}.

- As explained earlier in the course, the value at maturity T of a call option with strike K on a forward contract with price $\mathcal{P}^{Fwd}$ maturing at $T' > T$ is:

$$\max\left(0, \mathcal{P}^{Fwd}(T) - K\right) \mathcal{P}^{Z}(T, T').$$

- Similarly, for caplets, the value at maturity of a call option maturing at date t_i with strike price $\bar{\mathcal{R}}\Delta t$ on the forward contract with price $\mathcal{P}_i^{fwd}$ is:

$$\max\left(0, \mathcal{P}_i^{fwd}(t_i) - \bar{\mathcal{R}}\Delta t\right) \mathcal{P}^{Z}(t_i, t_{i+1}).$$

- From Equation (8.14), $\mathcal{P}_i^{fwd}(t_i) = \mathcal{R}_i \Delta t$; thus, the above expression equals

$$\max\left(0, \mathcal{R}_i - \bar{\mathcal{R}}\right) \Delta t \; \mathcal{P}^{Z}(t_i, t_{i+1}), \tag{8.15}$$

  which is also the value of the caplet at date t_i.

- Thus, it follows that the value of the caplet at any date $t < t_i$ is equal to the value of the call option on the forward contract.

- To apply the formula from Black (1976), we need the forward price to have a constant (or, non-randomly varying) volatility.

  ⋆ As we showed in Equation (8.14), at dates $t < t_i$,

$$\mathcal{P}_i^{fwd}(t) = \mathcal{R}_i^{fwd}(t)\, \Delta t, \qquad \text{where } \mathcal{R}_i^{fwd}(t) \text{ is the forward rate.}$$

  ⋆ So,
    ∗ the volatility of the forward price $\mathcal{P}_i^{fwd}(t)$
    ∗ is equal to the volatility of the forward rate, $\mathcal{R}_i^{fwd}(t)$.

  ⋆ So, we will assume the forward rate $\mathcal{R}_i^{fwd}(t)$ has a constant volatility σ.

- Earlier (in the first part of the course), we derived the following expression for options on forwards.

- Recall that the Black (1976) formula for call options on a forward contract (from Part 1 of the course) is:

$$\text{Call Price} = \mathcal{P}^Z(0,T')\,\mathcal{P}^{Fwd}(0)\,N(d_1) - \mathcal{P}^Z(0,T')\,K\,N(d_2),$$

$$d_1 = \frac{\log\left(\frac{\mathcal{P}^Z(0,T')\,\mathcal{P}^{Fwd}(0)}{\mathcal{P}^Z(0,T')\,K}\right) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}} = \frac{\log\left(\frac{\mathcal{P}^{Fwd}(0)}{K}\right) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}.$$

- Assuming the forward rate $\mathcal{R}_i^{fwd}(t)$ has constant volatility σ:

The value at $0 < t_i$ of a caplet with reset date t_i and payment date t_{i+1} is

$$\mathcal{P}^Z(0,t_{i+1})\mathcal{R}_i^{fwd}(0)\,\Delta t\,N(d_1) - \mathcal{P}^Z(0,t_{i+1})\bar{\mathcal{R}}\,\Delta t\,N(d_2);$$

where

$$d_1 = \frac{\log\left(\frac{\mathcal{R}_i^{fwd}(0)}{\bar{\mathcal{R}}}\right) + \frac{1}{2}\sigma^2 t_i}{\sigma\sqrt{t_i}},$$

$d_2 = d_1 - \sigma\sqrt{t_i}$, where σ is the constant volatility of $\mathcal{R}_i^{fwd}(t)$.

8.11.5 Put-Call Parity Relation for Caplets and Floorlets

- The put-call parity relation for caplets and floorlets can be established as follows.
- If we add $\bar{\mathcal{R}}\Delta t$ to the caplet payoff, we get

$$\bar{\mathcal{R}}\Delta t + \max\left(0, \mathcal{R}_i - \bar{\mathcal{R}}\right)\Delta t = \max\left(\bar{\mathcal{R}}, \mathcal{R}_i\right)\Delta t.$$

- On the other hand, if we add $\mathcal{R}_i\Delta t$ to the floorlet payoff, we get the same payoff:

$$\mathcal{R}_i\Delta t + \max\left(0, \bar{\mathcal{R}} - \mathcal{R}_i\right)\Delta t = \max\left(\mathcal{R}_i, \bar{\mathcal{R}}\right)\Delta t.$$

- Thus,
  - ⋆ the value of a caplet plus the value at t of receiving $\bar{\mathcal{R}}\Delta t$ at t_{i+1}
  - ⋆ must equal value of floorlet plus the value at t of receiving $\mathcal{R}_i\Delta t$ at t_{i+1}.

- The value at any date $t < t_i$ of receiving $\bar{\mathcal{R}}\Delta t$ at t_{i+1} is simply its present value, $[\bar{\mathcal{R}}\Delta t] \times \mathcal{P}^Z(t, t_{i+1})$.
- Similarly, the value at any date $t < t_i$ of receiving $\mathcal{R}_i\Delta t$ at t_{i+1} is, from our earlier results, $\mathcal{P}^Z(t, t_i) - \mathcal{P}^Z(t, t_{i+1})$.
- Therefore, the put-call parity relation for caplets and floorlets is:

$$\text{Value of Caplet} + [\bar{\mathcal{R}}\Delta t] \times \mathcal{P}^Z(t, t_{i+1}) = \text{Value of Floorlet} + [\mathcal{P}^Z(t, t_i) - \mathcal{P}^Z(t, t_{i+1})].$$

8.11.6 Valuing Caplets as Puts on Discount Bonds

- Earlier, we considered a caplet as a call option on the forward rate and thus a "bet" on higher interest rates.
- This, of course, is equivalent to a bet on lower bond prices.
- We use this equivalence to show that a caplet with payment date t_{i+1} is equivalent to
  - $1 + \bar{\mathcal{R}}\Delta t$ put options
  - on the discount bond with maturity t_{i+1}
  - with the options maturing at the reset date t_i of the caplet, and
  - with a strike price equal to $1/(1 + \bar{\mathcal{R}}\Delta t)$.

- To see this equivalence, note that the value at the maturity date t_i of $(1 + \bar{\mathcal{R}}\Delta t)$ such options with strike price $1/(1 + \bar{\mathcal{R}}\Delta t)$ is

$$\begin{aligned}
&\left[1 + \bar{\mathcal{R}}\Delta t\right] \max\left(0, \frac{1}{1 + \bar{\mathcal{R}}\Delta t} - \mathcal{P}^Z(t_i, t_{i+1})\right) \\
&= \max\left(0, 1 - [1 + \bar{\mathcal{R}}\Delta t]\mathcal{P}^Z(t_i, t_{i+1})\right) && \ldots \text{multiplying inside parenthesis by } 1 + \bar{\mathcal{R}}\Delta t \\
&= \mathcal{P}^Z(t_i, t_{i+1}) \max\left(0, \frac{1}{\mathcal{P}^Z(t_i, t_{i+1})} - 1 - \bar{\mathcal{R}}\Delta t]\right) && \ldots \text{multiplying \& dividing by } \mathcal{P}^Z(t_i, t_{i+1}) \\
&= \mathcal{P}^Z(t_i, t_{i+1}) \max\left(0, \mathcal{R}_i - \bar{\mathcal{R}}\right) \Delta t && \ldots \text{using the result that } 1/\mathcal{P}^Z(t_i, t_{i+1}) = 1 + \mathcal{R}_i\Delta t.
\end{aligned}$$

- This is exactly the expression in (8.15) on Page 97 for the value at date t_i of the caplet with payment date t_{i+1}.
- Thus, the caplet and the $(1 + \bar{\mathcal{R}}\Delta t)$ put options must have the same value at any date prior to t_i.

8.12 Fixed-Income Derivatives: European Swaptions

- In this section, we study the market model for swaptions.
- The owner of a European swaption has an option to enter into a swap at the maturity date of the swaption.
- A payer swaption gives the owner of the option the right to enter into the swap as the payer of the fixed rate, for a given swap rate.
- A receiver swaption gives the owner of the option the right to enter into the swap as a receiver of the fixed rate.
- The value of payer and receiver swaptions are linked by put-call parity, similar to put-call parity for caplets & floorlets.

- A payer swaption has similarities to a cap.
  - ⋆ The owner of a cap has the right to receive the floating rate and pay the fixed rate, and will do so in each period that the floating rate is higher than the fixed rate.
  - ⋆ Similarly, the owner of a payer swaption has the right to receive floating and pay fixed.
  - ⋆ However,
    - ∗ the owner of a cap chooses each period whether to exercise her option,
    - ∗ whereas the owner of a swaption makes a once-and-for-all decision whether to exercise, at the maturity date of the swaption.
- A cap is therefore a portfolio of options, whereas the swaption is an option on a portfolio.
- Generally, all else being equal, a portfolio of options is worth more than an option on a portfolio.

- Consider a payer swaption with maturity date T and swap rate $\bar{\mathcal{R}}$, where the underlying swap has payment dates t_1, t_2, t_N, with the first reset date of the swap being $t_0 = t_1 - \Delta t > T$. Assume that the notional principal is \$1.

- In Equation (8.9) on Page 79, which is re-written below, we already found the value of a swap to a fixed-rate payer:

$$[\text{Value of payer swap}]_t = \underbrace{[\mathcal{P}^Z(t,t_0) - \mathcal{P}^Z(t,t_N)]}_{V^{\text{floating}}(t)} - \underbrace{[\bar{\mathcal{R}}\Delta t \sum_{i=1}^{N} \mathcal{P}^Z(t,t_i)]}_{V^{\text{fixed}}(t)}. \qquad (8.9)$$

- From the above, the value at T of a swap to the fixed-rate payer is

$$[\text{Value of swap to fixed-rate payer}]_T = V^{\text{floating}}(T) - V^{\text{fixed}}(T).$$

- Thus, the value of a swaption at its maturity T is:

$$\max\left(0, V^{\text{floating}}(T) - V^{\text{fixed}}(T)\right).$$

- We can value the swaption using Margrabe's formula for exchange options, **if** the ratio of $V^{\text{floating}}/V^{\text{fixed}}$ has a constant volatility.

- So, we proceed in two steps:

  1. First, we show that the volatility of $V^{\text{floating}}/V^{\text{fixed}}$ is the same as the volatility of the forward swap rate.
  2. Second, we assume that the volatility of the forward swap rate is constant, which then allows us to use Margrabe's formula.

- To see that the volatility of $V^{\text{floating}}/V^{\text{fixed}}$ is the same as the volatility of the forward swap rate:
  - ⋆ Recall that on Page 86 the forward swap rate $\mathcal{R}^{swap}(t)$ was defined to be the rate such that the value of the swap would be zero value if it were initiated at this rate:

$$\mathcal{P}^Z(t,t_0) - \mathcal{P}^Z(t,t_N) = \mathcal{R}^{swap}(t)\Delta t \sum_{i=1}^{N} \mathcal{P}^Z(t,t_i). \tag{8.12}$$

  - ⋆ We start with the ratio of the value of the floating- and fixed-rate legs:

$$\begin{aligned}
\frac{V^{\text{floating}}(t)}{V^{\text{fixed}}(t)} &= \frac{\mathcal{P}^Z(t,t_0) - \mathcal{P}^Z(t,t_N)}{\bar{\mathcal{R}}\Delta t \sum_{i=1}^{N} \mathcal{P}^Z(t,t_i)} && \text{. . . using definitions of } V^{\text{floating}} \text{ and } V^{\text{fixed}} \\
&= \frac{\mathcal{R}^{swap}(t)}{\bar{\mathcal{R}}} \frac{\Delta t \sum_{i=1}^{N} \mathcal{P}^Z(t,t_i)}{\Delta t \sum_{i=1}^{N} \mathcal{P}^Z(t,t_i)} && \text{. . . substituting into numerator result in (8.12)} \\
&= \frac{\mathcal{R}^{swap}(t)}{\bar{\mathcal{R}}} && \text{. . . canceling common term in numerator \& denominator}
\end{aligned}$$

- So, if the forward swap rate $\mathcal{R}^{swap}(t)$ has a constant volatility σ, then Margrabe's formula tells us that:

  - Value of European payer swaption at date 0 is

$$= \underbrace{\left[\mathcal{P}^Z(0,t_0) - \mathcal{P}^Z(0,t_N)\right]}_{V^{\text{floating}}(0)} N(d_1) - \underbrace{\left[\bar{\mathcal{R}}\Delta t \sum_{i=1}^{N} \mathcal{P}^Z(0,t_i)\right]}_{V^{\text{fixed}}(0)} N(d_2)$$

  - Value of European receiver swaption at date 0 is

$$= \left[\bar{\mathcal{R}}\Delta t \sum_{i=1}^{N} \mathcal{P}^Z(0,t_i)\right] N(-d_2) - \left[\mathcal{P}^Z(0,t_0) - \mathcal{P}^Z(0,t_N)\right] N(-d_1)$$

where

$$d_1 = \frac{\log\left(\frac{V^{\text{floating}}(0)}{V^{\text{fixed}}(0)}\right) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}; \qquad d_2 = d_1 - \sigma\sqrt{T}.$$

8.13 Inconsistency in Assuming Constant Volatilities for Forward Interest Rates and Forward Swap Rate

- It is inconsistent to assume that both the forward interest rates $\mathcal{R}_i^{fwd}(t)$ and the forward swap rate $\mathcal{R}^{swap}(t)$ have constant volatilities.

- To see this, recall the definitions on Pages 75 and 86, reproduced below:

$$\mathcal{R}_i^{fwd}(t)\,\Delta t = \frac{\mathcal{P}^Z(t,t_i) - \mathcal{P}^Z(t,t_{i+1})}{\mathcal{P}^Z(t,t_{i+1})}, \tag{8.8}$$

$$\mathcal{R}^{swap}(t)\,\Delta t = \frac{\mathcal{P}^Z(t,t_0) - \mathcal{P}^Z(t,t_N)}{\sum_{i=0}^{N-1}\mathcal{P}^Z(t,t_{i+1})}. \tag{8.12}$$

- Consider the numerator of the term on the right-hand-side of Equation (8.8), and sum this over $i = \{0, 1, \ldots, N-1\}$:

$$\left[\mathcal{P}^Z(t,t_0) - \mathcal{P}^Z(t,t_1)\right] + \left[\mathcal{P}^Z(t,t_1) - \mathcal{P}^Z(t,t_2)\right] + \ldots \left[\mathcal{P}^Z(t,t_{N-1}) - \mathcal{P}^Z(t,t_N)\right] = \mathcal{P}^Z(t,t_0) - \mathcal{P}^Z(t,t_N).$$

- Therefore,

$$\sum_{i=0}^{N-1}\left[\mathcal{R}_i^{fwd}(t)\Delta t\,\mathcal{P}^Z(t,t_{i+1})\right] = \sum_{i=0}^{N-1}\left[\mathcal{P}^Z(t,t_i) - \mathcal{P}^Z(t,t_{i+1})\right]$$
$$= \mathcal{P}^Z(t,t_0) - \mathcal{P}^Z(t,t_N).$$

- Substituting the above in the numerator of the term on the right-hands-side of Equation (8.12), we get:

$$\mathcal{R}^{swap}(t)\,\Delta t = \frac{\mathcal{P}^Z(t,t_0) - \mathcal{P}^Z(t,t_N)}{\sum_{i=0}^{N-1}\mathcal{P}^Z(t,t_{i+1})} = \frac{\sum_{i=0}^{N-1}\left[\mathcal{R}_i^{fwd}(t)\Delta t\,\mathcal{P}^Z(t,t_{i+1})\right]}{\sum_{i=0}^{N-1}\mathcal{P}^Z(t,t_{i+1})}$$

which, upon canceling the Δt term on both sides, simplifies to

$$\mathcal{R}^{swap}(t) = \frac{\sum_{i=0}^{N-1}\left[\mathcal{R}_i^{fwd}(t)\,\mathcal{P}^Z(t,t_{i+1})\right]}{\sum_{i=0}^{N-1}\mathcal{P}^Z(t,t_{i+1})}.$$

- We can now write the last expression in terms of a weighted average:

$$\mathcal{R}^{swap}(t) = \sum_{i=0}^{N-1} w_i(t) \left[\mathcal{R}_i^{fwd}(t)\right] \quad \text{where} \quad w_i(t) = \frac{\mathcal{P}^Z(t, t_{i+1})}{\sum_{i=0}^{N-1} \mathcal{P}^Z(t, t_{i+1})}.$$

- Thus, the swap rate $\mathcal{R}^{swap}(t)$ is a weighted average of forward rates $\mathcal{R}_i^{fwd}(t)$.
- But, a sum or average of lognormal variables is not lognormal; so,
  - ⋆ if the forward rates have constant volatilities,
  - ⋆ then the forward swap rate will not (and vice versa).

- So, if one wished to be perfectly consistent, one should not simultaneously
  - ⋆ use the Black (1976) formula for valuing caps and floors, and
  - ⋆ use the Margrabe (1978) formula for valuing swaptions.
- However, empirical evidence suggests that the error introduced by doing so is small, and, in practice, the two models are often used in conjunction.
- In the next class, we will study other models of the term structure of interest rates that do not suffer from this type of inconsistency.

8.14 Summary

1. The yield of a zero-coupon bond with \$1 face value having $\tau = T - t$ years to maturity and price $\mathcal{P}^Z(t,T)$ is defined to be the rate $y(\tau)$ such at:

$$\mathcal{P}^Z(t,T) = \frac{\text{amount received}}{\text{discount factor}} = \frac{1_\tau}{e^{y\,\tau}} = e^{-y\,\tau} \quad \Leftrightarrow \quad y = -\frac{1}{\tau}\log \mathcal{P}^Z(t,T);$$

2. The yield curve is the collection of yields for different maturities.

3. For a bond with cashflows $C_1, C_2, \ldots, C_N$ at dates $u_1 < u_2 < \ldots < u_N$ and price P at date $t < u_1$, the continuously compounded yield to maturity of a bond at date t is defined to be the rate $\mathbf{y}$ such that

$$\mathcal{P}^B = \sum_{j=1}^{N} e^{-\mathbf{y}\tau_j} C_j,$$

where $\tau_j = u_j - t$ denotes the time remaining until j-th cash flow is paid.

4. "Macaulay duration" of the bond, or simply duration is the term

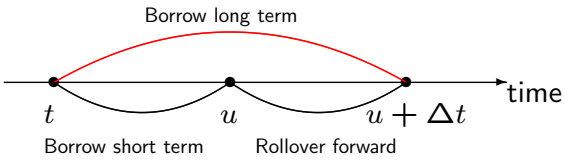
$$-\frac{\frac{d\mathcal{P}^B}{\mathcal{P}^B}}{d\mathbf{y}} = \sum_{j=1}^{N} \tau_j \times \frac{e^{-\mathbf{y}\tau_j} C_j}{P},$$

which is a weighted average of the times-to-maturity τ_j of the cashflows, with the weights being the fraction of bond value that the cashflow constitutes.

5. The relation between the simple (spot) interest rate $\mathcal{R}_i$ and the price at t of a zero-coupon discount bond with payment of $\$1$ at maturity u is:

$$\mathcal{P}^Z(t,u) = \frac{1}{1+\mathcal{R}_i\,\Delta t}, \quad \text{or equivalently:} \quad 1+\mathcal{R}_i\,\Delta t = \frac{1}{\mathcal{P}^Z(t,u)}.$$

6. The forward interest rate allows you to lock in at date t a rate for a loan or deposit that starts at $u > t$ for a period Δt:



The forward interest rate $\mathcal{R}^{fwd}$ is given by:

$$1+\mathcal{R}^{fwd}(u,u+\Delta t)\Delta t = \frac{1+\mathcal{R}(t,u+\Delta t)(u+\Delta t-t)}{1+\mathcal{R}(t,u)(u-t)} = \frac{\mathcal{P}^Z(t,u)}{\mathcal{P}^Z(t,u+\Delta t)}.$$

7. A standard (plain vanilla) interest rate swap is the exchange of fixed for floating interest rate payments on a given notional principal.

- Value of swap to fixed-rate payer is

$$\text{Value}_t = \underbrace{[\mathcal{P}^Z(t,t_0) - \mathcal{P}^Z(t,t_N)]}_{\text{value of floating leg}} - \underbrace{[\bar{\mathcal{R}}\Delta t \sum_{i=1}^{N} \mathcal{P}^Z(t,t_i)]}_{\text{value of fixed leg}}.$$

8. Buyers of caps and floors pay a premium to the seller of the option; all future cashflows are paid by the seller to the buyer.

For each $\$1$ of notional principal,

- the owner/purchaser of a cap with cap rate $\bar{\mathcal{R}}$ receives at t_{i+1}

$$\max(0, \mathcal{R}_i - \bar{\mathcal{R}})\Delta t;$$

- the owner of a floor receives with floor rate $\bar{\mathcal{R}}$ receives at t_{i+1}

$$\max(0, \bar{\mathcal{R}} - \mathcal{R}_i)\Delta t.$$

9. The value at $0 < t_i$ of a caplet with reset date t_i and payment date t_{i+1} is

$$\mathcal{P}^Z(0, t_{i+1})\mathcal{R}_i^{fwd}(0)\,\Delta t\,N(d_1) - \mathcal{P}^Z(0, t_{i+1})\bar{\mathcal{R}}\,\Delta t\,N(d_2);$$

where

$$d_1 = \frac{\log\left(\frac{\mathcal{R}_i^{fwd}(0)}{\bar{\mathcal{R}}}\right) + \frac{1}{2}\sigma^2 t_i}{\sigma\sqrt{t_i}},$$

$d_2 = d_1 - \sigma\sqrt{t_i}$, where σ is the constant volatility of $\mathcal{R}_i^{fwd}(t)$.

10. A payer swaption gives the owner of the option the right to enter into the swap as the payer of the fixed rate, for a given swap rate;
a receiver swaption gives the owner of the option the right to enter into the swap as a receiver of the fixed rate.

- Note that the owner of a swaption makes a once-and-for-all decision whether to exercise, at the maturity date of the swaption.
- In contrast, the owner of a cap chooses each period whether to exercise her option.

- Value of swap to fixed-rate payer: $\underbrace{[\mathcal{P}^Z(t,t_0)-\mathcal{P}^Z(t,t_N)]}_{V^{\text{floating}}(t)} - \underbrace{[\bar{\mathcal{R}}\Delta t\sum_{i=1}^{N}\mathcal{P}^Z(t,t_i)]}_{V^{\text{fixed}}(t)}.$

- Thus, value of European payer swaption at date 0 is

$$= \underbrace{\left[\mathcal{P}^Z(0,t_0)-\mathcal{P}^Z(0,t_N)\right]}_{V^{\text{floating}}(0)}N(d_1) - \underbrace{\left[\bar{\mathcal{R}}\Delta t\sum_{i=1}^{N}\mathcal{P}^Z(0,t_i)\right]}_{V^{\text{fixed}}(0)}N(d_2)$$

$$d_1 = \frac{\log\left(\frac{V^{\text{floating}}(0)}{V^{\text{fixed}}(0)}\right)+\frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}; \qquad d_2 = d_1-\sigma\sqrt{T},$$

where σ is the constant volatility of the forward swap rate $\mathcal{R}^{swap}(t)$.

8.15 Appendix: Estimating the Yield Curve

- In the real world, zero-coupon bonds for all maturities are not traded; that is, zero-coupon bonds for only a limited number of maturities are traded.

- Thus, one needs to infer the yield curve (or the prices of bonds that are not being currently traded) from:
  - ⋆ the prices of zero-coupon bonds being traded;
  - ⋆ the prices of coupon-paying bonds being traded;
  - ⋆ swap rates quoted in the market (we will study this in the next chapter).

- There are at least three approaches for estimating the yield curve and finding prices of zero-coupon bonds:

1. Bootstrapping
   - ⋆ obtaining prices of zero-coupon bond from prices of coupon bonds;
   - ⋆ used historically but has limitations, and so is not used much these days.

2. Polynomial and exponential splines
   - ⋆ fitted to the spot curve;
   - ⋆ identify different segments of the yield curve and knot together the segments;
   - ⋆ used by the U.S. and Japanese central banks

3. Nelson-Siegel-Svensson approach
   - ⋆ fit a single function to the forward curve;
   - ⋆ used by central banks in U.K., Canada, and several European countries.

8.15.1 Bootstrapping Approach

- In the bootstrapping approach,
  - ⋆ we use the shortest maturity bonds to identify the prices of zero-coupon bonds for short maturities, and
  - ⋆ then we use
    - ∗ the prices of zero-coupon bonds that we have identified
    - ∗ along with prices of longer-maturity coupon bonds

    to identity the prices of longer-maturity zero-coupon bonds.

Bootstrapping Via Matrix Inversion

- If we have N equations and we wish to find the prices of N zero-coupon bonds, then this can be done via matrix inversion (as long as the bonds are linearly independent).
- To see this, denote the N points in time as t_j, with $j = \{1, 2, \ldots, N\}$.
- We wish to find the prices of zero-coupon bonds, $P(0, t_j)$.
- Denote the time t_j cashflow from the kth bond by C_j^k.
- Denote the price of the kth coupon-bond by P^{C_k}.

- Then, using matrix notation, we can write the prices of coupon bonds as:

$$\begin{bmatrix} P^{C_1} \\ P^{C_2} \\ \vdots \\ P^{C_N} \end{bmatrix} = \begin{bmatrix} C_1^1 & \cdots & C_N^1 \\ C_1^2 & \cdots & C_N^2 \\ \vdots & \vdots & \vdots \\ C_1^N & \cdots & C_N^N \end{bmatrix} \cdot \begin{bmatrix} P(0,t_1) \\ P(0,t_2) \\ \vdots \\ P(0,t_N) \end{bmatrix}$$

$$[\text{Prices of coupon bonds}] = [\text{Coupons from each bond}] \cdot [\text{Prices of zero-coupon bonds}]$$

- Multiplying both sides of the equation by the inverse of the matrix of coupon payments, we get:

$$\begin{bmatrix} C_1^1 & \cdots & C_N^1 \\ C_1^2 & \cdots & C_N^2 \\ \vdots & \vdots & \vdots \\ C_1^N & \cdots & C_N^N \end{bmatrix}^{-1} \cdot \begin{bmatrix} P^{C_1} \\ P^{C_2} \\ \vdots \\ P^{C_N} \end{bmatrix} = \begin{bmatrix} P(0,t_1) \\ P(0,t_2) \\ \vdots \\ P(0,t_N) \end{bmatrix} \tag{8.16}$$

$$[\text{Coupons from each bond}]^{-1} \cdot [\text{Prices of coupon bonds}] = [\text{Prices of zero-coupon bonds}]$$

- Thus, to obtain the prices of zero-coupon bonds, all we need to do is:
  - ⋆ Invert the matrix of coupon payments;
  - ⋆ Multiply this inverse by the vector of prices for the coupon bonds.

8.15.2 Limitations of Bootstrapping

1. All the bonds that we use in bootstrapping must have cashflows on the same set of dates.

2. Matrix of coupon payments must be invertible; that is, the bond cashflows need to be linearly independent.

3. Knowing $P(0,t_j)$ does not tell us how to value cashflows that fall in-between the coupon payment dates.

4. The prices of discount bonds that you get are sensitive to your choice of coupon bonds used.

5. The number of coupon bonds that are liquid and reflect current market conditions are limited.

- Ideally, we want a methodology that uses the information present in all bonds regardless of whether or not the cashflow dates coincide.

8.15.3 Cubic Splines

- The most common way of estimating the yield curve from bond prices is to fit a cubic spline, using the prices of the bonds being traded.
- Let
  - ⋆ P_i denote the price of a particular bond i,
  - ⋆ N_i denote the number of dates on which this bond pays a coupon or its face value, and
  - ⋆ τ_{ij} with $j = \{1, 2, \ldots, N_i\}$ denote the dates on which these payments are made.
- Then the price of the coupon bond P^{C_k} should be the discounted sum of its payments:

$$P^{C_k} = \sum_{j=1}^{N_i} e^{-y(\tau_j)\tau_{ij}} C_{ij}.$$

- In practice, we will usually be unable to find the yields $y(\tau_{ij})$ such that the above expression holds exactly. This may be because of
  - ⋆ zero-coupon bonds not being traded for a particular maturity τ_{ij};
  - ⋆ bid-ask spreads;
  - ⋆ non-synchronous trading.
- The cubic spline is one way to resolve this problem.
- A cubic spline consists of several cubic polynomials "spliced" together at a set of "knot points."

- Specifically, it consists of maturities $\tau_1, \tau_2, \ldots, \tau_n$ (the "knot points") and coefficients (a_i, b_i, c_i, d_i) for $i = 0, 1, \ldots, n$, with the yield curve being modeled as:

$$y(\tau) = \begin{cases} a_0\tau^3 + b_0\tau^2 + c_0\tau + d_0 & \text{for } 0 < \tau \leq \tau_1, \\ a_1(\tau - \tau_1)^3 + b_1(\tau - \tau_1)^2 + c_1(\tau - \tau_1) + d_1 & \text{for } \tau_1 < \tau \leq \tau_2, \\ a_2(\tau - \tau_2)^3 + b_2(\tau - \tau_2)^2 + c_2(\tau - \tau_2) + d_2 & \text{for } \tau_2 < \tau \leq \tau_3, \\ \ldots & \\ a_n(\tau - \tau_n)^3 + b_n(\tau - \tau_n)^2 + c_n(\tau - \tau_n) + d_n & \text{for } \tau_n < \tau \leq T, \end{cases}$$

  where T is the maximum maturity considered.

- To "**splice**" these polynomials together at the knot points means to choose the coefficient so that the two polynomials that meet at a knot point have
  - ⋆ the same value,
  - ⋆ the same first derivative, and
  - ⋆ the same second derivatives

  at the knot point.

- For example, given $y(\tau)$ on the previous page, which we re-write below:

$$y(\tau) = \begin{cases} a_0\tau^3 + b_0\tau^2 + c_0\tau + d_0 & \text{for } 0 < \tau \leq \tau_1, \\ a_1(\tau - \tau_1)^3 + b_1(\tau - \tau_1)^2 + c_1(\tau - \tau_1) + d_1 & \text{for } \tau_1 < \tau \leq \tau_2, \\ \ldots \end{cases}$$

at the first knot point $\tau = \tau_1$, we want the adjacent polynomials to satisfy:

Equality of yields: $$a_0\tau_1^3 + b_0\tau_1^2 + c_0\tau_1 + d_0 = d_1, \quad (8.17)$$

Equality of first derivatives: $$3a_0\tau_1^2 + 2b_0\tau_1 + c_0 = c_1, \quad (8.18)$$

Equality of second derivatives: $$6a_0\tau_1 + 2b_0 = 2b_1. \quad (8.19)$$

- So, given the coefficients a_0, b_0, c_0, d_0, the Equations (8.17), (8.18), and (8.19) specify d_1, c_1, b_1, so that the only free coefficient for the next polynomial is a_1.

- Similarly, given the expression for $y(\tau)$, which we re-write one more time:

$$y(\tau) = \begin{cases} \ldots \\ a_1(\tau - \tau_1)^3 + b_1(\tau - \tau_1)^2 + c_1(\tau - \tau_1) + d_1 & \text{for } \tau_1 < \tau \leq \tau_2, \\ a_2(\tau - \tau_2)^3 + b_2(\tau - \tau_2)^2 + c_2(\tau - \tau_2) + d_2 & \text{for } \tau_2 < \tau \leq \tau_3, \\ \ldots \end{cases}$$

at the second knot point $\tau = \tau_2$, we want the adjacent polynomials to satisfy:

Yield: $$a_1(\tau_2 - \tau_1)^3 + b_1(\tau_2 - \tau_1)^2 + c_1(\tau_2 - \tau_1) + d_1 = d_2, \quad (8.20)$$

First derivative: $$3a_1(\tau_2 - \tau_1)^2 + 2b_1(\tau_2 - \tau_1) + c_1 = c_2, \quad (8.21)$$

Second derivative: $$6a_1(\tau_2 - \tau_1) + 2b_1 = 2b_2, \quad (8.22)$$

- So, Equations (8.20), (8.21), and (8.22) specify d_2, c_2, b_2, with the only free coefficient for the next polynomial being a_2.

- Continuing in this way, we see that the cubic spline is defined by
  - ⋆ the knot points $\tau_1, \tau_2, \ldots, \tau_n$;
  - ⋆ the coefficients for the shortest maturity yield: a_0, b_0, c_0, d_0; and,
  - ⋆ the free coefficients for the remaining maturity yields: $a_1, a_2, \ldots, a_n$.
- We wish to
  - ⋆ choose the knot points and these coefficients
  - ⋆ so that we have the "best possible fit" (say using regression methods) for the pricing relations below, for the set of k bonds that we have:

$$P^{C_k} = \sum_{j=1}^{N_i} e^{-y(\tau_j)\tau_{ij}} C_{ij}.$$

8.15.4 Exponential Splines

- As the name suggests, instead of fitting polynomials, one can also fit an exponential function to estimate the yield curve.
- That is, instead of fitting a cubic function of the form below (for each segment of the yield curve)

$$y(\tau) = a_0\tau^3 + b_0\tau^2 + c_0\tau + d_0 \quad \text{for } 0 < \tau \leq \tau_1,$$

  we can fit an exponential function of the form:

$$y(\tau) = a_0\left(1 - e^{-3m\,\tau}\right) + b_0\left(1 - e^{-2m\,\tau}\right) + c_0\left(1 - e^{-m\,\tau}\right) + d_0.$$

- Here, $m > 0$ is an additional free parameter.
- It turns out that m can be interpreted to be the long forward rate:

$$m = \lim_{T\to\infty} r^{fwd}(T).$$

8.15.5 Implementation Issues with Splines

- To use splines, one needs to decide:
  1. How many knot points to have
  2. Where to place the knot points
- In making the above choices, there is a tradeoff between fit and smoothness.
- As you increase the number of knot points,
  - ⋆ fit improves
  - ⋆ smoothness decreases (yield curve is more jagged)
- The solution is to
  - ⋆ use a large number of knot points where there is lots of data (typically the short end of the yield curve);
  - ⋆ use fewer knot points at the long end of the yield curve.

- One may also want to adjust bond prices for
  - ⋆ liquidity,
  - ⋆ coupons,
  - ⋆ tax issues, and
  - ⋆ embedded options.

8.15.6 Nelson-Siegel-Svensson Method for Estimating Term Structure

- In contrast to spline methods, the Nelson and Siegel (1987) approach, and its extension by Svensson (1994):
  - ⋆ model the forward curve,
    instead of the spot curve (or prices of zero-coupon bonds);
  - ⋆ fit the entire forward curve by a single function,
    instead of breaking the curve into several segments as spline methods do,
- Nelson-Siegel approach is to use the following function for the forward rate:

$$r^{fwd}(T) = \beta_0 + \beta_1 \exp\left(\frac{-T}{\theta}\right) + \beta_2 \left(\frac{T}{\theta}\right) \exp\left(\frac{-T}{\theta}\right).$$

- Svensson adds one more term to this functional form:

$$r^{fwd}(T) = \beta_0 + \beta_1 \exp\left(\frac{-T}{\theta}\right) + \beta_2 \left(\frac{T}{\theta}\right) \exp\left(\frac{-T}{\theta}\right) + \beta_3 \left(\frac{T}{\nu}\right) \exp\left(\frac{-T}{\nu}\right).$$

- This functional form generates a smooth forward curve with no jagged bits.
- The parameters of the model map directly into the three key characteristics of the forward curve: its
  1. Level;
  2. Slope; and,
  3. Curvature.

- We now study the function specified by the Nelson-Siegel-Svensson model:

$$r^{fwd}(T) = \beta_0 + \beta_1 \exp\left(\frac{-T}{\theta}\right) + \beta_2 \left(\frac{T}{\theta}\right) \exp\left(\frac{-T}{\theta}\right) + \beta_3 \left(\frac{T}{\nu}\right) \exp\left(\frac{-T}{\nu}\right).$$

- β_0:
  - ⋆ As $T \to \infty$, $r^{fwd}(T) \to \beta_0$, which implies that β_0 can be interpreted as the "long forward rate."
  - ⋆ Changes in β_0 result in parallel shifts in the forward curve.
  - ⋆ Hence, β_0 is also called the **level** parameter.

- β_1:
  - ⋆ At $T = 0$, $r^{fwd} = \beta_0 + \beta_1$.
  - ⋆ Therefore, $\beta_0 + \beta_1$ is the "short forward rate."
  - ⋆ And, β_1 is the difference between the short and long forward rates, which is called the **slope** of the forward curve.

- We re-write the function specified by the Nelson-Siegel-Svensson model:

$$r^{fwd}(T) = \beta_0 + \beta_1 \exp\left(\frac{-T}{\theta}\right) + \beta_2 \left(\frac{T}{\theta}\right) \exp\left(\frac{-T}{\theta}\right) + \beta_3 \left(\frac{T}{\nu}\right) \exp\left(\frac{-T}{\nu}\right).$$

- β_2 determines the magnitude and direction of the "hump" in $r^{fwd}(T)$.
  - ⋆ If $\beta_2 > 0$, then $r^{fwd}(T)$ has a hump at θ;
  - ⋆ If $\beta_2 < 0$, then $r^{fwd}(T)$ has a U-shape at θ;

  Therefore, β_2 is called the **curvature** parameter.

- θ determines the **location** of the hump or U-shape.

- β_3 determines magnitude and direction of second hump/U-shape in $r^{fwd}(T)$.

- ν determines location of the second hump or U-shape.

8.16 Suggestions for Further Readings

8.16.1 Source material for the lecture notes

- Most of these lecture notes are based on Back (2005, Chapters 11 & 12)
- The early parts of Back (2017, Chapter 18) and Dumas and Luciano (2017, Chapter 16) also discuss this material.
- Sundaram and Das (2015, Chapter 27) contains a very gentle and clear introduction to the material on the estimation of the term structure.

8.16.2 Potential applications and extensions of lecture notes

- There is a very large literature on the estimation of the yield curve.
  - ⋆ A good description of estimating the yield curve using bootstrapping is described in the Wikipedia article https://en.wikipedia.org/wiki/Bootstrapping_(finance). The limitations of this approach are described in McCulloch (1971).
  - ⋆ The polynomial and exponential spline methods are developed in McCulloch (1971, 1975) and Vasicek and Fong (1982).
  - ⋆ The approach of modeling using exponential functions the forward curve (instead of the spot yield curve) is developed in Nelson and Siegel (1987) and extended in Svensson (1994).
  - ⋆ A good review of these different approaches for estimating the yield curve is in Gibson, Lhabitant, and Talay (2010) and in Nawalkha and Soto (2017).
  - ⋆ Nymand-Andersen (2018) describes how the European Central Bank uses these models, and you can download this article from https://www.ecb.europa.eu/pub/pdf/scpsps/ecb.sps27.en.pdf.

- ⋆ The model for the term structure of interest rates that was popular in the early 1990s was the Heath, Jarrow, and Morton (1992) model that, as we will see in the next chapter (but only briefly), is in terms of the forward rates, which are difficult to estimate.
- ⋆ Therefore, there was a desire to develop a model for market-observed rates, such as LIBOR.
- ⋆ The market model was developed by Miltersen, Sandmann, and Sondermann (1997) and Brace, Gątarek, and Musiela (1997); the model is sometimes referred to as the "BGM" model.
- ⋆ These models were followed by the development of the Jamshidian (1997) model based on swap rates; see also Musiela and Rutkowski (1997) and the book Musiela and Rutkowski (2005).
- ⋆ Rebonato (2012) provides an excellent description of these LIBOR-based models.

- There are a large number of books on the term structure of interest rates and options on interest rates.
  - ⋆ A relatively simple and short book is Britten-Jones (1998).
  - ⋆ More advanced books are Brigo and Mercurio (2007), Filipovic (2009), Cairns (2018), and Jarrow (2019).
- Rebonato (1996) is a book with lots of institutional detail.
- Veronesi (2010) is a comprehensive book on fixed-income securities.
- Other papers on the term structure of interest rates include Diebold, Piazzesi, and Rudebusch (2005), Diebold and Li (2006), and Christensen, Diebold, and Rudebusch (2011).

- **Credit risk** is a topic that we did not discuss, but:
  - ⋆ Is discussed in the following books: Brigo and Mercurio (2007, Chapters 21–23), Filipovic (2009, Chapter 12), and Cairns (2018, Chapter 11).
  - ⋆ Some excellent books dedicated entirely to credit risk are: Schönbucher (2003), Lando (2009), Duffie and Singleton (2012), and Bielecki and Rutkowski (2013).
  - ⋆ See also the review article Jarrow, R. A. 2009. Credit risk models. *Annual Review of Financial Economics* 1 (1): 37–68.

8.16.3 Ph.D. elective courses offered on related topics

- Frank Diebold has taught several elective PhD courses for EDHEC.
  - ⋆ He taught the course titled "*The Yield Curve*," in September 2010.
- Riccardo Rebonato has taught the course "*Term Structure Modelling in the P and Q measures*" in January 2017.
- Pietro Veronesi taught a course on "*Fixed Income Securities*" in 2021 and in 2024.

Lecture Notes for 2025

Continuous Time Financial Economics

Chapter 9

Models of the Term Structure of Interest Rates

Raman Uppal

EDHEC Business School

Chapter contents

9.1 Plan for the Day . . . 153
9.2 Background . . . 154
9.3 Main Points for Today . . . 155
9.4 Motivation for Dynamic Models of the Term Structure of Interest Rates . . . 156
9.4.1 Differences in Modeling Movements in Equity and Bond Prices . . . 157
9.4.2 State variables and factors . . . 160
9.4.3 An Example of Arbitrage in a Badly Specified Dynamic Model of the Term Structure . . . 161
9.5 Overview and Background of Material To be Covered . . . 166
9.5.1 Plan: Focus on the Vasicek Model . . . 167
9.5.2 List of Topics For Each Model of the Term Structure . . . 168
9.6 The Short Rate and Discount Bond Prices . . . 169
9.7 The Basic Vasicek Model . . . 173
9.7.1 The Short Rate and Its Distribution for the Vasicek Process . . . 174
9.7.2 Bond Prices under the Vasicek Process . . . 187
9.7.3 Hedging in the Vasicek Model . . . 199
9.7.4 Estimating the Vasicek Model . . . 210
9.8 Extensions of Vasicek Model: Hull & White Model and Ho & Lee Model . . . 214
9.8.1 Interest Rates in the Hull-White and Ho-Lee Models . . . 218
9.8.2 Bond Prices in the Extended Vasicek model . . . 220
9.8.3 Bond Returns in the Extended Vasicek Model . . . 223
9.8.4 Fitting Discount Bond Prices and Forward Rates . . . 225
9.8.5 Aside: The Instantaneous Forward Interest Rate . . . 230
9.9 Options on Discount Bonds & Interest Rates (Caps/Floors) . . . 233
9.9.1 Valuation of Options on Discount Bonds . . . 234
9.9.2 Valuation of Caps and Floors . . . 240
9.10 Other Continuous-Time Models of the Term Structure of Interest Rates . . . 244
9.10.1 General Hull and White (1990) Model . . . 246
9.10.2 Cox, Ingersoll, and Ross (1985a) Model . . . 247
9.10.3 Multifactor Cox, Ingersoll, and Ross (1985a) and Longstaff and Schwartz (1992) Models . . . 248
9.10.4 Affine Models of Duffie and Kan (1996) . . . 249
9.10.5 Essentially Affine Model of Duffee (2002) . . . 253
9.10.6 Quadratic Models . . . 254
9.10.7 Heath, Jarrow, and Morton (1992) Model . . . 255
9.11 Summary . . . 257
9.12 Appendix: Discrete-Time Models of Term Structure of Interest Rates . . . 267
9.12.1 The Black, Derman, and Toy (1990) of Dynamics of the Term Structure . . . 269
9.12.2 The Volatility Equation . . . 271
9.12.3 Building the Short-rate Tree for Black, Derman, and Toy (1990) Model . . . 272
9.12.4 Numerical Example from Black, Derman, and Toy (1990) . . . 273
9.13 Suggestions for Further Readings . . . 286
9.13.1 Source material for the lecture notes . . . 286
9.13.2 Potential applications and extensions of lecture notes . . . 287
9.13.3 Ph.D. elective courses offered on related topics . . . 289

9.1 Plan for the Day

Last Time

- Fixed-income concepts and valuation of derivatives in the Market model

Today

- **How to model term structure of interest rates and value derivatives?**
  - ⋆ The Vasicek model
  - ⋆ Valuation of derivatives in the Vasicek model
  - ⋆ Other term structure models: Hull & White; Longstaff & Schwartz; Cox, Ingersoll, Ross; Multifactor; Affine; Quadratic; Heath, Jarrow, Morton.

For Next Time

- Review the material covered in class.
- Read Back, Chapters 18 and Dumas & Luciano, Chapter 16.
- Do the practice problems.

9.2 Background

Recall, that in this course we wish to answer three questions, which are nested:

1. How to value redundant securities?

   Least general question: we are given prices of fundamental assets and the desired cashflows; need to determine only the replicating portfolio strategy.

2. How to determine optimal consumption and portfolio policies of investors?

   More general than the previous question: we are given prices of the fundamental assets; need to determine both the desired cashflows and the portfolio strategy generating these cashflows.

3. How to determine the equilibrium values in a market with investors who are optimizing their consumption and portfolio policies?

   Most general question: need to determine prices of fundamental assets, desired cashflows, and the portfolio strategy that replicates these cashflows.

9.3 Main Points for Today

1. The short rate and prices of discount bonds
2. Detailed discussion of Vasicek model
   - Estimating the Vasicek (1977) model
   - Hedging in the Vasicek model
   - Valuation of options on discount bonds, caps, and floors
3. Brief discussion of other models of the term structure
   - The continuous-time Ho and Lee (1986) and Hull and White (1990) models
   - The Cox, Ingersoll, and Ross (1985a) model
   - The Longstaff and Schwartz (1992) model
   - Multifactor models (Cox, Ingersoll, and Ross 1985a); Affine models (Duffie and Kan 1996); Completely affine models (Dai and Singleton 2000); Quadratic models (Ahn, Dittmar, and Gallant 2002)
   - The Heath, Jarrow, and Morton (1992) model

9.4 Motivation for Dynamic Models of the Term Structure of Interest Rates

- In the last class, we studied the valuation of caps, floors, and swaps under the assumption that certain forward rates have constant volatilities.
- As noted at the end of the last class, the assumptions made to value caps and floors are inconsistent with the assumptions made to value swaptions.
- In this chapter, we
  - ⋆ develop a consistent framework by modeling how the term structure will evolve over time, and
  - ⋆ then use this model to find the values of fixed-income derivatives.

9.4.1 Differences in Modeling Movements in Equity and Bond Prices

- Recall the following assumptions of the Black and Scholes (1973) model for valuing options on stocks:

  1. Price of the stock follows a lognormal distribution.
  2. The stock return has constant volatility.
  3. The interest rate is constant.

- All three of these assumptions are inconsistent with the requirements of a model for the term structure of interest rates.

1. Assumption of a lognormal distribution implies that price of the underlying asset grows indefinitely over time; but:
   - the price of a bond cannot grow indefinitely because it must revert to par at maturity;
   - moreover, price of the bond at maturity is known.
2. Assumption that volatility is constant implies that variance of stock's future price distribution grows with time; but
   - bond at maturity reverts to par, so bond price cannot have constant volatility.
3. Assumption of constant interest rate does not make sense of modeling options on interest rates (such as caps and floors).

- The above discussion suggests that term-structure modeling must be built on different foundations from that for equities and currencies.
  - ⋆ Even the choice of the underlying variable is an open question:
  - ⋆ It may make more sense to model interest rates instead of bond prices, because interest rates can have constant volatilities.
  - ⋆ May be more convenient to model the forward interest rate curve, instead of the spot curve.

- **Key concern when modeling interest rates:**
  - ⋆ When modeling a stock price, only a single quantity to keep track of;
  - ⋆ When modeling interest rates, we are modeling the entire spot curve so extra care is required to
    - ∗ limit the dimensionality of the tree;
    - ∗ ensure that the model is internally consistent and free of arbitrage.

9.4.2 State variables and factors

- When modeling interest rates, we are modeling the entire spot curve, so there are as many variables as maturities.

- To keep the stochastic dynamic model simple, we typically assume that there are only a small number (one, two, three, or at most, four) factors that influence all interest rates.
  - ⋆ These factors are often called "state variables."

- When specifying the model, it is important that the model be arbitrage free.

9.4.3 An Example of Arbitrage in a Badly Specified Dynamic Model of the Term Structure

- Below, we construct an example to demonstrate how seemingly harmless assumptions lead to models that permit arbitrage, and therefore, are not appropriate.

- Then, in the rest of the class, we study arbitrage-free term structure models in continuous time.

  ⋆ In the appendix, we describe the Black-Derman-Toy model, which is an arbitrage-free term structure models in discrete time.

- In this example, we have a binomial model of a single-factor term-structure model, so:

  ⋆ either: the entire term structure moves up (all rates move up together);

  ⋆ or: the entire term structure moves down (all rates move down together).

- That is, there are parallel shifts in the term structure.

- Time is discrete, and each period is one year long.

- The yield curve (spot interest rate curve) consists of one, two, and three-year zero-coupon yields.

- The spot curve at $t = 0$ is given by:

$$[\text{Spot interest rate curve}]_0 = \begin{bmatrix} 0.10 \\ 0.11 \\ 0.12 \end{bmatrix}.$$

- At $t = 1$, the spot curve can either move up or down.
  - ⋆ If the spot curve moves up, then all yields increase by 1%, so the new spot curve is:

$$[\text{Spot interest rate curve}]_u = \begin{bmatrix} 0.11 \\ 0.12 \\ 0.13 \end{bmatrix}.$$

  - ⋆ If the spot curve moves down, then all yields decrease by 1%, so the new spot curve is:

$$[\text{Spot interest rate curve}]_d = \begin{bmatrix} 0.09 \\ 0.10 \\ 0.11 \end{bmatrix}.$$

- We now show that this binomial model with parallel shifts in the spot curve (yield curve) results in an internally inconsistent model with arbitrage opportunities.

- Assume that zero-coupon bonds have a face value of $\$100$ at maturity.

- Consider the following trading strategy:

  At $t = 0$:
  - ∗ Buy a two-year zero-coupon bond, which costs $\frac{100}{1.11^2} = \$81.16$;
  - ∗ Finance this purchase by borrowing $\$81.16$ for one year at 10%.

  At $t = 1$:
  - ∗ Sell the two-year zero-coupon bond;
  - ∗ Repay the loan; that is, pay $\$81.16 \times 1.10 = \$89.276$.

- Now, let us look at the cashflows of this trading strategy.

- At $t = 0$, there are no cashflows because purchase of two-year zero-coupon bond is financed by borrowing money.

- At $t = 1$, the two-year zero-coupon bond is a one-year zero-coupon bond, and the one-year spot interest rate is either 0.11 or 0.09:

$$[\text{Spot curve}]_u = \begin{bmatrix} 0.11 \\ 0.12 \\ 0.13 \end{bmatrix}; \qquad [\text{Spot curve}]_d = \begin{bmatrix} 0.09 \\ 0.10 \\ 0.11 \end{bmatrix}.$$

- So, at $t = 1$ the price of the two-year zero-coupon bond, which is now a one-year zero-coupon bond, is

$$\text{in the "up" state}: \quad \frac{100}{1.11} = \$90.10; \qquad \text{in the "down" state}: \quad \frac{100}{1.09} = \$91.74.$$

- But, the amount to be repaid (see previous page) is only $\$89.276$.

- So, at $t = 1$, both in the up and in down state, we have a net cash inflow.

- That is, there is an arbitrage opportunity!

9.5 Overview and Background of Material To be Covered

- As mentioned in the last class, we will study only default-free discount bonds.

- There are two quantities to study:

  1. Prices of discount bonds,
     for which it suffices to work entirely under the risk-neutral measure.
  2. Expected returns (or risk premia) on discount bonds,
     for which, one needs the physical measure.

- Recall that these two probability measures are related via the SDF process (and Girsanov's theorem).

9.5.1 Plan: Focus on the Vasicek Model

⋆ We will focus on the basic Vasicek model.

⋆ We will then discuss extensions of the basic Vasicek models
  ∗ to allow for time-varying parameters
  ∗ that will allow for a better fit of the model to the yield curve and other market quantities.

⋆ We will conclude by briefly studying other models of the term structure that allow for multiple factors (state variables).

9.5.2 List of Topics For Each Model of the Term Structure

1. Specify the process for the interest rate, $dr(t)$;
2. Solve for the level of the short-rate, $r(t)$;
3. Use the short rate to find discount-bond price, $\mathcal{P}^Z(t,u)$;
4. Find the process (return) for discount-bond prices, $d\mathcal{P}^Z(t,u)$;
5. Use these results to value options on discount bonds and interest rates.
6. Understand how to hedge under these models.
7. Discuss how to estimate the parameters of these models.

⋆ Valuation of options and hedging depend only on the volatility of the bond-return process.

⋆ Therefore, for term-structure models that are
  ∗ more general than Vasicek model, but have same bond-return volatility,
  ∗ the *approach* for valuation of options and hedging will stay the same (but the actual answer will be different).

9.6 The Short Rate and Discount Bond Prices

- An assumption of the Vasicek and other term-structure models we discuss is that there is an instantaneously risk-free rate, denoted at time s by $r(s)$.
- This assumption means there is an asset (bank deposit) with price process

$$\mathcal{P}^D(t) = \exp\left(\int_0^t r(s)\,ds\right). \tag{9.1}$$

- The instantaneous risk-free rate of return on this asset is

$$\frac{d\mathcal{P}^D(t)}{\mathcal{P}^D(t)} = r(t)\,dt.$$

  Note that there is no random term (no Brownian), implying that we know the return at t.

- The rate $r(t)$ is called the "short rate" and $\mathcal{P}^D(t)$ is the value of the bank deposit (money-market account) or accumulation factor.

- The probability measure associated with $\mathcal{P}^D$ being the numeraire is called the risk-neutral measure.
- From the first half of the course, we recall that deflated prices are martingales under the measure corresponding to the deflator.
- Thus, under the risk neutral measure, the price $\mathcal{P}^Z(t,u)$ at date t of a discount (zero-coupon) bond maturing at u must be:

$$\frac{\mathcal{P}^Z(t,u)}{\mathcal{P}^D(t)} = \mathbb{E}_t^{\mathbb{Q}}\left[\frac{\mathcal{P}^Z(u,u)}{\mathcal{P}^D(u)}\right] \quad \ldots \text{ using the martingale property}$$

- Upon simplification of the above expression, we get:

$$\frac{\mathcal{P}^Z(t,u)}{\mathcal{P}^D(t)} = \mathbb{E}_t^{\mathbb{Q}}\left[\frac{\mathcal{P}^Z(u,u)}{\mathcal{P}^D(u)}\right] \quad \ldots \text{ martingale property} \tag{9.2}$$

$$= \mathbb{E}_t^{\mathbb{Q}}\left[\frac{1}{\mathcal{P}^D(u)}\right], \quad \ldots \text{ at maturity } \mathcal{P}^Z(u,u) = 1$$

$$\mathcal{P}^Z(t,u) = \mathcal{P}^D(t)\,\mathbb{E}_t^{\mathbb{Q}}\left[\frac{1}{\mathcal{P}^D(u)}\right] \quad \ldots \text{ multiplying both sides by } \mathcal{P}^D(t)$$

$$= \left[\exp\left(\int_0^t r(s)\,ds\right)\right]\,\mathbb{E}_t^{\mathbb{Q}}\left[\frac{1}{\exp\left(\int_0^u r(s)\,ds\right)}\right] \quad \ldots \text{ using definition of } \mathcal{P}^D \text{ in (9.1)}$$

$$= \mathbb{E}_t^{\mathbb{Q}}\left[\exp\left(-\int_t^u r(s)\,ds\right)\right] \quad \ldots \text{ canceling out integral from 0 to } t \text{ in numer. and denom.} \tag{9.3}$$

- Assume that there is an SDF process M such that $M\,\mathcal{P}^D$ is a martingale and that $M\,\mathcal{P}^Z(T)$ is a martingale. Then,

$$\mathcal{P}^Z(t,u) = \mathbb{E}_t^{\mathbb{P}}\left[\frac{M_T}{M_t}\,1\right] = \mathbb{E}_t^{\mathbb{Q}}\left[\exp\left(-\int_t^u r(s)\,ds\right)\right].$$

- Thus, the two expressions above show that prices of discount bonds can be obtained in two ways:
  - ⋆ either by modeling SDF process M under physical probability measure $\mathbb{P}$,
  - ⋆ or by modeling the short rate under the risk-neutral measure $\mathbb{Q}$.
- And, once we have prices of discount bonds, we can obtain the price of any (default-free) coupon bond.

9.7 The Basic Vasicek Model

- In the basic Vasicek (1977) model, it is assumed that the short rate follows the process:

$$dr(t) = \kappa \left[\theta - r(t)\right] dt + \sigma dB^{\mathbb{Q}}(t), \tag{9.4}$$

where:

- ⋆ $B^{\mathbb{Q}}$ is a Brownian motion under the risk-neutral measure $\mathbb{Q}$;
- ⋆ $\kappa \geq 0$, θ, and σ are constants;
- ⋆ κ is the rate of mean reversion;
- ⋆ θ is the long-run mean of the short rate, and so when $r(t) < \theta$, the drift term will be positive and will push r toward θ;
- ⋆ σ is the "volatility" of the short rate.

9.7.1 The Short Rate and Its Distribution for the Vasicek Process

- On the previous slide, we are given

$$dr(t) = \kappa \left[\theta - r(t)\right] dt + \sigma dB^{\mathbb{Q}}(t). \tag{9.4}$$

- We now wish to derive the expression for the:
  1. level of $r(t)$,
  2. mean of $r(t)$, and
  3. variance of $r(t)$.

- We will then use these results to identify the price of discount bonds:

$$\mathcal{P}^Z(t,u) = \mathbb{E}_t^{\mathbb{Q}}\left[\exp\left(-\int_t^u r(s)\, ds\right)\right].$$

Deriving the level of the short rate $r(t)$

- Define a new variable that will simplify the derivation: $Y(t, r_t) = e^{\kappa t} r_t$.

- We want to apply Itô's formula to $Y(t, r_t) = e^{\kappa t} r_t$, we can first compute the partial derivatives we will need:

$$\frac{\partial Y}{\partial t} = r_t \frac{\partial e^{\kappa t}}{\partial t} = r_t \kappa e^{\kappa t};$$

$$\frac{\partial Y}{\partial r} = e^{\kappa t} \frac{\partial r}{\partial r} = e^{\kappa t} 1;$$

$$\frac{\partial^2 Y}{\partial r^2} = 0.$$

- We now use Itô's formula to get the process for the interest rate, dr_t:

$$\begin{aligned}
dY(t,r) &= \frac{\partial Y}{\partial t} dt + \frac{\partial Y}{\partial r} dr + \frac{1}{2} \frac{\partial^2 Y}{\partial r^2} (dr)^2 && \text{... using Itô's formula} \\
&= r_t \frac{\partial e^{\kappa t}}{\partial t} dt + e^{\kappa t} \frac{\partial r}{\partial r} dr + 0 && \text{... differentiating } Y(t,r) \\
&= r_t \kappa e^{\kappa t} dt + e^{\kappa t} 1 \left(\kappa \left[\theta - r_t \right] dt + \sigma \, dB^{\mathbb{Q}}(t) \right) && \text{... substituting for } dr \\
&= r_t \kappa e^{\kappa t} dt - r_t \kappa e^{\kappa t} dt + \kappa e^{\kappa t} \theta \, dt + e^{\kappa t} \sigma dB^{\mathbb{Q}}(t) && \text{... expanding parenthesis} \\
&= e^{\kappa t} \left[\kappa \theta \, dt + \sigma dB^{\mathbb{Q}}(t) \right] && \text{... canceling out terms}
\end{aligned}$$

- Starting with the last equation on the previous slide, integrate and simplify to get the expression for the interest rate level:

$$dY(t, r_t) = e^{\kappa t}\left[\kappa\,\theta\,dt + \sigma dB_t^{\mathbb{Q}}\right] \quad \text{. . . from previous page}$$

$$Y_t - Y_0 = \kappa\,\theta\int_0^t e^{\kappa s}\,ds + \sigma\int_0^t e^{\kappa s} dB_s^{\mathbb{Q}} \quad \text{. . . integrating both sides}$$

$$e^{\kappa t}\,r_t - r_0 = \kappa\,\theta\int_0^t e^{\kappa s}\,ds + \sigma\int_0^t e^{\kappa s} dB_s^{\mathbb{Q}} \quad \text{. . . using definition } Y(t,r_t) = e^{\kappa t} r_t$$

$$e^{\kappa t}\,r_t - r_0 = \theta\left[e^{\kappa t} - 1\right] + \sigma\int_0^t e^{\kappa s} dB_s^{\mathbb{Q}} \quad \text{. . . simplifying the first integral}$$

$$e^{\kappa t}\,r_t = r_0 + \theta\left[e^{\kappa t} - 1\right] + \sigma\int_0^t e^{\kappa s} dB_s^{\mathbb{Q}} \quad \text{. . . adding } r_0 \text{ to both sides}$$

$$r_t = r_0\,e^{-\kappa t} + \theta\left[1 - e^{-\kappa t}\right] + \sigma\int_0^t e^{\kappa(s-t)} dB_s^{\mathbb{Q}} \quad \text{. . . dividing both sides by } e^{\kappa t}$$

- Reproducing the result on the previous page

$$r_t = \underbrace{r_0\,e^{-\kappa t} + \theta\left[1 - e^{-\kappa t}\right]}_{\text{deterministic}} + \underbrace{\sigma\int_0^t e^{\kappa(s-t)} dB_s^{\mathbb{Q}}}_{\text{stochastic}}. \tag{9.5}$$

- The result above gives us a nice expression for the level of the short rate, r_t.
- We now use this result to derive the mean and variance of the short rate.

Deriving the mean of the short rate $r(t)$

- We start by reproducing the result we just derived:

$$r_t = \underbrace{r_0\, e^{-\kappa t} + \theta\left[1 - e^{-\kappa t}\right]}_{\text{deterministic}} + \underbrace{\sigma \int_0^t e^{\kappa\,(s-t)} dB_s^{\mathbb{Q}}}_{\text{stochastic}}. \tag{9.5}$$

- To obtain the mean,
  - ⋆ we take expectations, and
  - ⋆ use the fact that the stochastic integral with respect to a Brownian motion is a martingale starting at zero, (that is, the expected value of $dB_s^{\mathbb{Q}} = 0$)
  - ⋆ which leaves us with just the deterministic part of the expression above:

$$\mathbb{E}^{\mathbb{Q}}[r_t] = r_0\, e^{-\kappa t} + \theta\left(1 - e^{-\kappa t}\right). \tag{9.6}$$

Deriving the variance of the short rate $r(t)$

- We start by reproducing the result for the level of the short rate:

$$r_t = \underbrace{r_0\, e^{-\kappa t} + \theta\left[1 - e^{-\kappa t}\right]}_{\text{deterministic}} + \underbrace{\sigma \int_0^t e^{\kappa\,(s-t)} dB_s^{\mathbb{Q}}}_{\text{stochastic}}. \tag{9.5}$$

- To obtain the variance, we take expectations of the square, and then use a result called Itô isometry:

$$\begin{aligned}
\text{var}(r_t) &= \sigma^2 \mathbb{E}\left\{\left(\int_0^t e^{\kappa\,(s-t)} dB_s^{\mathbb{Q}}\right)^2\right\} && \text{... variance equals expectation of the square of } r_t \\
&= \sigma^2 \mathbb{E}\left\{\int_0^t e^{2\kappa\,(s-t)}\, ds\right\} && \text{... switch order of squaring and integrating using Itô isometry} \\
&= \frac{\sigma^2}{2\kappa}\left(1 - e^{-2\kappa t}\right) && \text{... find the definite integral}
\end{aligned} \tag{9.7}$$

Aside: Understanding Itô isometry (from Munk, p. 47)

- We wish to prove the following result that we used above, which says that
  - ⋆ instead of first integrating and then squaring the integral, one can first square the integrand and then integrate (this result applied for any measure, so we write B_s instead of for $B_s^{\mathbb{Q}}$):

$$\text{var}_t\left[\int_t^T \sigma_s\, dB_s\right] = \mathbb{E}_t\left[\left(\int_t^T \sigma_s\, dB_s\right)^2\right] \quad \ldots \text{ because variance = expected value of square}$$

$$= \mathbb{E}_t \int_t^T \left[\sigma_s^2\right] ds \quad \ldots (dB_s)^2 = ds \text{ (usually denoted } dt)$$

$$= \int_t^T \mathbb{E}_t\left[\sigma_s^2\right] ds. \quad \ldots \text{ interchanging order of integration and expectation}$$

- Now,
  - ⋆ suppose that σ is piecewise constant and
  - ⋆ we divide the time interval $[t, T]$ into subintervals defined by the time points $t \equiv t_0 < t_1 < \cdots < \cdots t_n \equiv T$
  - ⋆ so that σ is constant on each subinterval $[t_{i,t_{i+1}})$ with a value σ_i that is known at time t_i (this is our usual assumption, needed for Itô's calculus).
- Then, we get the following:

$$\mathbb{E}_t\left[\left(\int_t^T \sigma_s\, dB_s\right)^2\right] \quad \text{. . . starting point to prove Itô isometry}$$

$$= \mathbb{E}_t\left[\sum_{i=0}^{n-1}\sum_{j=0}^{n-1}\sigma_{t_i}\sigma_{t_j}(B_{t_{i+1}} - B_{t_i})(B_{t_{j+1}} - B_{t_j})\right] \quad \text{. . . expressing the integral squared as a double summation}$$

$$= \sum_{i=0}^{n-1}\sum_{j=0}^{n-1}\mathbb{E}_t\left[\sigma_{t_i}\sigma_{t_j}(B_{t_{i+1}} - B_{t_i})(B_{t_{j+1}} - B_{t_j})\right] \quad \text{. . . moving the expectations operator inside the summations}$$

$$= \sum_{i=0}^{n-1}\mathbb{E}_t\left[\sigma_{t_i}^2(B_{t_{i+1}} - B_{t_i})^2\right] \quad \text{. . . separating the variance terms (i.e., terms on this line)}$$

$$+ 2\sum_{i=0}^{n-1}\sum_{j=i+1}^{n-1}\mathbb{E}_t\left[\sigma_{t_i}\sigma_{t_j}(B_{t_{i+1}} - B_{t_i})(B_{t_{j+1}} - B_{t_j})\right] \quad \text{. . . from the covariance terms (i.e., terms on this line)}$$

$$= \sum_{i=0}^{n-1}\mathbb{E}_t\left[\sigma_{t_i}^2(t_{i+1} - t_i)\right] + 2\sum_{i=0}^{n-1}\sum_{j=i+1}^{n-1}\mathbb{E}_t\left[\sigma_{t_i}\sigma_{t_j}0\right] \quad \text{. . . } (B_{t_{i+1}} - B_{t_i})^2 = (t_{i+1} - t_i)$$

$$= \int_t^T \mathbb{E}_t\left[\sigma_s^2\right] ds. \quad \text{. . . END of aside on Itô isometry.}$$

Summarizing our main results so far

- Re-writing the main results derived so far:

$$\begin{aligned} r_t &= r_0\, e^{-\kappa t} + \theta\left[1 - e^{-\kappa t}\right] + \sigma\int_0^t e^{\kappa(s-t)}dB_s^{\mathbb{Q}} \\ &= \theta - e^{-\kappa t}\left[\theta - r_0\right] + \sigma\int_0^t e^{\kappa(s-t)}dB_s^{\mathbb{Q}} \end{aligned} \tag{9.5}$$

$$\mathbb{E}^{\mathbb{Q}}[r_t] = \theta - e^{-\kappa t}\left[\theta - r_0\right]; \tag{9.6}$$

$$\text{var}(r_t) = \frac{\sigma^2}{2\kappa}\left(1 - e^{-2\kappa t}\right). \tag{9.7}$$

- From Equation (9.5), we see that
  - ⋆ the interest rate has a Normal distribution (because $dB_s^{\mathbb{Q}}$ is Normal), with mean given in Equations (9.6), and variance given in (9.7);
  - ⋆ As $t \to \infty$, the mean converges to θ, and the variance converges to $\sigma^2/2\kappa$ (because as $t \to \infty$, the term $e^{-\kappa t} \to 0$).

- On the previous slide, we said that the interest rate in the Vasicek model is Normally distributed with

$$\text{var}(r_t) = \frac{\sigma^2}{2\kappa}\left(1 - e^{-2\kappa t}\right). \tag{9.7}$$

- Because the short rate is Normally distributed, there is a finite probability that the interest rate can become negative at some date $t > 0$,
  - ⋆ which (most people believed) should not happen for nominal interest rates
  - ⋆ but in the last few years several countries have had slightly negative nominal interest rates.
- The probability of negative interest rates will be small if the variance is small, which we see from (9.7) will be the case if
  - ⋆ either σ is small,
  - ⋆ or κ is large.

- For the special case where $\kappa = 0$, then $e^{-\kappa t} = 1$, and we get that:

$$\begin{aligned} r_t &= \theta - e^{-\kappa t}\left[\theta - r_0\right] + \sigma \int_0^t e^{\kappa(s-t)} dB_s^{\mathbb{Q}} \\ &= r_0 + \sigma\left[B^{\mathbb{Q}}(t) - B^{\mathbb{Q}}(0)\right]; \end{aligned} \tag{9.8}$$

$$\begin{aligned} \mathbb{E}^{\mathbb{Q}}[r_t] &= \theta - e^{-\kappa t}\left[\theta - r_0\right] \\ &= r_0; \end{aligned} \tag{9.9}$$

$$\begin{aligned} \text{var}(r_t) &= \frac{\sigma^2}{2\kappa}\left(1 - e^{-2\kappa t}\right) = \sigma^2\left(\frac{1 - e^{-2\kappa t}}{2\kappa}\right) \\ &= \sigma^2 t \quad \ldots \text{using L'Hôpital's rule: } \lim_{\kappa \to 0} \frac{1 - e^{-2\kappa t}}{2\kappa} = t. \end{aligned} \tag{9.10}$$

- So, for the special case where $\kappa = 0$, the variance $\sigma^2 t$ increases as time increases.

9.7.2 Bond Prices under the Vasicek Process

- We now wish to use the results derived above to obtain the price of a zero-coupon bond in the Vasicek model:

$$\mathcal{P}^Z(t,u) = \underbrace{\mathbb{E}_t^{\mathbb{Q}}\underbrace{\left[\exp\underbrace{\left(-\int_t^u \underbrace{r(s)}_{(1)}\,ds\right)}_{(2)}\right]}_{(3)}}_{(4)}.$$

- This will involve the following four steps:

(1) Start with the expression for the level of the interest rate, $r(s)$;

(2) Integrate this;

(3) Take the exponential of the negative of the resulting expression;

(4) Find the expectation of that exponential.

- Below, and also later in the course: $\tau = u - t$.

Derivation of the Bond Price for the Special Case of $\kappa = 0$

- **Step 1:** We start with (9.8), the expression for the short rate when $\kappa = 0$:

$$r(s) = r(t) + \sigma \int_t^s dB^{\mathbb{Q}}(y), \quad \ldots \text{where } y \text{ is the dummy variable for time}$$

- **Step 2:** Integrating from t to u the expression for level of the interest rate:

$$\int_t^u r(s)\,ds = \int_t^u \left[r(t) + \sigma \int_t^s dB^{\mathbb{Q}}(y) \right] ds \qquad \ldots \text{integrating the expression for } r(s)$$

$$= \int_t^u r(t) ds + \int_t^u \sigma \int_t^s dB^{\mathbb{Q}}(y) ds \qquad \ldots \text{separating the two terms to be integrated}$$

We start by simplifying the first term:

$$= r(t)\, s|_t^u + \sigma \int_t^u \left[\int_t^s dB^{\mathbb{Q}}(y) \right] ds \qquad \ldots \text{first term is constant, so its integral is } s$$

$$= r(t)(u-t) + \sigma \int_t^u \left[\int_t^s dB^{\mathbb{Q}}(y) \right] ds \qquad \ldots \text{using the limits of integration}$$

$$= \tau\, r(t) + \sigma \int_t^u \left[\int_t^s dB^{\mathbb{Q}}(y) \right] ds \qquad \ldots \text{substituting } \tau = (u-t)$$

- Next, we simplify the second term:

$$\int_t^u r(s)\,ds = \tau\, r(t) + \sigma \int_t^u \left[\int_t^s dB^{\mathbb{Q}}(y) \right] ds$$

$$= \tau\, r(t) + \sigma \int_t^u \left[\int_y^u ds \right] dB^{\mathbb{Q}}(y) \qquad \ldots \text{changing order of integration (and limits of integration)}$$

$$= \tau\, r(t) + \sigma \int_t^u (u-y) dB^{\mathbb{Q}}(y) \qquad \ldots \text{integrate the inner term (i.e. the term in red)}$$

- Looking at just the second term, $\sigma \int_t^u (u-y) dB^{\mathbb{Q}}(y)$, we see that, conditional on information at date t, this term is
  - ⋆ a Normally-distributed random variable, with
  - ⋆ mean equal to zero, and
  - ⋆ variance equal to $\sigma^2 \int_t^u (u-y)^2 [dB^{\mathbb{Q}}(y)]^2 = \sigma^2 \int_t^u (u-y)^2 dy = \frac{\sigma^2 \tau^3}{3}$.

- Therefore, the entire term,

$$\int_t^u r(s)ds = \tau\, r(t) + \sigma \int_t^u \left[\int_t^s dB^{\mathbb{Q}}(y)\right] ds$$

  is Normally distributed with

  ⋆ mean given by $\tau\, r(t)$, and

  ⋆ variance given by $\frac{\sigma^2\tau^3}{3}$.

- **Step 3:** Recall that we wish to evaluate the expectation of the exponential of the negative of this expression, shown below:

$$\mathcal{P}^Z(t,u) = \mathbb{E}_t^{\mathbb{Q}}\left[\exp\left(-\int_t^u r(s)\, ds\right)\right]$$

- **Step 4:** To evaluate this expectation, we use the fact that

  ⋆ if a variable x is distributed Normally with mean μ and variance σ^2,

  ⋆ then the expectation of e^x is $e^{\mu+\frac{1}{2}\sigma^2}$.

- On the previous slide, we showed that $\int_t^u r(s)ds$ is Normally distributed with

  ⋆ mean given by $\tau r(t)$, and

  ⋆ variance given by $\frac{\sigma^2\tau^3}{3}$.

- Thus, $-\int_t^u r(s)ds$ is Normally distributed with

  ⋆ mean given by $-\tau\, r(t)$, and

  ⋆ variance given by $\frac{\sigma^2\tau^3}{3}$.

- So, the expectation of $e^{-\int_t^u r(s)ds}$ is: $e^{-\tau\, r(t)+\frac{1}{2}\frac{\sigma^2\tau^3}{3}} = e^{-\tau\, r(t)+\frac{\sigma^2\tau^3}{6}}$.

Final expression for the price of a zero-coupon bond

- Thus, in the special case in which $\kappa = 0$:

$$\begin{aligned}\mathcal{P}^Z(t,u) &= \mathbb{E}_t^{\mathbb{Q}}\Big[\exp\Big(-\int_t^u r(s)\,ds\Big)\Big] \\ &= \exp\Big[-\tau\, r(t) + \frac{\sigma^2\tau^3}{6}\Big] \\ &= \exp\Big[-a(\tau) - b(\tau)\, r(t)\Big]\end{aligned}$$

where

$$\begin{aligned}a(\tau) &= \frac{-\sigma^2\tau^3}{6}, \\ b(\tau) &= \tau.\end{aligned}$$

- And, for the more general case in which $\kappa > 0$:

$$\mathcal{P}^Z(t,u) = \exp\Big[-a(\tau) - b(\tau) r(t)\Big], \tag{9.11}$$

where

$$a(\tau) = \theta\,\tau - \frac{\theta}{\kappa}\left(1 - e^{-\kappa\tau}\right) - \frac{\sigma^2}{4\kappa^3}\left(2\kappa\tau - e^{-2\kappa\tau} + 4e^{-\kappa\tau} - 3\right), \tag{9.12}$$

$$b(\tau) = \frac{1}{\kappa}\left(1 - e^{-\kappa\tau}\right). \tag{9.13}$$

Aside: Example of changing order of integration in Newtonian calculus

- Suppose we wish to integrate the function $\exp(y^2)$ over the region

$$0 \leq x \leq 2 \quad \text{and} \quad \frac{x}{2} \leq y \leq 1;$$

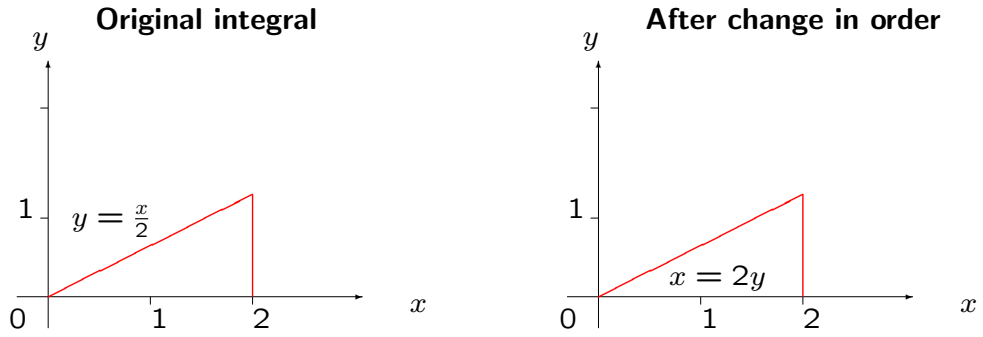
that is, we wish to compute

$$\int_0^2 \int_{x/2}^1 \exp(y^2)\, dy\, dx.$$

- Unfortunately, it is not possible (that is, the fundamental theorem of calculus does not tell us how) to compute the integral:

$$\int_{x/2}^1 \exp(y^2)\, dy.$$

- So, one approach for solving this problem is to change the order of integration.



- To do this, we need to integrate over the region

$$0 \leq y \leq 1 \quad \text{and} \quad 0 \leq x \leq 2y,$$

which means that the integral to be computed is given by:

$$\int_0^1 \int_0^{2y} \exp(y^2)\, dx\, dy.$$

- This integral can be computed in a straightforward manner after changing the order of integration:

$$\int_0^1 \int_0^{2y} \exp(y^2)\, dx\, dy$$

$$= \int_0^1 \exp(y^2) \int_0^{2y} 1\, dx\, dy \quad \ldots \exp(y^2) \text{ can be taken out of the "inner" integral because it does not depend on } x$$

$$= \int_0^1 \exp(y^2)\ x\Big|_0^{2y} dy \quad \ldots \text{integrating 1 against } x \text{ gives } x$$

$$= \int_0^1 2y \exp(y^2)\ dy \quad \ldots \text{evaluating } x \text{ at the upper limit gives } 2y\text{; and at the lower limit gives 0}$$

$$= \exp(y^2)\Big|_0^1 \quad \ldots \text{integral of } 2y \exp(y^2) \text{ against } y \text{ is } \exp(y^2)\text{; can verify by differentiating } \exp(y^2)$$

$$= \exp(1^2) - \exp(0^2) \quad \ldots \text{evaluating } \exp(y^2) \text{ at upper limit gives } \exp(1^2)\text{; at lower limit gives } \exp(0^2)$$

$$= e - 1.$$

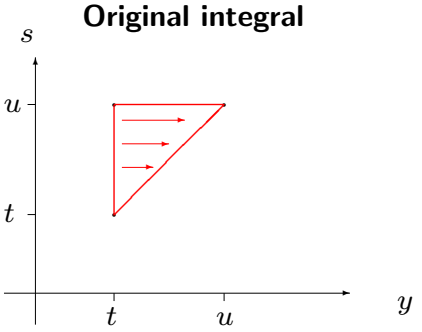
Understanding the change in the order of integration for our problem

- Re-writing the second term, the change in the order of integration gives

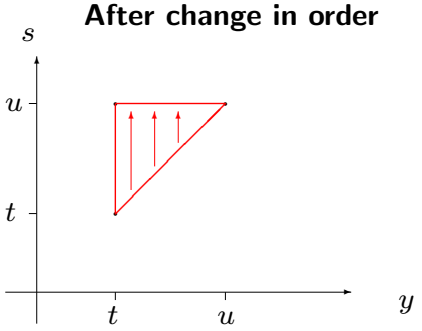
$$\int_t^u \left[\int_t^s dB(y)\right] ds = \int_t^u \left[\int_y^u ds\right] dB(y)$$

$$= \int_t^u (u-y) dB(y).$$

Original integral



After change in order



9.7.3 Hedging in the Vasicek Model

- The Vasicek model has a single Brownian motion driving the short interest rate.

- Thus, there is only a single Brownian motion driving all bond prices, and so the bond prices are perfectly correlated in the Vasicek model.

  ⋆ This will be true in any model in which there is only a single factor, such as the short rate, that determines all bond prices.

- Therefore, the hedge ratio for hedging one bond with another will depend only on the relative volatilities of the bond returns.

- We show this formally on the next few pages.

- Using the expression for the bond price in (9.11), $\mathcal{P}^Z(t,u) = e^{-a(u-t)-b(u-t)r(t)}$, one can show that for any date t and any fixed maturity date $u > t$:

$$\frac{d\mathcal{P}^Z(t,u)}{\mathcal{P}^Z(t,u)} = r(t)dt - \sigma\, b(u-t)dB^{\mathbb{Q}}(t). \tag{9.14}$$

- We will derive (9.14) for the special (simpler) case in which $\kappa = 0$ on the next page.

- Note that in Equation (9.14)

  ⋆ $b(u-t)$ is the same as $b(\tau)$, just written explicitly for $\tau = u - t$;

  ⋆ the drift term is the short rate because under the risk-neutral measure all assets must earn the risk-free rate of return;

  ⋆ the volatility is a non-random function of the date t, given the maturity u.

Derivation of the bond return in (9.14) for the case where $\kappa = 0$

- Defining $\tau = u - t$, we have already shown in (9.11) that for the case $\kappa = 0$,

$$\mathcal{P}^Z(t,u) = g(t,r) = \exp\left[\frac{\sigma^2}{6}(u-t)^3 - (u-t)r(t)\right]. \qquad (9.11)$$

- We first work out the partial derivatives that we will need for Itô's formula:

$$\begin{aligned} g_t &= g(t,r) \times \left[\frac{-\sigma^2}{2}(u-t)^2 + r(t)\right]; \\ g_r &= g(t,r) \times [-(u-t)]; \\ g_{rr} &= g(t,r) \times \left[(u-t)^2\right]; \end{aligned}$$

- And, we see from (9.4) reproduced below, that for the case of $\kappa = 0$,

$$dr(t) = \kappa\left[\theta - r(t)\right]dt + \sigma dB^{\mathbb{Q}}(t) = \sigma\, dB^{\mathbb{Q}}(t). \quad \ldots \text{because } \kappa = 0 \qquad (9.4)$$

- Then:
  - ⋆ applying Itô's formula to $\mathcal{P}^Z$ and dividing both sides by $\mathcal{P}^Z(t,u) = g(t,r)$
  - ⋆ substituting into Itô's formula the partial derivatives derived above, and
  - ⋆ simplifying by canceling out the terms (in red) involving $(u-t)^2$,

  we get the following expression for the return on the bond:

$$\begin{aligned} \frac{d\mathcal{P}^Z}{\mathcal{P}^Z} &= \frac{1}{g(t,r)}\left(g_t dt + g_r dr(t) + \frac{1}{2}g_{rr}\left[dr(t)\right]^2\right) \\ &= \left[\frac{-\sigma^2}{2}(u-t)^2 + r(t)\right]dt + \left[-(u-t)\right]\sigma dB^{\mathbb{Q}}(t) + \frac{\sigma^2}{2}\left[(u-t)^2\right]dt \\ &= r(t)dt - (u-t)\sigma dB^{\mathbb{Q}}(t). \qquad (9.15) \end{aligned}$$

- Thus, the volatility of the return on the bond depends on $(u-t)\sigma$.
- Thus, the ratio of volatilities of bonds of different maturities will equal the ratio of their maturities.

Hedging a short two-year bond with a long one-year bond

- Consider the problem of hedging one unit of a two-year bond that we have sold, by going long a one-year bond.
- Suppose we hold x units of the one-year bond, to hedge the one unit of the two-year bond that we have sold.
- Finally, we finance the difference in the value of the two positions, by borrowing at the instantaneously risk-free rate, the amount

$$x\,\mathcal{P}^Z(t,t+1) - 1\,\mathcal{P}^Z(t,t+2).$$

- We now wish to calculate the change in the value of our portfolio.

- We re-write the general expression we have derived for the change in the price of a bond in Equation (9.14) on Page 200:

$$d\mathcal{P}^Z(t,u) = \mathcal{P}^Z(t,u)\,r(t)dt - \mathcal{P}^Z(t,u)\,\sigma\,b(u-t)dB^{\mathbb{Q}}(t). \qquad (9.14)$$

- According to this, the change in the value of the short two-year bond will be

$$-1\,d\mathcal{P}^Z(t,t+2) = -\left[\mathcal{P}^Z(t,t+2)\,r(t)dt - \mathcal{P}^Z(t,t+2)\,\sigma\,b(2)dB^{\mathbb{Q}}(t)\right].$$

- Similarly, the change in the value of x one-year bonds will be

$$x\,d\mathcal{P}^Z(t,t+1) = x\left[\mathcal{P}^Z(t,t+1)\,r(t)dt - \mathcal{P}^Z(t,t+1)\,\sigma\,b(1)dB^{\mathbb{Q}}(t)\right].$$

- Finally, the interest that has to be paid on the amount borrowed is

$$\left[x\mathcal{P}^Z(t,t+1) - \mathcal{P}^Z(t,t+2)\right] r(t)dt.$$

- Then, the change in the value of this position will be given by

$$\Big[x\,\mathcal{P}^Z(t,t+1)\,r(t)dt - x\,\mathcal{P}^Z(t,t+1)\,\sigma\,b(1)dB^{\mathbb{Q}}(t)\Big]$$
$$-\Big[\mathcal{P}^Z(t,t+2)\,r(t)dt - \mathcal{P}^Z(t,t+2)\,\sigma\,b(2)dB^{\mathbb{Q}}(t)\Big]$$
$$-\Big[x\mathcal{P}^Z(t,t+1) - \mathcal{P}^Z(t,t+2)\Big]\,r(t)dt$$

- We first look at the terms multiplying $r(t)$ in the expression above, and we see that they add up to zero (they cancel out).

- We would like the terms multiplying the Brownian $dB^{\mathbb{Q}}(t)$ to also be zero:

$$-x\,\mathcal{P}^Z(t,t+1)\,\sigma\,b(1)dB(t) + \mathcal{P}^Z(t,t+2)\,\sigma\,b(2)dB(t) = 0.$$

- So, we need to pick x to be:

$$x = \frac{b(2)}{b(1)}\frac{\mathcal{P}^Z(t,t+2)}{\mathcal{P}^Z(t,t+1)}.$$

- For the special case where $\kappa = 0$, we have that $b(u-t) = u-t$, so the above simplifies to

$$x = \frac{b(2)}{b(1)}\frac{\mathcal{P}^Z(t,t+2)}{\mathcal{P}^Z(t,t+1)} = \frac{2}{1}\frac{\mathcal{P}^Z(t,t+2)}{\mathcal{P}^Z(t,t+1)},$$

  which means that the dollar value of $x\,\mathcal{P}^Z(t,t+1)$ of the holding in the one-year bond is twice that of the short position in the two-year bond.

- The intuition for this is simple: one holds twice as much of the one-year bond because it is half as volatile as the two year bond in this model.

- By the same reasoning, one can compute a perfect hedge for any discount bond by using a discount bond of another maturity.

- In fact, because of the simplicity of the model, one can hedge any term structure derivative in this model by investing the right amount in a bond of any maturity.

Relation Between Duration Hedging and Hedging in Vasicek Model

- There is a close connection between hedging in the Vasicek model and duration hedging, as discussed earlier.

- To see this, consider a coupon bond that at date t has price $\mathcal{P}^B(t)$, and remaining cash flows C_1, C_2, C_n at dates $t+\tau_1 < t+\tau_2 < \cdots < t+\tau_n$.

- Its price should satisfy:

$$\mathcal{P}^B(t) = \sum_{j=1}^{n} C_j \, \mathcal{P}^Z(t, t+\tau_j).$$

- In the Vasicek model, the random part of the return on the coupon bond is

$$-\sigma \left(\sum_{j=1}^{n} b(\tau_j) \frac{C_j \, \mathcal{P}^Z(t, t+\tau_j)}{\mathcal{P}^B(t)} \right) dB^{\mathbb{Q}}(t). \tag{9.16}$$

- For the simple case where $\kappa = 0$, we have that $b(\tau_j) = \tau_j$, so the random part of the return of the coupon bond is

$$-\sigma \left(\sum_{j=1}^{n} \tau_j \times \underbrace{\frac{C_j \, \mathcal{P}^Z(t, t+\tau_j)}{\mathcal{P}^B(t)}}_{\text{weights}} \right) dB^{\mathbb{Q}}(t).$$

- The factor in parenthesis is DurationYields that we defined earlier in Equation (8.3) on Page 62 of Chapter 8.

- Denoting it now as $D(t)$, and recognizing that under the risk-neutral measure the expected rate of return must be the short rate, we have that in the Vasicek model with $\kappa = 0$, the return on a coupon bond is

$$\frac{d\mathcal{P}^B(t)}{\mathcal{P}^B(t)} = r(t)\, dt - \sigma D(t)\, dB^{\mathbb{Q}}(t).$$

- It follows that in this model one can hedge any fixed-income liability by holding enough of any coupon-bond such that the dollar value multiplied by its duration equals the dollar value of the liability multiplied by its duration.

- Recall that we saw in our earlier discussion that duration hedging works for parallel shifts in the yield curve.
  - ⋆ In the Vasicek model with $\kappa = 0$, only parallel shifts in the yield curve are possible.
  - ⋆ This is the reason that duration matching works in this model.

- For the general case where $\kappa > 0$, one can interpret the volatility of the coupon bond similarly.
  - ⋆ The analogue to duration (that is, the weighted average of τ_j) in the more general model is the weighted average of $b(\tau_j)$, as shown in Equation (9.16) on Page 207.

9.7.4 Estimating the Vasicek Model

- One way to estimate the parameters $\{\kappa, \theta, \sigma\}$ of the Vasicek model is to imply them from market bond prices using the formula in Equation (9.11).

$$\mathcal{P}^Z(t,u) = \exp\left[-a(\tau) - b(\tau)r(t)\right], \tag{9.11}$$

$$a(\tau) = \theta\tau - \frac{\theta}{\kappa}\left(1 - e^{-\kappa\tau}\right) - \frac{\sigma^2}{4\kappa^3}\left(2\kappa\tau - e^{-2\kappa\tau} + 4e^{-\kappa\tau} - 3\right),$$

$$b(\tau) = \frac{1}{\kappa}\left(1 - e^{-\kappa\tau}\right).$$

  - ⋆ This is analogous to implied volatility for stock options.
  - ⋆ It can also be viewed as an alternative to fitting a cubic spline.
  - ⋆ To implement this one needs to choose some proxy in the real world that corresponds to the theoretical short rate; for example,
    - ∗ the Federal Funds rate,
    - ∗ or the yield of a one-month or three-month Treasury bill.

- An alternative is to estimate the parameters from historical short rate data.
  - ⋆ For this approach, too, one needs to choose a proxy for the short rate.
  - ⋆ A second problem is that the Vasicek Equation (9.4), reproduced below,

$$dr(t) = \kappa \Big[\theta - r(t)\Big]\, dt + \sigma dB^{\mathbb{Q}}(t), \tag{9.4}$$

    characterizes the evolution of the short rate under the risk-neutral measure whereas the historical data is under the actual (physical) measure $\mathbb{P}$.
  - ⋆ So, one needs to understand how to go from the historical data to the model under the risk-neutral measure.
  - ⋆ We see how to do this on the next slide.

- Start with Equation (9.4), which is under the risk-neutral measure, $B^{\mathbb{Q}}$:

$$dr(t) = \kappa \Big[\theta - r(t)\Big]\, dt + \sigma dB^{\mathbb{Q}}(t). \tag{9.4}$$

- Vasicek actually wrote the original model under the physical measure, $B^{\mathbb{P}}$.

$$dr(t) = \kappa^{\mathbb{P}} \Big[\theta^{\mathbb{P}} - r(t)\Big]\, dt + \sigma^{\mathbb{P}} dB^{\mathbb{P}}(t). \tag{9.17}$$

- Vasicek assumed a constant market price of risk, λ, which implies that the relation between the actual and risk-neutral measures is:

$$dB^{\mathbb{P}} = \lambda\, dt + dB^{\mathbb{Q}}. \tag{9.18}$$

- Substituting Equation (9.18) in (9.17), gives:

$$dr(t) = \kappa^{\mathbb{P}} \Big[\theta^{\mathbb{P}} - r(t)\Big]\, dt + \sigma^{\mathbb{P}} \Big(\lambda\, dt + dB^{\mathbb{Q}}(t)\Big). \tag{9.19}$$

- We start by writing (9.4) and (9.19):

$$dr(t) = \kappa\left[\theta - r(t)\right]dt + \sigma dB^{\mathbb{Q}}(t) \tag{9.4}$$

$$dr(t) = \kappa^{\mathbb{P}}\left[\theta^{\mathbb{P}} - r(t)\right]dt + \sigma^{\mathbb{P}}\left(\lambda\,dt + dB^{\mathbb{Q}}(t)\right) \quad \ldots \text{from (9.18): } dB^{\mathbb{P}} = \lambda\,dt + dB^{\mathbb{Q}}$$

$$= \left(\kappa^{\mathbb{P}}\left[\theta^{\mathbb{P}} + \sigma^{\mathbb{P}}\lambda/\kappa^{\mathbb{P}} - r(t)\right]\right)dt + \sigma^{\mathbb{P}}dB^{\mathbb{Q}}(t). \quad \ldots \text{collecting } dt \text{ and } dB^{\mathbb{Q}} \text{ terms} \tag{9.19}$$

- Note that the left-hand side of Equations (9.4) and (9.19) are the same. The right-hand sides will be the same only if:
  - ⋆ $\sigma = \sigma^{\mathbb{P}}$ (in order to match the coefficient of $dB^{\mathbb{Q}}(t)$),
  - ⋆ $\kappa = \kappa^{\mathbb{P}}$ (in order to match the coefficient of $r(t)$), and
  - ⋆ $\theta = \theta^{\mathbb{P}} + \sigma^{\mathbb{P}}\lambda/\kappa^{\mathbb{P}}$ (so that $\kappa\,\theta = \kappa^{\mathbb{P}}\,\theta^{\mathbb{P}} + \sigma^{\mathbb{P}}\lambda$).
- So, one could
  - ⋆ estimate from historical data $\kappa^{\mathbb{P}}$, $\sigma^{\mathbb{P}}$, $\theta^{\mathbb{P}}$, which would give κ and σ; and
  - ⋆ then choose θ (equivalent to choosing λ) to best fit market bond prices.

9.8 Extensions of Vasicek Model: Hull & White Model and Ho & Lee Model

- To fit the model to current market conditions, the parameters κ, θ, and σ can be taken to be time-dependent.
- The model with time-dependent parameters is usually called the Hull and White (1990) model.
- In this section, we will study how the Vasicek (1977) model can be extended in a straightforward manner to allow for time-dependent parameters.

- Recall, from Equation (9.4), the short-rate process under the Vasicek model

$$\begin{aligned} dr(t) &= \kappa \Big[\theta - r(t)\Big]\, dt + \sigma dB^{\mathbb{Q}}(t), \\ &= \kappa\, \theta\, dt - \kappa\, r(t)\, dt + \sigma\, dB^{\mathbb{Q}}(t). \end{aligned}$$

- Writing this with time-dependent parameters, and using $\theta(t)$ to now denote the product $\kappa\,\theta$ for convenience, we get the general Hull and White 1990 model:

$$dr(t) = \theta(t)\, dt - \kappa(t)\, r(t)\, dt + \sigma(t)\, dB^{\mathbb{Q}}(t).$$

- We will investigate two special cases of the general Hull and White model.

- In the first special case, we assume that κ and σ are constants and that only $\theta(t)$ is a (non-random) function of time:

$$dr(t) = \theta(t)\, dt - \kappa\, r(t)\, dt + \sigma\, dB^{\mathbb{Q}}(t), \tag{9.20}$$

  which one can re-write as

$$dr(t) = \kappa \left(\frac{\theta(t)}{\kappa} - r(t) \right) dt + \sigma\, dB^{\mathbb{Q}}(t),$$

  where we can now interpret $\theta(t)/\kappa$ as a time-varying long run mean of the short-rate process.

- We will refer to this as the Hull and White model.

- The second special case is when we take the Hull and White model described on the previous page in (9.20), and re-written below:

$$dr(t) = \theta(t)\,dt - \kappa\, r(t)\,dt + \sigma\, dB^{\mathbb{Q}}(t), \tag{9.20}$$

and we set $\kappa = 0$ to get:

$$dr(t) = \theta(t)\,dt + \sigma\, dB^{\mathbb{Q}}(t), \tag{9.21}$$

which is the continuous time Ho and Lee (1986) model.

- Note that the Ho and Lee model has no mean reversion.

9.8.1 Interest Rates in the Hull-White and Ho-Lee Models

- The interest rate in the Hull and White model is:

$$r(s) = \phi(s) + \widehat{r}(s), \tag{9.22}$$

where (note: here y denotes the dummy variable representing time, over which we integrate)

$$\phi(s) = \int_0^s e^{-\kappa(s-y)}\theta(y)\,dy, \tag{9.23}$$

$$\widehat{r}(s) = e^{-\kappa s}r(0) + \sigma\int_0^s e^{-\kappa(s-y)}dB^{\mathbb{Q}}(y). \tag{9.24}$$

- In the Ho and Lee model, where $\kappa = 0$, the interest rate is given by:

$$r(s) = \phi(s) + \widehat{r}(s), \quad \text{where} \tag{9.25}$$

$$\phi(s) = \int_0^s \theta(y)\,dy, \tag{9.26}$$

$$\widehat{r}(s) = r(0) + \sigma\, B^{\mathbb{Q}}(s). \tag{9.27}$$

- Note that (9.24), which is re-written below,

$$\widehat{r}(s) = e^{-\kappa s} r(0) + \sigma \int_0^s e^{-\kappa(s-y)} dB^{\mathbb{Q}}(y), \tag{9.24}$$

implies that

$$d\widehat{r}(t) = -\kappa\, \widehat{r}(t)\, dt + \sigma\, dB^{\mathbb{Q}}(t), \tag{9.28}$$

which can be interpreted as a Vasicek short-rate process having a long-run mean $\theta = 0$.

- Similarly, (9.27) implies that for the case with $\kappa = 0$:

$$d\widehat{r}(t) = \sigma\, dB^{\mathbb{Q}}(t), \tag{9.29}$$

which can be interpreted as a Vasicek short-rate process having no drift and only the Brownian term with a volatility of σ.

9.8.2 Bond Prices in the Extended Vasicek model

- One can find the bond price in the Hull and White model using the expression in (9.22) for the interest rate, which is re-written below:

$$r(s) = \phi(s) + \widehat{r}(s), \tag{9.22}$$

- To find the bond price:

$$\begin{aligned}
\mathcal{P}^Z(t,u) &= \mathbb{E}_t^{\mathbb{Q}}\left[\exp\left(-\int_t^u r(s)\, ds\right)\right] \\
&= \mathbb{E}_t^{\mathbb{Q}}\left[\exp\left(-\int_t^u [\phi(s) + \widehat{r}(s)] ds\right)\right] \\
&= \mathbb{E}_t^{\mathbb{Q}}\left[\exp\left(-\int_t^u \phi(s) ds\right) \exp\left(-\int_t^u \widehat{r}(s) ds\right)\right] \\
&= \exp\left(-\int_t^u \phi(s) ds\right) \mathbb{E}_t^{\mathbb{Q}}\left[\exp\left(-\int_t^u \widehat{r}(s) ds\right)\right]. \qquad (9.30)
\end{aligned}$$

- We re-write the last expression on the previous slide:

$$\mathcal{P}^Z(t,u) = \exp\left(-\int_t^u \phi(s)ds\right)\mathbb{E}_t^{\mathbb{Q}}\left[\exp\left(-\int_t^u \hat{r}(s)ds\right)\right]. \tag{9.30}$$

- But, we have already calculated the expectation in the equation above, which is the price of a discount bond in the Vasicek model with the long-run mean $\theta = 0$.

- So, we can take advantage of the calculations we have already done.

- Thus, for $t < u$ and defining $\tau = u - t$, the price at date t of a discount bond maturing at date u in the **extended Vasicek model** is

$$\mathcal{P}^Z(t,u) = \exp\left(-\underbrace{\int_t^u \phi(s)ds}_{\text{new term}} \underbrace{-a(\tau) - b(\tau)\hat{r}(t)}_{\text{from Vasicek model}}\right), \tag{9.31}$$

where for the continuous-time Ho and Lee model in which $\kappa = 0$

$$a(\tau) = \frac{-\sigma^2\tau^3}{6}, \tag{9.32}$$

$$b(\tau) = \tau \tag{9.33}$$

and for the Hull and White model, in which $\kappa > 0$,

$$a(\tau) = -\frac{\sigma^2}{4\kappa^3}\left(2\kappa\tau - e^{-2\kappa\tau} + 4e^{-\kappa\tau} - 3\right) \tag{9.34}$$

$$b(\tau) = \frac{1}{\kappa}\left(1 - e^{-\kappa\tau}\right). \tag{9.35}$$

9.8.3 Bond Returns in the Extended Vasicek Model

- To get the return on the bond, we put together the following three results:

  1. the expression for $\mathcal{P}^Z(t,u)$ in (9.31), re-written below, is:

$$\mathcal{P}^Z(t,u) = \exp\left(\underbrace{-\int_t^u \phi(s)ds}_{\text{new term}} \quad \underbrace{-a(\tau) - b(\tau)\widehat{r}(t)}_{\text{from Vasicek model}} \right), \tag{9.31}$$

  2. the expected rate of return on a discount bond under the risk-neutral measure must be the short rate, $r(t)$, and

  3. the process for $d\widehat{r}(t)$ in (9.28), which we re-write below, is:

$$d\,\widehat{r}(t) = -\kappa\,\widehat{r}(t)\,dt + \sigma\,dB^{\mathbb{Q}}(t), \tag{9.28}$$

  we get:

$$\frac{d\mathcal{P}^Z(t,u)}{\mathcal{P}^Z(t,u)} = r(t)\,dt - \sigma\,b(\tau)\,dB^{\mathbb{Q}}(t). \tag{9.36}$$

- From the expression for bond returns,

$$\frac{d\mathcal{P}^Z(t,u)}{\mathcal{P}^Z(t,u)} = r(t)\,dt - \sigma\,b(\tau)\,dB^{\mathbb{Q}}(t),$$

  we see that the volatilities of bond returns are determined by the functions $b(\tau)$, *just as in the basic Vasicek model*.

  ⋆ The volatilities of discount bond returns in the Hull and White model are the same as in the Vasicek model with $\kappa > 0$;

  ⋆ The volatilities of discount bond returns in the continuous time Ho and Lee model are the same as in the Vasicek model with $\kappa = 0$.

- Thus, the computation of the hedge ratios is analogous to the computations in Section 9.7.3.

9.8.4 Fitting Discount Bond Prices and Forward Rates

- It is straightforward to fit the Hull and White and the continuous-time Ho & Lee model to market prices of discount bonds by choosing the function ϕ.

- Denote by $t = 0$ the date we are fitting the model. Then, we would like

$$[\mathcal{P}^Z]^{mkt}(0,u) = [\mathcal{P}^Z]^{model}(0,u),$$

for all maturities u, where $[\mathcal{P}^Z]^{mkt}(0,u)$ is the market price and $[\mathcal{P}^Z]^{model}$ is the model price for the extended Vasicek model, given in (9.31):

$$\mathcal{P}^Z(0,u) = \exp\left(-\int_0^u \phi(s)ds - a(u) - b(u)r(0)\right) \quad \ldots \text{ note: } \hat{r}(0) = r(0)$$

- The functions a and b depend on κ and σ, but we regard them now as having already been chosen.

- Therefore, to match the model prices to market prices we need to solve the equation below for a finite number of maturities u:

$$\begin{aligned}
[\mathcal{P}^Z]^{model}(0,u) &= [\mathcal{P}^Z]^{mkt}(0,u) \\
\exp\left(-\int_0^u \phi(s)ds - a(u) - b(u)r(0)\right) &= [\mathcal{P}^Z]^{mkt}(0,u) \\
\exp\left(-\int_0^u \phi(s)ds\right) &= \left(\exp\left[a(u) + b(u)r(0)\right]\right)[\mathcal{P}^Z]^{mkt}(0,u)
\end{aligned}$$

where

⋆ $\phi(s) = \int_0^s e^{-\kappa(s-y)}\theta(y)\,dy$, and

⋆ $\theta(y)$ is the long-run mean at date y.

- However, in principle, one could solve this equation for each maturity u as follows:

$$\exp\left(-\int_0^u \phi(s)ds\right) = \left(\exp\left[a(u)+b(u)r(0)\right]\right)[\mathcal{P}^Z]^{mkt}(0,u)$$

$$-\int_0^u \phi(s)ds = \Big[a(u)+b(u)r(0)\Big] + \log[\mathcal{P}^Z]^{mkt}(0,u) \quad \text{... take logs of both sides}$$

$$\int_0^u \phi(s)ds = \Big[-a(u)-b(u)r(0)\Big] - \log[\mathcal{P}^Z]^{mkt}(0,u) \quad \text{... multiply both sides by } -1$$

$$\phi(u) = -\frac{\partial a(u)}{\partial u} - \frac{\partial b(u)}{\partial u}r(0) - \frac{\partial \log P^{mkt}(0,u)}{\partial u} \quad \text{... differentiate both sides wrt } u$$

which we can re-arrange as

$$\phi(u) + \frac{\partial a(u)}{\partial u} + \frac{\partial b(u)}{\partial u}r(0) = -\frac{\partial \log[\mathcal{P}^Z]^{mkt}(0,u)}{\partial u}. \tag{9.37}$$

- The term on the:
  - ⋆ right-hand-side, $-\frac{\partial \log[\mathcal{P}^Z]^{mkt}(0,u)}{\partial u}$,
    is the market instantaneous forward rate at 0 for maturity u.
  - ⋆ left-hand side, $-\frac{\partial \log[\mathcal{P}^Z]^{model}(0,u)}{\partial u}$,
    is the model instantaneous forward rate.

- Therefore,
  - ⋆ fitting the model to the yield curve is
  - ⋆ equivalent to fitting the model forward rates to market forward rates.

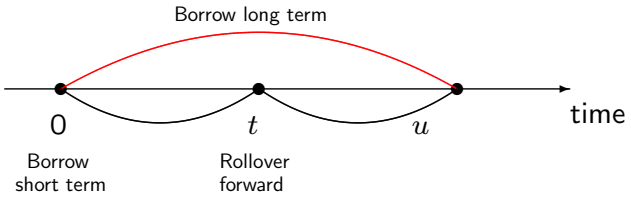
- Note that if we choose the function $\phi(t)$ at date 0 to match the market, then when we want to recalibrate the model at some later date $u > t > 0$ to match the model at that date, we will have to select a different ϕ function.
- The ϕ functions, as well as κ and σ functions in the general Hull and White model, are continuously discarded and refitted to the model.
- Thus, we use a model today knowing we will discard it tomorrow as incorrect.
- This is similar to changing implied volatility in the Black-Scholes model.

9.8.5 Aside: The Instantaneous Forward Interest Rate

- In this section, we demonstrate that the forward rate at t for an instantaneous loan is the derivative of $-\log \mathcal{P}^Z(0,t)$; that is,

$$r_t^{Fwd} = -\frac{\partial \log \mathcal{P}^Z(0,t)}{\partial t}.$$

- Recall that the forward interest rate allows you to lock in at date 0 a rate for a loan or deposit that starts at $t > 0$ for a maturity date of u.



- Consider two ways of borrowing \$1 from date 0 to date u.
  1. Borrow long term from 0 to u at a cost of: $\exp[r(0,u)u]$.
  2. Alternatively, borrow from 0 to t, and then borrow forward from t to u. The cost of these loans is $\exp[r(0,t)t]$ and $\exp[r^{Fwd}(t,u)(u-t)]$.

- Because the forward rate is set at date 0, no-arbitrage requires that the cost of the two ways of doing the transaction be the same:

$$\underbrace{\exp[r(0,u)u]}_{\text{borrow long term}} = \underbrace{\exp[r(0,t)t]}_{\text{borrow short term}} \times \underbrace{\exp[r^{Fwd}(t,u)(u-t)]}_{\text{rollover forward}}. \tag{9.38}$$

- Thus, the forward rate at 0 for a loan at t that matures at u is given by:

$$\exp[r^{Fwd}(t,u)(u-t)] = \frac{\exp[r(0,u)u]}{\exp[r(0,t)t]} = \frac{\mathcal{P}^Z(0,t)}{\mathcal{P}^Z(0,u)} \quad \text{. . . rearranging equation (9.38)}$$

$$r^{Fwd}(t,u)(u-t) = \log \mathcal{P}^Z(0,t) - \log \mathcal{P}^Z(0,u) \quad \text{. . . taking logs on both sides}$$

$$r^{Fwd}(t,u) = -\frac{\left[\log \mathcal{P}^Z(0,u) - \log \mathcal{P}^Z(0,t)\right]}{u-t} \quad \text{. . . dividing by } u-t \text{ on both sides}$$

- We start by reproducing the last equation on the previous page:

$$r^{Fwd}(t,u) = -\frac{\left[\log \mathcal{P}^Z(0,u) - \log \mathcal{P}^Z(0,t)\right]}{u-t}.$$

- As we shorten the maturity, with $u \to t$, the limit of the above shows that the forward rate at t for an instantaneous loan is the derivative of $-\log \mathcal{P}^Z(0,t)$:

$$\lim_{u\to t} r^{Fwd}(t,u) = \lim_{u\to t} -\frac{\left[\log \mathcal{P}^Z(0,u) - \log \mathcal{P}^Z(0,t)\right]}{u-t}$$

$$r_t^{Fwd} = -\frac{\partial \log \mathcal{P}^Z(0,t)}{\partial t}.$$

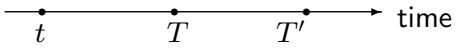
9.9 Options on Discount Bonds & Interest Rates (Caps/Floors)

- In the Vasicek model and its extensions, the volatility of the forward price of a discount bond is changing over time but non-randomly.
- Thus, we can use Black's formula to value options on discount bonds.
- Caps and floors are equivalent to portfolios of options on discount bonds, and so the values of caps (floors) can be obtained by summing the value of individual caplets (floorlets).

9.9.1 Valuation of Options on Discount Bonds

- Consider valuing at date 0 an option
  - ⋆ maturing at date T
  - ⋆ on a discount bond maturing at date $T' > T$.
- Because at maturity the forward equals the spot, the price of an option written on a forward contract maturing at T is equivalent to the price of an option on the spot—we have used this trick before.
- Recall also that to use this trick we need to identify the volatility of the forward price, which is what we do on the next few slides.

- We start with Equation (9.38), which gives the relation between spot and forward interest rates, but we now write it at date t, with the discount bond maturing at date $T' > T$:



$$\exp[r(t,T')(T'-t)] = \exp[r(t,T)(T-t)] \times \exp[r^{Fwd}(T,T')(T'-T)]. \qquad (9.38)$$

- Next, we express the above equation in terms of bond prices:

$$\frac{1}{\mathcal{P}^Z(t,T')} = \frac{1}{\mathcal{P}^Z(t,T)} \times \frac{1}{\mathcal{P}^{Fwd}(t;T,T')}$$

$$\mathcal{P}^{Fwd}(t) = \frac{\mathcal{P}^Z(t,T')}{\mathcal{P}^Z(t,T)}, \qquad (9.39)$$

where $\mathcal{P}^{Fwd}(t) = \mathcal{P}^{Fwd}(t;T,T')$ is the forward price of the discount bond at date t for a contract maturing at T on a bond maturing at T'.

- Recall from (9.36) on Page 223 that:

$$\frac{d\mathcal{P}^Z(t,T')}{\mathcal{P}^Z(t,T')} = r(t)\,dt - \sigma\, b(T'-t)\, dB^{\mathbb{Q}}(t). \qquad (9.36)$$

- Therefore, using Itô's formula and applying the rule for ratios to the expression for $\mathcal{P}^{Fwd}(t;T,T') = \frac{\mathcal{P}^Z(t,T')}{\mathcal{P}^Z(t,T)}$, we have:

$$\begin{aligned}
\frac{d\mathcal{P}^{Fwd}(t)}{\mathcal{P}^{Fwd}(t)} &= \text{something}\, dt + \frac{d\mathcal{P}^Z(t,T')}{\mathcal{P}^Z(t,T')} - \frac{d\mathcal{P}^Z(t,T)}{\mathcal{P}^Z(t,T)} \\
&= \text{something}\, dt - \sigma\, b(T'-t)\, dB^{\mathbb{Q}}(t) + \sigma\, b(T-t)\, dB^{\mathbb{Q}}(t) \\
&= \text{something}\, dt - \sigma \left[b(T'-t) - b(T-t) \right] dB(t).
\end{aligned}$$

- Thus, the volatility of the forward price $\mathcal{P}^{Fwd}(t)$ depends on the non-random function $b(\tau)$.

- In the continuous-time **Ho and Lee model** where $\kappa = 0$:

$$b(T'-t) - b(T-t) = (T'-t) - (T-t) = T' - T,$$

implying that the forward price has a constant volatility of $(T'-T)\sigma$.

- In the **Hull and White model** where $\kappa > 0$:

$$b(T'-t) - b(T-t) = \frac{1}{\kappa}\left[e^{-\kappa(T-t)} - e^{-\kappa(T'-t)}\right] = \frac{\left(e^{-\kappa T} - e^{-\kappa T'}\right) e^{\kappa t}}{\kappa},$$

so the volatility is time varying (it changes with t), so we compute the average volatility as discussed in the first part of the course:

$$\begin{aligned}\sigma_{avg} &= \sigma \frac{\left(e^{-\kappa T} - e^{-\kappa T'}\right)}{\kappa}\sqrt{\frac{1}{T}\int_0^T e^{2\kappa t}\, dt} \quad \text{... remember: it is } \textit{variance} \text{ that scales with time} \\ &= \sigma \frac{\left(e^{-\kappa T} - e^{-\kappa T'}\right)}{\kappa}\sqrt{\frac{e^{2\kappa T} - 1}{2\kappa T}}. \end{aligned} \tag{9.40}$$

- Substituting $[\mathcal{P}^Z]^{mkt}(0,T')/[\mathcal{P}^Z]^{mkt}(0,T)$ as the price of the discount bond maturing at $T' > T$ in Black's formula leads to the following result.

- In the extended Vasicek model, the value at date 0 of a call option with exercise price K maturing at date T on a discount bond maturing at date $T' > T$ is:

$$\text{Call Price}_0 = [\mathcal{P}^Z]^{mkt}(0,T')N(d_1) - [\mathcal{P}^Z]^{mkt}(0,T)\,KN(d_2), \tag{9.41}$$

where

$$d_1 = \frac{\log\left(\frac{[\mathcal{P}^Z]^{mkt}(0,T')}{[\mathcal{P}^Z]^{mkt}(0,T)K}\right) + \frac{1}{2}\sigma_{avg}^2\, T}{\sigma_{avg}\sqrt{T}} \tag{9.42}$$

$$d_2 = d_1 - \sigma_{avg}\sqrt{T}, \tag{9.43}$$

with σ_{avg} for the continuous-time Ho and Lee model and the Hull and White model given on Page 237.

- Observe that because the volatilities are the same
  - ⋆ in the continuous-time Ho and Lee model as in the basic Vasicek model with $\kappa = 0$, and
  - ⋆ in the Hull and White model as in the basic Vasicek model with $\kappa > 0$,

  the formulas for options on discount bonds are the same in these models.
- The extensions to the Vasicek model
  - ⋆ only permit one to match better model bond prices to market bond prices,
  - ⋆ but otherwise have no effect on the value of discount bond options.

9.9.2 Valuation of Caps and Floors

- Recall from Section 8.11.6 on Page 103 of the Chapter 8 that a caplet with reset date t_i and payment date T_{i+1} is equivalent to
  - ⋆ $(1 + \bar{\mathcal{R}}\Delta t)$ put options
  - ⋆ maturing at t_i
  - ⋆ on the discount bond maturing at t_{i+1},
  - ⋆ with the exercise price of each option being $1/(1 + \bar{\mathcal{R}}\Delta t)$.
- Making these substitutions in the above formulas (and analogously for floorlets), we get the following result.

- Consider a caplet and a floorlet with reset date t_i and payment date t_{i+1}.
- Define $\Delta t = t_{i+1} - t_i$ and let $\bar{\mathcal{R}}$ denote the cap and floor rate.
- Then, in the extended Vasicek model:

$$\text{Floorlet Price}_0 = (1 + \bar{\mathcal{R}}\Delta t)P^{mkt}(0, t_{i+1})N(d_1) - P^{mkt}(0, t_i)N(d_2) \quad (9.44)$$

$$\text{Caplet Price}_0 = P^{mkt}(0, t_i)N(-d_2) - (1 + \bar{\mathcal{R}}\Delta t)P^{mkt}(0, t_{i+1})N(-d_1) \quad (9.45)$$

where

$$d_1 = \frac{\log\left(\frac{(1+\bar{\mathcal{R}}\Delta t)P^{mkt}(0,t_{i+1})}{P^{mkt}(0,t_i)}\right) + \frac{1}{2}\sigma_{avg}^2 t_i}{\sigma_{avg}\sqrt{t_i}} \quad (9.46)$$

$$d_2 = d_1 - \sigma_{avg}\sqrt{t_i}, \quad (9.47)$$

where for the continuous-time Ho and Lee model $\sigma_{avg} = \sigma\Delta t$ and for the Hull and White model is $\sigma_{avg} = \frac{\sigma\, e^{-\kappa t_i}\left(1-e^{-\kappa\Delta t}\right)}{\kappa}\sqrt{\frac{e^{2\kappa t_i}-1}{2\kappa\, t_i}}$.

- We used Black's formula to value caplets and floorlets also in Section 8.11.4 of the previous class.
- However, the formulas here and those in Section 8.11.4 will give different values because they are based on different assumptions.
- In the Vasicek model and its extensions, the valuation of caplets/floorlets is based on the assumption that the forward price of the discount bond corresponding to the caplet with payment date t_{i+1} has non-random volatility:

$$\mathcal{P}^{Fwd}(t) = \frac{\mathcal{P}^Z(t, t_{i+1})}{\mathcal{P}^Z(t, t_i)}$$

- In Sec 8.11.4, we assumed a non-random volatility for LIBOR forward rate:

$$\mathcal{R}_i^{Fwd}(t) = \frac{\mathcal{P}^Z(t, t_i)}{\mathcal{P}^Z(t, t_{i+1})} - 1 = \frac{1}{\mathcal{P}^{Fwd}(t)} - 1.$$

- So, if the forward price has a non-random volatility, then the forward rate, which is its inverse, will have a random volatility, and vice versa.

- We could use similar methods to study the valuation of
  - ⋆ Options on coupon bonds
  - ⋆ Swaptions
  - ⋆ Captions
  - ⋆ Floortions

  but we will leave this for you to study on your own.

9.10 Other Continuous-Time Models of the Term Structure of Interest Rates

- So far, we have studied the Vasicek model for the short-rate,

$$dr(t) = \kappa \Big[\theta - r(t)\Big] dt + \sigma dB^{\mathbb{Q}}(t),$$

  and two extensions to this model that allow θ to be time dependent.

- These two extensions are
  - ⋆ the continuous-time Ho and Lee model, in which $\kappa = 0$, and
  - ⋆ the basic Hull and White model, in which $\kappa > 0$.

- For each term structure model, we
  1. Specified the process for the interest rate, $dr(t)$;
  2. We solved for the short-rate, $r(t)$;
  3. We used the short rate to find discount-bond prices, $\mathcal{P}^Z(t,u)$;
  4. We also found the process for discount-bond prices, $d\mathcal{P}^Z(t,u)$;
  5. We used these results to value options on discount bonds and options on discount rates (caps, and floors).
  6. We discussed how to estimate the parameters of these models and how to hedge under these models.

- We now discuss some other models of the term structure.

- For each of these models of the term structure, we contrast it with the models that we have already studied, but we do not repeat the analysis that we have done above for the Vasicek model and its extensions.

9.10.1 General Hull and White (1990) Model

- In the General Hull and White (1990) model, that we already mentioned above, recall that θ, κ, and σ are allowed to be functions of time:

$$dr(t) = \theta(t)\,dt - \kappa(t)\,r(t)\,dt + \sigma(t)\,dB^{\mathbb{Q}}(t). \tag{9.48}$$

- Under this model, forward prices of discount bonds continue to have volatilities that are non-random functions of time. This means that the valuation of derivatives can be done the same way as we did above.

- The main advantage of this model, in contrast to the ones we have studied above, is that because there are three time-dependent parameters, there is more flexibility in fitting the model to current market conditions.

9.10.2 Cox, Ingersoll, and Ross (1985a) Model

- Cox, Ingersoll, and Ross (1985a) specified the following model for the short rate:

$$dr(t) = \kappa \left[\theta - r(t)\right] dt + \sigma \sqrt{r(t)} dB^{\mathbb{Q}}(t),$$

  where κ, θ, and σ are positive constants, and $B^{\mathbb{Q}}$ is a Brownian motion under the risk-neutral measure $\mathbb{Q}$.

- Like the Vasicek (1977) model, this short rate process has a long-run mean of θ.

- The main innovation is that, in contrast to the Vasicek model, in this model the volatility is proportional to the square root of the short rate rather than being constant; thus, the short rate can never be negative.

- This model, just like the Vasicek model, delivers bond prices that are exponential affine.

9.10.3 Multifactor Cox, Ingersoll, and Ross (1985a) and Longstaff and Schwartz (1992) Models

- Cox, Ingersoll, and Ross (1985a) suggest adding two independent square-root processes in order to get a two-factor model of the short rate. That is:

$$\begin{aligned} r(t) &= x_1(t) + x_2(t) \\ dx_i(t) &= \kappa_i \left[\theta_i - x_i(t)\right] dt + \sigma_i \sqrt{x_i(t)} dB_i^{\mathbb{Q}}(t), \qquad i = \{1, 2\}, \end{aligned}$$

  where B_1 and B_2 are independent Brownian motions under the risk-neutral measure and κ_i, θ_i, and σ_i for $i = \{1, 2\}$ are positive constants.

- Longstaff and Schwartz (1992) investigate this model further, including providing an equilibrium foundation, deriving prices of options on discount bonds, and estimating the coefficients.

- An important insight is that the model can be rewritten so that the two factors are the level of the short rate and its volatility, rather than some unobservable variables x_1 and x_2.

9.10.4 Affine Models of Duffie and Kan (1996)

- Linear function of x:

$$f(x) = \alpha_1 x$$

- Affine function of x is the sum of a constant and a linear function:

$$f(x) = \alpha_0 + \alpha_1 x$$

- All the models we have studied are particular examples of affine models of the term structure:
  - ⋆ Vasicek (1977);
  - ⋆ Cox, Ingersoll, and Ross (1985a); and,
  - ⋆ Longstaff and Schwartz (1992).

- An affine models of the term structure is defined by a set of factors $x_1, x_2, \ldots, x_n$, where:
  - ⋆ The short rate is an affine function of the factors

$$r(t) = \alpha_0 + \sum_{i=1}^{n} \alpha_i x_i, \quad \text{for constants } \alpha_i;$$

  - ⋆ The drift of each factor is an affine function of the factors;
  - ⋆ The instantaneous variance of each factor is an affine function of the factors;
  - ⋆ The instantaneous covariance of each pair of factors is an affine function of the factors.
  - ⋆ Discount bonds are exponential-affine functions:

$$\mathcal{P}^Z(t,u) = \exp\left[-a(\tau) - \sum_{i=1}^{n} b_i(\tau)\, x_i(t)\right], \text{ for deterministic functions } a \text{ and } b_i.$$

- In any single-factor affine model, the short rate can be used as a factor.
- Thus, the general affine one-factor model is of the form:

$$dr(t) = \kappa \Big[\theta - r(t)\Big]\, dt + \sqrt{b_0 + b_1\, r}\; dB^{\mathbb{Q}}(t), \tag{9.49}$$

where $\kappa, \theta, \alpha, \beta$ are constants (or deterministic functions of time), and where $B^{\mathbb{Q}}$ is a Brownian motion under the risk-neutral measure $\mathbb{Q}$.

  - ⋆ The Vasicek (1977) model is the special case where $b_0 = \sigma^2$ and $b_1 = 0$.
  - ⋆ The Cox, Ingersoll, and Ross (1985a) is the special case where $b_0 = 0$, $b_1 = \sigma^2$.

- **Yields of discount bonds** are affine functions of the factors, as we see from (9.49), and therefore the short rate along with yields at $n - 1$ fixed time-to-maturities τ_i can be chosen to be the n factors.
  - ⋆ This transformation from factors to yields is analogous to the transformation from factors to the short rate and its volatility in the Longstaff and Schwartz (1992) model.
- For more on affine models, see
  - ⋆ Duffie and Kan (1996), and
  - ⋆ Dai and Singleton (2000).

9.10.5 Essentially Affine Model of Duffee (2002)

- Duffee (2002) introduces a class of models that is called "completely affine" or "essentially affine."

- In these models, the price of risk is specified in a such a way that
  - ⋆ if it is affine under the physical probability measure,
  - ⋆ then it is affine also under the risk-neutral probability measure.

9.10.6 Quadratic Models

- A quadratic model is specified by a vector of Gaussian factors, with the short rate being a quadratic function of the factors:

$$r(t) = \alpha_0 + \alpha_1^\top X + X^\top \alpha_2 X,$$

  where α_0 is a constant scalar, α_1 is a constant vector, α_2 is a constant matrix, and $\top$ denotes the transpose operator for a vector (or matrix).

- Under the appropriate assumptions, the prices of discount bonds in such a model are exponential-quadratic.

- If one assumes that the market price of risk are affine in the state variables, then risk premia are quadratic functions of the factors (with the coefficients depending on the time to maturity of the bond).

- See Ahn, Dittmar, and Gallant (2002) and Leippold and Wu (2002) for further details.

9.10.7 Heath, Jarrow, and Morton (1992) Model

- It is equivalent to model prices of discount bonds, yields of discount bonds, or forward rates, because:

$$-\log \mathcal{P}^Z(s,u) = (u-s)\, y(s,u) = \int_s^u f(s,t)dt.$$

- Heath, Jarrow, and Morton (1992) propose a different approach from the ones we have discussed so far.

- Instead of modeling the evolution of the short rate,
  - ⋆ they propose modeling the evolution of the instantaneous forward rates,
  - ⋆ taking the prevailing market forward rates as the initial conditions.

- The advantage of this approach is that by using the initial forward rate curve as an input, one automatically fits the current yield curve.

- Heath, Jarrow & Morton show that the drifts of instantaneous forward rates under the risk-neutral measure are related to the volatilities of forward rates.

- Thus, once you specify the volatilities of the forward rates,
  - ⋆ the drifts are determined, and
  - ⋆ so the model is completely defined by specifying just the volatilities.

- The main *disadvantage* of the Heath, Jarrow, and Morton (1992) model is that
  - ⋆ bond prices and prices of fixed-income instruments are path-dependent;
  - ⋆ that is, they depend on the entire history of the forward rate processes, rather than depending only on the current values of a small set of factors.

- All the other continuous-time models we discussed above
  - ⋆ can be written as special cases of the Heath, Jarrow and Morton model;
  - ⋆ this is shown elegantly in the book by Baxter and Rennie (1996, Chapter 5).

9.11 Summary

1. Under the risk neutral measure, the price $\mathcal{P}^Z(t,u)$ at date t of a discount bond maturing at u must be:

$$\mathcal{P}^Z(t,u) = \mathbb{E}_t^{\mathbb{Q}}\left[\exp\left(-\int_t^u r(s)\,ds\right)\right].$$

2. The forward rate at t for an instantaneous loan is given by

$$r_t^{Fwd} = -\frac{\partial \log \mathcal{P}^Z(0,t)}{\partial t}.$$

3. We study a number of term structure models. For each model we:

3.1. Specify the process for the interest rate, $dr(t)$;

3.2. Solve for the short-rate, $r(t)$;

3.3. Use the short rate to find discount-bond prices, $\mathcal{P}^Z(t,u)$;

3.4. Find the process (return) for discount-bond prices, $d\mathcal{P}^Z(t,u)$;

3.5. Use these results to value options on discount bonds and options on discount rates (caps, and floors).

3.6. Understand how to hedge in these models.

3.7. Discuss how to estimate the parameters of these models and how to hedge under these models.

4. In the basic Vasicek (1977) model, with $B^{\mathbb{Q}}$ denoting a Brownian motion under the risk-neutral measure

- It is assumed that the short rate follows the process:
$$dr(t) = \kappa\left[\theta - r(t)\right]dt + \sigma dB^{\mathbb{Q}}(t),$$
- The level of the interest rate is:
$$r_t = \theta - e^{-\kappa t}\left[\theta - r_0\right] + \sigma\int_0^t e^{\kappa(s-t)}dB_s^{\mathbb{Q}}$$
- The mean of the interest rate is
$$\mathbb{E}^{\mathbb{Q}}[r_t] = r_0\, e^{-\kappa t} + \theta\left(1 - e^{-\kappa t}\right).$$
- The variance of the interest rate is
$$\text{var}(r_t) = \sigma^2\mathbb{E}\left\{\int_0^t e^{2\kappa(s-t)}\,ds\right\} = \frac{\sigma^2}{2\kappa}\left(1 - e^{-2\kappa t}\right).$$
- The price of discount bonds is:
$$\begin{aligned}\mathcal{P}^Z(t,u) &= \exp\left[-a(\tau) - b(\tau)r(t)\right],\\ a(\tau) &= \theta\tau - \frac{\theta}{\kappa}\left(1 - e^{-\kappa\tau}\right) - \frac{\sigma^2}{4\kappa^3}\left(2\kappa\tau - e^{-2\kappa\tau} + 4e^{-\kappa\tau} - 3\right)\\ b(\tau) &= \frac{1}{\kappa}\left(1 - e^{-\kappa\tau}\right).\end{aligned}$$

5. Hedging in the Vasicek Model

- The Vasicek model has a single Brownian driving the short interest rate:
  - ⋆ Thus, there is only a single Brownian motion driving all bond prices
  - ⋆ So the bond prices are perfectly correlated in the Vasicek model.
  - ⋆ Therefore, the hedge ratio for hedging one bond with another will depend only on the relative volatilities of the bond returns.

6. The general Hull and White model is:

$$dr(t) = \theta(t)\,dt - \kappa(t)\,r(t)\,dt + \sigma(t)\,dB^{\mathbb{Q}}(t).$$

- In the first special case, which we refer to as the Hull and White model, κ and σ are constants and only $\theta(t)$ is a (non-random) function of time:

$$\begin{aligned} & dr(t) = \theta(t)\,dt - \kappa\,r(t)\,dt + \sigma\,dB^{\mathbb{Q}}(t), \\ \text{or:} \quad & dr(t) = \kappa\left(\frac{\theta(t)}{\kappa} - r(t)\right)dt + \sigma\,dB^{\mathbb{Q}}(t), \end{aligned}$$

  where $\theta(t)/\kappa$ is the time-varying long run mean of the short-rate process.

- In the second special case we set $\kappa = 0$ to get:
  $dr(t) = \theta(t)\,dt + \sigma\,dB^{\mathbb{Q}}(t)$,
  which is the continuous time Ho and Lee (1986) model that has zero mean reversion.

- The interest rate in the Hull and White model is:

$$\begin{aligned} r(s) &= \phi(s) + \widehat{r}(s), \\ \phi(s) &= \int_0^s e^{-\kappa(s-y)}\theta(y)\,dy, \\ \widehat{r}(s) &= e^{-\kappa s}r(0) + \sigma\int_0^s e^{-\kappa(s-y)}dB^{\mathbb{Q}}(y). \end{aligned}$$

- The interest rate in the Ho and Lee model, where $\kappa = 0$, is:

$$\begin{aligned} r(s) &= \phi(s) + \widehat{r}(s), \quad \text{where} \\ \phi(s) &= \int_0^s \theta(y)\,dy, \\ \widehat{r}(s) &= r(0) + \sigma\,B^{\mathbb{Q}}(s). \end{aligned}$$

- The price at t of a discount bond in the extended Vasicek model is

$$\mathcal{P}^Z(t,u) = \exp\Big(\underbrace{-\int_t^u \phi(s)ds}_{\text{new term}}\ \underbrace{-a(\tau) - b(\tau)\widehat{r}(t)}_{\text{from Vasicek model}}\Big).$$

7. Because

- the volatilities of discount bond returns in the Hull and White model are the same as in the Vasicek model with $\kappa > 0$; and,
- the volatilities of discount bond returns in the continuous time Ho and Lee model are the same as in the Vasicek model with $\kappa = 0$;

Thus, the computation of the hedge ratios in these models is analogous to the computations for the hedge ratio in the Vasicek model.

8. In the extended Vasicek model, the value at date 0 of a call option with exercise price K maturing at date T on a discount bond maturing at date $T' > T$ is:

$$\text{Call Price}_0 = [\mathcal{P}^Z]^{mkt}(0,T')N(d_1) - [\mathcal{P}^Z]^{mkt}(0,T)\,KN(d_2),$$

where

$$d_1 = \frac{\log\left(\frac{[\mathcal{P}^Z]^{mkt}(0,T')}{[\mathcal{P}^Z]^{mkt}(0,T)K}\right) + \frac{1}{2}\sigma_{avg}^2\,T}{\sigma_{avg}\sqrt{T}}$$

$$d_2 = d_1 - \sigma_{avg}\sqrt{T},$$

with

$$\sigma_{avg} = \sigma\,\frac{\left(e^{-\kappa T} - e^{-\kappa T'}\right)}{\kappa}\sqrt{\frac{e^{2\kappa T}-1}{2\kappa\,T}}.$$

9. Consider a floorlet with reset date t_i and payment date t_{i+1}. Define $\Delta t = t_{i+1} - t_i$ and let $\bar{R}$ denote the floor rate.

Then, in the extended Vasicek model, at date 0 the values of the floorlet is:

$$\text{Floorlet Price}_0 = (1 + \bar{\mathcal{R}}\Delta t)P^{mkt}(0, t_{i+1})N(d_1) - P^{mkt}(0, t_i)N(d_2)$$

where

$$d_1 = \frac{\log\left(\frac{(1+\bar{\mathcal{R}}\Delta t)P^{mkt}(0,t_{i+1})}{P^{mkt}(0,t_i)}\right) + \frac{1}{2}\sigma_{avg}^2 t_i}{\sigma_{avg}\sqrt{t_i}}$$

$$d_2 = d_1 - \sigma_{avg}\sqrt{t_i}\,,$$

where

- for the continuous-time Ho and Lee model $\sigma_{avg} = \sigma\Delta t$ and,
- for the Hull and White model

$$\sigma_{avg} = \frac{\sigma\, e^{-\kappa t_i}\left(1 - e^{-\kappa\Delta t}\right)}{\kappa}\sqrt{\frac{e^{2\kappa t_i} - 1}{2\kappa\, t_i}}.$$

10. We also discussed briefly the following continuous-time term structure models:

- General Hull and White (1990);
- Cox, Ingersoll, and Ross (1985a);
- Longstaff and Schwartz 1992;
- Duffie and Kan (1996);
- Heath, Jarrow, and Morton (1992).

9.12 Appendix: Discrete-Time Models of Term Structure of Interest Rates

- So far, the models of the term structure that we have discussed were all set in continuous time.
- But, just like the discrete-time binomial model for stocks, one can also model interest rates in discrete time.
- The Ho and Lee (1986) model is a binomial version of the Vasicek model without mean reversion, in which the one-period interest rate is assumed to have a deterministic drift.

- The Black, Derman, and Toy (1990) model is also a binomial model of the one-period interest rate.
  - ⋆ In contrast to the Vasicek (1977) and Ho and Lee (1986) models, this model specifies a process for $\log[r(t)]$, and so interest rates are always nonnegative.
  - ⋆ This model also allows volatility to be time-dependent;
  - ⋆ And, the drift is allowed to vary across the nodes for a particular date t_i so that we get a recombining tree.
  - ⋆ This model is described in detail below.
- The Black and Karasinski (1991) model is similar to the Black, Derman, and Toy (1990) model in that it specifies a process for the log of the interest rate.
  - ⋆ But the model removes the linkage between the mean-reversion parameter and volatility.
  - ⋆ It does this by allowing the lengths of the time step on the tree to vary, in order to get a recombining tree.

9.12.1 The Black, Derman, and Toy (1990) of Dynamics of the Term Structure

- Black, Derman, and Toy (1990) develop a model of the dynamics of the term structure that is consistent with the current
  - ⋆ term structure of interest rates and
  - ⋆ volatilities.
- The Black, Derman, and Toy (1990) model is:
  - ⋆ Developed under the risk-neutral measure;
  - ⋆ Therefore, arbitrage is not possible in the model;
  - ⋆ Thus, the only challenge is to ensure that model is consistent with given
    - ∗ term structure of interest rates, and
    - ∗ volatilities.

- Black, Derman, and Toy (1990) have a single factor model, with the factor being the short rate. Therefore:
  - ⋆ all yields go up and down together, and
  - ⋆ all movements in bond prices are perfectly correlated.
- The short interest rate is assumed to follow
  - ⋆ a lognormal distribution (to ensure interest rates are always positive)
  - ⋆ on a binomial tree.

9.12.2 The Volatility Equation

- If σ denotes volatility of the short rate the current node, and if r_u and r_d denote the two possible values of the short rate one period hence, then the volatility is

$$\sigma = \frac{1}{2} \log\left(\frac{r_u}{r_d}\right) \quad \ldots \quad \text{this is called the volatility equation,} \tag{9.50}$$

  because if we assume risk-neutral probability of up and down equals 0.50, then the standard deviation of the log of one-period future short rate is σ.

- Note that log denotes the natural logarithm (ln).

- While this appears similar to the Cox, Ross, and Rubinstein (1979) tree for equities, note that
  - ⋆ in Black, Derman, and Toy (1990) tree, the volatility will not be constant;
  - ⋆ thus, the branching process will vary node-to-node.

9.12.3 Building the Short-rate Tree for Black, Derman, and Toy (1990) Model

- Maturity of the tree extends to that of the longest underlying bond.

- Recursive procedure (bootstrap) is used to build the tree:
  - ⋆ One year bond is used to determine the short rate, r_0.
  - ⋆ Short rate (r_0) and the two-year zero-coupon bond are used to determine r_u and r_d.
  - ⋆ Short rate (r_0), r_u, r_d and the three-year zero-coupon bond are used to determine r_{uu}, r_{ud}, r_{dd}.

- Volatility depends on time, but not on state.

- We look at the following example, adapted from the paper by Black, Derman and Toy, in which we assume annual compounding.

9.12.4 Numerical Example from Black, Derman, and Toy (1990)

- Assume that you are given the following data:

Maturity in years	Spot interest rate per year	Volatility of yield per year
1	10%	20%
2	11%	19%
3	12%	18%

Step 1: Find $t = 0$ Prices of Zero-Coupon Bond Maturing at T_1, T_2, T_3

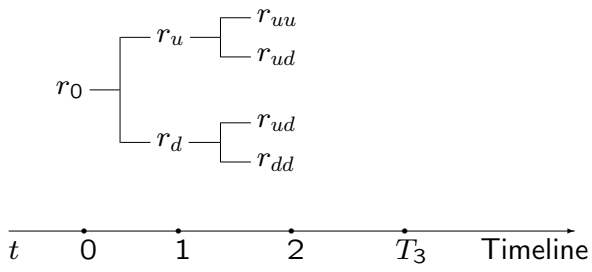
$$\mathcal{P}^Z(0, T_1) = \frac{100}{1.10} = 90.909;$$

$$\mathcal{P}^Z(0, T_2) = \frac{100}{1.11^2} = 81.162;$$

$$\mathcal{P}^Z(0, T_3) = \frac{100}{1.12^3} = 71.178.$$

Step 2: Find the tree for the short rate

Tree for the short rate



Step 2.1: Find r_0

- From the data given to us, we know that: $r_0 = 0.10$.

Step 2.2: Find r_u and r_d

- After one year has elapsed, a two-year zero-coupon bond becomes a one-year zero-coupon bond.
- The value of the two-year zero-coupon bond after one year has elapsed is:
  - ⋆ in the "up" state: $\frac{100}{1+r_u}$;
  - ⋆ in the "down" state: $\frac{100}{1+r_d}$.
- The gross return from investing in a two-year zero-coupon bond from $t = 0$ to $t = 1$ is:

$$1 + \text{return}_{0,1} = \frac{\text{expected value at } T_1}{\text{initial investment at } t_0}$$

$$1.10 = \frac{\left(\frac{1}{2} \times \frac{100}{1+r_u}\right) + \left(\frac{1}{2} \times \frac{100}{1+r_d}\right)}{81.162}. \tag{9.51}$$

- The above is one equation; we now need a second equation.

- For the second equation, we use the information that the two-year zero-coupon bond has a yield volatility of 19%.
- After one year, the two-year zero-coupon bond is a one-year zero-coupon bond, and its yield is either r_u or r_d.
- Therefore, the yield volatility of the two-year zero-coupon bond is:

$$\frac{1}{2} \log\left(\frac{r_u}{r_d}\right) = 0.19 \tag{9.52}$$

- Solving Equations (9.51) and (9.52) simultaneously, we obtain:

$$r_u = 14.32\%; \tag{9.53}$$

$$r_d = 9.79\%. \tag{9.54}$$

Step 2.3: Find r_{uu}, r_{ud}, and r_{dd}

- We use r_0, r_u, r_d and three-year zero-coupon bond, to identify r_{uu}, r_{ud}, r_{dd}.
- At $t = 2$, the possible values of the three-year bond are:

$$\mathcal{P}^Z(uu, T_3) = \frac{100}{1 + r_{uu}}; \tag{9.55}$$

$$\mathcal{P}^Z(ud, T_3) = \frac{100}{1 + r_{ud}}; \tag{9.56}$$

$$\mathcal{P}^Z(dd, T_3) = \frac{100}{1 + r_{dd}}. \tag{9.57}$$

- At $t = 1$, we know from risk-neutral pricing that:

$$\mathcal{P}^Z(u, T_3) = \frac{1}{1 + r_u}\left[\frac{1}{2}\mathcal{P}^Z(uu, T_3) + \frac{1}{2}\mathcal{P}^Z(ud, T_3)\right]; \tag{9.58}$$

$$\mathcal{P}^Z(d, T_3) = \frac{1}{1 + r_d}\left[\frac{1}{2}\mathcal{P}^Z(ud, T_3) + \frac{1}{2}\mathcal{P}^Z(dd, T_3)\right]. \tag{9.59}$$

- At $t = 0$, applying risk-neutral pricing one more time:

$$\mathcal{P}^Z(0, T_3) = 71.178 = \frac{1}{1 + r_0}\left[\frac{1}{2}\mathcal{P}^Z(u, T_3) + \frac{1}{2}\mathcal{P}^Z(d, T_3)\right]. \tag{9.60}$$

- Substituting
  - ⋆ Equations (9.55), (9.56) and (9.57) into Equations (9.58) and (9.59);
  - ⋆ And, then substituting the resulting Equations (9.58) and (9.59) into Equation (9.60)
- leads to our first equation in the three unknowns r_{uu}, r_{ud}, r_{dd}:

$$\begin{aligned}
&\mathcal{P}^Z(0, T_3) \\
&= 71.178 \\
&= \frac{1}{1 + r_0}\left[\frac{1}{2}\mathcal{P}^Z(u, T_3) + \frac{1}{2}\mathcal{P}^Z(d, T_3)\right] \\
&= \frac{1}{1 + r_0}\left\{\frac{1}{2}\cdot\frac{1}{1 + r_u}\left[\frac{1}{2}\mathcal{P}^Z(uu, T_3) + \frac{1}{2}\mathcal{P}^Z(ud, T_3)\right] + \frac{1}{2}\cdot\frac{1}{1 + r_d}\left[\frac{1}{2}\mathcal{P}^Z(ud, T_3) + \frac{1}{2}\mathcal{P}^Z(dd, T_3)\right]\right\} \\
&= \frac{1}{1 + r_0}\left\{\frac{1}{2}\cdot\frac{1}{1 + r_u}\left[\frac{1}{2}\cdot\frac{100}{1 + r_{uu}} + \frac{1}{2}\cdot\frac{100}{1 + r_{ud}}\right] + \frac{1}{2}\cdot\frac{1}{1 + r_d}\left[\frac{1}{2}\cdot\frac{100}{1 + r_{ud}} + \frac{1}{2}\cdot\frac{100}{1 + r_{dd}}\right]\right\}.
\end{aligned} \tag{9.61}$$

- We still need two more equations because we need to identify three unknowns.
- We get these from the volatility equation.
- Consider the three-year zero-coupon bond at $t = 1$.
- To use the volatility equation, we need to figure out the yield on the three-year zero-coupon bond.
- If at $t = 1$ we are in the "up" state, then:

$$\mathcal{P}^Z(u, T_3) = \frac{100}{(1 + y_u)^2}$$

which implies: $$y_u = \sqrt{\frac{100}{\mathcal{P}^Z(u, T_3)}} - 1.$$

- Similarly, if at $t = 1$ we are in the "down" state, then:

$$y_d = \sqrt{\frac{100}{\mathcal{P}^Z(d, T_3)}} - 1.$$

- From the data given to us on Page 273, we know that the per annum yield volatility of the three-year zero-coupon bond is 18%.
- Therefore, we have our second equation:

$$0.18 = \frac{1}{2} \log \left(\frac{y_u}{y_d} \right). \tag{9.62}$$

  ⋆ Note that y_j depends on $\mathcal{P}^Z(j, T_3)$, which depends on $\mathcal{P}^Z(jk, T_3)$, which depends on r_{jk}, where $j = \{u, d\}$ and $k = \{u, d\}$.

- Finally, for our third equation we use the condition that the volatility of the short rate depends only on time and is the same across states:

$$\frac{1}{2} \log \left(\frac{r_{uu}}{r_{ud}} \right) = \frac{1}{2} \log \left(\frac{r_{ud}}{r_{dd}} \right)$$

$$(r_{ud})^2 = r_{dd}\, r_{uu}. \tag{9.63}$$

- Solving the three equations (9.61), (9.62), and (9.63) gives:

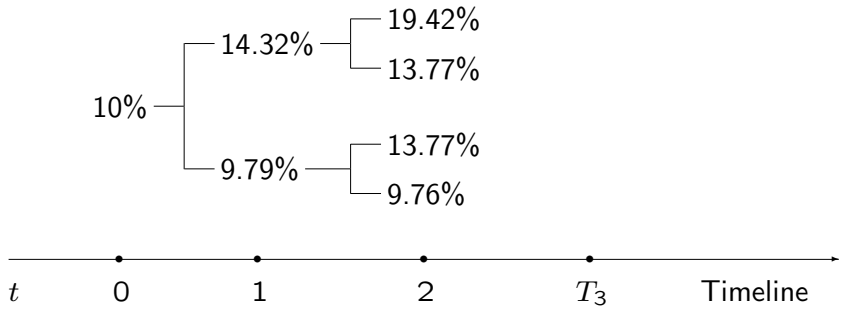
  $r_{uu} = 19.42\%$;

  $r_{ud} = 13.77\%$;

  $r_{dd} = 9.76\%$.

- So, the tree for the short rate is:

Tree for the short rate



Step 3: Find the tree for the prices of zero-coupon bonds

- From the tree for the short rate, we can find the trees for the price of zero-coupon bonds maturing at T_1, T_2 and T_3.

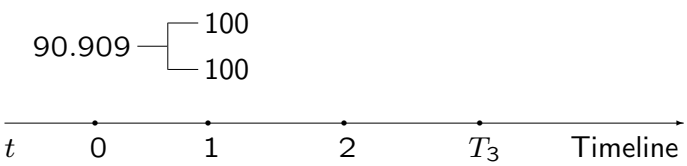
Step 3.1: Find the tree for the price of the one-year zero-coupon bond

- We know that the short rate at $t = 0$ is 10%.

- Thus, the price of the one-year zero-coupon bond at $t = 0$ is

$$\mathcal{P}^Z(0, T_1) = \frac{100}{1.10} = 90.909.$$

Tree for the price of the one-year zero-coupon bond



Step 3.2: Find the tree for the price of the two-year zero-coupon bond

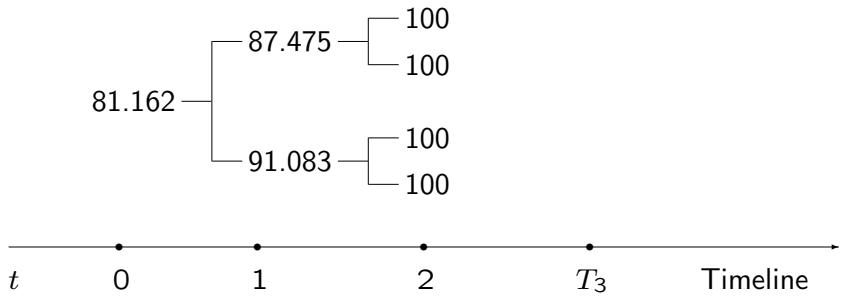
- From the data on Page 273, we know the two-year spot rate is 11%, so:

$$\mathcal{P}^Z(0,T_2) = \frac{100}{1.11^2} = 81.162.$$

- We also know $r_u = 14.32\%$ and $r_d = 9.79\%$ from Equations (9.53) and (9.54), respectively. This implies:

$$\mathcal{P}^Z(u,T_2) = \frac{100}{1.1432} = 87.475;$$
$$\mathcal{P}^Z(d,T_2) = \frac{100}{1.0979} = 91.083.$$

Tree for the price of the two-year zero-coupon bond



Step 3.3: Find the tree for the price of the three-year zero-coupon bond

- From the data on Page 273, we know the two-year spot rate is 11%, so:

$$\mathcal{P}^Z(0,T_3) = \frac{100}{1.12^3} = 71.178.$$

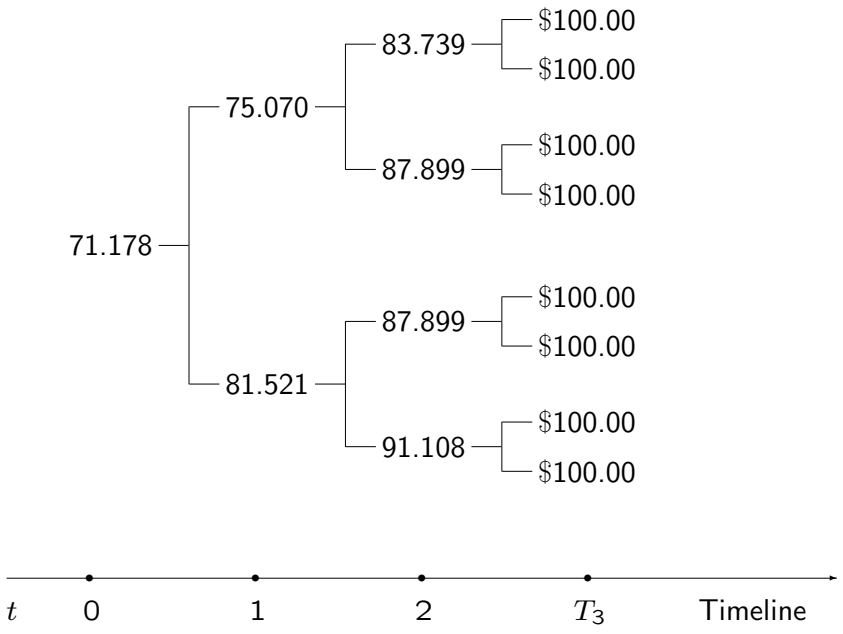
- From Equations (9.55), (9.56), and (9.57):

$$\mathcal{P}^Z(uu,T_3) = \frac{100}{1+r_{uu}} = \frac{100}{1.1932} = 83.739;$$
$$\mathcal{P}^Z(ud,T_3) = \frac{100}{1+r_{ud}} = \frac{100}{1.1377} = 87.899;$$
$$\mathcal{P}^Z(dd,T_3) = \frac{100}{1+r_{dd}} = \frac{100}{1.0976} = 91.108.$$

- Finally, from Equations (9.58) and (9.59), we have:

$$\mathcal{P}^Z(u,T_3) = \frac{1}{1+r_u}\frac{\left[\mathcal{P}^Z(uu,T_3)+\mathcal{P}^Z(ud,T_3)\right]}{2} = \frac{1}{1.1432}\frac{[83.739+87.899]}{2} = 75.070;$$
$$\mathcal{P}^Z(d,T_3) = \frac{1}{1+r_d}\frac{\left[\mathcal{P}^Z(ud,T_3)+\mathcal{P}^Z(dd,T_3)\right]}{2} = \frac{1}{1.0979}\frac{[87.899+91.108]}{2} = 81.521.$$

Tree for price of three-year zero-coupon bond



9.13 Suggestions for Further Readings

9.13.1 Source material for the lecture notes

- These lecture notes are based on Back (2005, Chapters 13 &14) and Back (2017, Chapter 18). See also Dumas and Luciano (2017, Chapter 16).
- The models we discussed in this chapter are from the following articles:
  - ⋆ Vasicek (1977)
  - ⋆ Cox, Ingersoll, and Ross (1985a)
  - ⋆ Ho and Lee (1986)
  - ⋆ Hull and White (1990)
  - ⋆ Black, Derman, and Toy (1990) and Black and Karasinski (1991)
  - ⋆ Longstaff and Schwartz (1992)
  - ⋆ Duffie and Kan (1996) and Dai and Singleton (2000)
  - ⋆ Ahn, Dittmar, and Gallant (2002)
  - ⋆ Heath, Jarrow, and Morton (1992).

9.13.2 Potential applications and extensions of lecture notes

- There are a large number of books that discuss factor models of the term structure of interest rates: Jarrow (2002), Brigo and Mercurio (2007), Filipovic (2009) and Cairns (2018). For a gentle introduction, see Sundaram and Das (2015).
- See the review paper by Piazzesi (2010).
- Rebonato (1996) is a book with lots of institutional details.
- Veronesi (2010) is a comprehensive book on fixed-income securities.

- The topic of credit risk, which we did not touch on,
  - ⋆ is discussed in the following books:
    - ∗ Brigo and Mercurio (2007, Chapters 21–23),
    - ∗ Filipovic (2009, Chapter 12), and
    - ∗ Cairns (2018, Chapter 11).
  - ⋆ Some excellent books dedicated entirely to this topic are
    - ∗ Schönbucher (2003),
    - ∗ Lando (2009),
    - ∗ Duffie and Singleton (2012), and
    - ∗ Bielecki and Rutkowski (2013).
  - ⋆ See also the review article Jarrow (2009).

9.13.3 Ph.D. elective courses offered on related topics

- Riccardo Rebonato has taught the course "*Term Structure Modelling in the P and Q measures*" in January 2017.

- Pietro Veronesi taught a course on "*Fixed Income Securities*" in September 2021.

- Frank Diebold has taught several elective PhD courses for EDHEC.

  ⋆ He taught the course titled "*The Yield Curve*," in September 2010.

Blank page.

Lecture Notes for 2025

Continuous Time Financial Economics

Chapter 10

Portfolio Choice: Mean-Variance and Two-Date Models

Raman Uppal

EDHEC Business School

Chapter contents

10.1 Plan for the Day . . . 297

10.2 Main Points for Today . . . 298

10.3 Background . . . 300

10.3.1 Valuation of Redundant Assets . . . 301

10.3.2 Optimal Consumption and Portfolio Policies . . . 302

10.3.3 Plan for Understanding Optimal Portfolio Selection . . . 304

10.4 Understanding Portfolio Choice Via Nobel Pictures . . . 305

10.4.1 A Summary of Main Insights from Modern Portfolio Theory . . . 314

10.5 Mean-Variance Portfolio Choice in a One-Date Model . . . 316

10.5.1 Guide to Notation . . . 318

10.5.2 Finding the Mean-Variance Efficient Portfolio in the Absence of a Riskfree Asset . . . 326

10.5.3 Global Minimum-Variance Portfolio . . . 329

10.5.4 Mean-Variance Efficient Portfolio in Presence of a Risk-Free Asset . . . 331

10.5.5 Characterizing the Tangency Portfolio and the Sharpe Ratio . . . 334

10.5.6 Identifying the Weights of the Tangency Portfolio: Mean-Variance Utility Maximization . . . 338

10.5.7 Relation Between Returns of Individual Assets and Frontier Portfolios: CAPM . . . 341

10.6 Measures of Portfolio Performance . . . 350

10.6.1 Sharpe Ratio . . . 350

10.6.2 The Treynor Ratio . . . 351

10.6.3 Jensen's Alpha . . . 352

10.6.4 Multifactor Alpha . . . 353

10.7 Consumption-Portfolio Choices in a Two-Date Model . . . 357

10.7.1 Understanding the First-Order Condition for Optimal Portfolios . . . 365

10.7.2 Understanding First-Order Condition for Intertemporal Allocation . . . 367

10.7.3 Elasticity of Intertemporal Substitution (EIS) . . . 372

10.7.4 Understanding relation between the SDF and risky asset prices . . . 377

10.7.5 Understanding relation between the SDF and the risk-free rate . . . 378

10.8 Summary . . . 379

10.9 Suggestions for Further Readings . . . 383

10.9.1 Source material for the lecture notes . . . 383

10.9.2 Potential applications and extensions of lecture notes . . . 385

10.9.3 Ph.D. elective courses offered on related topics . . . 392

10.1 Plan for the Day

Last Time

- Models of the Term Structure of Interest Rates

Today

- **How do we find the optimal portfolio in a mean-variance setting?**
  - ⋆ Overview of main insights of modern portfolio theory and asset pricing

For Next Time

- Review the material covered in class.
- Read Back, Chapters 2, 5, and 6 and Dumas & Luciano, Chapter 4.
- Do the practice problems.

10.2 Main Points for Today

- Understanding the main insights of modern portfolio theory:
  - ⋆ Diversifying optimally reduces the risk of a portfolio;
  - ⋆ Diversification eliminates unsystematic risk, so that a diversified portfolio has only systematic risk;
  - ⋆ Two-fund separation in the two-date CAPM model:
    - ∗ In the one period (from date 0 to date 1) CAPM equilibrium model, the two funds that investors hold are the risk-free asset and the market portfolio.
  - ⋆ Two *plus* k fund separation in multiperiod models
    - ∗ In the multiperiod Intertemporal CAPM model, investors desire these two funds, and an additional k funds, where k is the number of state variables in the model.

- And, after that:
  - ⋆ In the next chapter, we will study optimal portfolio choice in discrete and in continuous time using the martingale approach.
  - ⋆ In the next-to-next chapter, we will study optimal portfolio choice in continuous time using the dynamic-programming approach.
  - ⋆ Of course, both approaches lead to the same solution, but
    - ∗ when markets are complete, the martingale approach is easier to implement, and
    - ∗ when markets are not complete, then the dynamic programming approach is easier.

10.3 Background

Recall, that in this course we wish to answer three questions, which are nested:

1. How to value redundant securities?

Least general question: we are given prices of fundamental assets and the desired cashflows; need to determine only the replicating portfolio strategy.

2. How to determine optimal consumption and portfolio policies of investors?

More general than the previous question: we are given prices of the fundamental assets; need to determine both the desired cashflows and the portfolio strategy generating these cashflows.

3. How to determine the equilibrium values in a market with investors who are optimizing their consumption and portfolio policies?

Most general question: need to determine prices of fundamental assets, desired cashflows, and the portfolio strategy that replicates these cashflows.

10.3.1 Valuation of Redundant Assets

- For the valuation of redundant assets,
  - ⋆ we are given the cashflows to be valued
    (for example, the cashflows generated by an option), and
  - ⋆ we are also given the prices of fundamental securities
    (for example, the stock and the bond), and
  - ⋆ all that we are asked to do is to find the value of the given cashflow in terms of the value of the fundamental securities.

- We have finished studying the valuation of redundant assets
  whose payoff depends on
  - ⋆ the value of a stock, currency, index, etc.
  - ⋆ the value of a bond or an interest rate.

10.3.2 Optimal Consumption and Portfolio Policies

- Second question: determining optimal consumption and portfolio policies:
  - ⋆ we are given prices of fundamental securities (bond and N stocks), and
  - ⋆ we are asked to find both
    - ∗ the optimal cashflows (the consumption policy) and
    - ∗ the portfolio that replicates this cashflow (the portfolio policy).
  - ⋆ That is, in contrast to the previous problem, we are not given the cashflows to be valued, and so we need to identify both:
    1. the desired cashflows, and
    2. also the trading strategy that replicates this cashflow.

- So, the additional step is to find the investor's desired pattern of cashflows.

- To determine the desired pattern of cashflows, we will need to specify the investor's objective function.

- If markets are complete, this is a straightforward problem to solve.

  In complete markets, the problem can be decomposed into two steps, and each step can be solved separately:

  - ⋆ In the first step, we determine the optimal "consumption" policy.
  - ⋆ In the second step, given that we know the cashflow we wish to generate, we use the approach discussed for replicating redundant claims.

- If markets are not complete, then:

  - ⋆ one **cannot** decompose the consumption-portfolio problem into two distinct steps; the optimal consumption and portfolio must be identified simultaneously.
  - ⋆ We will study how to do this using the "dynamic programming" approach. (We will do this in the next session.)

- Equilibrium asset pricing

  - ⋆ Once we have identified the portfolio demanded by investors, we will equate this to the supply of risky securities, which will then give us the asset prices in equilibrium. (We will do this in the next-to-next class.)

10.3.3 Plan for Understanding Optimal Portfolio Selection

1. Study portfolio selection at an intuitive level, using only pictures.

2. Study a one-date model of portfolio choice.

3. Study a two-date model of portfolio choice.

   - You have already seen this in your earlier course on financial economics set in discrete time.

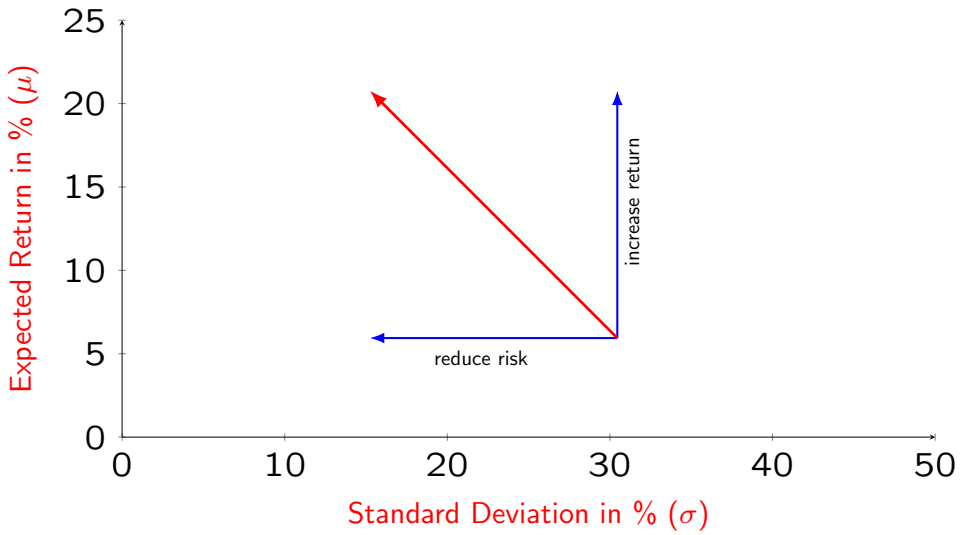
4. Then, in the next chapter, study a multi-date portfolio choice problem when markets are complete using the Martingale approach, first in discrete time and then in continuous-time.

5. In the following chapter, study portfolio choice in continuous time for incomplete markets using the Dynamic Programming approach.

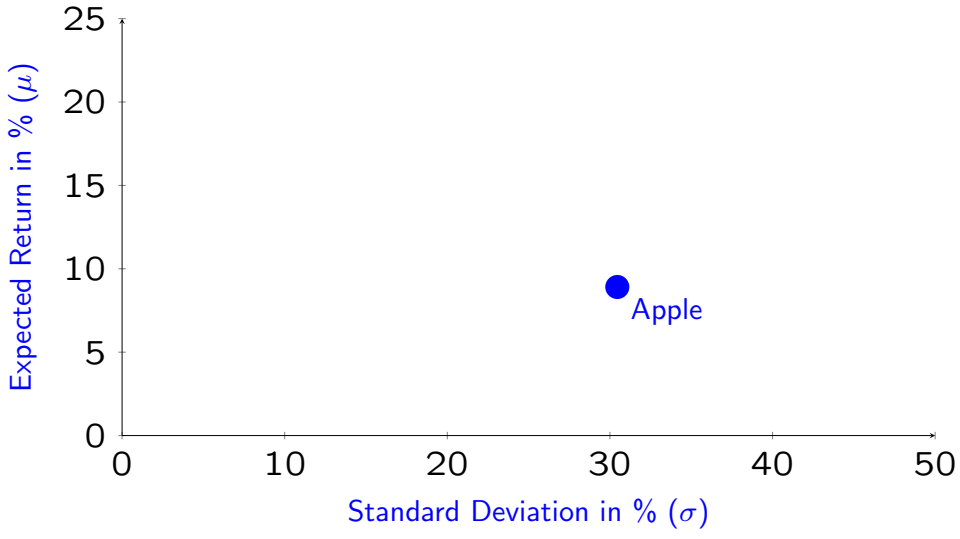
6. Finally, study implications of optimal portfolio selection for asset prices.

10.4 Understanding Portfolio Choice Via Nobel Pictures

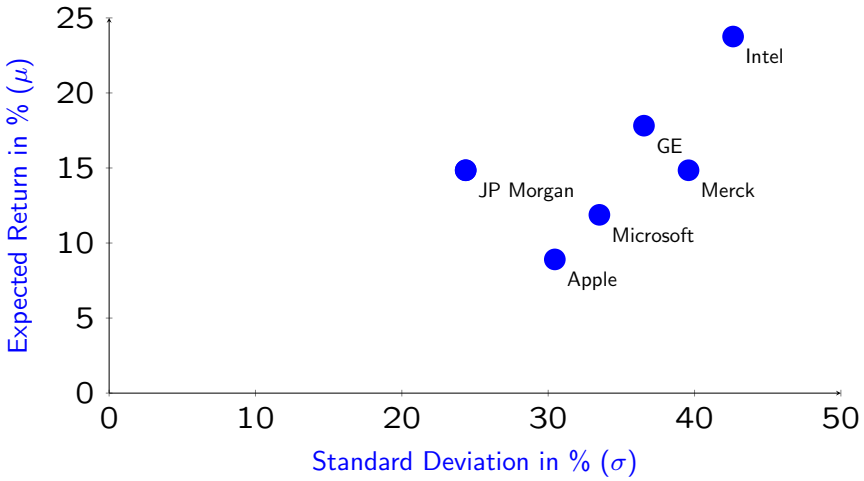
- Modern portfolio theory says that:
  "investors wish to balance their risk and return"



- Therefore, should not look at risk and return for each stock in isolation.



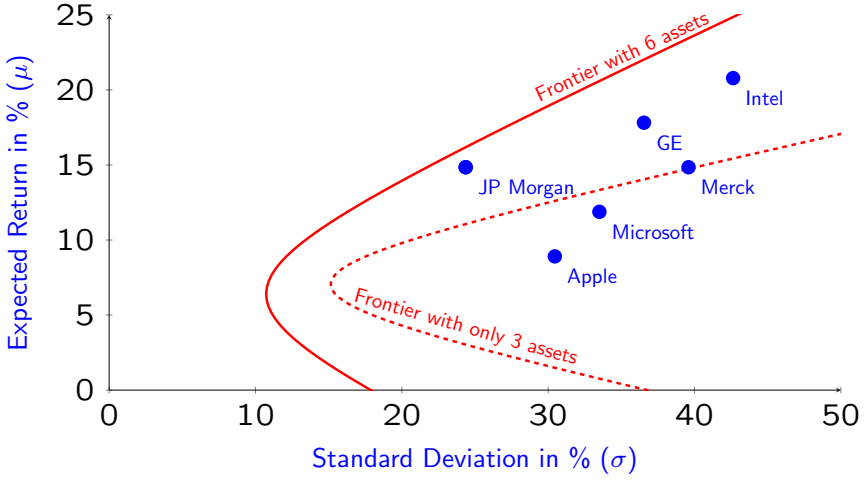
- Should consider risk and return for several stocks together.



Markowitz (Nobel Prize, 1990)

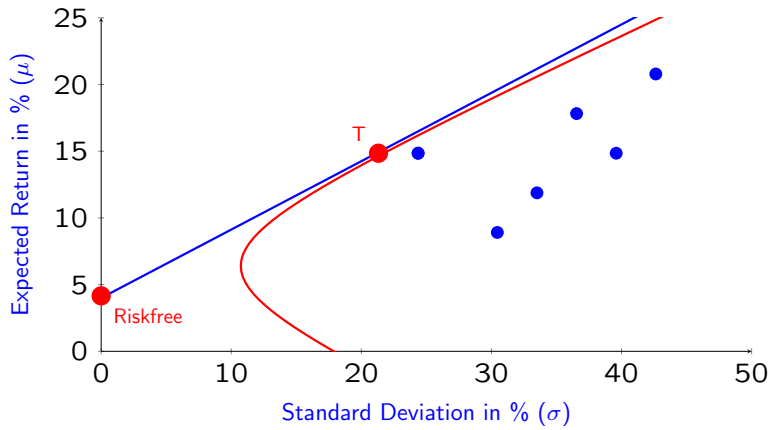
- Look at not just expected return and risk of individual stocks, but, consider also the "interaction" between stocks: correlation

- Reap the benefits of optimal diversification: Efficient frontier

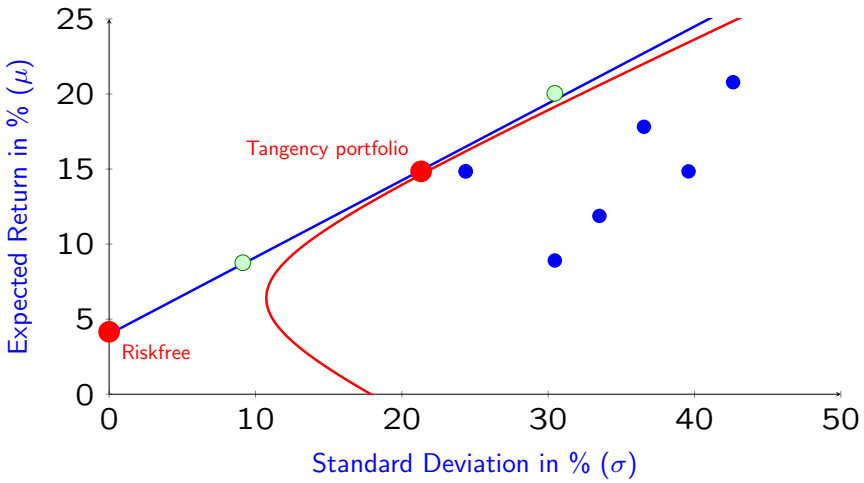


James Tobin (Nobel Prize, 1981)

- Simplified portfolio problem by introducing a riskless asset
- If a riskless asset (bank account) is available then:
  Frontier portfolios are combinations of
  - ⋆ Riskless asset , and
  - ⋆ Portfolio of only-risky assets.



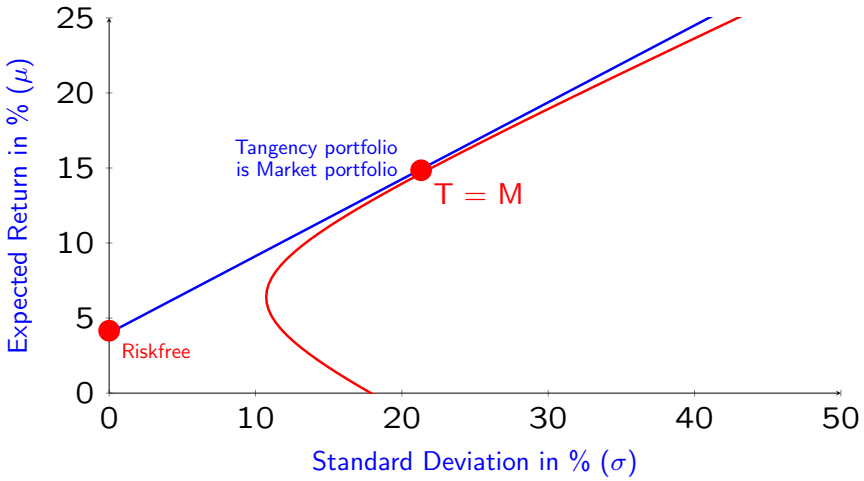
- Two-fund separation



- Thus, every investor's portfolio consists of only two funds:
  - ⋆ the riskless asset
  - ⋆ the tangency portfolio T, consisting of only-risky assets.

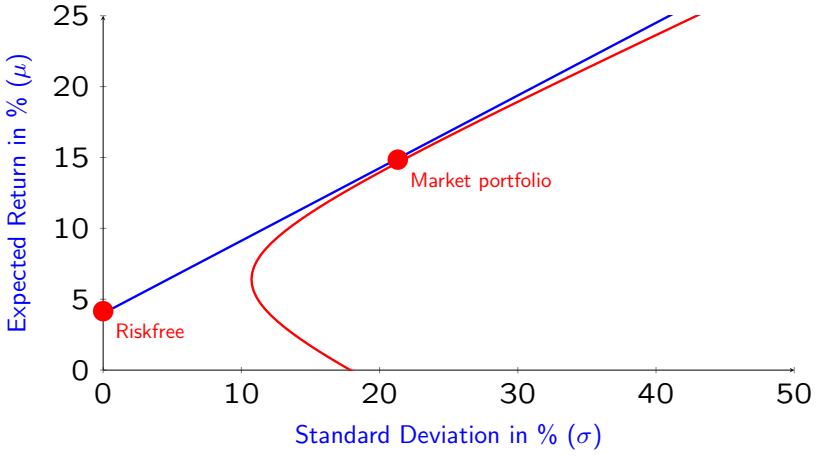
William Sharpe (Nobel Prize, 1990)

- Identified the tangency portfolio as the market portfolio: T = M
- This led to the **Capital Asset Pricing Model (CAPM)**: only systematic risk matters



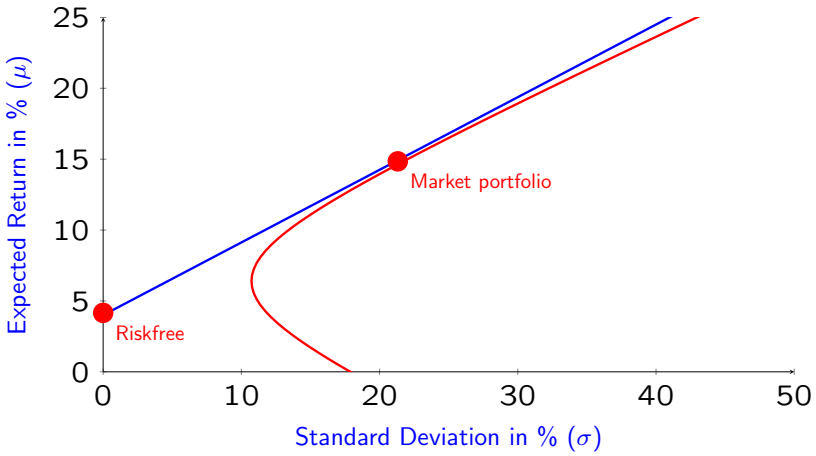
Paul Samuelson (Nobel Prize, 1970)

- What is the optimal portfolio in a **multiperiod** setting?
  - ⋆ If riskless rate does **not** change
  - ⋆ If slope of tangency line (Sharpe ratio) does **not** change
  - ⋆ Then, optimal portfolio does not change



Robert Merton (Nobel Prize, 1990)

- What is the optimal portfolio in a **multiperiod** setting?
  - ⋆ If riskless rate and slope of tangency line **can** change
  - ⋆ Then, optimal portfolio is equal to:
    - ∗ Markowitz portfolio + hedges against these changes



10.4.1 A Summary of Main Insights from Modern Portfolio Theory

Based on the results described above, we make the following observations:

1. Risks in individual asset returns have two components:
   - Systematic risks—common to many assets; therefore, non-diversifiable
   - Non-systematic risks—specific to individual assets; therefore, diversifiable.
2. Forming diversified portfolios can eliminate non-systematic risks.
3. Investors (should) hold diversified portfolios instead of single assets.
4. Consequently, investors (should) care only about systematic risks.
5. Thus, return on an asset (should) compensate only for systematic risks.

6. When there is risk-free asset, frontier portfolios are linear combinations of the risk-free asset and the tangent portfolio; that is, we have two-fund separation.

7. An individual asset's risk is its contribution to portfolio risk.

- Variance of each asset contributes little to portfolio risk; covariances between assets determine portfolio risk.

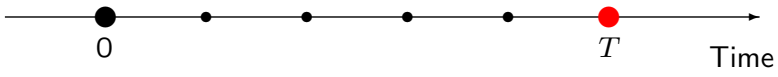
8. For any asset i:

- Its risk can be measured by its beta with respect to a frontier portfolio.
- Its return per unit of risk is the premium on the frontier portfolio.

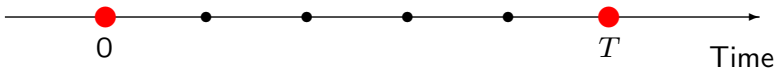
$\star$ These results are true not just for one- or two-date models, but also for multiperiod models set in discrete time and in continuous time.

10.5 Mean-Variance Portfolio Choice in a One-Date Model

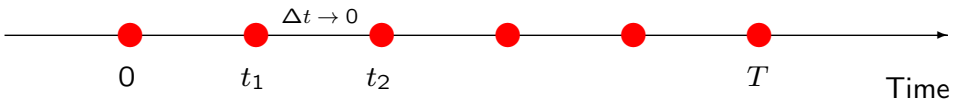
- We start by analyzing optimal portfolio choice in a one-date model where the investor chooses her portfolio at $t = 0$ and cares only about wealth at date T.



- We then analyze consumption and portfolio choice in a static model where the investor can trade and consume at the dates 0 and T.



- Finally, we will consider dynamic continuous-time models of portfolio choice where the investor can rebalance her portfolio at each date (instant):
  - ⋆ Using the Martingale approach if markets are complete;
  - ⋆ Using the Dynamic Programming approach if markets are incomplete.



10.5.1 Guide to Notation

- The main objects that we wish to study are the
  - ⋆ a portfolio;
  - ⋆ return of a portfolio; and
  - ⋆ risk of a portfolio.
- A portfolio is defined by the weights allocated to the assets in the portfolio.
  - ⋆ The weight of asset i in the portfolio is denoted by: w_i.
  - ⋆ We denote by N the total number of assets in a portfolio.
  - ⋆ So, a portfolio is defined by the weights of the N assets in it:

$$\{w_1, w_2, \cdots, w_N\}.$$

- w denotes the N-dimensional column vector of portfolio weights:

$$w = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_N \end{bmatrix}.$$

- $w^\top$ is a row vector, which is the transpose of w, the N-dimensional column-vector of portfolio weights:

$$w^\top = [w_1, \ w_2, \ \cdots, \ w_N].$$

- r denotes the N-vector of returns on the N risky assets.

$$r = \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_N \end{bmatrix}.$$

- μ denotes the N-dimensional vector of expected (or *mean*) returns.

$$\mu = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_N \end{bmatrix}.$$

- $\mathbf{V}$ denotes the corresponding $N \times N$ variance-covariance matrix of returns.

$$\mathbf{V} = \begin{bmatrix} \sigma_1^2 & \sigma_{12} & \cdots & \sigma_{1N} \\ \sigma_{21} & \sigma_2^2 & \cdots & \sigma_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{N1} & \sigma_{N2} & \cdots & \sigma_N^2 \end{bmatrix}$$

- This is an important matrix, that we will refer to throughout the course.

  In the literature, this matrix is often denoted using the Greek letter $\mathbf{\Sigma}$.

- We will often need the inverse of the $\mathbf{V}$ matrix, which we denote as $\mathbf{V}^{-1}$.

- $\mathbf{1}$ denotes an N-dimensional vector of ones:

$$\mathbf{1} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$$

- I_N to indicate the $N \times N$ identity matrix.

$$I_N = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

Guide to Notation for Return and Risk of Portfolio with Multiple Assets

- The returns on the N assets, $\{r_1, r_2, \ldots, r_N\}$, are characterized by their mean, variance and covariances.

- Mean returns:

Asset	1	2	$\cdots$	n
Mean Return	μ_1	μ_2	$\cdots$	μ_N

- Variances and covariances:

	r_1	r_2	$\cdots$	r_N
r_1	σ_1^2	σ_{12}	$\cdots$	σ_{1N}
r_2	σ_{21}	σ_2^2	$\cdots$	σ_{2N}
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
r_N	σ_{N1}	σ_{N2}	$\cdots$	σ_N^2

Consider a portfolio of N assets in which the proportion of total value invested in asset i is denoted by w_i.

1. The portfolio weights sum to one:

$$\sum_{i=1}^{N} w_i = w^\top \mathbf{1} = 1.$$

2. The portfolio return is:

$$r_p = w_1 r_1 + w_2 r_2 + \cdots + w_N r_N = \sum_{i=1}^{N} w_i r_i = w^\top r.$$

3. The portfolio expected return is:

$$\mu_p = \mathbb{E}^{\mathbb{P}}[r_p] = w_1 \mu_1 + w_2 \mu_2 + \cdots + w_N \mu_N = \sum_{i=1}^{N} w_i \mu_i = w^\top \mu.$$

4. The variance of portfolio return is:

$$\sigma_p^2 = \text{Var}[r_p] = \sum_{i=1}^{N}\sum_{j=1}^{N} w_i w_j \sigma_{ij} = w^\top \mathbf{V}\, w, \qquad \text{where } \sigma_{ii} = \sigma_i^2.$$

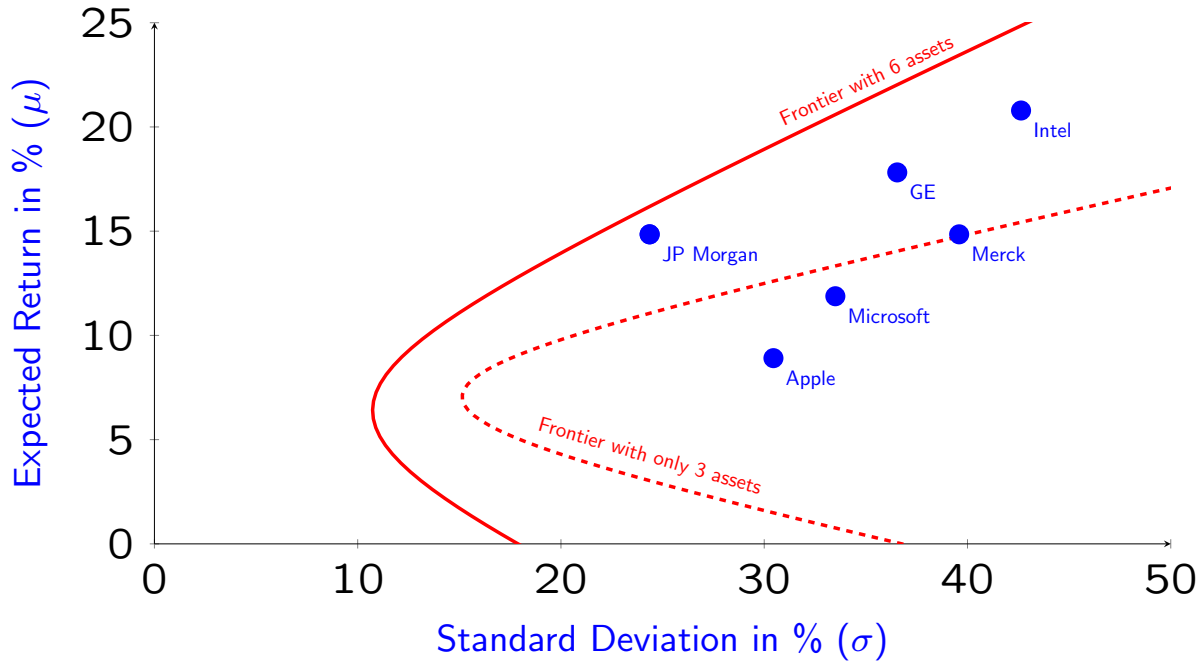
5. The risk or volatility (StD) of the portfolio return is:

$$\sigma_p = \sqrt{\text{Var}[r_p]} = \sqrt{\sigma_p^2} = \sqrt{w^\top \mathbf{V}\, w}.$$

10.5.2 Finding the Mean-Variance Efficient Portfolio in the Absence of a Riskfree Asset

- How to identify the mean-variance efficient portfolio frontier when the investor can choose from only risky assets?
  - ⋆ Minimize risk for a given expected return; or
  - ⋆ Maximize expected return for a given risk.
- Definition: Given an expected return, the portfolio that minimizes risk (measured by StD) is a *mean-StD frontier portfolio*.
- Definition: The locus of all frontier portfolios in the mean-StD plane is called *portfolio frontier*. The upper part of the portfolio frontier gives *efficient* frontier portfolios.

The portfolio frontier in the absence of a risk-free asset



Non-Linear Optimization Problem that Gives Efficient Frontier

- To obtain the frontier portfolios, we need to solve the following problem:

$$\underset{\{w_1,\ldots,w_N\}}{\text{Minimize}} \quad \text{Portfolio risk} = \sigma_p^2 = \sum_{i=1}^{N}\sum_{j=1}^{N} w_i\, w_j\, \sigma_{ij} = w^\top \mathbf{V}\, w$$

subject to the following two constraints:

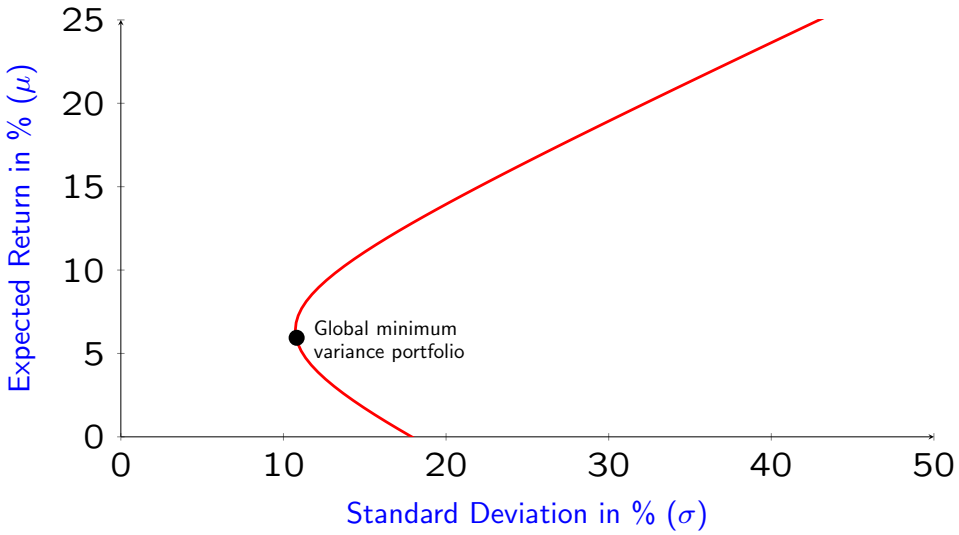
$$(1) \quad \sum_{i=1}^{N} w_i = w^\top \mathbf{1} = 1 \qquad \ldots \text{ weights sum to 1}$$

$$(2) \quad \sum_{i=1}^{N} w_i\, \mu_i = w^\top \mu = \mu_p \qquad \ldots \text{ portfolio's expected return is } \mu_p$$

- The inputs to this problem need to be estimated from the data:
  - ⋆ expected returns: μ_i, that is, the vector μ;
  - ⋆ risk (variances and covariances): σ_{ij}, that is, the matrix $\mathbf{V}$.

10.5.3 Global Minimum-Variance Portfolio

- There is a point on the efficient frontier that has the lowest volatility, which corresponds to the global minimum-variance portfolio.



- To identify the global minimum-variance ("MINV") portfolio, we choose weights that minimize the variance of portfolio returns; that is,

$$\min_{w} \quad \underbrace{w^\top \mathbf{V} w}_{\text{portfolio variance}} \quad , \qquad \text{subject to:} \quad \mathbf{1}^\top w = 1.$$

- Observe that the optimization problem above uses as an input only the estimate of the covariance matrix of asset returns, and does not use estimates of expected returns.

- The minimum-variance weights are a limiting case of the mean-variance problem if a mean-variance investor:

  ⋆ either ignores expected returns,

  ⋆ or equivalently, restricts expected returns to be identical across all assets.

10.5.4 Mean-Variance Efficient Portfolio in Presence of a Risk-Free Asset

- With a risk-free asset, with return r_F, the optimal portfolio problem is:

  Minimize Portfolio variance $= \sigma_p^2 = \sum_{i=1}^{N} \sum_{j=1}^{N} w_i \, w_j \, \sigma_{ij}, = w^\top \mathbf{V} \, w$

  $\{w_1, \ldots, w_N\}$

  subject to the (single) constraint that the portfolio's expected return is μ_p,

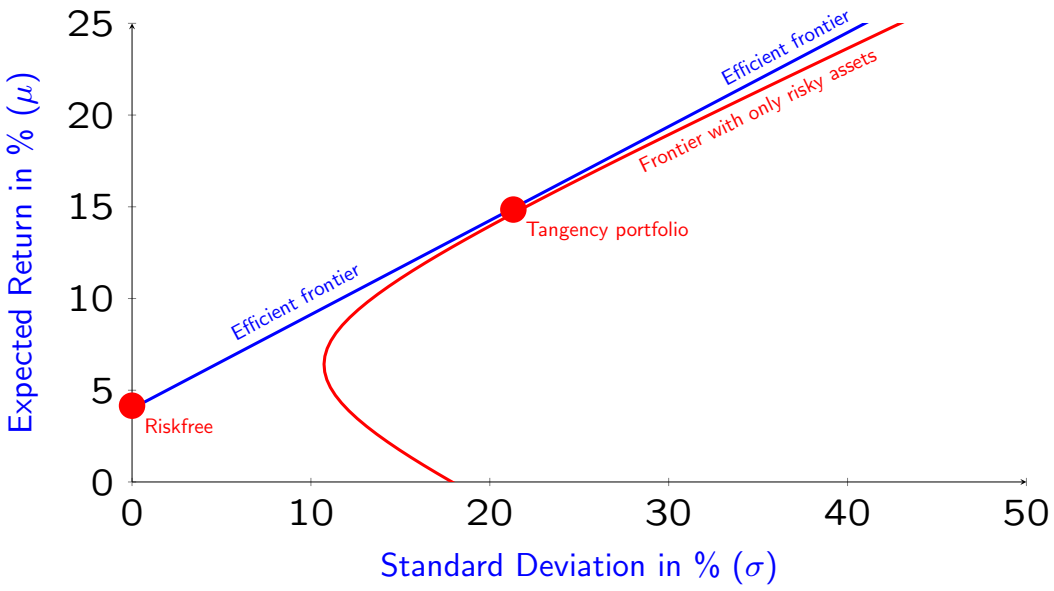
  $$\underbrace{\left(1 - \sum_{i=1}^{N} w_i\right)}_{\text{constraint: weights sum to 1}} \; r_F + \sum_{i=1}^{N} w_i \, \mu_i = \mu_p$$

  or, using vector notation, the constraint is: $(1 - w^\top \mathbf{1}) r_F + w^\top \mu = \mu_p$.

- Note that the constraint that the weights have to be sum to one has been included in the constraint requiring one to achieve a target mean return.

Solution to above problem gives the portfolio frontier with a safe asset

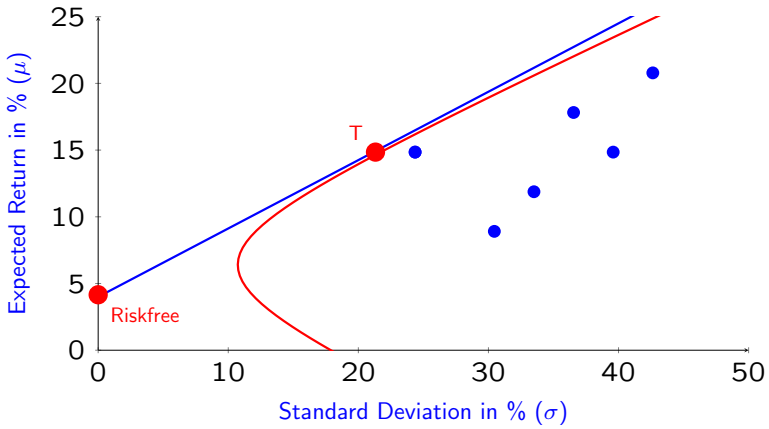
Portfolio frontier with six risky assets and also a risk-free asset



Two-fund separation

Observation: A portfolio of risk-free and risky assets can be viewed as a portfolio of two sub-portfolios:

1. a subportfolio of only the risk-free asset, and
2. a subportfolio of only risky assets, called the Tangency portfolio.



10.5.5 Characterizing the Tangency Portfolio and the Sharpe Ratio

- Consider a portfolio that has
  - w^{Tangency} invested in the tangency portfolio, with return r^{Tangency};
  - $(1 - w^{\text{Tangency}})$ invested in the risk-free asset, which has the return r_F;
- Then, the return on this portfolio, denoted r_p, is:

$$r_p = w^{\text{Tangency}}\, r^{\text{Tangency}} + (1 - w^{\text{Tangency}}) r_F.$$

- The expected return on this portfolio is

$$\mu_p = w^{\text{Tangency}}\, \mu^{\text{Tangency}} + (1 - w^{\text{Tangency}}) r_F = w^{\text{Tangency}} (\mu^{\text{Tangency}} - r_F) + r_F$$

- The variance of the return on this portfolio is: $\sigma_p^2 = (w^{\text{Tangency}})^2\, (\sigma^{\text{Tangency}})^2$
- The volatility or risk of the return on this portfolio is: $\sigma_p = w^{\text{Tangency}}\, \sigma^{\text{Tangency}}$.

- We start by re-writing the expressions for the expected return and volatility:

$$\text{Expected return:} \quad \mu_p = w^{\text{Tangency}}(\mu^{\text{Tangency}} - r_F) + r_F;$$

$$\text{Volatility:} \quad \sigma_p = w^{\text{Tangency}}\,\sigma^{\text{Tangency}}, \quad \text{which implies:}$$

$$w^{\text{Tangency}} = \frac{\sigma_p}{\sigma^{\text{Tangency}}}.$$

- Substituting the expression for w^{Tangency} from the equation for volatility into the equation for expected return gives:

$$\mu_p = r_F + \sigma_p\left(\frac{\mu^{\text{Tangency}} - r_F}{\sigma^{\text{Tangency}}}\right),$$

which is the equation of a straight line with

⋆ intercept given by r_F

⋆ slope given by $\left(\frac{\mu^{\text{Tangency}} - r_F}{\sigma^{\text{Tangency}}}\right)$, which is Sharpe ratio.

- Note that the tangency portfolio is the portfolio that attains the highest Sharpe ratio.

- Finally, note that the last equation on the previous page

$$\mu_p = r_F + \sigma_p\left(\frac{\mu^{\text{Tangency}} - r_F}{\sigma^{\text{Tangency}}}\right),$$

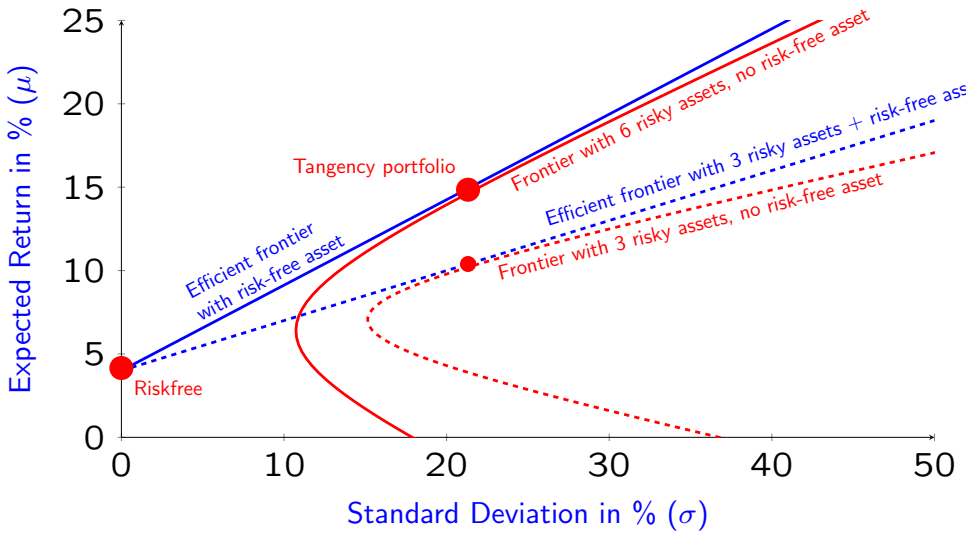
can be re-written as:

$$\frac{\mu_p - r_F}{\sigma_p} = \left(\frac{\mu^{\text{Tangency}} - r_F}{\sigma^{\text{Tangency}}}\right),$$

which says that the Sharpe ratio of portfolio p must equal the Sharpe ratio for the tangency portfolio.

- Thus, all portfolios on the straight-line efficient frontier have the highest Sharpe ratio that is attainable.

- Observation: As in the case with only risky assets, diversification is still useful.
  - ⋆ When more risky assets assets are included, the portfolio frontier with only risky assets improves (from the dashed red curve to the solid red curve),
  - ⋆ implying that the frontier with the risk-free asset also improves (from dashed blue line to solid blue line).



10.5.6 Identifying the Weights of the Tangency Portfolio: Mean-Variance Utility Maximization

- Recall that to find the mean-variance efficient portfolio in the presence of a risk-free asset, we
  - ⋆ minimize the portfolio variance: $\sigma_p^2 = w^\top \mathbf{V} w$,
  - ⋆ subject to the constraint that the expected return of the portfolio is equal to μ_p; that is, $(1 - w^\top \mathbf{1}) r_F + w^\top \mu = \mu_p$.

- Markowitz shows that the above problem can be re-written as maximizing the difference between
  - ⋆ the expected return on the portfolio, and
  - ⋆ the variance of the portfolio, adjusted for the risk-aversion γ of the investor.

$$\max_w \quad \mu_p - \frac{\gamma}{2}\sigma_p^2 \qquad \ldots \text{mean-variance utility.}$$

- We start by re-writing the problem on the last page:

$$\begin{aligned} &\max_w \quad \mu_p - \frac{\gamma}{2}\sigma_p^2 \\ \Rightarrow &\max_w \quad \left[(1 - w^\top \mathbf{1})r_F + w^\top \mu\right] - \frac{\gamma}{2} w^\top \mathbf{V} w \\ \Rightarrow &\max_w \quad \left[r_F + w^\top (\mu - r_F \mathbf{1})\right] - \frac{\gamma}{2} w^\top \mathbf{V} w \\ \Rightarrow &\max_w \quad w^\top (\mu - r_F \mathbf{1}) - \frac{\gamma}{2} w^\top \mathbf{V} w. \end{aligned}$$

- The solution to this problem (obtained by differentiating the above expression with respect to w and solving the resulting first-order condition) is:

$$w = \frac{1}{\gamma} \mathbf{V}^{-1} (\mu - r_F \mathbf{1}) \quad \ldots \text{optimal portfolio.} \tag{10.1}$$

- We see that the investment in the risk-free asset will be:

$$w_F = 1 - w^\top \mathbf{1}.$$

- The vector of relative portfolio weights, ω^{Tangency}, of the N risky assets in the tangency portfolio is:

$$\omega^{\text{Tangency}} = \frac{\text{optimal weight}}{\text{sum of weights in all risky assets}} = \frac{\mathbf{V}^{-1} (\mu - r_F \mathbf{1})}{\mathbf{1}^\top \mathbf{V}^{-1} (\mu - r_F \mathbf{1})}. \tag{10.2}$$

- Note that the investor's risk aversion, γ, does not appear in ω^{Tangency}, the expression for the weights in the tangency portfolio.

- Thus, all investors will hold risky assets in the same proportion; that is, the proportion in which they are present in the tangency portfolio.

- That is, all investors will demand the same portfolio of risky assets: the tangency portfolio.

- In equilibrium, the supply of risky assets (market portfolio) must match demand for risky assets; thus, the tangency portfolio is the market portfolio.

10.5.7 Relation Between Returns of Individual Assets and Frontier Portfolios: CAPM

- So far, we have observed that:
  - ⋆ Investors hold portfolios to reduce risk.
    - ∗ "Unsystematic risk" of individual assets doesn't matter.
    - ∗ Only "systematic risk" matter.
  - ⋆ Investors hold only frontier portfolios.
- The natural questions to ask next are:
  - ⋆ How does an individual asset contribute to the risk of portfolios, especially the frontier portfolios?
  - ⋆ Can we be more specific about what is "systematic risk"?
  - ⋆ How is an asset's systematic risk related to its expected return?

10.5.7.1 Contribution of an Asset to a Portfolio's Return

- We consider the case where one can invest in a risk-free asset.
- The expected portfolio return is: $\mu_p = r_F + \sum_{i=1}^{N} w_i (\mu_i - r_F)$.
- Definition: "Marginal contribution of X to Y" means the incremental change in Y when X changes by a small amount.
- The marginal contribution of risky asset i to the expected portfolio return is its risk premium:

$$\frac{\partial \mu_p}{\partial w_i} = \mu_i - r_F.$$

10.5.7.2 Contribution of an Asset to Portfolio Risk

- To identify the contribution of an asset to the risk of a portfolio, we proceed in three steps:

1. Find the total contribution of an asset to portfolio variance;
2. Find the marginal contribution of an asset to portfolio variance;
3. Find the marginal contribution of an asset to portfolio risk.

10.5.7.3 Total Contribution of an Asset to Portfolio Variance

- Recall that the variance of portfolio return is the sum of all entries of the following table:

	$w_1 r_1$	$w_2 r_2$	$\cdots$	$w_N r_N$
$w_1 r_1$	$w_1^2 \sigma_1^2$	$w_1 w_2 \sigma_{12}$	$\cdots$	$w_1 w_N \sigma_{1n}$
$w_2 r_2$	$w_2 w_1 \sigma_{21}$	$w_2^2 \sigma_2^2$	$\cdots$	$w_2 w_N \sigma_{2n}$
$\cdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
$w_N r_N$	$w_N w_1 \sigma_{n1}$	$w_N w_2 \sigma_{n2}$	$\cdots$	$w_N^2 \sigma_N^2$

- The sum of the entries of the i-th-row and the i-th column is the total contribution of asset i to portfolio variance:

$$w_i^2 \sigma_i^2 + 2 \sum_{j \neq i}^{n} w_i \, w_j \sigma_{ij}.$$

10.5.7.4 Marginal Contribution of an Asset to Portfolio Variance: $\partial\sigma_p^2/\partial w_i$

- The marginal contribution of asset i to portfolio variance is (twice) its covariance with the portfolio:

$$\begin{aligned}
\frac{\partial\sigma_p^2}{\partial w_i} &= \frac{\partial}{\partial w_i}\left(w_i^2\sigma_i^2 + 2\sum_{j\neq i}^{n} w_i w_j \sigma_{ij}\right) \\
&= 2w_i\sigma_i^2 + 2\sum_{j\neq i}^{n} w_j\sigma_{ij} \\
&= 2\sum_{j=1}^{n} w_j\sigma_{ij} \\
&= 2\,\mathsf{Cov}[r_i, r_p] \quad \ldots \quad \sum_{j=1}^{n} w_j\sigma_{ij} = \sum_{j=1}^{n} w_j \mathsf{Cov}[r_i, r_j] = \mathsf{Cov}[r_i, \sum_{j=1}^{n} w_j r_j] = \mathsf{Cov}[r_i, r_p] \\
&= 2\sigma_{ip}.
\end{aligned}$$

10.5.7.5 Marginal Contribution of an Asset to Portfolio Risk: $\partial\sigma_p/\partial w_i$

- On the previous slide, we have shown that:

$$\frac{\partial\sigma_p^2}{\partial w_i} = 2\mathsf{Cov}[r_i, r_p] = 2\sigma_{ip}.$$

- But, using basic calculus, we also know that:

$$\frac{\partial\sigma_p^2}{\partial w_i} = 2\sigma_p\frac{\partial\sigma_p}{\partial w_i}.$$

- The left-hand sides of the two expressions are the same, implying that:

$$\begin{aligned}
2\sigma_p\frac{\partial\sigma_p}{\partial w_i} &= 2\sigma_{ip} \\
\frac{\partial\sigma_p}{\partial w_i} &= \frac{2\sigma_{ip}}{2\sigma_p} = \frac{\sigma_{ip}}{\sigma_p} = \frac{\text{covariance between return of asset } i \text{ and portfolio } p}{\text{volatility of return of portfolio } p}.
\end{aligned} \tag{10.3}$$

10.5.7.6 CAPM: Relation between Individual Asset and Frontier Portfolio Returns

- Definition: The (marginal) return-to-risk ratio (RRR) of risky asset i in a portfolio p is:

$$\text{RRR}_i = \frac{\text{marginal return}}{\text{marginal risk}} = \frac{\partial \mu_p / \partial w_i}{\partial \sigma_p / \partial w_i} = \frac{\mu_i - r_F}{(\sigma_{ip}/\sigma_p)}.$$

- Claim: For any frontier portfolio p, the return-to-risk ratio of all risky assets must be the same:

$$\text{RRR}_i = \frac{\mu_i - r_F}{(\sigma_{ip}/\sigma_p)} = \text{RRR}_p = \frac{\mu_p - r_F}{(\sigma_{pp}/\sigma_p)} = \frac{\mu_p - r_F}{(\sigma_p^2/\sigma_p)} = \frac{\mu_p - r_F}{\sigma_p}.$$

- Intuition: The RRR of a frontier portfolio cannot be improved.

- We start by re-writing the last result:

$$\frac{\mu_i - r_F}{(\sigma_{ip}/\sigma_p)} = \frac{\mu_p - r_F}{\sigma_p},$$

which implies the following important result:

$$\mu_i - r_F = \frac{\sigma_{ip}}{\sigma_p^2}\Big(\mu_p - r_F\Big) = \beta_{ip}\Big(\mu_p - r_F\Big),$$

where $\beta_{ip} = \sigma_{ip}/\sigma_p^2$ is the beta of asset i with respect to frontier portfolio p.

- We start by re-writing our last result:

$$\mu_i - r_F = \frac{\sigma_{ip}}{\sigma_p^2}\Big(\mu_p - r_F\Big) = \beta_{ip}\Big(\mu_p - r_F\Big),$$

- The interpretation of the above result is that given any frontier portfolio p (except the risk-free asset)
  - ⋆ β_{ip} gives a measure of asset i's systematic risk.
  - ⋆ $\mu_p - r_F$ gives the premium (price) per unit of systematic risk.
  - ⋆ The risk premium on asset i equals the amount of its systematic risk (β_{ip}) times the premium per unit of the risk ($\mu_p - r_F$).
- If the frontier portfolio is the market portfolio, that is $p = mkt$, then the above equation gives us the Capital Asset Pricing Model (CAPM):

$$\mu_i - r_F = \frac{\sigma_{im}}{\sigma_m^2}\Big(\mu_m - r_F\Big) = \beta_{im}\Big(\mu_m - r_F\Big), \qquad \text{CAPM}.$$

10.6 Measures of Portfolio Performance

10.6.1 Sharpe Ratio

- We have already seen that one measure of portfolio performance is

$$\text{Sharpe ratio} = \frac{\mu_p - r_F}{\sigma_p}.$$

- This measure tells us about the "excess return-to-risk" ratio of a portfolio, where
  - ⋆ excess return is measured in terms of the portfolio return in excess of the risk free rate;
  - ⋆ risk is measured as the volatility of the portfolio.

10.6.2 The Treynor Ratio

- We start by re-writing the equation for the CAPM:

$$\mu_i - r_F = \beta_{im}\Big(\mu_m - r_F\Big), \quad \text{which, on dividing both sides by } \beta_{im}, \text{ leads to:}$$

$$\text{Treynor ratio} = \frac{\mu_i - r_F}{\beta_{im}} = \Big(\mu_m - r_F\Big), \quad \text{note that the right-hand side is the same for all assets}$$

- In contrast, recall that:

$$\text{Sharpe ratio} = \frac{\mu_p - r_F}{\sigma_p}.$$

- Note that the:
  - ⋆ Sharpe ratio applies only to efficient portfolios in the presence of a risk-free asset, and the risk measure is total risk of the portfolio, σ_p;
  - ⋆ Treynor ratio applies to individual assets, and the risk measure is systematic risk, β_{im}.

10.6.3 Jensen's Alpha

- We start by re-writing the equation for the CAPM:

$$(\mu_i - r_F) = \beta_{im}\Big(\mu_m - r_F\Big).$$

- Jensen's alpha (α) is defined as the difference between the realized excess return and the excess return according to the CAPM:

$$\alpha = \underbrace{(r_i - r_F)}_{\text{realized return}} - \underbrace{\beta_{im}\Big(r_m - r_F\Big)}_{\text{CAPM return}}.$$

- Thus, Jensen's alpha is the marginal return associated with a unit exposure to a particular strategy; that is, it is the additional return that is not explained by the market factor.

10.6.4 Multifactor Alpha

- Following the development of the single-factor CAPM, it was found that returns were better described by models that allowed for multiple factors.
  - ⋆ Fama and French (1993) propose a three-factor model for returns, where the three factors are the market, value, and size.
  - ⋆ Carhart (1997) proposes a fourth factor, momentum:
- Thus, the four-factor model of returns is:

$$(\mu_i - r_F) = \beta_{i,m}\Big(\mu_m - r_F\Big) + \beta_{smb}\,\mu_{i,smb} + \beta_{i,hml}\,\mu_{hml} + \beta_{i,umd}\,\mu_{umd}$$

  - ⋆ smb = small market capitalization minus big (size);
  - ⋆ hml = high book-to-market ratio minus low (value);
  - ⋆ umd = up positive momentum minus down momentum.

- So, one can also measure the realized return that is not explained by this four-factor model, in which case we get the four-factor alpha:

$$\alpha_4 = \underbrace{(r_i - r_F)}_{\text{realized return}} - \underbrace{\Big[\beta_{im}\Big(\mu_m - r_F\Big) + \beta_{smb}\,\mu_{smb} + \beta_{hml}\,\mu_{hml} + \beta_{umd}\,\mu_{umd}\Big]}_{\text{four-factor return}}.$$

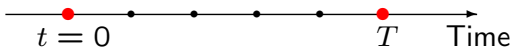
- For more recent work on factor models and anomalies, see Fama and French (2015) and the series of very interesting papers: Hou, Xue, and Zhang (2015, 2017a, 2017b) and Hou, Mo, Xue, and Zhang (2019).

End of **single-date** mean-variance portfolio choice model

10.7 Consumption-Portfolio Choices in a Two-Date Model

Notation

- W_t denotes wealth at date t;
- C_t denotes consumption at date t;
- So, the initial wealth of the investor is W_0, and at date T, the investor consumes all of her wealth: $C_T = W_T$.
  - ⋆ Alternatively, one could assume that there is bequest motive at date T.
- $u(C_t)$ denotes the utility from consuming C_t.
  - ⋆ $u(C)$ increases with consumption, but at a decreasing rate because of diminishing marginal utility.
- We now analyze consumption and portfolio choice in a static model where the investor can trade and consume at only two dates: t and T.



Assumptions About Utility Function

- The investor's lifetime utility is defined over beginning- and end-of-period consumption:

$$\mathcal{U}(C_0, C_T).$$

- We assume that utility is additively separate over time; that is:

$$\mathcal{U}(C_0, C_T) = u(C_0) + e^{-\delta}\,\mathbb{E}^{\mathbb{P}}\left[u(\tilde{C}_T)\right], \qquad \text{where} \tag{10.4}$$

  - ⋆ δ represents a subjective discount rate which reflects the rate of time preference (that is, preference for consuming earlier rather than later);
  - ⋆ $\mathbb{E}[\cdot]$ is the expectations operator conditional on information at date 0, because future consumption is risky.

Assumptions About Asset Prices and Returns

- Suppose that the investor can choose to invest in n different assets.
- Let $\mathcal{P}_{0,i}$ be the date 0 price per share of asset i.
- Let $\tilde{X}_{T,i}$ be the payoff of the asset on the terminal date, T. For example:
  - ⋆ A dividend-paying stock would have a random payoff of $\tilde{X}_{T,stock} = \tilde{\mathcal{P}}_{T,stock} + \tilde{D}_{T,stock}$
  - ⋆ A discount bond would have a non-random payoff of $X_{T,bond} = 1$.
- Given this definition, we can define the gross return on asset i:

$$\tilde{R}_i = \frac{\tilde{X}_{T,i}}{\mathcal{P}_{0,i}}.$$

Assumptions about Labor income

- We also allow the agent to receive labor income (wages) at dates 0 and T, which we denote as y_0 and $\tilde{y}_T$.
- There is a difference between tangible wealth, W, and wage income, y:
  - ⋆ The present value of wage income, which is referred to as human capital is assumed to be a nontradable asset.
  - ⋆ In contrast, tangible wealth is "marketable" and so its holdings can be changed.

Intertemporal Budget Constraint

- Denoting by w_i the proportion of date 0 savings that the investor chooses to invest in Asset i, then his intertemporal budget constraint is:

$$C_T = \underbrace{y_T}_{\text{wages at } T} + \underbrace{\left(W_0 + y_0 - C_0\right)}_{\text{savings at date 0}} \times \underbrace{\sum_{i=1}^{n} w_i R_i}_{\text{return on savings}} \tag{10.5}$$

Investor's Objective Function

- The individual's maximization problem can then be stated as

$$\max_{C_0,\{w_i\}} u(C_0) + e^{-\delta}\,\mathbb{E}^{\mathbb{P}}\left[u(C_T)\right] \tag{10.6}$$

subject to

1. the budget constraint in (10.5), rewritten below:

$$C_T = y_T + \left(W_0 + y_0 - C_0\right) \times \sum_{i=1}^{n} w_i R_i \tag{10.5}$$

2. the constraint that the investment proportions add up to 1:

$$1 = \sum_{i=1}^{n} w_i.$$

The Lagrangian

- Substituting into $\mathbb{E}^{\mathbb{P}}[u(C_T)]$ the budget constraint in Equation (10.5) that $C(T)$ must satisfy, and
- denoting by λ_{temp} the Lagrange multiplier on the constraint that the weights sum to 1,
- we can write the Lagrangian as:

$$\mathcal{L} = \max_{C_0,\{w_i\}} u(C_0) + e^{-\delta}\,\mathbb{E}^{\mathbb{P}}\Bigg[\underbrace{u\left(y_T + (W_0 + y_0 - C_0)\sum_{i=1}^{n} w_i R_i\right)}_{C_T}\Bigg] + \lambda_{temp}\left(1 - \sum_{i=1}^{n} w_i\right).$$

- So, we need to choose C_0 and w_i in order to maximize the Lagrangian $\mathcal{L}$.

The First-Order Conditions

- The first-order condition with respect to optimal consumption C_0 is:

$$\begin{aligned} 0 &= \frac{\partial \mathcal{L}}{\partial C_0} \\ &= u'(C_0) - e^{-\delta}\,\mathbb{E}^{\mathbb{P}}\left[u'(C_T)\sum_{i=1}^{n} w_i R_i\right]; \end{aligned} \tag{10.7}$$

- The first-order condition with respect to the optimal portfolio weight w_i is

$$\begin{aligned} 0 &= \frac{\partial \mathcal{L}}{\partial w_i} \\ &= e^{-\delta}\,(W_0 + y_0 - C_0)\mathbb{E}^{\mathbb{P}}\left[u'(C_T)R_i\right] - \lambda_{temp} \\ \lambda_{weight} &= e^{-\delta}\,\mathbb{E}^{\mathbb{P}}\left[u'(C_T)R_i\right], \end{aligned} \tag{10.8}$$

where: $\lambda_{weight} \equiv \lambda_{temp}/\left(W_0 + y_0 - C_0\right).$

10.7.1 Understanding the First-Order Condition for Optimal Portfolios

- Observe that the first-order condition for the optimal portfolio weight for asset i is:

$$e^{-\delta}\,\mathbb{E}^{\mathbb{P}}\left[u'(C_T)R_i\right] = \lambda_{weight}.$$

- Similarly, the first-order condition for the optimal weight for asset j is

$$e^{-\delta}\,\mathbb{E}^{\mathbb{P}}\left[u'(C_T)R_j\right] = \lambda_{weight}.$$

- Because the right-hand side is the same for all n assets, we get:

$$\mathbb{E}^{\mathbb{P}}\left[u'(C_T)R_i\right] = \mathbb{E}^{\mathbb{P}}\left[u'(C_T)R_j\right], \quad \text{for any two assets } i \text{ and } j. \tag{10.9}$$

- The above equation says that the investor trades-off investing in asset i for asset j until their expected marginal utility-weighted returns are equal.

- **How does the investor achieve this equality?**
  - ⋆ Recall from (10.5), rewritten below, that C_T depends on R_i and R_j:

$$C_T = y_T + \left(W_0 + y_0 - C_0\right) \times \sum_{i=1}^{n} w_i R_i. \tag{10.5}$$

  - ⋆ Thus, as w_i increases, C_T becomes more positively correlated with R_i.
  - ⋆ Thus, the greater is asset i's portfolio weight, the lower will be $u'(C_T)$ when R_i is high (because the utility function is concave, which implies diminishing marginal utility).
  - ⋆ Hence, as w_i becomes large, smaller marginal utility weights multiply the higher realization of asset i's return, and $\mathbb{E}^{\mathbb{P}}[u'(C_T)]$ drops.
  - ⋆ Intuitively, as w_i becomes large, the portfolio becomes less diversified.
  - ⋆ The first-order condition says that the investor chooses the weights so that the effect on the random distribution of C_T is such that $\mathbb{E}^{\mathbb{P}}\left[u'(C_T)R_i\right]$ is equalized across all assets, implying the optimal level of diversification.

10.7.2 Understanding First-Order Condition for Intertemporal Allocation

- Observe that the first-order condition for consumption in (10.7) can be written as

$$u'(C_0) = e^{-\delta}\,\mathbb{E}^{\mathbb{P}}\left[u'(C_T)\sum_{i=1}^{n} w_i R_i\right], \tag{10.10}$$

which says that there must be equality between

⋆ marginal utility from consumption at date 0, $u'(C_0)$; and,

⋆ marginal utility from consuming at date T,
  ∗ the expected return on the portfolio, $\sum_{i=1}^{n} w_i R_i$,
  ∗ evaluated in terms of (expected) marginal utility at date T, $e^{-\delta}\,\mathbb{E}^{\mathbb{P}}[u'(C_T)]$.

Implication of First-Order Conditions for Intertemporal Allocation

- The first-order condition for consumption can be re-written as follows:

$$\begin{aligned}
u'(C_0) &= e^{-\delta}\,\mathbb{E}^{\mathbb{P}}\left[u'(C_T)\sum_{i=1}^{n} w_i R_i\right] && \text{. . . from equation (10.10)}\\
&= \sum_{i=1}^{n} w_i \underbrace{\left(e^{-\delta}\,\mathbb{E}^{\mathbb{P}}\left[u'(C_T)R_i\right]\right)}_{\lambda_{weight}} && \text{. . . moving constants outside the expectations operator, } E\\
&= \sum_{i=1}^{n} w_i\,\lambda_{weight} && \text{. . . using the result in equation (10.8) on page 364}\\
&= \lambda_{weight}\sum_{i=1}^{n} w_i && \text{. . . moving the constant outside the summation operator}\\
&= \lambda_{weight}. && \text{. . . because the weights sum to 1} \qquad (10.11)
\end{aligned}$$

- Substituting the result in (10.11), $u'(C_0) = \lambda_{weight}$, into (10.8),

$$0 = e^{-\delta}\,\mathbb{E}^{\mathbb{P}} u'[C_T] R_i - \lambda_{weight} \tag{10.8}$$

we get

$$\begin{aligned} u'(C_0) &= e^{-\delta}\,\mathbb{E}^{\mathbb{P}}\left[u'(C_T) R_i\right] \\ u'(C_0) &= e^{-\delta}\,\mathbb{E}^{\mathbb{P}}\left[u'(C_T)\frac{X_i}{\mathcal{P}_i}\right] \\ \mathcal{P}_i\, u'(C_0) &= \mathbb{E}^{\mathbb{P}}\left[e^{-\delta}\; u'(C_T) X_i\right]. \end{aligned} \tag{10.12}$$

- The last equation says that an optimizing investor will invest in Asset i
  - ⋆ until the loss in marginal utility of giving up $\mathcal{P}_i$ dollars at date 0
  - ⋆ equals exactly the expected marginal utility of receiving X_i at T.
- This interpretation generalizes to multiperiod consumption-portfolio choice problems in discrete and continuous time.

- To see this more clearly, consider the special case where Asset i is the risk-free asset.

$$\begin{aligned} \mathcal{P}_i\, u'(C_0) &= e^{-\delta}\,\mathbb{E}^{\mathbb{P}}\left[u'(C_T) X_i\right] \\ \frac{1}{R_f} u'(C_0) &= e^{-\delta}\,\mathbb{E}^{\mathbb{P}}\left[u'(C_T)\right] \\ u'(C_0) &= R_f\, e^{-\delta}\,\mathbb{E}^{\mathbb{P}}\left[u'(C_T)\right]. \end{aligned} \tag{10.13}$$

- The above equation states that the investor trades off date 0 for date T consumption until
  - ⋆ the marginal utility of giving up $\$1$ of date 0 consumption just equals
  - ⋆ the marginal utility of receiving $\$R_f$ of date T consumption.

Relative Risk Aversion

- Suppose that the utility function is of the constant relative risk aversion form:

$$u(C) = \frac{C^{1-\gamma}}{1-\gamma}, \quad \text{where } \gamma > 0 \text{ and } \text{“}\textbf{relative risk aversion}\text{”} = \gamma.$$

- For the above utility function, observe that

$$u'(C) = C^{-\gamma}; \quad u''(C) = -\gamma\, C^{-\gamma-1}.$$

- Relative risk aversion is defined as:

$$\frac{-C\, u''(C)}{u'(C)} = \gamma.$$

10.7.3 Elasticity of Intertemporal Substitution (EIS)

- Using the utility function with constant relative risk aversion:

$$\begin{aligned} u'(C_0) &= R_f\, e^{-\delta}\, \mathbb{E}^{\mathbb{P}}\left[u'(C_T)\right] \\ \frac{1}{R_f} &= e^{-\delta}\, \mathbb{E}^{\mathbb{P}}\left[\frac{u'(C_T)}{u'(C_0)}\right] \\ \frac{1}{R_f} &= e^{-\delta}\, \mathbb{E}^{\mathbb{P}}\left[\left(\frac{C_T}{C_0}\right)^{-\gamma}\right]. \end{aligned} \tag{10.14}$$

- Now, consider the special case where there is no labor income ($y_t = 0$ for all t), and the only asset available to the investor is the risk-free asset.

- For this special case, C_T becomes non-stochastic, and thus we can simplify Equation (10.14) from the previous page,

$$\frac{1}{R_f} = e^{-\delta}\,\mathbb{E}^{\mathbb{P}}\left[\left(\frac{C_T}{C_0}\right)^{-\gamma}\right], \tag{10.14}$$

$$\frac{1}{R_f} = e^{-\delta}\left[\left(\frac{C_T}{C_0}\right)^{-\gamma}\right], \quad \text{. . . drop the expectations operator}$$

$$R_f = e^{\delta}\left(\frac{C_T}{C_0}\right)^{\gamma} \quad \text{. . . raise both sides to the power } -1$$

$$\log R_f = \delta + \gamma \log\left(\frac{C_T}{C_0}\right) \quad \text{. . . taking logs on both sides}$$

- We start by re-writing the last expression on the previous page:

$$\log R_f = \delta + \gamma \log\left(\frac{C_T}{C_0}\right) \tag{10.14}$$

- Because $\log R_f$ is the continuously compounded risk-free interest rate and $\log(C_T/C_0)$ is the growth rate of consumption, we can define the elasticity of intertemporal substitution (EIS) to be:

$$\text{EIS} = \frac{\partial \log(C_T/C_0)}{\partial \log R_f} = \frac{1}{\gamma}.$$

- Re-writing the definition of EIS from the previous slide, we see that there are three possible cases:

$$\text{EIS} = \frac{\partial \log(C_T/C_0)}{\partial \log R_f} = \frac{1}{\gamma}.$$

1. If $\gamma < 1$, EIS > 1, in which case
   - ⋆ a higher interest rate leads to an even greater increase in C_T,
   - ⋆ implying an increase in savings at date 0.

2. Conversely, if $\gamma > 1$, then EIS < 1, in which case
   - ⋆ a higher interest rate leads to a relative decrease in C_T,
   - ⋆ implying a decrease in savings at date 0

3. For the intermediate case where $\gamma = 1$, then EIS $= 1$, in which case
   - ⋆ a change in the interest rate
   - ⋆ has no effect on savings.

- These results can be interpreted as the consequence of two effects from an increase in interest rates.
  - ⋆ A substitution effect that increases the return from transforming current consumption into future consumption; this higher return on savings provides an incentive to increase savings.
  - ⋆ An income effect due to the greater return that is earned on a given amount of savings. This makes the investor better off, and all else being equal, would raise consumption in both periods, hence reducing savings at date 0.

- If
  - ⋆ EIS > 1 the substitution effect outweighs the income effect;
  - ⋆ EIS < 1 the income effect outweighs the substitution effect;
  - ⋆ EIS $= 1$ the income effect equals the substitution effect.

10.7.4 Understanding relation between the SDF and risky asset prices

- We end our discussion of two-date models by looking at the implication for asset pricing of equation (10.12) that we derived on Page 369:

$$\mathcal{P}_i\, u'(C_0) = \mathbb{E}^{\mathbb{P}}\left[e^{-\delta}\; u'(C_T) X_i\right]. \tag{10.12}$$

- Dividing both sides of this equation by $u'(C_0)$ leads to

$$\mathcal{P}_i = \mathbb{E}^{\mathbb{P}}\Big[\underbrace{\frac{e^{-\delta}\; u'(C_T)}{u'(C_0)}}_{\text{SDF}} X_i\Big], \tag{10.15}$$

  which shows that the SDF given by $\frac{e^{-\delta}\; u'(C_T)}{u'(C_0)}$ adjusts next period's risky payoff X_i to give us this period's price $\mathcal{P}_i$.

10.7.5 Understanding relation between the SDF and the risk-free rate

- Similarly, we can use equation (10.13) that we derived on Page 370 to show the relation between the risk-free interest rate and the SDF:

$$u'(C_0) = R_f\, e^{-\delta}\, \mathbb{E}^{\mathbb{P}}\left[u'(C_T)\right]. \tag{10.13}$$

- Dividing both sides of this equation by $u'(C_0)\, R_f$ leads to

$$\frac{1}{R_f} = \mathbb{E}^{\mathbb{P}}\Big[\underbrace{\frac{e^{-\delta}\; u'(C_T)}{u'(C_0)}}_{\text{SDF}}\Big],$$

  which shows that the mean (expected value) of the SDF is the inverse of the risk-free rate, exactly as we had demonstrated in Part 1 of the course.

10.8 Summary

1. Insights of modern portfolio theory

- Investors hold frontier portfolios; large asset base improves the frontier.
- An individual asset's risk is its contribution to portfolio risk.
  - ⋆ Variance of each asset contributes little to portfolio risk; covariances between assets determine portfolio risk.
- When there is risk-free asset, frontier portfolios are linear combinations of the risk-free asset and the tangent portfolio.
- The optimal mean-variance portfolio is given by

$$w = \frac{1}{\gamma} \mathbf{V}^{-1} (\mu - r_F \mathbf{1}). \tag{10.16}$$

- For any asset i:
  - ⋆ Its risk can be measured by its beta with respect to a frontier portfolio.
  - ⋆ Its return per unit of risk is the premium on the frontier portfolio.

2. Implications of the Capital Asset Pricing Model (CAPM):

- The market portfolio is an efficient frontier portfolio.
- The portfolio frontier consists of portfolios of the market portfolio and the risk-free asset.
- Systematic risk is that part of total risk that is correlated with the market portfolio and cannot be diversified away, and so needs to be rewarded; unsystematic risk is not rewarded:

$$\mathbb{E}^{\mathbb{P}}[r_i] - r_F = \beta_{im} \Big(\mathbb{E}^{\mathbb{P}}[r_m] - r_F \Big), \quad \text{where} \quad \beta_{im} = \sigma_{im} / \sigma_m^2.$$

  - ⋆ Two securities with the same total risk can have very different exposure to systematic risk.
- The Capital Market Line gives tradeoff between **portfolio** risk and return; the price of one unit risk, i.e. market price of risk for an efficient portfolio is the risk premium on market portfolio per unit of its volatility: $\frac{\mathbb{E}[r_m] - r_F}{\sigma_m}$.

3. Consumption-portfolio choice in a two-date model

- First-order condition for assets i and j in an optimal portfolio:

$$\mathbb{E}^{\mathbb{P}} U'[C_T] R_i = \mathbb{E}^{\mathbb{P}} U'[C_T] R_j, \quad \text{for any two assets } i \text{ and } j.$$

- First-order condition for optimal intertemporal consumption:

$$P_i \, U'(C_0) = \mathbb{E}^{\mathbb{P}} \left[\delta \, U'(C_T) X_i \right]$$

- The equation above implies that the price of any asset must be

$$P_i = \mathbb{E}^{\mathbb{P}} \left[\left(\frac{\delta \, U'(C_T)}{U'(C_0)} \right) X_i \right] = \mathbb{E}^{\mathbb{P}} \left[m_{0,T} \, X_i \right],$$

where $m_{0,T}$ is the marginal rate of substitution between initial and end-of-period consumption.

- We also have the result that:

$$\mathbb{E}^{\mathbb{P}} [R_i] - R_f = (-) \frac{\text{Cov} \left(U'(C_T), R_i \right)}{\mathbb{E}^{\mathbb{P}} \left[U'(C_T) \right]},$$

which says that:

⋆ the risk-premium decreases as the covariance between marginal utility and return increases;

⋆ that is, investors are willing to accept a lower risk premium from those assets whose returns come in states with high marginal utility (that is, low consumption).

- All the results listed in this summary are true also in continuous time models.

10.9 Suggestions for Further Readings

10.9.1 Source material for the lecture notes

- In this chapter, we have covered material from several parts of our two textbooks.
  - ⋆ The Markowitz portfolio problem is discussed in Back (2017, Chapter 5).
  - ⋆ The CAPM and factor models are in Back (2017, Chapter 6) and Dumas and Luciano (2017, Chapter 4).
  - ⋆ The two-date problem is discussed in Back (2017, Chapters 2) and Dumas and Luciano (2017, Chapter 3).

- You can read about the main ideas we have discussed in the following:
  - ⋆ The mean-variance model was developed in Markowitz (1952, 1959).
  - ⋆ However, De Finetti (1940) had, many years earlier, already developed a model similar to that by Markowitz; for a discussion of this, see Rubinstein (2006a, 2006b), Markowitz (2006), and Pressacco and Serafini (2007).
  - ⋆ The two-fund separation result is developed in Tobin (1958). See also Cass and Stiglitz (1970).
  - ⋆ Sharpe (1964) developed the single-date CAPM, with similar models developed by Treynor (1961a, 1961b), Mossin (1966), and Lintner (1969).
  - ⋆ Following the award in 1990 of the Nobel Prize to Markowitz and Sharpe (and Miller), Varian (1993) describes the importance of their work.
  - ⋆ An excellent analysis of the efficient frontier is given in Roll (1980) and the book by Huang and Litzenberger (1988). See also Merton (1972).
  - ⋆ Campbell (2018) is an excellent source for discrete-time portfolio choice.

10.9.2 Potential applications and extensions of lecture notes

- Almost all of finance relies on the risk-return tradeoff that captured so well in the Markowitz model. For example, the following important papers all build on the Markowitz model:
  - ⋆ Sharpe (1964),
  - ⋆ Samuelson (1969),
  - ⋆ Merton (1969, 1971, 1973a, 1990).
- Rubinstein (2002) provides a retrospective of this literature and Rubinstein (2006a) provides additional details.
- For a recent review of the literature that builds on the Markowitz model, see Zhang, Li, and Guo (2018).

- **Factor models of the cross-section of stock returns:**
  - ⋆ There is a very large literature on factor models of returns.
  - ⋆ The popular model in industry is based on Fama and French (2015).
  - ⋆ Recent work on factor models and anomalies is contained in Hou, Xue, and Zhang (2015, 2017a, 2017b) and Hou, Mo, Xue, and Zhang (2019).
  - ⋆ For a paper that studies how transaction costs affect the number of significant factors, see DeMiguel, Martin-Utrera, Nogales, and Uppal (2019) and the references therein.

- **Empirical evidence on diversification:**
  - ⋆ There is a large literature that documents that investors do not hold diversified portfolios: the median investor holds only 3 stocks: see, Guiso, Haliassos, and Jappelli (2002), Polkovnichenko (2005), Campbell (2006), Calvet, Campbell, and Sodini (2007), and Guiso and Sodini (2013).
  - ⋆ Lack of diversification is a phenomenon across the world, including: Australia (Worthington 2009), France (Arrondel and Lefebvre 2001), Germany (Börsch-Supan and Eymann 2002; Barasinska, Schäfer, and Stephan 2008), India (Campbell, Ramadorai, and Ranish 2012), Italy (Guiso and Jappelli 2002), Netherlands (Alessie, Hochguertel, and Van Soest 2002), Norway (Fagereng, Gottlieb, and Guiso 2017), and United Kingdom (Banks and Smith 2002).
  - ⋆ Typically, the few risky assets held are ones with which households are familiar; see Huberman (2001), Grinblatt and Keloharju (2001), Bhamra, Uppal, and Walden (2020).
  - ⋆ The most striking example of investing in familiar assets is investment in "own-company stock," that is, stock of the company in which the person is employed; see Haliassos (2002), Mitchell and Utkus (2004), Benartzi, Thaler, Utkus, and Sunstein (2007).
  - ⋆ Bhamra and Uppal (2019) show that not only is underdiversification costly for the household, but it also imposes a significant cost on society as a whole.

- **Robust portfolio choice:**
  - ⋆ For a "robust portfolio" interpretation in the sense of Hansen and Sargent (2007), see Garlappi, Uppal, and Wang (2007) and Boyle, Garlappi, Uppal, and Wang (2012).
- **Behavioral finance:**
  - ⋆ Shefrin and Statman (1985) and Shefrin and Statman (2000) look at the implications of behavioral finance for the Markowitz portfolio choice model. See also the book by Shefrin (2010).
  - ⋆ Empirical evidence on behavioral biases in portfolio rebalancing is presented in Odean (1998), Odean (1999), and Barber and Odean (2000, 2001, 2003) and the review paper Barber and Odean (2013).
  - ⋆ Another interesting paper on the naïve behavior of investors is Benartzi and Thaler (2001).
  - ⋆ Comprehensive reviews are provided by Vissing-Jorgensen (2003), Barberis and Thaler (2003), and Barberis (2013).

- **Implementation challenges:**
  - ⋆ DeMiguel, Garlappi, and Uppal (2009) show that the out-of-sample performance of mean-variance optimal portfolios is very poor, much worse than even that of the naive $1/N$ rule that entails no optimization.
  - ⋆ The reason for this is the substantial error in estimating the mean vector (μ) and variance-covariance matrix ($\mathbf{V}$) of returns in the expression for the optimal portfolio weights:

$$w = \frac{1}{\gamma}\mathbf{V}^{-1}(\mu - r_F\mathbf{1})$$

    - ∗ In the equation above, expected returns μ are estimated with large error. There is error also in estimating $\mathbf{V}^{-1}$.
    - ∗ Moreover, the inverse of the variance-covariance matrix is "ill conditioned," which makes its determinant very large (for instance, the order of 10^6) that when multiplied by μ magnifies the estimation error in expected returns. Thus, the estimated portfolio weights have large error.

- ⋆ There is a substantial literature that tries to address this problem, such as:
  - ∗ models that improve the estimation of the variance-covariance matrix: Ledoit and Wolf (2003, 2004a, 2004b, 2017) and DeMiguel, Garlappi, Nogales, and Uppal (2009);
  - ∗ models that account explicitly for estimation error though a third fund, such as Kan and Zhou (2007);
  - ∗ model that use lasso-type constrained regressions to obtain weights with superior out-of-sample portfolio performance: Ao, Li, and Zheng (2019);
  - ∗ models that are based on the SDF: Ghosh, Julliard, and Taylor (2019).
  - ∗ approach that uses the Arbitrage Pricing Theory of Ross (1976) to determine mean-variance weights that are robust to model misspecification: Raponi, Uppal, and Zaffaroni (2020).
- ⋆ For a textbook treatment of portfolio choice, see Meucci (2009).
- ⋆ For a book devoted to estimating expected returns, see Ilmanen (2011).

- **Dynamic mean-variance portfolio choice**
  - ⋆ The mean-variance model we studied in this chapter is for the setting where the investor wishes to optimize the objective function at a single date.
  - ⋆ For an extension of how to determine mean-variance portfolios in a time-consistent way in a multiperiod setting, see the very nice and important paper by Basak and Chabakauri (2010).
  - ⋆ See also Basak and Chabakauri (2012) that studies the problem of multi-period hedging.

10.9.3 Ph.D. elective courses offered on related topics

- Michael Brandt taught a course on "*Portfolio Choice*" in 2011 and 2013.
- Jianqing Fan taught the course on "*Portfolio Problems with High-Dimension Covariance Matrices*" in 2012.
- Harrison Hong has a taught courses on "*Behavioral Finance and Asset Management*" in 2013, 2017, and 2022.
- Vikas Agarwal taught a course in 2016 and 2022 on "*Asset Management*" that focuses on the Mutual Fund and Hedge Fund asset classes.
- Ludovic Phalippou taught a course in 2016 that focuses on the "*Private Equity*" asset class.
- Tarun Ramadorai taught a course on *Household Finance* in 2024.
- See also the courses listed in the next two chapters, which are on the topic of portfolio choice and asset allocation.

Lecture Notes for 2025

Continuous Time Financial Economics

Chapter 11

Portfolio Choice Using the Martingale Approach

Raman Uppal

EDHEC Business School

Chapter contents

11.1 Plan for the Day . . . 397
11.2 Main Points for Today . . . 398
11.3 Martingale Approach to Consumption and Portfolio Choice in a Multiperiod Discrete-Time Model 399
11.3.1 Steps to Identify Optimal Consumption and Portfolio . . . 402
11.3.2 Summary of Martingale Approach to Consumption-Portfolio Problem . . . 420
11.4 Martingale Approach to Consumption and Portfolio Choice in Continuous Time . . . 423
11.5 Special Case: Single Risky Asset, No Bequest, and Constant Investment Opportunity Set . . . 427
11.5.1 Description of the Setting for this Special Case . . . 427
11.5.2 Expression for the Optimal Portfolio Weight . . . 428
11.5.3 Identifying the Expression $W_M M/W$ . . . 439
11.5.4 Showing that $W(M_t, t) = f(t)\, M_t^{\frac{-1}{\gamma}}$ . . . 441
11.6 General Case: Many risky assets, Bequest, and Stochastic Investment Opportunity Set . . . 446
11.7 Summary . . . 461
11.8 Appendix: Case with Multiple Risky Assets and Power Utility But Constant Investment Opportunity Set . . . 464
11.8.1 Expression for the Optimal Portfolio . . . 465
11.8.2 Showing that $-W_M M/W = 1/\gamma$ . . . 466
11.8.3 Explicit Expression for the Optimal Portfolio . . . 467
11.8.4 Interpreting the Expression for the Optimal Portfolio . . . 468
11.8.5 Showing that $W(M_t, t) = f(t)\, M_t^{\frac{-1}{\gamma}}$ . . . 469
11.9 Suggestions for Further Readings . . . 474
11.9.1 Source material for the lecture notes . . . 474
11.9.2 Potential applications and extensions of lecture notes . . . 475
11.9.3 Ph.D. elective courses offered on related topics . . . 479

11.1 Plan for the Day

Last Time

- Portfolio Choice in Mean-Variance and Two-Date Models

Today

- **How to find the optimal portfolio in a complete-market setting?**
  - ⋆ Optimal portfolio selection using the martingale approach

For Next Time

- Review the material covered in class.
- Read Back, Chapters 14 and Dumas & Luciano, Chapter 10.
- Do the practice problems.

11.2 Main Points for Today

- Optimal portfolio choice in continuous time using the martingale approach.
  - ⋆ The static budget constraint;
  - ⋆ The dynamic budget constraint;
  - ⋆ Identifying the optimal consumption policy and the value of the Lagrange multiplier for the static budget constraint;
  - ⋆ Identifying the optimal portfolio policy.
- In the next chapter, we will study optimal portfolio choice in continuous time using the dynamic-programming approach.
- Of course, both approaches lead to the same solution, but when markets are complete, the martingale approach is easier to implement, and when markets are not complete, then the dynamic programming approach is easier.

11.3 Martingale Approach to Consumption and Portfolio Choice in a Multiperiod Discrete-Time Model

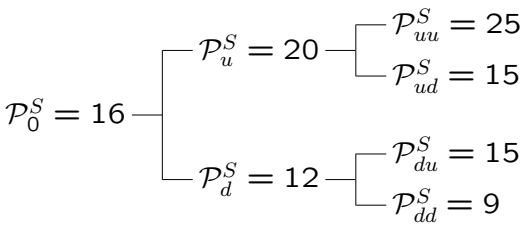
Description of the Problem and Solution Approach

- We wish to find the optimal consumption and portfolio of the investor
- We are **given**
  - ⋆ Prices of bond and stock(s)
  - ⋆ Objective of the investor: lifetime utility function
  - ⋆ Initial wealth of the investor.
- We will explain the method using a simple example.
- We will break up the computation of the optimal portfolio into six steps; the continuous time model will then be solved using the same six steps.

Example

Given: Prices of Bond and Stock

- Consider an economy where
  - ⋆ the interest rate is zero (bond price $\mathcal{P}^Z(t) = 1$ for all t), and
  - ⋆ the price of the risky stock, $\mathcal{P}^S(t)$, is as follows:



  - ⋆ with the true (objective) probability of
    - ∗ 0.60 of an up move, and
    - ∗ 0.40 of a down move.

Given: Objective of the Investor

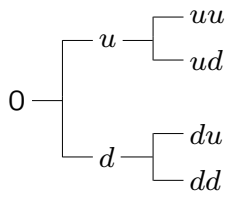
- In this economy, there is a single agent who wishes to maximize the sum of lifetime expected utility from consumption at dates $t = 0, 1, 2$.
- Also, to simplify matters, we assume that the utility function is (natural) log, $u(C_t) = \ln C_t$ and that the subjective discount rate (time preference) is $\delta = 0$.
- Finally, the initial wealth of the investor is $W_0 = 10$.
- Formally, the problem of the investor is to

$$\max_{C_0, C_1, C_2} \left\{ u(C_0) + \mathbb{E}^{\mathbb{P}} \Big[u(C_1) + u(C_2) \Big] \right\}, \quad \text{subject to } W_0 = 10.$$

11.3.1 Steps to Identify Optimal Consumption and Portfolio

1. Find at $t = 0$ prices of Arrow-Debreu securities, that is, securities that pay one unit of consumption in a particular state and nothing in the other states.
2. Now, write the portfolio problem of the investor in a static form: that is,
   - the decisions are all made at $t = 0$, and
   - the budget equation is also of a static form.
3. Solve for the optimal C_0, C_1, C_2 in terms of λ_{BC}, and also the optimal number of state-contingent claims to hold at $t = 0$.
4. Identify λ_{BC} from the budget constraint.
5. Solve for the optimal C_0, C_1, C_2 using λ_{BC} that we have identified in the previous step.
6. Find optimal trading strategy in the *bond* and *stock* that finances (or replicates) the optimal consumption at dates 0, 1 and 2.

Step 1: Identify Arrow-Debreu State Prices



- From the figure above, we see that
  - ⋆ at date 1, there are two possible states: $\{u, d\}$, and
  - ⋆ at date 2 there are four possible states: $\{uu, ud, du, dd\}$.
- So, we need to find the date-0 prices of the following Arrow-Debreu securities:
  - ⋆ The Arrow-Debreu securities that pay off 1 unit at $t = 1$ in the up or down states: $\{\mathcal{P}^{A_u}, \mathcal{P}^{A_d}\}$.
  - ⋆ The Arrow-Debreu securities that pay off 1 unit at $t = 2$ in the four possible states: $\{\mathcal{P}^{A_{uu}}, \mathcal{P}^{A_{ud}}, \mathcal{P}^{A_{du}}, \mathcal{P}^{A_{dd}}\}$.

- To find the price of each state-contingent claim, we need to find the portfolio of stock and bond that replicates the appropriate state-contingent payoff, and then find the price of this replicating portfolio.
- For the claim that delivers 1 unit in state u, if α_0 is the number of shares of the stock to be held and β_0 is the number of bonds to be held at $t = 0$, then the system to be solved is:

$$\begin{aligned} \alpha_0 \mathcal{P}_u^S + \beta_0 \mathcal{P}_u^Z = 1 \quad &\Longleftrightarrow \quad \alpha_0 20 + \beta_0 1 = 1; \\ \alpha_0 \mathcal{P}_d^S + \beta_0 \mathcal{P}_d^Z = 0 \quad &\Longleftrightarrow \quad \alpha_0 12 + \beta_0 1 = 0. \end{aligned}$$

- Solving this system of equations gives: $\alpha_0 = \frac{1}{8}$ and $\beta_0 = -\frac{3}{2}$.
- Denoting the price of this Arrow-Debreu state security by $\mathcal{P}_0^{A_u}$, we have from the solution of this system of two equations that:

$$\mathcal{P}_0^{A_u} = \frac{1}{8} \times \mathcal{P}_0^S - \frac{3}{2} \times \mathcal{P}_0^Z = \frac{1}{8} \times 16 - \frac{3}{2} \times 1 = 0.50.$$

- Similarly, for the claim that pays 1 unit in state d, the system to be solved is:

$$\begin{aligned} \alpha_0 \mathcal{P}_u^S + \beta_0 \mathcal{P}_u^Z &= 0; \\ \alpha_0 \mathcal{P}_d^S + \beta_0 \mathcal{P}_d^Z &= 1, \end{aligned}$$

- Solving these equations gives us: $\mathcal{P}_0^{A_d} = 0.50$.

- To value Arrow-Debreu securities paying off at $t = 2$, two steps are required:
  - ⋆ we first need to find the value at $t = 1$, and
  - ⋆ then find the value at $t = 0$.
- To find the value at $t = 1$ of a claim that pays 1 unit in state uu,
  - ⋆ in state u the replicating equations are:
$$\begin{aligned} \alpha_u \mathcal{P}_{uu}^S + \beta_u \mathcal{P}_{uu}^Z &= 1; \\ \alpha_u \mathcal{P}_{ud}^S + \beta_u \mathcal{P}_{du}^Z &= 0. \end{aligned}$$
  - ⋆ So the value at u is $\alpha_u \mathcal{P}_u^S + \beta_u \mathcal{P}_u^Z = 0.50$.
  - ⋆ And, the value at d is 0 (because all future payoffs from this state are zero).
- To find the value at $t = 0$ of a contract that pays 0.50 in state u and 0 in state d, we use our previous result about $\mathcal{P}_0^{A_u} = 0.50$ to get:

$$\mathcal{P}_0^{A_{uu}} = 0.50 \times \mathcal{P}_0^{A_u} = 0.50 \times 0.50 = 0.25.$$

- Similarly, the cost of the claim that delivers 1 unit in state ud is obtained by first solving:

$$\begin{aligned}\alpha_u \mathcal{P}^S_{uu} + \beta_u \mathcal{P}^Z_{uu} &= 0;\\ \alpha_u \mathcal{P}^S_{ud} + \beta_u \mathcal{P}^Z_{du} &= 1,\end{aligned}$$

- The cost of which at u is $\alpha_u \mathcal{P}^S_u + \beta_u \mathcal{P}^Z_u = 0.50$;
- The cost of this at $t = 0$ is: $\mathcal{P}_0^{A_{ud}} = 0.50 \times \mathcal{P}_0^{A_u} = 0.50 \times 0.50 = 0.25$.
- Similarly, one can find that the cost of the claim that delivers
  - ⋆ 1 unit in state du is: $\mathcal{P}_0^{A_{du}} = 0.25$.
  - ⋆ 1 unit in state du is: $\mathcal{P}_0^{A_{dd}} = 0.25$.

Step 2: Write the Investor's Problem in a Static Form

- Denote by $\text{prob}^{\mathbb{P}}_{ts}$ the probability of state (t, s).
- The consumption-portfolio problem written using the static formulation is:

$$\begin{aligned}\max \Big\{ & u(C_0)\\ & + \left[\text{prob}^{\mathbb{P}}_u u(C_u) + \text{prob}^{\mathbb{P}}_d u(C_d)\right]\\ & + \left[\text{prob}^{\mathbb{P}}_{uu} u(C_{uu}) + \text{prob}^{\mathbb{P}}_{ud} u(C_{ud}) + \text{prob}^{\mathbb{P}}_{du} u(C_{du}) + \text{prob}^{\mathbb{P}}_{dd} u(C_{dd})\right] \Big\},\end{aligned}$$

- subject to the static budget constraint that the present value of consumption must equal initial wealth:

$$\begin{aligned}C_0 + & \left(\mathcal{P}_0^{A_u} C_u + \mathcal{P}_0^{A_d} C_d\right)\\ & + \left(\mathcal{P}_0^{A_{uu}} C_{uu} + \mathcal{P}_0^{A_{ud}} C_{ud} + \mathcal{P}_0^{A_{du}} C_{du} + \mathcal{P}_0^{A_{dd}} C_{dd}\right) = W_0 = 10,\end{aligned}$$

where the cost of consumption at date 0 is 1 unit.

- Thus, the Lagrangian is:

$$\mathcal{L} = \max \left[\text{objective function}\right] - \lambda_{BC}\left(\text{budget constraint}\right)$$

$$\mathcal{L} = \max \Bigg\{ \Big[u(C_0) + \text{prob}_u^{\mathbb{P}} u(C_u) + \text{prob}_u^{\mathbb{P}} u(C_d) + \text{prob}_{uu}^{\mathbb{P}} u(C_{uu}) + \text{prob}_{ud}^{\mathbb{P}} u(C_{ud}) + \text{prob}_{du}^{\mathbb{P}} u(C_{du}) + \text{prob}_{dd}^{\mathbb{P}} u(C_{dd}) \Big] + \lambda_{BC} \left(W_0 - \left[C_0 + \mathcal{P}_0^{A_u} C_u + \mathcal{P}_0^{A_d} C_d + \mathcal{P}_0^{A_{uu}} C_{uu} + \mathcal{P}_0^{A_{ud}} C_{ud} + \mathcal{P}_0^{A_{du}} C_{du} + \mathcal{P}_0^{A_{dd}} C_{dd} \right] \right) \Bigg\}$$

where λ_{BC} is the Lagrange multiplier on the budget constraint (BC).

- Differentiating the Lagrangian with respect to consumption, the first-order conditions for consumption are of the general form:

$$\text{prob}_s^{\mathbb{P}} \frac{\partial u(C_{ts})}{\partial C_{ts}} = \lambda_{BC}\, \mathcal{P}_0^{A_s}, \qquad \text{where}$$

  - ⋆ $t = \{0, 1, 2\}$ denotes time,
  - ⋆ $s = \{0, u, d, uu, ud, du, dd\}$ denotes the states at $t = \{0, 1, 2\}$, and
  - ⋆ $\text{prob}_s^{\mathbb{P}}$ is the probability of that state.

- Note that the price of an Arrow-Debreu claim that pays 1 unit at date $t = 0$ is 1; that is, $\mathcal{P}_0^{A_0} = 1$.

Step 3: Solve for Optimal Consumption in Terms of λ_{BC}

- Re-arranging the first-order condition for consumption, we get:

$$\frac{\partial u(C_{ts})}{\partial C_{ts}} = \frac{1}{\text{prob}_s^{\mathbb{P}}} \lambda_{BC} \mathcal{P}_0^{A_s},$$

- Because the utility function is log, the above first-order condition is:

$$\frac{1}{C_{ts}} = \frac{1}{\text{prob}_s^{\mathbb{P}}} \lambda_{BC} \mathcal{P}_0^{A_s},$$

- Inverting the above, we get an explicit expression in terms of consumption:

$$C_{ts} = \frac{\text{prob}_s^{\mathbb{P}}}{\lambda_{BC} \mathcal{P}_0^{A_s}}.$$

- Next, we use the above expression to write the optimal consumption for each particular state:

$$C_0 = \frac{\text{prob}_0^{\mathbb{P}}}{\lambda_{BC} \mathcal{P}_0^{A_0}} = \frac{1}{\lambda_{BC}\, 1};$$

$$C_{1,u} = \frac{\text{prob}_u^{\mathbb{P}}}{\lambda_{BC} \mathcal{P}_0^{A_u}} = \frac{0.60}{\lambda_{BC}\, 0.5};$$

$$C_{1,d} = \frac{\text{prob}_d^{\mathbb{P}}}{\lambda_{BC} \mathcal{P}_0^{A_d}} = \frac{0.40}{\lambda_{BC}\, 0.5};$$

$$C_{2,uu} = \frac{\text{prob}_{uu}^{\mathbb{P}}}{\lambda_{BC} \mathcal{P}_0^{A_{uu}}} = \frac{0.36}{\lambda_{BC}\, 0.25};$$

$$C_{2,ud} = \frac{\text{prob}_{ud}^{\mathbb{P}}}{\lambda_{BC} \mathcal{P}_0^{A_{ud}}} = \frac{0.24}{\lambda_{BC}\, 0.25};$$

$$C_{2,du} = \frac{\text{prob}_{du}^{\mathbb{P}}}{\lambda_{BC} \mathcal{P}_0^{A_{du}}} = \frac{0.24}{\lambda_{BC}\, 0.25};$$

$$C_{2,dd} = \frac{\text{prob}_{dd}^{\mathbb{P}}}{\lambda_{BC} \mathcal{P}_0^{A_{dd}}} = \frac{0.16}{\lambda_{BC}\, 0.25}.$$

Step 4: Identify the Lagrange Multiplier Using the Budget Constraint

- Now substitute optimal consumption in the static budget constraint to identify λ_{BC}, which is the only unknown ($\text{prob}_s^{\mathbb{P}}$ is given and we found $\mathcal{P}_0^{A_s}$):

$$W_0 = C_0 + \left(\mathcal{P}_0^{Au} C_u + \mathcal{P}_0^{Ad} C_d\right) + \left(\mathcal{P}_0^{Auu} C_{uu} + \mathcal{P}_0^{Aud} C_{ud} + \mathcal{P}_0^{Adu} C_{du} + \mathcal{P}_0^{Add} C_{dd}\right)$$

$$10 = \frac{1}{\lambda_{BC}} + \left(\frac{0.60}{\lambda_{BC}} + \frac{0.40}{\lambda_{BC}}\right) + \left(\frac{0.36}{\lambda_{BC}} + \frac{0.24}{\lambda_{BC}} + \frac{0.24}{\lambda_{BC}} + \frac{0.16}{\lambda_{BC}}\right)$$

$$10 = \frac{1}{\lambda_{BC}}\Big[1 + (0.60 + 0.40) + (0.36 + 0.24 + 0.24 + 0.16)\Big]$$

$$10 = \frac{3}{\lambda_{BC}}$$

$$\lambda_{BC} = \frac{3}{10} = 0.30.$$

Step 5: Identify Optimal Consumption Using λ_{BC}

- Substituting $\lambda_{BC} = 0.30$ into the first-order conditions for consumption

$$C_{ts} = \frac{\text{prob}_s^{\mathbb{P}}}{\lambda_{BC}\,\mathcal{P}_0^{A_s}},$$

then gives us the optimal consumption for each state:

$$C_0 = \frac{1}{0.30 \times 1} = \frac{10}{3};$$

$$C_u = \frac{0.60}{0.30 \times 0.5} = 4;$$

$$C_d = \frac{0.40}{0.30 \times 0.5} = \frac{8}{3};$$

$$C_{uu} = \frac{0.36}{0.30 \times 0.25} = \frac{24}{5};$$

$$C_{ud} = \frac{0.24}{0.30 \times 0.25} = \frac{16}{5};$$

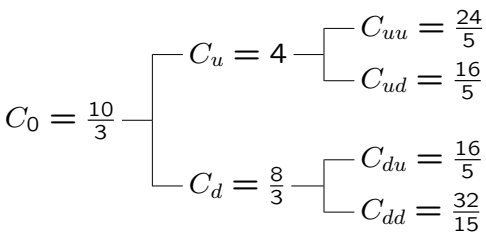
$$C_{du} = \frac{0.24}{0.30 \times 0.25} = \frac{16}{5};$$

$$C_{dd} = \frac{0.16}{0.30 \times 0.25} = \frac{32}{15}.$$

Step 6: Identify Portfolio that "Replicates" Optimal Consumption

- To identify the trading strategy in the bond and stock that replicates/finances optimal consumption,
  - ⋆ we interpret consumption as an "option" and
  - ⋆ solve the following system of equations for (α_s, β_s),
  - ⋆ which then gives us the strategy that replicate the payoff to the "option."
- Note that
  - ⋆ in Step 1 we were replicating the payoff to Arrow-Debreu securities;
  - ⋆ in Step 6, we are replicating the optimal consumption profile desired by the investor.

- That is, we wish to identify
  - ⋆ the number of stocks and bonds to hold at each state
  - ⋆ so that the value of that portfolio replicates the consumption stream desired by the investor.



- At state u, we need to identify (α_u, β_u) to replicate consumption in the two possible states in the next period:

$$\alpha_u \mathcal{P}^S_{uu} + \beta_u \mathcal{P}^Z_{uu} = C_{uu} \iff \alpha_u 25 + \beta_u 1 = \frac{24}{5}$$
$$\alpha_u \mathcal{P}^S_{ud} + \beta_u \mathcal{P}^Z_{du} = C_{ud} \iff \alpha_u 15 + \beta_u 1 = \frac{16}{5}.$$

  The solution is $\alpha_u = 4/25$ and $\beta_u = 4/5$, with the cost of this portfolio

$$V_u = \alpha_u \mathcal{P}^S_u + \beta_u \mathcal{P}^Z_u = \frac{4}{25} \times 20 + \frac{4}{5} \times 1 = 4.$$

- At state d, we need to identify (α_d, β_d) to replicate consumption in the two possible states in the next period:

$$\alpha_d \mathcal{P}^S_{ud} + \beta_d \mathcal{P}^Z_{ud} = C_{dd} \iff \alpha_d 15 + \beta_d 1 = \frac{16}{5};$$
$$\alpha_d \mathcal{P}^S_{dd} + \beta_d \mathcal{P}^Z_{dd} = C_{dd} \iff \alpha_d 9 + \beta_d 1 = \frac{32}{15};$$

  The solution is $\alpha_d = 8/45$ and $\beta_d = 8/15$, with the cost of this portfolio

$$V_d = \alpha_d \mathcal{P}^S_d + \beta_d \mathcal{P}^Z_d = \frac{8}{45} \times 12 + \frac{8}{15} \times 1 = \frac{8}{3}.$$

- And, then at $t = 0$, we need to solve for (α_0, β_0) that finance
  - ⋆ the optimal consumption in states C_u and C_d, **and** also
  - ⋆ V_u and V_d, the cost of the portfolios that gives the desired consumption at $t = 2$):

$$\alpha_0 \mathcal{P}^S_u + \beta_0 \mathcal{P}^Z_u = C_u + V_u \iff \alpha_0 20 + \beta_0 1 = 4 + 4$$
$$\alpha_0 \mathcal{P}^S_d + \beta_0 \mathcal{P}^Z_d = C_d + V_d \iff \alpha_0 12 + \beta_0 1 = \frac{8}{3} + \frac{8}{3}$$

- The solution is $\alpha_0 = 1/3$ and $\beta_0 = 4/3$, with the portfolio costing

$$V_0 = \alpha_0 \mathcal{P}^S_0 + \beta_0 \mathcal{P}^Z_0 = \frac{1}{3} \times 16 + \frac{4}{3} \times 1 = \frac{20}{3}.$$

- Putting it all together, the optimal dynamic portfolio strategy is:
  - ⋆ At $t = 0$, the investor needs to
    - ∗ consume $C_0 = 10/3$ units and use the remaining wealth, $20/3$, to
    - ∗ buy $\alpha_0 = 1/3$ shares and $\beta_0 = 4/3$ bonds.
  - ⋆ In state u, the investor needs to
    - ∗ consume $C_u = 4$ units and use the remaining wealth, 4 units, to
    - ∗ revise the portfolio so that it consists of $\alpha_u = 4/25$ shares and $\beta_0 = 4/5$ bonds.
  - ⋆ In state d, the investor needs to
    - ∗ consume $C_d = 8/3$ units and use the remaining wealth, $8/3$ units, to
    - ∗ revise the portfolio so that it consists of $\alpha_d = 8/45$ shares and $\beta_0 = 8/15$ bonds.

11.3.2 Summary of Martingale Approach to Consumption-Portfolio Problem

1. Determine optimal consumption:

- Use state prices to express the lifetime budget constraint as a single (static) constraint
- Write the multiperiod problem as a static optimization problem, with the:
  - ⋆ objective being to maximize lifetime consumption,
  - ⋆ constraint being the static budget constraint.
- Identify optimal consumption and the Lagrange multiplier on the budget constraint from the first-order conditions.

2. Determine optimal portfolio

- Identify optimal portfolio as the trading strategy that replicates the optimal consumption stream.
- This step can be interpreted as an "option replication" problem.

End of multi-date consumption-portfolio choice model
in discrete time

11.4 Martingale Approach to Consumption and Portfolio Choice in Continuous Time

- The main difference between discrete-time and continuous-time models is
  - ⋆ continuous trading allows the market to be dynamically complete with a small number of securities.
  - ⋆ continuous time mathematics also allows for cleaner and sharper results.
  - ⋆ an investor's optimal portfolio in the intertemporal setting consists of
    - ∗ myopic portfolio, which is the optimal in a "single-period" mean-variance setting, and
    - ∗ hedge portfolios to hedge against changes in investment opportunities.

- The main advantage of the martingale-pricing approach over the dynamic programming approach is that:
  - ⋆ under the martingale-pricing approach, to obtain the solution one needs to solve only a linear partial differential equation;
  - ⋆ under the dynamic programming approach, one needs to solve a nonlinear differential equation.
- A disadvantage of the martingale-pricing approach is that it requires markets to be complete; when markets are not complete, the martingale approach can still be applied, but it is no longer so easy.
- For a nice exposition on how to apply the martingale approach in a discrete-time setting with markets that are not complete, see Pliska (1997, Chapter 5.8).

Description of the General Setting

- The objective of the investor is to maximize a time-separable expected utility function that depends on the rate of consumption at all future instants and on bequest.
  - ⋆ The model can be extended to allow for non-separable preferences, which you have seen in the *Financial Economics* course set in discrete time.
- We examine this problem in a setting where the prices of the n assets are subject to continuous random changes; that is, returns follow diffusion processes.
  - ⋆ The model can be extended to allow for discontinuous changes (jumps).
- The investor can adjust consumption and portfolio allocations continuously.
  - ⋆ The model can be extended to relax this assumption.

How the notes are arranged

- To make it easier to understand the main ideas:

1. We first look at a special case of the consumption-portfolio problem with:
- ⋆ only a single risky asset,
- ⋆ utility from consumption, but no bequest motive, and
- ⋆ constant investment opportunity set.

2. Then, we look at the general case with:
- ⋆ n risky assets,
- ⋆ utility from consumption and also bequest, and
- ⋆ a stochastic investment opportunity set.

11.5 Special Case: Single Risky Asset, No Bequest, and Constant Investment Opportunity Set

11.5.1 Description of the Setting for this Special Case

- The objective of the investor is to maximize a time-separable expected utility function that depends on the rate of consumption at all future instants, but there is no bequest motive.

- There is a risk-free asset with return r.

- There is also 1 risky asset whose price $\mathcal{P}^S(t)$ follows the process:

$$\frac{d\mathcal{P}^S(t)}{\mathcal{P}^S(t)} = \mu\, dt + \sigma\, dB^{\mathbb{P}}, \tag{11.1}$$

  where r, μ and σ are constant and $dB^{\mathbb{P}}$ is a Brownian motion under the physical measure $\mathbb{P}$.

11.5.2 Expression for the Optimal Portfolio Weight

Step 1: The Stochastic Discount Factor

- Given the structure described on the previous slide, in the absence of arbitrage there is a unique stochastic discount factor that follows the process:

$$\frac{dM}{M} = -r dt - \lambda\, dB^{\mathbb{P}}, \tag{11.2}$$

  where λ is the market price of risk associated with the Brownian motion, and the market price of risk is:

$$\lambda = \frac{\mu - r}{\sigma}. \tag{11.3}$$

- Note that here λ denotes the market prices of risk, and is different from λ_{BC} and λ_{weight}, which denote Lagrange multipliers.

Step 2: The Static Budget Constraint

- The key to the martingale pricing approach for finding the optimal consumption and portfolio of the investor is to view the optimally invested wealth as an asset that provides a dividend equal to the consumption of the investor.
- This means that the "return on wealth" can be "priced" using the stochastic discount factor.
- That is, current wealth must be equal to the expected discounted value of the stream of future consumption ("dividends"):

$$W_t = \mathbb{E}_t^{\mathbb{P}}\left[\int_t^T \frac{M_s}{M_t} C_s ds\right], \tag{11.4}$$

or, equivalently:

$$W_t M_t = \mathbb{E}_t^{\mathbb{P}}\left[\int_t^T M_s C_s ds\right] \qquad \ldots \text{ multiplying both sides by } M_t. \tag{11.5}$$

- The above equation is the static budget constraint.

Step 3: The Investor's Optimization Problem in Static Form

- The objective of the investor, subject to the static budget constraint, is:

$$\max_{C_s} \mathbb{E}_t^{\mathbb{P}}\left[\int_t^T U(C_s, s) ds\right] \tag{11.6}$$

$$+ \lambda_{BC}\left(M_t W_t - \mathbb{E}_t^{\mathbb{P}}\left[\int_t^T M_s C_s ds\right]\right). \tag{11.7}$$

- Note that in the problem we choose only C_s for s going from t to T; the optimal portfolio will be identified later as the one that "replicates" the investor's optimal consumption.
- Just as before, the first-order condition for the above problem is:

$$\partial U(C_s, s)/\partial C_s = \lambda_{BC} M_s \quad \Longleftrightarrow \quad C_s = g(\lambda_{BC} M_s, s). \tag{11.8}$$

where $g(\cdot)$ is the inverse of the marginal-utility function.

- So, if we knew the value of λ_{BC}, this equation would give us optimal C_s.

Step 4: Identifying the Value of the Lagrange Multiplier

- To identify the Lagrange multiplier, we use the static budget constraint (just as we did in the discrete-time setting), which says that current wealth has to be equal to the discounted value of future consumption.

$$W_t = \mathbb{E}_t^{\mathbb{P}}\left[\int_t^T \frac{M_s}{M_t} C_s \, ds\right]. \tag{11.9}$$

- Specifically, we substitute the solution $C_s = g(\lambda_{BC} M_s, s)$ in (11.8) on the previous page, into the static budget constraint, and solve for λ_{BC}:

$$W_t = \mathbb{E}_t^{\mathbb{P}}\left[\int_t^T \frac{M_s}{M_t} g(\lambda_{BC} M_s, s) ds\right]. \tag{11.10}$$

- Solving the above equation gives us λ_{BC}, just as it did in discrete-time.

Step 5: Identifying Optimal Consumption

- Once we know λ_{BC}, then just as we did in the discrete-time example, we can obtain C_s by substituting the value of λ_{BC} into the first-order condition, which we re-write below from Page 430:

$$\partial U(C_s)/\partial C_s = \lambda_{BC} M_s \quad \Longleftrightarrow \quad C_s = g(\lambda_{BC} M_s, s) \tag{11.8}$$

- Now the only remaining task is to identify the optimal portfolio that finances/replicates the optimal consumption stream we have identified, C_s.

Step 6: Identifying the Optimal Portfolio

- We have identified optimal consumption, the discounted value of which equals current wealth.
- There are two ways in which we can express the process for wealth:
  1. Using the dynamic budget constraint.
  2. Using Itô's formula, we can obtain the process for "optimal wealth."
- By equating the two different ways of writing down the process for wealth, we can identify the portfolio that replicates the desired consumption.
- This is exactly what we did in the context of replicating financial options; it is also what we did for the discrete-time example earlier in this chapter.

The Dynamic Budget Constraint—First Way

- The wealth dynamics are given by the dynamic budget constraint:

$$
\begin{aligned}
dW &= W\Bigg[\overbrace{w\frac{d\mathcal{P}^S}{\mathcal{P}^S}}^{\text{return from risky asset}} + \overbrace{(1-w)\,r\,dt}^{\text{return from risk-free asset}}\Bigg] - C\,dt \\
&= W\left[w\Big(\mu\,dt + \sigma\,dB^{\mathbb{P}}\Big) + (1-w)\,r\,dt\right] - C\,dt \\
&= Ww\,(\mu - r)\,dt + (r\,W - C)\,dt + Ww\sigma\,dB^{\mathbb{P}} \qquad (11.11)
\end{aligned}
$$

$$
\frac{dW}{W} = \Bigg[\underbrace{w\,(\mu - r) + r}_{\text{expected return on invested wealth}} - \underbrace{\frac{C}{W}}_{\text{consumption}}\Bigg]dt + \underbrace{w\,\sigma\,dB^{\mathbb{P}}}_{\text{risk of invested wealth}}
$$

The Process for Optimally-Invested Wealth—Second Way

- Recall our assumption that the investment opportunity set is constant.
- Now, we know from (11.10) on Page 431, which we re-write below,

$$W_t = \mathbb{E}_t^{\mathbb{P}}\left[\int_t^T \frac{M_s}{M_t} g(\lambda_{BC} M_s, s) ds\right], \tag{11.10}$$

  that optimally invested wealth is a function only of M_t and the investor's horizon; that is, $W = W(M_t, t)$.

- And, we know from Equation (11.2) on Page 428, reproduced below, that:

$$dM = -M\, r\, dt - M\, \lambda\, dB^{\mathbb{P}}. \tag{11.2}$$

- So, applying Itô's formula to $W(M_t, t)$ gives:

$$dW = [\text{some drift terms}]\, dt - W_M\, M\, \lambda\, dB^{\mathbb{P}}, \quad \text{where } W_M = \frac{\partial W}{\partial M}. \tag{11.12}$$

Matching the Diffusion Terms (coefficients of the Brownian motion)

- We now collect the two expressions for optimally-invested wealth:
  1. From the dynamic budget constraint in (11.11) on Page 434:

$$dW = [\text{some drift terms}]\, dt + W\, w\, \sigma\, dB^{\mathbb{P}}. \tag{11.11}$$

  2. From Itô's formula applied to $W(M_t, t)$ on the previous slide:

$$dW = [\text{some drift terms}]\, dt - W_M\, M\, \lambda\, dB^{\mathbb{P}}. \tag{11.12}$$

- Equating the diffusion terms in (11.11) and (11.12), we get:

$$W\, w\, \sigma = -W_M\, M\, \lambda. \tag{11.13}$$

- We re-write the last equation on the previous page:

$$W\, w\, \sigma = -W_M\, M\, \lambda. \tag{11.13}$$

- Solving the above equation for w (dividing by $W\sigma$), we get:

$$w = -\frac{W_M\, M}{W}\, \sigma^{-1}\, \lambda = -\frac{W_M\, M}{W}\, \frac{1}{\sigma}\, \lambda. \tag{11.14}$$

- Using the no-arbitrage condition in (11.3) on Page 428 that $\lambda = \frac{\mu - r}{\sigma}$, we can substitute for λ to get the optimal portfolio weight

$$w = -\frac{W_M\, M}{W} \times \underbrace{\frac{(\mu - r)}{\sigma^2}}_{\text{mean-variance portfolio}}. \tag{11.15}$$

- We start by re-writing the last equation on the previous page:

$$w = -\frac{W_M\, M}{W} \times \underbrace{\frac{(\mu - r)}{\sigma^2}}_{\text{mean-variance portfolio}} \tag{11.16}$$

- The last step is to identify the term $\frac{W_M\, M}{W}$. In general, we do this by solving the linear partial differential equation for $W(M_t, t)$.

- As we will show below, $-\frac{W_M\, M}{W} = \frac{1}{\gamma}$, implying that the optimal portfolio in the case of a single risky asset is:

$$w = \frac{1}{\gamma} \times \underbrace{\frac{(\mu - r)}{\sigma^2}}_{\text{mean-variance portfolio}} = \text{investment in the risky asset}, \tag{11.17}$$

with $1 - w$ being the investment in the risk-free asset.

11.5.3 Identifying the Expression $W_M M/W$

- On the last line of the previous slide, we have agreed that all we need to do is to identify $-\frac{W_M\, M}{W}$.

- Assume that the investor has power utility function, given by: $u(c,t) = e^{-\delta\, t}\, \frac{c^{1-\gamma}}{1-\gamma}$.

- Suppose that I tell you that $W = W(M_t, t)$ can be written as follows:

$$W(M_t,t) = f(t)\, M_t^{\frac{-1}{\gamma}}, \quad \text{... we will prove this below in Section 11.5.4} \tag{11.18}$$

  ⋆ where $f(t)$ is some function of time (to be determined below), and

  ⋆ γ is the relative risk aversion for the utility function: $u(c,t) = e^{-\delta\, t}\, \frac{c^{1-\gamma}}{1-\gamma}$.

- Then, differentiating the above definition of $W(M_t, t)$ with respect to M_t:

$$\begin{aligned} W_M(M_t,t) &= \left(\frac{-1}{\gamma}\right) f(t)\, M_t^{\frac{-1}{\gamma}-1} \\ M_t \times W_M(M_t,t) &= \left(\frac{-1}{\gamma}\right) f(t)\, M_t^{\frac{-1}{\gamma}} \\ -\frac{W_M\, M}{W} &= \frac{1}{\gamma}, \end{aligned} \tag{11.19}$$

  where γ is relative risk aversion and $1/\gamma$ is relative risk tolerance.

11.5.4 Showing that $W(M_t, t) = f(t)\, M_t^{\frac{-1}{\gamma}}$

Step 1: A Preliminary Result About $\mathbb{E}_t M_s^{1-\frac{1}{\gamma}}$ that We Will Need

- We start with the dynamics of the stochastic discount factor, M_t:

$$\frac{dM}{M} = -r\,dt - \lambda\,dB^{\mathbb{P}}, \quad \text{... from (11.2) on page 428}$$

$$d\log M_t = \left[-r - \frac{1}{2}\lambda^2\right] dt - \lambda\,dB^{\mathbb{P}} \quad \text{... using Itô's formula for logs}$$

$$\log M_s - \log M_t = \left[-r - \frac{1}{2}\lambda^2\right] (s-t) - \lambda\left[B^{\mathbb{P}}(s) - B^{\mathbb{P}}(t)\right] \quad \text{... integrating}$$

$$\log M_s = \log M_t + \left[-r - \frac{1}{2}\lambda^2\right] (s-t) - \lambda\left[B^{\mathbb{P}}(s) - B^{\mathbb{P}}(t)\right] \quad \text{... adding } \log M_t \text{ to both sides}$$

$$M_s = M_t \exp\left[\left[-r - \frac{1}{2}\lambda^2\right] (s-t) - \lambda\left[B^{\mathbb{P}}(s) - B^{\mathbb{P}}(t)\right]\right] \quad \text{... taking exponentials}$$

$$M_s^{1-\frac{1}{\gamma}} = M_t^{1-\frac{1}{\gamma}} \exp\left[\left[-r - \frac{1}{2}\lambda^2\right] (s-t)\left(1-\frac{1}{\gamma}\right) - \lambda\left[B^{\mathbb{P}}(s) - B^{\mathbb{P}}(t)\right]\left(1-\frac{1}{\gamma}\right)\right]$$

where in the last step we raise both sides of the equation to the power $1 - \frac{1}{\gamma}$.

- We start by re-writing the last equation on the previous page:

$$M_s^{1-\frac{1}{\gamma}} = M_t^{1-\frac{1}{\gamma}} \exp\left[\left[-r - \frac{1}{2}\lambda^2\right] (s-t)\left(1-\frac{1}{\gamma}\right) - \lambda\left[B^{\mathbb{P}}(s) - B^{\mathbb{P}}(t)\right]\left(1-\frac{1}{\gamma}\right)\right].$$

- Observe that on the right-hand side, $B^{\mathbb{P}}(s) - B^{\mathbb{P}}(t)$ is distributed Normally.

- Thus, the right-hand side is the exponential of a Normally distributed variable.

- And, the expectation of the exponential of a Normally distributed variable, recalling earlier results, is the exponential of the "mean plus half variance".

$$\mathbb{E}_t^{\mathbb{P}} M_s^{1-\frac{1}{\gamma}} = M_t^{1-\frac{1}{\gamma}} \exp\left[\underbrace{\left[-r - \frac{1}{2}\lambda^2\right]\left(1-\frac{1}{\gamma}\right)(s-t)}_{\text{mean}} + \underbrace{\frac{1}{2}\lambda^2\left(1-\frac{1}{\gamma}\right)^2(s-t)}_{\text{variance}}\right]$$

$$\mathbb{E}_t^{\mathbb{P}} M_s^{1-\frac{1}{\gamma}} = M_t^{1-\frac{1}{\gamma}} \exp\left[\alpha(s-t)\right], \quad \text{... this is the preliminary result we wanted to prove} \tag{11.20}$$

$$\text{where} \quad \alpha \equiv \left[-r - \frac{1}{2}\lambda^2\right]\left(1-\frac{1}{\gamma}\right) + \frac{1}{2}\lambda^2\left(1-\frac{1}{\gamma}\right)^2. \tag{11.21}$$

Step 2: First-order Conditions for Consumption for Power Utility

- We start by re-writing the first-order condition for consumption given in equation (11.8) on Page 430:

$$\frac{\partial U(C_s, s)}{\partial C_s} = \lambda_{BC} M_s \quad \Longleftrightarrow \quad C_s = g(\lambda_{BC} M_s, s) \tag{11.8}$$

- For the case of power utility, $u(c,t) = e^{-\delta\, t}\, \frac{c^{1-\gamma}}{1-\gamma}$, this can be expressed by

$$\frac{\partial U(C_s, s)}{\partial C_s} = e^{-\delta\, s}\, C_s^{-\gamma} = \lambda_{BC}\, M_s \quad \Longleftrightarrow \quad C_s = \left(e^{\delta\, s}\, \lambda_{BC}\, M_s\right)^{-1/\gamma} \tag{11.22}$$

Step 3: Identify $W_t = M_t^{\frac{-1}{\gamma}}\, f(t)$ Using the Static Budget Constraint

- We start by re-writing the static budget-constraint from Page 431:

$$W_t = \mathbb{E}_t^{\mathbb{P}}\left[\int_t^T \frac{M_s}{M_t}\, C_s\, ds\right] \tag{11.9}$$

$$= \frac{1}{M_t}\mathbb{E}_t^{\mathbb{P}}\left[\int_t^T M_s\, C_s\, ds\right] \quad \text{. . . pulling out } \frac{1}{M_t}\text{, which is known at } t$$

$$= \frac{1}{M_t}\mathbb{E}_t^{\mathbb{P}}\left[\int_t^T M_s \left(e^{\delta\, s}\, \lambda_{BC}\, M_s\right)^{-1/\gamma} ds\right] \quad \text{. . . substitute for } C_s$$

$$= \frac{1}{M_t}\mathbb{E}_t^{\mathbb{P}}\left[\int_t^T M_s^{1-\frac{1}{\gamma}} \left(e^{\delta\, s}\, \lambda_{BC}\right)^{-1/\gamma} ds\right] \quad \text{. . . collect powers of } M_s$$

$$= \frac{1}{M_t}\left[\int_t^T \left(e^{\delta\, s}\, \lambda_{BC}\right)^{-1/\gamma} \mathbb{E}_t^{\mathbb{P}} M_s^{1-\frac{1}{\gamma}}\, ds\right] \quad \text{. . . apply expectation operator}$$

- We start by re-writing the last equation on the previous page:

$$W_t = \frac{1}{M_t}\left[\int_t^T \left(e^{\delta\, s}\,\lambda_{BC}\right)^{-1/\gamma} \mathbb{E}_t^{\mathbb{P}} M_s^{1-\frac{1}{\gamma}}\, ds\right]$$

$$= \frac{1}{M_t}\left[\int_t^T \left(e^{\delta\, s}\,\lambda_{BC}\right)^{-1/\gamma} M_t^{1-\frac{1}{\gamma}} \exp\left[\alpha(s-t)\right] ds\right] \quad \ldots \text{using "preliminary result" in (11.20)}$$

$$= M_t^{\frac{-1}{\gamma}}\left[\int_t^T \left(e^{\delta\, s}\,\lambda_{BC}\right)^{-1/\gamma} \exp\left[\alpha(s-t)\right] ds\right] \quad \ldots \text{collecting powers of } M_t$$

$$W_t = M_t^{\frac{-1}{\gamma}}\, f(t), \quad \ldots \text{which is the result we wanted to prove} \tag{11.23}$$

where

$$f(t) = \left[\int_t^T \left(e^{\delta\, s}\,\lambda_{BC}\right)^{-1/\gamma} \exp\left[\alpha(s-t)\right] ds\right].$$

11.6 General Case: Many risky assets, Bequest, and Stochastic Investment Opportunity Set

- We now look at the general case with:
  - ⋆ n risky assets,
  - ⋆ a stochastic investment opportunity set, and
  - ⋆ utility from consumption and also bequest;
- The analysis is similar to that for the special case, except that now we have n risky assets, so scalar quantities for the single risky asset in the special case are replaced by vectors/matrices for the n risky assets.
- The notes for the general case are written to match exactly the notes for the special case (with all subsections having the same titles), so if something is not clear for the general case, you can refer to the corresponding notes for the special case.

Description of the Setting for the General Case

- There is a risk-free asset with return $r(x,t)$ and n risky assets whose prices $\mathcal{P}^{S_i}(t)$ follow the following process

$$\frac{d\mathcal{P}^{S_i}(t)}{\mathcal{P}^{S_i}(t)} = \mu_i(x,t)\,dt + \boldsymbol{\sigma}_i(x,t)\,d\mathbf{B}^{\mathbb{P}}, \quad i = \{1,2,\ldots,n\}, \quad \text{where,} \quad (11.24)$$

  - ⋆ $\boldsymbol{\sigma}_i = (\sigma_{i1}, \sigma_{i2}, \ldots, \sigma_{in})$ is a $1 \times n$ vector of volatility components;
  - ⋆ $\boldsymbol{\sigma}$ is a $n \times n$ matrix whose i-th row equals $\boldsymbol{\sigma}_i$; and
  - ⋆ $d\mathbf{B}^{\mathbb{P}} = (dB_1^{\mathbb{P}}, \ldots, dB_n^{\mathbb{P}})$ is $n \times 1$ vector of independent Brownian motions.

- Furthermore, it is assumed that the n assets are nonredundant in the sense that their instantaneous covariance matrix $\mathbf{V} = \boldsymbol{\sigma}\boldsymbol{\sigma}'$ is nonsingular.

- The scalar $\mu_i(x,t)$, the elements of $\boldsymbol{\sigma}_i(x,t)$, and $r(x,t)$ are allowed to be functions of state variables x that are driven by the vector of Brownians $d\mathbf{B}^{\mathbb{P}}$.

- The **main assumption** is that
  - ⋆ all the state variables $x(t)$ are driven by the same Brownian motions $d\mathbf{B}^{\mathbb{P}}$ that are driving the stock returns $\mathcal{P}^{S_i}(t)$, and
  - ⋆ hence, changes in the state variables can be hedged using the n risky assets (that is, markets are complete).

- In the next chapter, where we will allow for incomplete markets,
  - ⋆ stock returns will be driven by a set of n Brownian motions,
  - ⋆ while state variables will be driven by a different set of k Brownian motions.

Step 1: The Stochastic Discount Factor

- Given the structure described on the previous slide, in the absence of arbitrage there is a unique stochastic discount factor that follows the process:

$$\frac{dM}{M} = -rdt - \lambda^\top d\mathbf{B}^{\mathbb{P}}, \tag{11.25}$$

where $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_n)'$ is a $n \times 1$ vector of market prices of risk associate with each Brownian motion, and where λ satisfies:

$$\boldsymbol{\mu} - r\mathbf{1} = \boldsymbol{\sigma}\,\lambda. \tag{11.26}$$

- Note that the λ in the above expressions is the vector of market prices of risk, and is different from λ_{BC} and λ_{weight}, which denote Lagrange multipliers.

Step 2: The Static Budget Constraint

- The key to the martingale pricing approach for finding the optimal consumption and portfolio of the investor is to view the optimally invested wealth as an asset that provides a dividend equal to the consumption of the investor.

- This means that the "return on wealth" can be priced using the stochastic discount factor.

- That is, current wealth must be equal to the expected discounted value of the stream of future consumption (dividends) plus discounted terminal wealth:

$$W_t = \mathbb{E}_t^{\mathbb{P}}\left[\int_t^T \frac{M_s}{M_t} C_s ds + \frac{M_T}{M_t} W_T\right], \tag{11.27}$$

or, equivalently:

$$W_t M_t = \mathbb{E}_t^{\mathbb{P}}\left[\int_t^T M_s C_s ds + M_T W_T\right] \quad \ldots \text{multiplying both sides by } W_t. \tag{11.28}$$

- The above equation is the static budget constraint.

Step 3: The Investor's Optimization Problem in Static Form

- The objective of the investor, subject to the static budget constraint, is:

$$\max_{C_s, W_T} \mathbb{E}_t^{\mathbb{P}} \left[\int_t^T U(C_s, s) ds + \text{Beq}(W_T, T) \right] \tag{11.29}$$

$$+ \lambda_{BC} \left(M_t W_t - \mathbb{E}_t^{\mathbb{P}} \left[\int_t^T M_s C_s ds + M_T W_T \right] \right). \tag{11.30}$$

- Note that in the problem above we do not explicitly address the portfolio choice decision. The optimal portfolio is identified later as the one that replicates the investor's optimal consumption and terminal wealth.

- Just as before, the first-order conditions for the above problem are:

$$\partial U(C_s, s)/\partial C_s = \lambda_{BC} M_s \iff C_s = g(\lambda_{BC} M_s, s) \tag{11.31}$$

$$\partial \text{Beq}(W_T, T)/\partial W_T = \lambda_{BC} M_T \iff W_T = G(\lambda_{BC} M_T, T). \tag{11.32}$$

- So, if we knew λ_{BC}, these two equations would give us optimal C_s and W_T.

Step 4: Identifying the Value of the Lagrange Multiplier

- To identify the Lagrange multiplier, we use the static budget constraint (just as we did in the discrete-time setting) which says that current wealth has to be equal to the discounted value of consumption and terminal wealth.

$$W_t = \mathbb{E}_t^{\mathbb{P}} \left[\int_t^T \frac{M_s}{M_t} C_s \, ds + \frac{M_T}{M_t} W_T \right]. \tag{11.33}$$

- Specifically, we substitute the solutions to C_s and W_T in (11.31) and (11.32) on the previous page, into the static budget constraint, and solve for λ_{BC}:

$$W_t = \mathbb{E}_t^{\mathbb{P}} \left[\int_t^T \frac{M_s}{M_t} g(\lambda_{BC} M_s, s) ds + \frac{M_T}{M_t} G(\lambda_{BC} M_T, T) \right]. \tag{11.34}$$

- Solving the above equation gives us λ_{BC}.

Step 5: Identifying Optimal Consumption

- Once we know λ_{BC} then, just as we did in the discrete-time example,
  - ⋆ we can obtain C_s and W_T by substituting the value of λ_{BC} in the first-order conditions, which were expressed in terms of λ_{BC}, which we re-write below from Page 451:

$$\partial U(C_s)/\partial C_s = \lambda_{BC} M_s \quad \Longleftrightarrow \quad C_s = g(\lambda_{BC} M_s, s) \qquad (11.31)$$
$$\partial \text{Beq}(W_T)/\partial W_T = \lambda_{BC} M_T \quad \Longleftrightarrow \quad W_T = G(\lambda_{BC} M_T, T). \qquad (11.32)$$

- Now, the only task left is to identify the optimal portfolio.

Step 6: Identifying the Optimal Portfolio

- We have identified optimal consumption and terminal wealth.
- Using Itô's formula, we can write down the process that this "optimal wealth" must follow.
- We can also write down the process for wealth using the dynamic budget constraint (which we do below).
- By equating the two different ways of writing down the process for wealth, we can identify the portfolio that replicates the desired consumption and terminal wealth.
- This is exactly what we did
  - ⋆ when replicating financial options
  - ⋆ for the discrete-time example earlier in this chapter in Section 11.3, and
  - ⋆ for the special case considered in the previous section.

The Dynamic Budget Constraint—First Way

- The wealth dynamics are given by the dynamic budget constraint:

$$
\begin{aligned}
dW &= W\left[\overbrace{\sum_{i=1}^{n} w_i \frac{d\mathcal{P}^{S_i}}{\mathcal{P}^{S_i}}}^{\text{investment in risky assets}} + \overbrace{\left(1-\sum_{i=1}^{n} w_i\right) r\, dt}^{\text{investment in risk-free asset}}\right] - C\, dt \\
&= W\left[\sum_{i=1}^{n} w_i\left[\mu_i\, dt + \sigma_i\, dB_i^{\mathbb{P}}\right] + \left(1-\sum_{i=1}^{n} w_i\right) r\, dt\right] - C\, dt \\
&= W\sum_{i=1}^{n} w_i\left(\mu_i - r\right) dt + \left(r\, W - C\right) dt + W\sum_{i=1}^{n} w_i \sigma_i\, dB_i^{\mathbb{P}} \qquad (11.35)
\end{aligned}
$$

$$
\frac{dW}{W} = \left[\underbrace{\sum_{i=1}^{n} w_i\left(\mu_i - r\right) + r}_{\text{expected return on invested wealth}} - \underbrace{\frac{C}{W}}_{\text{consumption}}\right] dt + \underbrace{\sum_{i=1}^{n} w_i \sigma_i\, dB_i^{\mathbb{P}}}_{\text{risk of invested wealth}}
$$

The Process for Optimally-Invested Wealth—Second Way

- Consider the setting where there is only a single state variable x that affects the distribution of returns, and the process for this is:

$$
dx = a(x,t)dt + \boldsymbol{\sigma}_x(x,t)^{\top} d\mathbf{B}^{\mathbb{P}}, \qquad (11.36)
$$

  where $\boldsymbol{\sigma}_x(x,t)$ is an $n \times 1$ vector of volatilities.

- One could, of course, consider the case with multiple state variables, but to keep the mathematical notation simple, it is better to start with the case of a single state variable.

- Now, we know from (11.34), which we re-write below,

$$
W_t = \mathbb{E}_t^{\mathbb{P}}\left[\int_t^T \frac{M_s}{M_t} g(\lambda_{BC} M_s, s) ds + \frac{M_T}{M_t} G(\lambda_{BC} M_T, T)\right] \qquad (11.34)
$$

  that optimally invested wealth is a function only of M_t, x_t, and the investor's horizon; that is, $W = W(M_t, x_t, t)$:

Matching the Diffusion Terms (coefficients of the Brownian motion)

- From the dynamic budget constraint in (11.35) on Page 455:

$$\begin{aligned} dW &= [\text{some drift terms}]\, dt + W \sum_{i=1}^{n} w_i \sigma_i \, dB_i^{\mathbb{P}} \\ &= [\text{some drift terms}]\, dt + W \mathbf{w}^\top \boldsymbol{\sigma} \, d\mathbf{B}^{\mathbb{P}}. \end{aligned} \tag{11.37}$$

- From Itô's formula applied to $W(M_t, x_t, t)$:

$$dW = [\text{some drift terms}]\, dt - W_M \, M \, \lambda^\top \, d\mathbf{B}^{\mathbb{P}} + W_x \, \boldsymbol{\sigma}_x^\top \, d\mathbf{B}^{\mathbb{P}}. \tag{11.38}$$

- Equating the diffusion terms in (11.37) and (11.38), we get:

$$W \mathbf{w}^\top \boldsymbol{\sigma} = -W_M \, M \, \lambda^\top + W_x \, \boldsymbol{\sigma}_x^\top. \tag{11.39}$$

- We re-write the last equation on the previous page:

$$W \mathbf{w}^\top \boldsymbol{\sigma} = -W_M \, M \, \lambda^\top + W_x \, \boldsymbol{\sigma}_x^\top. \tag{11.39}$$

- Solving the above equation for $\mathbf{w}$, we get:

$$\mathbf{w} = -\frac{W_M \, M}{W} (\boldsymbol{\sigma}^\top)^{-1} \lambda + \frac{W_x}{W} (\boldsymbol{\sigma}^\top)^{-1} \boldsymbol{\sigma}_x. \tag{11.40}$$

- Using the no-arbitrage condition in (11.26) $\boldsymbol{\mu} - r\mathbf{1} = \boldsymbol{\sigma} \, \lambda$, we can substitute for λ to get the optimal portfolio weight

$$\begin{aligned} \mathbf{w} &= -\frac{W_M \, M}{W} \boldsymbol{\sigma}^{-1} (\boldsymbol{\sigma}^\top)^{-1} (\boldsymbol{\mu} - r\mathbf{1}) + \frac{W_x}{W} (\boldsymbol{\sigma}^\top)^{-1} \boldsymbol{\sigma}_x \\ &= -\frac{W_M \, M}{W} \underbrace{\mathsf{V}^{-1} (\boldsymbol{\mu} - r\mathbf{1})}_{\text{mean-variance portfolio}} + \frac{W_x}{W} \underbrace{(\boldsymbol{\sigma}^\top)^{-1} \boldsymbol{\sigma}_x}_{\substack{\text{intertemporal hedging} \\ \text{portfolio}}} . \quad \ldots \text{because } \boldsymbol{\sigma}^{-1} (\boldsymbol{\sigma}^\top)^{-1} = \mathsf{V}^{-1} \end{aligned} \tag{11.41}$$

Interpreting the Expression for Optimal Portfolio Weights

- We start by re-writing the last equation on the previous page:

$$\mathbf{w} = -\frac{W_M\, M}{W} \underbrace{\mathbf{V}^{-1}(\boldsymbol{\mu} - r\mathbf{1})}_{\text{mean-variance portfolio}} + \frac{W_x}{W} \underbrace{(\boldsymbol{\sigma}^\top)^{-1}\boldsymbol{\sigma}_x}_{\text{intertemporal hedging portfolio}} \quad . \tag{11.41}$$

- Note that the mean-variance portfolio is also called
  - ⋆ the myopic portfolio, because it is the portfolio of an investor who ignores the future when optimizing;
  - ⋆ the tactical portfolio, because it depends only on the expected return and risk over the next period.
- The intertemporal-hedging portfolio is also called the strategic portfolio of long-term investors.

- We start by re-writing the last equation on the previous page:

$$\mathbf{w} = -\frac{W_M\, M}{W} \underbrace{\mathbf{V}^{-1}(\boldsymbol{\mu} - r\mathbf{1})}_{\text{mean-variance portfolio}} + \frac{W_x}{W} \underbrace{(\boldsymbol{\sigma}^\top)^{-1}\boldsymbol{\sigma}_x}_{\text{intertemporal hedging portfolio}} \quad . \tag{11.41}$$

- So, to find the optimal portfolio weights, the last step is to identify the terms $\frac{W_M\, M}{W}$ and $\frac{W_x}{W}$.
- We do this by solving the linear partial differential equation for $W(M_t, x_t, t)$.
- In the appendix, we demonstrate how one can do this for the special case of a constant investment opportunity set and n risky assets.

11.7 Summary

1. Under the martingale approach to consumption and portfolio choice:

   - We view the optimally invested wealth as an asset that provides a dividend equal to the consumption of the investor.

   - Thus, current wealth must equal the expected discounted value of the stream of future consumption (dividends) plus discounted terminal wealth:

$$W_t = \mathbb{E}_t^{\mathbb{P}}\left[\int_t^T \frac{M_s}{M_t} C_s ds + \frac{M_T}{M_t} W_T\right],$$

     which is the static budget constraint.

   - The objective of the investor, subject to the static budget constraint, is:

$$\max_{C_s, W_T} \mathbb{E}_t^{\mathbb{P}}\left[\int_t^T U(C_s, s) ds + \text{Beq}(W_T, T)\right] + \lambda_{BC}\left(M_t W_t - \mathbb{E}_t^{\mathbb{P}}\left[\int_t^T M_s C_s ds + M_T W_T\right]\right).$$

   - The first-order conditions for C_s and W_T are:

$$\begin{aligned} \partial U(C_s, s)/\partial C_s &= \lambda_{BC} M_s \quad \Longleftrightarrow \quad C_s = g(\lambda_{BC} M_s, s) \\ \partial \text{Beq}(W_T, T)/\partial W_T &= \lambda_{BC}\, M_T \quad \Longleftrightarrow \quad W_T = G(\lambda_{BC}\, M_T, T). \end{aligned}$$

   - In the expressions above, λ_{BC} can be identified by substituting the expressions for optimal consumption and optimal terminal wealth into the static budget constraint.

- The first-order condition for the optimal portfolio weight vector is:

$$\mathbf{w} = -\frac{W_M\, M}{W}\mathbf{V}^{-1}(\boldsymbol{\mu} - r\mathbf{1}) + \frac{W_x}{W}(\boldsymbol{\sigma}^\top)^{-1}\boldsymbol{\sigma}_x,$$

  which is the same as the one we will obtain using dynamic programming when markets are complete.

- In the expression above, wealth (and hence also its derivatives, W_M and W_x) can be identified by solving a linear partial differential equation.

11.8 Appendix: Case with Multiple Risky Assets and Power Utility But Constant Investment Opportunity Set

- In this section, we consider a setting where
  1. There are multiple risky assets.
  2. The investor has power utility function, given by: $u(c,t) = e^{-\delta\, t}\,\frac{c^{1-\gamma}}{1-\gamma}$ and also a bequest function that has the same power form.
  3. But, the investment opportunity set (defined by the capital market line) is assumed to be constant; that is, the interest rate is constant, and the market price of risk is constant.

- This case is in-between the special case that we studied in Section 11.5 and the general case that we studied in Section 11.6 because it
  - ⋆ allows for multiple instead of just one risky asset;
  - ⋆ but has a constant instead of stochastic investment opportunity set.

11.8.1 Expression for the Optimal Portfolio

- The general expression for the portfolio weight vector with a stochastic investment opportunity set, derived in Equation (11.41) on Page 458 is:

$$\mathbf{w} = -\frac{W_M\, M}{W} \underbrace{\mathbf{V}^{-1}(\boldsymbol{\mu} - r\mathbf{1})}_{\text{mean-variance portfolio}} + \frac{W_x}{W} \underbrace{(\boldsymbol{\sigma}^\top)^{-1}\boldsymbol{\sigma}_x}_{\text{intertemporal hedging portfolio}} \quad . \tag{11.41}$$

- Because we are now assuming that the investment opportunity set is constant, there is zero intertemporal hedging, so $W_x = 0$ and the above reduces to:

$$\mathbf{w} = -\frac{W_M\, M}{W} \underbrace{\mathbf{V}^{-1}(\boldsymbol{\mu} - r\mathbf{1})}_{\text{mean-variance portfolio}} . \tag{11.42}$$

- In the expression above, we are given:
  - ⋆ the risk-free interest rate, r; the expected return on risky assets, $\boldsymbol{\mu}$; and, the variance-covariance matrix for returns, $\mathbf{V}$.
- So, we need to identify only $-\frac{W_M\, M}{W}$, which we do as before.

11.8.2 Showing that $-W_M M/W = 1/\gamma$

- Suppose we assume that $W = W(M_t, t)$ can be written as follows:

$$W(M_t, t) = f(t)\, M_t^{\frac{-1}{\gamma}}, \quad \text{...we will prove this below in Section 11.8.5} \tag{11.43}$$

  - ⋆ where $f(t)$ is some function of time (to be determined below), and
  - ⋆ γ is the relative risk aversion for the utility function: $u(c,t) = e^{-\delta\, t}\, \frac{c^{1-\gamma}}{1-\gamma}$.

- Then, differentiating the above definition of $W(M_t, t)$ with respect to M_t:

$$\begin{aligned} W_M(M_t, t) &= \left(\frac{-1}{\gamma}\right) f(t)\, M_t^{\frac{-1}{\gamma}-1} \\ M_t \times W_M(M_t, t) &= \left(\frac{-1}{\gamma}\right) f(t)\, M_t^{\frac{-1}{\gamma}} \\ -\frac{W_M\, M}{W} &= \frac{1}{\gamma}, \end{aligned} \tag{11.44}$$

  where γ is relative risk aversion and $1/\gamma$ is relative risk tolerance.

11.8.3 Explicit Expression for the Optimal Portfolio

- Substituting the result we have just derived on the previous page

$$-\frac{W_M\, M}{W} = \frac{1}{\gamma}, \tag{11.44}$$

into the expression for optimal weights in Equation (11.42), re-written below:

$$\mathbf{w} = -\frac{W_M\, M}{W}\, \underbrace{\mathbf{V}^{-1}(\boldsymbol{\mu} - r\mathbf{1})}_{\text{mean-variance portfolio}}. \tag{11.42}$$

gives us the final expression for the optimal portfolio weights:

$$\mathbf{w} = \frac{1}{\gamma}\, \underbrace{\mathbf{V}^{-1}(\boldsymbol{\mu} - r\mathbf{1})}_{\text{mean-variance portfolio}}. \tag{11.45}$$

11.8.4 Interpreting the Expression for the Optimal Portfolio

- We can interpret the expression for the optimal portfolio as follows:

$$\mathbf{w} = \frac{1}{\gamma}\, \underbrace{\mathbf{V}^{-1}(\boldsymbol{\mu} - r\mathbf{1})}_{\text{mean-variance portfolio}}, \tag{11.46}$$

- The weight in each risky asset depends only on expected returns and variance-covariances of returns: $\mathbf{V}^{-1}(\boldsymbol{\mu} - r\mathbf{1})$.
  - ⋆ Thus, this "tangency portfolio" depends only on asset-return moments, and not on investor preferences; hence, the composition of this "tangency" portfolio is the same across all investors.
- Investment in this portfolio increases with the investor's risk tolerance, $1/\gamma$.
- The wealth not invested in risky assets is invested in the risk-free asset.
- Thus, all investors hold only two funds:
  - ⋆ The risk-free asset;
  - ⋆ The tangency portfolio, given by $\mathbf{V}^{-1}(\boldsymbol{\mu} - r\mathbf{1})$.

11.8.5 Showing that $W(M_t, t) = f(t)\, M_t^{\frac{-1}{\gamma}}$

Step 1: A Preliminary Result About $\mathbb{E}_t M_s^{1-\frac{1}{\gamma}}$ that We Will Need

- We start with the dynamics of the stochastic discount factor, M_t:

$$\frac{dM}{M} = -r\, dt - \lambda^\top\, d\mathbf{B}^{\mathbb{P}}, \tag{11.25}$$

$$d \log M_t = \left[-r - \frac{1}{2}\lambda^\top\lambda\right]\, dt - \lambda^\top\, d\mathbf{B}^{\mathbb{P}} \quad \text{... using Itô's formula for logs}$$

$$\log M_s - \log M_t = \left[-r - \frac{1}{2}\lambda^\top\lambda\right]\, (s-t) - \lambda^\top\, \left[\mathbf{B}^{\mathbb{P}}(s) - \mathbf{B}^{\mathbb{P}}(t)\right] \quad \text{... integrating}$$

$$\log M_s = \log M_t + \left[-r - \frac{1}{2}\lambda^\top\lambda\right]\, (s-t) - \lambda^\top\, \left[\mathbf{B}^{\mathbb{P}}(s) - \mathbf{B}^{\mathbb{P}}(t)\right] \quad \text{... adding } \log M_t \text{ to both side}$$

$$M_s = M_t \exp\left[\left[-r - \frac{1}{2}\lambda^\top\lambda\right]\, (s-t) - \lambda^\top\, \left[\mathbf{B}^{\mathbb{P}}(s) - \mathbf{B}^{\mathbb{P}}(t)\right]\right] \quad \text{... taking exponentials}$$

$$M_s^{1-\frac{1}{\gamma}} = M_t^{1-\frac{1}{\gamma}} \exp\left[\left[-r - \frac{1}{2}\lambda^\top\lambda\right]\, (s-t)\left(1-\frac{1}{\gamma}\right) - \lambda^\top\, \left[\mathbf{B}^{\mathbb{P}}(s) - \mathbf{B}^{\mathbb{P}}(t)\right]\left(1-\frac{1}{\gamma}\right)\right]$$

where in the last step we raise both sides of the equation to the power $1 - \frac{1}{\gamma}$.

- We start by re-writing the last equation on the previous page:

$$M_s^{1-\frac{1}{\gamma}} = M_t^{1-\frac{1}{\gamma}} \exp\left[\left[-r - \frac{1}{2}\lambda^\top\lambda\right]\, (s-t)\left(1-\frac{1}{\gamma}\right) - \lambda^\top\, \left[\mathbf{B}^{\mathbb{P}}(s) - \mathbf{B}^{\mathbb{P}}(t)\right]\left(1-\frac{1}{\gamma}\right)\right].$$

- Observe that on the right-hand side, $\mathbf{B}^{\mathbb{P}}(s) - \mathbf{B}^{\mathbb{P}}(t)$ is distributed Normally.

- Thus, the right-hand side is the exponential of a Normally distributed variable.

- Thus, the expectation of the exponential of a Normally distributed variable, recalling earlier results, is the exponential of the "mean plus half variance".

$$\mathbb{E}_t^{\mathbb{P}} M_s^{1-\frac{1}{\gamma}} = M_t^{1-\frac{1}{\gamma}} \exp\left[\left[-r - \frac{1}{2}\lambda^\top\lambda\right]\, \left(1-\frac{1}{\gamma}\right)(s-t) + \frac{1}{2}\lambda^\top\lambda\, \left(1-\frac{1}{\gamma}\right)^2 (s-t)\right]$$

$$\mathbb{E}_t^{\mathbb{P}} M_s^{1-\frac{1}{\gamma}} = M_t^{1-\frac{1}{\gamma}} \exp\left[\alpha(s-t)\right], \quad \text{this is the result we wanted to prove} \tag{11.47}$$

$$\text{where} \quad \alpha \equiv \left[-r - \frac{1}{2}\lambda^\top\lambda\right]\, \left(1-\frac{1}{\gamma}\right) + \frac{1}{2}\lambda^\top\lambda\, \left(1-\frac{1}{\gamma}\right)^2. \tag{11.48}$$

Step 2: First-order Conditions for Consumption for Power Utility

- We start by re-writing the first-order conditions for the general case, which from Page 451 are given by:

$$\partial U(C_s, s)/\partial C_s = \lambda_{BC} M_s \quad \Longleftrightarrow \quad C_s = g(\lambda_{BC} M_s, s) \tag{11.31}$$

$$\partial \text{Beq}(W_T, T)/\partial W_T = \lambda_{BC} M_T \quad \Longleftrightarrow \quad W_T = G(\lambda_{BC} M_T, T). \tag{11.32}$$

- For the case of power utility, $u(c,t) = e^{-\delta t}\frac{c^{1-\gamma}}{1-\gamma}$,

$$\frac{\partial U(C_s, s)}{\partial C_s} = e^{-\delta s} C_s^{-\gamma} = \lambda_{BC} M_s \quad \Longleftrightarrow \quad C_s = \left(e^{\delta s} \lambda_{BC} M_s\right)^{-1/\gamma} \tag{11.49}$$

$$\frac{\partial \text{Beq}(W_T, T)}{\partial W_T} = e^{-\delta T} W_T^{-\gamma} = \lambda_{BC} M_T \quad \Longleftrightarrow \quad W_T = \left(e^{\delta T} \lambda_{BC} M_T\right)^{-1/\gamma}. \tag{11.50}$$

Step 3: Identify $W_t = M_t^{\frac{-1}{\gamma}} f(t)$ Using the Static Budget Constraint

- We start by re-writing the static budget-constraint from Page 452:

$$W_t = \mathbb{E}_t^{\mathbb{P}}\left[\int_t^T \frac{M_s}{M_t} C_s \, ds + \frac{M_T}{M_t} W_T\right] \tag{11.33}$$

$$= \frac{1}{M_t}\mathbb{E}_t^{\mathbb{P}}\left[\int_t^T M_s \, C_s \, ds + M_T \, W_T\right] \quad \text{. . . pulling out } \frac{1}{M_t}$$

$$= \frac{1}{M_t}\mathbb{E}_t^{\mathbb{P}}\left[\int_t^T M_s \left(e^{\delta s} \lambda_{BC} M_s\right)^{-1/\gamma} ds + M_T \left(e^{\delta T} \lambda_{BC} M_T\right)^{-1/\gamma}\right] \quad \text{. . . substitute for } C_s \text{ \& } W_T$$

$$= \frac{1}{M_t}\mathbb{E}_t^{\mathbb{P}}\left[\int_t^T M_s^{1-\frac{1}{\gamma}} \left(e^{\delta s} \lambda_{BC}\right)^{-1/\gamma} ds + M_T^{1-\frac{1}{\gamma}} \left(e^{\delta T} \lambda_{BC}\right)^{-1/\gamma}\right] \quad \text{. . . collect powers of } M_s \text{ \& } M_T$$

$$= \frac{1}{M_t}\left[\int_t^T \left(e^{\delta s} \lambda_{BC}\right)^{-1/\gamma} \mathbb{E}_t^{\mathbb{P}} M_s^{1-\frac{1}{\gamma}} ds + \left(e^{\delta T} \lambda_{BC}\right)^{-1/\gamma} \mathbb{E}_t^{\mathbb{P}} M_T^{1-\frac{1}{\gamma}}\right] \quad \text{. . . apply expectation operator}$$

- We start by re-writing the last equation on the previous page:

$$W_t = \frac{1}{M_t}\left[\int_t^T \left(e^{\delta s}\lambda_{BC}\right)^{-1/\gamma} \mathbb{E}_t^{\mathbb{P}} M_s^{1-\frac{1}{\gamma}}\, ds + \left(e^{\delta T}\lambda_{BC}\right)^{-1/\gamma} \mathbb{E}_t^{\mathbb{P}} M_T^{1-\frac{1}{\gamma}}\right]$$

$$= \frac{1}{M_t}\left[\int_t^T \left(e^{\delta s}\lambda_{BC}\right)^{-1/\gamma} M_t^{1-\frac{1}{\gamma}} \exp\left[\alpha(s-t)\right] ds + \left(e^{\delta T}\lambda_{BC}\right)^{-1/\gamma} M_t^{1-\frac{1}{\gamma}} \exp\left[\alpha(T-t)\right]\right]$$

. . . using "preliminary result" in (11.47)

$$= M_t^{\frac{-1}{\gamma}}\left[\int_t^T \left(e^{\delta s}\lambda_{BC}\right)^{-1/\gamma} \exp\left[\alpha(s-t)\right] ds + \left(e^{\delta T}\lambda_{BC}\right)^{-1/\gamma} \exp\left[\alpha(T-t)\right]\right]$$

. . . collecting powers of M_t

$$W_t = M_t^{\frac{-1}{\gamma}} f(t), \quad \text{. . . which is the result we wanted to prove} \tag{11.51}$$

where $f(t) = \left[\int_t^T \left(e^{\delta s}\lambda_{BC}\right)^{-1/\gamma} \exp\left[\alpha(s-t)\right] ds + \left(e^{\delta T}\lambda_{BC}\right)^{-1/\gamma} \exp\left[\alpha(T-t)\right]\right]$.

11.9 Suggestions for Further Readings

11.9.1 Source material for the lecture notes

- The static approach to portfolio choice in discrete time is discussed in Back (2017, Chapters 14) and Dumas and Luciano (2017, Chapter 10).
  - ⋆ The multiperiod discrete-time example of using the static budget constraint to solve the portfolio problem was developed by Vasant Naik.
- The static approach for solving the portfolio choice problem in continuous time with complete markets was developed in Karatzas, Lehoczky, and Shreve (1987) and Cox and Huang (1989).
  - ⋆ For a textbook treatment, see Karatzas and Shreve (1998).
- Merton (1990, Chapter 6) shows the equivalence between the Cox and Huang (1989) approach described in this chapter and the dynamic programming approach used in Merton (1969, 1971) that we will study in the next chapter. This is explained also in Dumas and Luciano (2017, Section 9.5).

11.9.2 Potential applications and extensions of lecture notes

- **Strategic asset allocation**
  - ⋆ The articles by Campbell and Viceira (1999, 2001) and their book Campbell and Viceira (2002) provide excellent illustrations of the importance of the intertemporal hedging portfolio when doing strategic asset allocation using models set in discrete time.
  - ⋆ Wachter (2002) demonstrates how the martingale approach in continuous time can be used to identify the optimal portfolio in complete markets when expected returns are mean reverting.
  - ⋆ Uppal (1993) uses the martingale approach to determine the portfolio weights in a general equilibrium model of an international economy.

- **Stochastic Differential Utility**
  - ⋆ Schroder and Skiadas (1999) show how to use the martingale approach to identify the optimal portfolio when investors have Stochastic Differential Utility (Duffie and Epstein 1992), which is the counterpart to the discrete-time Epstein-Zin-Weil utility (Epstein and Zin 1989; Weil 1990).
  - ⋆ Dumas, Uppal, and Wang (2000) show how the problem of identifying the equilibrium in an economy with N agents who have Stochastic Differential Utility can be reduced to finding the solution to a single partial differential equation (instead of N differential equations).
  - ⋆ Dumas and Uppal (2001) use the martingale approach to identify optimal portfolios in an international setting when investors have Stochastic Differential Utility.

- **Incomplete markets:**
  - ⋆ The static approach for solving the portfolio choice problem in continuous time when markets are not complete was developed in
    - ∗ He and Pearson (1991),
    - ∗ Karatzas, Lehoczky, Shreve, and Xu (1991), and
    - ∗ Cvitanić and Karatzas (1992).
  - ⋆ For a textbook treatment, of portfolio choice in incomplete markets
    - ∗ see Pliska (1997, Chapter 5.8) for the discrete-time case;
    - ∗ see Karatzas and Shreve (1998) for the continuous-time case.
  - ⋆ The analysis of portfolio choice and equilibrium in incomplete markets is one of the key distinguishing features of the book by Dumas and Luciano (2017); see, in particular, Chapters 14 and 15.

- **Solution methods:**
  - ⋆ Usually, to obtain the solution to the portfolio problem, one needs to solve a linear partial differential equation; Pascucci (2011) and Lototsky and Rozovsky (2017) explain how to solve these PDEs.
  - ⋆ Detemple, Garcia, and Rindisbacher (2003) show how one can use Monte Carlo methods instead of PDEs to solve the portfolio problem; Glasserman (2004) is excellent reference on Monte Carlo methods.
- **Review papers:**
  - ⋆ Campbell (2000) – general review of asset pricing and portfolio choice
  - ⋆ Sundaresan (2000) – general review of continuous-time methods
  - ⋆ Brandt (2010) – focus is on portfolio-choice problems
  - ⋆ Wachter (2010) – focus is on asset allocation.
  - ⋆ Back (2010) – focus is on the SDF.

11.9.3 Ph.D. elective courses offered on related topics

- Lionel Martellini, a professor at EDHEC, has taught:
  - ⋆ "*Dynamic Asset Allocation*" in 2009
  - ⋆ "*Dynamic Asset Allocation-Long-term Investing with Short-term Constraints*" in 2012
  - ⋆ "*Dynamic Risk and Asset Allocation Decisions: From Investment Problems to Investment Solutions*" in 2014
- Pietro Veronesi taught the course "*Recent Developments in Portfolio Allocation and Macro-Finance Model*" in 2009.
- Michael Brandt has taught courses on "*Portfolio Choice*" in 2011, 2013.
- Jerome Detemple taught the course on "*Strategic Asset Allocation*" in 2013 and 2016.
- Rama Cont has taught courses on "*Monte Carlo Methods*" in 2010 and 2013.
- Marcel Rindisbacher has taught a course on "*Monte Carlo Methods*" in 2017.

Lecture Notes for 2025

Continuous Time Financial Economics

Chapter 12

Portfolio Choice Using Dynamic Programming

Raman Uppal

EDHEC Business School

Chapter contents

12.1 Plan for the Day . . . 485
12.2 Main Points for Today . . . 486
12.3 Background (repeated) . . . 487
12.3.1 Valuation of Redundant Assets . . . 488
12.3.2 Optimal Consumption and Portfolio Policies . . . 489
12.4 General Description of Consumption & Portfolio Choice in Continuous Time . . . 491
12.4.1 Description of the General Setting . . . 491
12.4.2 Assumptions of the General Model . . . 492
12.4.3 Investor's Dynamic Budget Constraint . . . 495
12.4.4 Investor's Objective . . . 498
12.5 Simple Case: Single Risky Asset and Constant Opportunity Set . . . 499
12.6 Special Case with Many Risky Assets and Constant Opportunity Set . . . 514
12.6.1 First-order condition for consumption . . . 516
12.6.2 First-order condition for portfolio choice . . . 517
12.6.3 Implications of expression for optimal portfolio weights . . . 519
12.7 Example of Portfolio Choice with Stochastic Risk-free Rate and Only One Risky Asset . . . 522
12.7.1 Utility Function . . . 522
12.7.2 Securities Market . . . 523
12.7.3 Wealth Process of the Investors . . . 524
12.7.4 State Variables and Bellman Equation . . . 524
12.7.5 First-Order Condition for Consumption and Portfolio . . . 525
12.7.6 Interpretation of the Optimal Portfolio Weight . . . 526

12.8 General Case with Stochastic Opportunity Set and Multiple Assets . . . 527
12.8.1 First-order condition for consumption . . . 529
12.8.2 First-order condition for portfolio choice . . . 530
12.8.3 Understanding the Intertemporal Hedging Portfolio . . . 532
12.8.4 Solving for optimal consumption and portfolio weights . . . 536
12.9 Summary . . . 537
12.10 Appendix: Special Case with Stochastic Investment Opportunities and Log Utility . . . 540
12.11 Suggestions for Further Readings . . . 543
12.11.1 Source material for the lecture notes . . . 543
12.11.2 Potential applications and extensions of lecture notes . . . 549
12.11.3 Ph.D. elective courses offered on related topics . . . 552

12.1 Plan for the Day

Last Time

- Portfolio selection using the martingale approach

Today

- **How do we find the optimal portfolio when markets are not complete?**
  - ⋆ Optimal portfolio selection using dynamic programming

For Next Time

- Review the material covered in class.
- Read Back, Chapters 9 and 14 and Dumas & Luciano, Chapter 9.
- Do the practice problems.

12.2 Main Points for Today

1. Applying dynamic programming to the consumption-portfolio problem.
2. Interpretation of the dynamic optimal portfolio.

12.3 Background (repeated)

Recall, that in this course we wish to answer three questions, which are nested:

1. How to value redundant securities?

   Least general question: we are given prices of fundamental assets and the desired cashflows; need to determine only the replicating portfolio strategy.

2. How to determine optimal consumption and portfolio policies of investors?

   More general than the previous question: we are given prices of the fundamental assets; need to determine both the desired cashflows and the portfolio strategy generating these cashflows.

3. How to determine the equilibrium values in a market with investors who are optimizing their consumption and portfolio policies?

   Most general question: need to determine prices of fundamental assets, desired cashflows, and the portfolio strategy that replicates these cashflows.

12.3.1 Valuation of Redundant Assets

- For the valuation of redundant assets,
  - ⋆ we are given the cashflows to be valued (for example, the cashflows generated by an option), and
  - ⋆ we are also given the prices of fundamental securities (for example, the stock and the bond), and
  - ⋆ all that we are asked to do is to find the value of the given cashflow in terms of the value of the fundamental securities.
- We have finished studying the valuation of redundant assets whose payoff depends on
  - ⋆ the value of a stock, currency, index, etc.
  - ⋆ the value of a bond or an interest rate.

12.3.2 Optimal Consumption and Portfolio Policies

- Second question: determining optimal consumption and portfolio policies:
  - ⋆ we are given prices of fundamental securities (bond and N stocks), and
  - ⋆ we are asked to find both
    - ∗ the optimal cashflows (the consumption policy) and
    - ∗ the portfolio that replicates this cashflow (the portfolio policy).
  - ⋆ That is, in contrast to the previous problem, we are not given the cashflows to be valued, and so we need to identify both:
    1. the **desired cashflows**, and
    2. also the **trading strategy** that replicates this cashflow.
- So, the additional step is to find the investor's desired pattern of cashflows.
- To determine the desired pattern of cashflows, we need to specify the investor's objective function.

- If markets are complete, this is a straightforward problem to solve.

  In complete markets, the problem can be decomposed into two steps, and each step can be solved separately:
  - ⋆ In the first step, we determine the optimal "consumption" policy.
  - ⋆ In the second step, given that we know the cashflow we wish to generate, we use the approach discussed for replicating/identifying redundant claims.
    (We did this in the last session.)
- If markets are not complete, then:
  - ⋆ one **cannot** decompose the consumption-portfolio problem into two distinct steps; the optimal consumption and portfolio must be identified simultaneously.
  - ⋆ We study today how to do this using the "dynamic programming" approach.
- Equilibrium asset pricing
  - ⋆ Once we have identified the portfolio demanded by investors, we will equate this to the supply of risky securities, which will then give us the asset prices in equilibrium.
    (We will do this in the next class.)

12.4 General Description of Consumption & Portfolio Choice in Continuous Time

12.4.1 Description of the General Setting

- We consider the intertemporal consumption-portfolio problem of an investor.
- The objective of the investor is to maximize a time-separable expected utility function that depends on the rate of consumption at all future dates.
  - ⋆ Model can be extended to allow for non-separable preferences.
- We examine this problem in a setting where asset prices are subject to continuous random changes; that is, the savings of this investor are allocated among assets whose returns follow diffusion processes.
  - ⋆ Model can be extended to allow for discontinuous changes (jumps).
- The investor can adjust consumption and portfolio allocations continuously.
  - ⋆ Model can be extended to relax this assumption.

12.4.2 Assumptions of the General Model

Assumptions: Financial Securities

- The investor can allocate wealth across n risky assets and a risk-free asset.
- Denote the instantaneous risk-free rate by $r(x,t)$.
- Define $\mathcal{P}^{S_i}(t)$ to be the price of the i-th risky asset at date t:

$$\frac{d\mathcal{P}_t^{S_i}}{\mathcal{P}_t^{S_i}} = \mu_i(x,t)\,dt + \sigma_i(x,t)\,dB_{s_i}^{\mathbb{P}}, \quad i = \{1,2,\ldots,n\}, \tag{12.1}$$

  - ⋆ where the correlation between Brownians is $(dB_{s_i}^{\mathbb{P}})(dB_{s_j}^{\mathbb{P}}) \equiv \rho_{s_i s_j} dt$,
  - ⋆ while the covariance between the returns on stock i and stock j is

$$\left(\frac{d\mathcal{P}_t^{S_i}}{\mathcal{P}_t^{S_i}}\right)\left(\frac{d\mathcal{P}_t^{S_j}}{\mathcal{P}_t^{S_j}}\right) = \left[\sigma_i(x,t)\,dB_{s_i}^{\mathbb{P}}\right]\left[\sigma_j(x,t)\,dB_{s_j}^{\mathbb{P}}\right] = \sigma_i(x,t)\,\sigma_j(x,t)\,\rho_{s_i s_j}\,dt$$
$$\equiv \sigma_{s_i s_j}(x,t)\,dt.$$

Assumptions: State Variables Driving Returns

- Note that μ_i, σ_i and r are allowed to be functions of time t and a $k \times 1$ vector of state variables, which we denote by $x(t) = (x_1, x_2, \ldots, x_k)$.

- If μ_i, σ_i or r are varying stochastically then the investor is said to face changing/stochastic investment opportunities;
  that is, the capital market line is not constant.

- We specify that the state variables follow diffusion processes.
  Let the i-th state variable x_i follow the process

$$dx_i = a_i(x,t)\,dt + b_i(x,t)\,dB^{\mathbb{P}}_{x_i}, \quad i = \{1, 2, \ldots, k\}. \tag{12.2}$$

- The correlation between state variables i and j is: $dB^{\mathbb{P}}_{x_i} \times dB^{\mathbb{P}}_{x_j} \equiv \rho_{x_i x_j} dt$.

- The correlation between security i & state-variable j is: $dB^{\mathbb{P}}_{s_i} \times dB^{\mathbb{P}}_{x_j} \equiv \rho_{s_i x_j} dt$.

- So, up to $(n + k)$ Brownian motions affect the distribution of the n stock returns. Therefore, financial markets are incomplete because
  - ⋆ we have $(n + k)$ Brownian motions (sources of risk)
  - ⋆ but only n risky assets (and one risk-free asset).

- Recall that in our discussion of the Martingale approach in the previous chapter, markets were complete because
  - ⋆ we had only n Brownian motions, and
  - ⋆ n risky assets (and one risk-free asset), and
  - ⋆ we had assumed that the k state variables $x(t)$ were driven by the same n Brownian motions that were driving stock returns.

12.4.3 Investor's Dynamic Budget Constraint

- Denote the investor's wealth at date t by W_t.
- Denote the proportion of total wealth allocated to risky asset i at t by w_{it}.
- Denote the investor's consumption rate per unit of time by C_t.
- To keep things simple, assume zero wage income, Y_t (but it could easily be added to the model by replacing C_t by $[C_t - Y_t]$ in the budget equation).

Case with just a single risky asset

- Wealth dynamics are given by the dynamic/intertemporal budget constraint:

$$dW = W\Bigg[\overbrace{w\frac{d\mathcal{P}^S}{\mathcal{P}^S}}^{\text{return from risky assets}} + \overbrace{(1-w)\,r\,dt}^{\text{return from risk-free asset}}\Bigg] - C\,dt \quad \ldots \text{change in wealth} = \text{return on wealth} - \text{cons.}$$

$$= W\left[w\left[\mu\,dt + \sigma\,dB_s^{\mathbb{P}}\right] + (1-w)\,r\,dt\right] - C\,dt \quad \ldots \text{substituting process for } \frac{d\mathcal{P}^S}{\mathcal{P}^S}$$

$$= W\,w\,(\mu - r)\,dt + (r\,W - C)\,dt + W\,w\,\sigma\,dB_s^{\mathbb{P}} \quad \ldots \text{collecting terms multiplied by } w \tag{12.3}$$

$$\frac{dW}{W} = \Bigg[\underbrace{w\,(\mu - r) + r}_{\text{expected return on invested wealth}} - \underbrace{\frac{C}{W}}_{\text{consumption}}\Bigg]dt + \underbrace{w\sigma\,dB_s^{\mathbb{P}}}_{\text{risk of invested wealth}} \quad \ldots \text{collecting } dt \text{ and } dB_s^{\mathbb{P}} \text{ terms} \tag{12.4}$$

Case with n risky assets

- Wealth dynamics are given by the dynamic/intertemporal budget constraint:

$$dW = W\Bigg[\overbrace{\sum_{i=1}^{n} w_i \frac{d\mathcal{P}^{S_i}}{\mathcal{P}^{S_i}}}^{\text{return from risky assets}} + \overbrace{\left(1 - \sum_{i=1}^{n} w_i\right) r\, dt}^{\text{return from risk-free asset}}\Bigg] - C\, dt \quad \text{... change in wealth = return on wealth − cons.}$$

$$= W\left[\sum_{i=1}^{n} w_i \left[\mu_i\, dt + \sigma_i\, dB^{\mathbb{P}}_{s_i}\right] + \left(1 - \sum_{i=1}^{n} w_i\right) r\, dt\right] - C\, dt \quad \text{... substituting process for } \frac{d\mathcal{P}^{S_i}}{\mathcal{P}^{S_i}}$$

$$= W \sum_{i=1}^{n} w_i\,(\mu_i - r)\, dt + (r\,W - C)\, dt + W \sum_{i=1}^{n} w_i \sigma_i\, dB^{\mathbb{P}}_{s_i} \quad \text{... collecting terms multiplied by } \sum_{i=1}^{n} w_i \tag{12.5}$$

$$\frac{dW}{W} = \Bigg[\underbrace{\sum_{i=1}^{n} w_i\,(\mu_i - r) + r}_{\text{expected return on invested wealth}} - \underbrace{\frac{C}{W}}_{\text{consumption}}\Bigg] dt + \underbrace{\sum_{i=1}^{n} w_i \sigma_i\, dB^{\mathbb{P}}_{s_i}}_{\text{risk of invested wealth}} \quad \text{... collecting } dt \text{ and } dB^{\mathbb{P}}_{s_i} \text{ terms} \tag{12.6}$$

12.4.4 Investor's Objective

- The objective of the investor is to
  - ⋆ maximize the expected lifetime utility from consumption and bequest,

$$\max_{C_s, \{w_{is}\}} \mathbb{E}^{\mathbb{P}}_t \left[\int_t^T U(C_s, s) ds + \text{Beq}(W_T, T)\right]; \tag{12.7}$$

  - ⋆ by choosing the optimal consumption and portfolio policy,
  - ⋆ subject to the intertemporal budget constraint in (12.5).
- The functions $U(C_t, t)$ for date $t < T$ and $\text{Beq}(W_T, T)$ for T are assumed to be strictly increasing and concave in C_t and W_T, respectively.
- The state variables at date s are W_s and x_s
- The control variables that the investor chooses are C_s and w_{is}, for $i = \{1, 2, \ldots n\}$ and s over the interval t to T.

12.5 Simple Case: Single Risky Asset and Constant Opportunity Set

- To illustrate the principles of dynamic programming, we first consider a simplified setting where
  - ⋆ there is only a single risky asset,
  - ⋆ the investment opportunity set is constant,
  - ⋆ the agent has time-separable power utility, $U(C,t) = e^{-\delta t}\frac{C^{1-\gamma}}{1-\gamma}$.
- We use this simple setting to understand the principle of dynamic programming.
- Then, for the rest of the class, we will apply this principle to solve more general models (without repeating each individual step of the derivation).

- We use the following steps to solve the consumption-portfolio problem:

1. Write down the objective function and dynamic budget constraint;
2. Identify the state variables, write the indirect utility function, and obtain the Hamilton-Bellman-Jacobi PDE;
3. Differentiate the Hamilton-Bellman-Jacobi equation to find optimal consumption and portfolio in terms of the indirect utility function.
4. Substitute the expressions for optimal consumption and portfolio into the Hamilton-Bellman-Jacobi PDE;
5. Solve the nonlinear PDE characterizing the indirect utility function to get an explicit solution for it (often, one needs to use numerical methods);
6. Use the explicit expression for the indirect utility function to get explicit expressions for optimal consumption and portfolio weights.

Step 1: Write consumption-portfolio problem

- The objective function of the investor for this special case is

$$\max_{C,w} \mathbb{E}_0^{\mathbb{P}} \left[\int_0^{\infty} U(C_s)\, ds \right] = \max_{C,w} \mathbb{E}_0^{\mathbb{P}} \left[\int_0^{\infty} e^{-\delta\, s} \frac{C^{1-\gamma}}{1-\gamma}\, ds \right]. \tag{12.8}$$

- From Equation (12.4) on Page 496 for the special case of a single risky asset, we have the dynamic budget constraint:

$$dW = \Big[W \left[w\ (\mu - r) + r \right] - C \Big] dt + W\, w\, \sigma\, dB^{\mathbb{P}}, \tag{12.9}$$

  where the choice variables w and C are highlighted in red.

Step 2: Obtain the Hamilton-Bellman-Jacobi equation

- The state variables are the quantities we need to keep track of in order to solve our problem.
  - ⋆ For our simple problem, the state variables are t and W_t.
- The indirect utility function is a function of the state variables.
  - ⋆ For partial equilibrium models, we usually denote it by $J(\cdot)$.
  - ⋆ For general equilibrium models, we usually denote it by $V(\cdot)$.
- For our problem, the indirect utility function is

$$J(W_t, t) = \max_{C,w} \mathbb{E}_t^{\mathbb{P}} \left[\int_t^{\infty} U(C_s) ds \right].$$

- Now, we apply the Bellman Principle of Optimality, which says that
  - ⋆ the optimal policy must be such that
    - for a given future realization of the state variable, $W_{t+\Delta t}$ (whose value may be affected by the control policy at date t and earlier),
    - any remaining decisions at date $t + \Delta t$ and later must be optimal with respect to $W_{t+\Delta t}$.
  - ⋆ That is, the optimal policy must be time consistent.

- To apply the Bellman Principle of Optimality, we write the indirect utility function as:

$$
\begin{aligned}
J(W_t, t) &= \max_{C,w} \mathbb{E}_t^{\mathbb{P}} \left[\int_t^{\infty} U(C_s) ds \right] && \text{... definition of indirect utility} \\
&= \max_{C,w} \mathbb{E}_t^{\mathbb{P}} \left[\int_t^{t+\Delta t} U(C_s) ds + \int_{t+\Delta t}^{\infty} U(C_s) ds \right] && \text{... breaking up integral into two par} \\
&= \max_{C,w} \mathbb{E}_t^{\mathbb{P}} \left[\int_t^{t+\Delta t} U(C_s) ds + \max_{C,w} \mathbb{E}_{t+\Delta t}^{\mathbb{P}} \int_{t+\Delta t}^{\infty} U(C_s) ds \right] && \text{... applying Bellman's Principle} \\
&= \max_{C,w} \mathbb{E}_t^{\mathbb{P}} \left[\int_t^{t+\Delta t} U(C_s) ds + J(W_{t+\Delta t}, t + \Delta t) \right]. && \text{... def. of indirect utility at } t + \Delta t
\end{aligned}
\tag{12.10}
$$

- The above equation has a nice recursive structure;
  that is, $J(W_t, t)$ is defined in terms of $J(W_t + \Delta t, t + \Delta t)$.

- We re-write the indirect utility function on the previous page:

$$J(W_t, t) = \max_{C,w} \mathbb{E}_t^{\mathbb{P}} \left[\int_t^{t+\Delta t} U(C_s) ds + J(W_{t+\Delta t}, t + \Delta t) \right]. \quad (12.10)$$

- To simplify this, subtract $J(W_t, t)$ from both sides:

$$0 = \max_{C,w} \mathbb{E}_t^{\mathbb{P}} \left[\int_t^{t+\Delta t} U(C_s) ds + J(W_{t+\Delta t}, t + \Delta t) - J(W_t, t) \right].$$

- Let Δt be very small, that is, dt:

$$\begin{aligned} 0 &= \max_{C,w} \mathbb{E}_t^{\mathbb{P}} \left[U(C_t) dt + J(W_{t+dt}, t + dt) - J(W_t, t) \right] \\ &= \max_{C,w} \mathbb{E}_t^{\mathbb{P}} \left[U(C_t) dt + dJ(W_t, t) \right]. \end{aligned}$$

- Start with the last expression on the previous page:

$$0 = \max_{C,w} \mathbb{E}_t^{\mathbb{P}} \left[U(C_t) dt + dJ(W_t, t) \right].$$

- Dividing both sides by dt, and noting that $U(C_t)$ is non-random, gives:

$$0 = \max_{C,w} \left[U(C_t) + \frac{1}{dt} \mathbb{E}_t^{\mathbb{P}} dJ(W_t, t) \right] \quad \ldots \text{we will use this result repeatedly} \quad (12.11)$$

- We start by re-writing the last equation on the previous slide:

$$0 = \max_{C,w} \left[U(C_t) + \frac{1}{dt} \mathbb{E}_t^{\mathbb{P}} \, dJ(W_t, t) \right] \tag{12.11}$$

- We now use Itô's formula to simplify the second term:

$$\mathbb{E}_t^{\mathbb{P}} \, dJ(W,t) = \mathbb{E}_t^{\mathbb{P}} \left[J_t \, dt + J_W (dW) + \frac{1}{2} J_{WW} (dW)^2 \right]. \tag{12.12}$$

- Substituting the processes for wealth in Equation (12.4), we get:

$$\frac{1}{dt} \mathbb{E}_t^{\mathbb{P}} \, dJ(W,t) = J_t + J_W \Big[W \, w \, (\mu - r) + (r \, W - C) \Big] + \frac{1}{2} J_{WW} \big(W^2 \, w^2 \, \sigma^2 \big) \tag{12.13}$$

- Substituting the expression for $(1/dt)\mathbb{E}_t^{\mathbb{P}} \, dJ(W,t)$ derived in (12.13) into (12.11) gives us the Hamilton-Bellman-Jacobi equation:

$$0 = \max_{C,w} \left[e^{-\delta t} \frac{C^{1-\gamma}}{1-\gamma} + J_t + J_W \Big[W \, w \, (\mu - r) + (r \, W - C) \Big] + \frac{1}{2} J_{WW} \big(W^2 \, w^2 \, \sigma^2 \big) \right]. \tag{12.14}$$

Step 3: Obtain expressions for optimal consumption and portfolio in terms of $J(W,t)$ and its derivatives

- We start by re-writing the last equation on the previous page:

$$0 = \max_{C,w} \left[e^{-\delta t} \frac{C^{1-\gamma}}{1-\gamma} + J_t + J_W \Big[W \, w \, (\mu - r) + (r \, W - C) \Big] + \frac{1}{2} J_{WW} \left(W^2 \, w^2 \, \sigma^2 \right) \right]. \tag{12.14}$$

- Differentiating the above expression with respect to C, we get:

$$e^{-\delta t} C^{-\gamma} = J_W, \quad \text{which implies that} \tag{12.15}$$

$$C = \left(e^{\delta t} \, J_W \right)^{\frac{-1}{\gamma}} \tag{12.16}$$

$$C^{1-\gamma} = \left(e^{\delta t} \, J_W \right)^{\frac{\gamma-1}{\gamma}}$$

$$\frac{1}{1-\gamma} C^{1-\gamma} = \frac{1}{1-\gamma} \left(e^{\delta t} \, J_W \right)^{\frac{\gamma-1}{\gamma}}. \tag{12.17}$$

- Differentiating (12.14) with respect to w, and simplifying, we get:

$$w = \left(\frac{-J_W}{W \, J_{WW}} \right) \frac{\mu - r}{\sigma^2}. \tag{12.18}$$

Step 4: Substitute expressions for optimal consumption and portfolio into Hamilton-Bellman-Jacobi equation

- We now need to substitute (12.16), (12.17), and (12.18) into (12.14):
- Substituting first only the expression for optimal w in (12.18), we get:

$$0 = \left[e^{-\delta t} \frac{C^{1-\gamma}}{1-\gamma} + J_t + J_W \Big(r\, W - C \Big) - \frac{1}{2} \frac{J_W^2}{J_{WW}} \left(\frac{\mu - r}{\sigma} \right)^2 \right]. \tag{12.19}$$

- Now, substitute for C from (12.16) and for $\frac{C^{1-\gamma}}{1-\gamma}$ from(12.17):

$$0 = \left[e^{-\delta t} \frac{1}{1-\gamma} \left(e^{\delta t}\, J_W \right)^{\frac{\gamma-1}{\gamma}} + J_t + J_W \Big(r\, W - \left(e^{\delta t}\, J_W \right)^{\frac{-1}{\gamma}} \Big) - \frac{1}{2} \frac{J_W^2}{J_{WW}} \left(\frac{\mu - r}{\sigma} \right)^2 \right].$$

- Collect terms to simplify the above:

$$0 = \left[e^{-\delta t} \left(\frac{1}{1-\gamma} - 1 \right) \left(e^{\delta t}\, J_W \right)^{\frac{\gamma-1}{\gamma}} + J_t + J_W \big(r\, W \big) - \frac{1}{2} \frac{J_W^2}{J_{WW}} \left(\frac{\mu - r}{\sigma} \right)^2 \right]. \tag{12.20}$$

- Observe that the above is a nonlinear partial differential equation for $J(W,t)$.

Step 5: Solve the partial differential equation for $J(W,t)$

- In general, it is difficult to solve nonlinear PDEs in closed-form, and we must resort to numerical methods (such as finite difference).
- In the case of our problem, we can guess a solution, and then verify that it satisfies the differential equation.
- Our guess is that the indirect utility function $J(W,t)$ inherits the functional form of the utility function, $u(c,t) = e^{-\delta t} \frac{c^{1-\gamma}}{1-\gamma}$:

$$J(W,t) = A\, e^{-\delta t} \frac{W^{1-\gamma}}{1-\gamma}, \tag{12.21}$$

where A is a constant whose value is to be determined.

- We now wish to confirm that our guess on the previous page

$$J(W,t) = A\, e^{-\delta\, t} \frac{W^{1-\gamma}}{1-\gamma}, \tag{12.21}$$

  does indeed satisfy the partial differential equation,

- To do this, we substitute the above guess into the PDE below,

$$0 = \left[e^{-\delta\, t} \left(\frac{1}{1-\gamma} - 1 \right) \left(e^{\delta\, t}\, J_W \right)^{\frac{\gamma-1}{\gamma}} + J_t + J_W (r\, W) - \frac{1}{2} \frac{J_W^2}{J_{WW}} \left(\frac{\mu - r}{\sigma} \right)^2 \right]. \tag{12.20}$$

  and then simplify the resulting equation.

- When we substitute (12.21) into (12.20), we find that the term $e^{-\delta\, t} W^{1-\gamma}$ cancels out, leaving us with the following result:

$$A = a^{-\gamma}, \quad \text{where} \tag{12.22}$$

$$a = \frac{1}{\gamma} \left[\delta - (1-\gamma) \left[r + \frac{1}{2} \frac{1}{\gamma} \left(\frac{\mu - r}{\sigma} \right)^2 \right] \right]. \tag{12.23}$$

Step 6: Use explicit expression for $J(W,t)$ to get explicit expressions for optimal consumption and portfolio

- Substituting the solution for J

$$J(W,t) = a^{-\gamma}\, e^{-\delta\, t} \frac{W^{1-\gamma}}{1-\gamma}, \tag{12.24}$$

  into the first-order condition for C in (12.16) on Page 508 gives:

$$C = \left(e^{\delta\, t}\, J_W \right)^{\frac{-1}{\gamma}} \tag{12.16}$$

$$= a\, W \quad \text{. . . consume constant proportion "}a\text{" of wealth.} \tag{12.25}$$

- Substituting the solution for J in (12.24) into the first-order condition for w

$$w = \left(\frac{-J_W}{W\, J_{WW}} \right) \frac{\mu - r}{\sigma^2} \tag{12.18}$$

$$= \frac{1}{\gamma} \frac{\mu - r}{\sigma^2} \quad \text{. . . invest constant proportion of wealth in the risky asset.} \tag{12.26}$$

- We see that the proportion of wealth in risky assets,

$$\% \text{ of wealth in risky assets} = w = \frac{1}{\gamma}\left(\frac{\mu - r}{\sigma^2}\right),$$

  - ⋆ increases with expected excess return $(\mu - r)$,
  - ⋆ increases with relative risk tolerance $1/\gamma$,
  - ⋆ decreases with risk σ^2, and
  - ⋆ is independent of time, the reason being that there is no wage income, and also the investment opportunity set is constant.

12.6 Special Case with Many Risky Assets and Constant Opportunity Set

- We now use the methodology described above to solve the problem of choosing the optimal consumption and portfolio policies.
- We assume, for now, that asset prices have a lognormal distribution; that is, μ_i, σ_i, r, and σ_{ij} are constants.
- This means that the expected rate of return and the variance of the rate of return do not change, implying a constant investment opportunity set.
- This means that the state variable x does not affect the choices of investors.
- Define the indirect utility-of-wealth function as:

$$J(W,t) = \max_{C_s,\{w_{is}\}} \mathbb{E}_t^{\mathbb{P}}\left[\int_t^T U(C_s,s)ds + \text{Beq}(W_T,T)\right]. \tag{12.27}$$

- Then, recall the result derived in (12.11) on Page 506:

$$0 = \max_{C_s, \{w_{is}\}} \left[U(C_t, t) + \frac{1}{dt} \mathbb{E}_t^{\mathbb{P}} \, dJ(W, t) \right]. \tag{12.28}$$

- Using Ito's formula we get:

$$\mathbb{E}_t^{\mathbb{P}} \, dJ(W, t) \; = \; \mathbb{E}_t^{\mathbb{P}} \left[J_t \, dt + J_W (dW) + \frac{1}{2} J_{WW} (dW)^2 \right].$$

- After substituting the processes for wealth in equation (12.5), we get:

$$\frac{1}{dt} \mathbb{E}_t^{\mathbb{P}} \, dJ(W, t) = J_t + J_W \Big(W \sum_{i=1}^{n} w_i \big(\mu_i - r \big) + \big(r \, W - C \big) \Big) \tag{12.29}$$

$$+ \frac{1}{2} J_{WW} \Big(W^2 \sum_{i=1}^{n} \sum_{j=1}^{n} w_i \, w_j \, \sigma_{ij} \Big) \qquad \ldots \text{ recall that } \mathbb{E}^{\mathbb{P}}[dB_{s_i}^{\mathbb{P}}] = 0.$$

12.6.1 First-order condition for consumption

- Substituting the expression for $E_t \, dJ(W, t)$ derived in (12.29) into (12.28):

$$0 = \max_{C_s, \{w_{is}\}} \Big[U(C_t, t) + J_t + J_W \Big(W \sum_{i=1}^{n} w_i \, (\mu_i - r) + (r \, W - C) \Big) + \frac{1}{2} J_{WW} \Big(W^2 \sum_{i=1}^{n} \sum_{j=1}^{n} w_i \, w_j \, \sigma_{ij} \Big) \Big]. \tag{12.30}$$

- Differentiating (12.30) with respect to consumption gives:

$$0 = U_c - J_W \qquad \text{or equivalently} \qquad U_C = J_W. \tag{12.31}$$

- This equation says that the investor consumes up to the point that
  - ⋆ the marginal utility from consumption is equal to
  - ⋆ the marginal (indirect) utility from saving.
- It is the exact analog of the equation we had in the two-date setting.

12.6.2 First-order condition for portfolio choice

- We re-write Equation (12.30):

$$0 = \max_{C_s,\{w_{is}\}} \Big[U(C_{t,t}) + J_t + J_W \Big(W \sum_{i=1}^{n} w_i \, (\mu_i - r) + (r\,W - C) \Big) \tag{12.30}$$

$$+ \frac{1}{2} J_{WW} \Big(W^2 \sum_{i=1}^{n} \sum_{j=1}^{n} w_i \, w_j \, \sigma_{ij} \Big) \Big]. \tag{12.32}$$

- The first-order condition for the optimal portfolio weight w_i is obtained by differentiating Equation (12.30) with respect to w_i:

$$0 \;=\; W \, J_W (\mu_i - r) + W^2 \, J_{WW} \sum_{j=1}^{n} w_j \, \sigma_{ij}. \tag{12.33}$$

- Note that $\sum_{j=1}^{n} w_j$ includes w_i.

- We re-write the first-order condition for the optimal portfolio weight:

$$0 = W \, J_W (\mu_i - r) + W^2 \, J_{WW} \sum_{j=1}^{n} w_j \, \sigma_{ij}. \tag{12.33}$$

- There will be n such linear equations, one for each risky asset, w_i, which need to be solved simultaneously.
- Denote by $\mathbf{V} \equiv [\sigma_{ij}]$ the $n \times n$ instantaneous covariance matrix whose i,j-th element is σ_{ij}; and v_{ij} is the i,j-the element of $\mathbf{V}^{-1} \equiv [v_{ij}]$.
- Then, (12.33) can be written as:

$$w_i = \underbrace{\left(\frac{-J_W}{W \, J_{WW}} \right)}_{\text{1/[relative risk aversion]}} \underbrace{\sum_{j=1}^{n} v_{ij} (\mu_j - r)}_{\text{mean-var portfolio}} \qquad \text{or} \qquad w = \frac{1}{\gamma} \underbrace{\mathbf{V}^{-1} (\boldsymbol{\mu} - r\mathbf{1})}_{\text{using vector notation}}. \tag{12.34}$$

- This is exactly the same equation as that in the Markowitz model.

12.6.3 Implications of expression for optimal portfolio weights

- There are several interesting implications of the expression for optimal portfolio weights in (12.34), re-written below:

$$w_i = \underbrace{\left(\frac{-J_W}{W\,J_{WW}}\right)}_{\text{1/[relative risk aversion]}} \underbrace{\sum_{j=1}^{n} v_{ij}(\mu_j - r)}_{\text{mean-var portfolio}}. \tag{12.34}$$

- All investors will hold risky assets in the same proportion:

$$\frac{w_i}{w_\ell} = \frac{\left(\frac{-J_W}{W\,J_{WW}}\right)\sum_{j=1}^{n} v_{ij}(\mu_j - r)}{\left(\frac{-J_W}{W\,J_{WW}}\right)\sum_{j=1}^{n} v_{\ell j}(\mu_j - r)} = \frac{\sum_{j=1}^{n} v_{ij}(\mu_j - r)}{\sum_{j=1}^{n} v_{\ell j}(\mu_j - r)}. \tag{12.35}$$

- All investors will hold the same proportion of risky asset i to all risky assets:

$$w_i^{\text{tangency}} \equiv \frac{w_i}{\sum_{\ell=1}^{n} w_\ell} = \frac{\sum_{j=1}^{n} v_{ij}(\mu_j - r)}{\sum_{\ell=1}^{n}\sum_{j=1}^{n} v_{\ell j}(\mu_j - r)}. \tag{12.36}$$

- Thus, all investors, no matter what their utility functions, invest in only
  1. the risk-free asset, and
  2. the same portfolio of only-risky assets.

- Thus, even in the continuous-time setting, we get two-fund separation as long as the investment opportunity set is constant.

- Of the two funds, one is the risk-free asset, which pays the rate r

- The second fund has a return of μ^{tangency} and variance of $(\sigma^{\text{tangency}})^2$, where

$$\mu^{\text{tangency}} = \sum_{i=1}^{n} w_i^{\text{tangency}}\,\mu_i, \tag{12.37}$$

$$(\sigma^{\text{tangency}})^2 = \sum_{i=1}^{n}\sum_{j=1}^{n} w_i^{\text{tangency}}\,w_j^{\text{tangency}}\,\sigma_{ij}. \tag{12.38}$$

- So, the result on portfolio selection in the static model and the continuous-time dynamic model is the same.
- The only difference is that
  - ⋆ in the static model, we assumed returns have a Normal distribution;
  - ⋆ in continuous-time, we assume returns have a Lognormal distribution.
- The intuition why we get the single-period Markowitz results in a continuous-time setting with Lognormally distributed asset returns is that
  - ⋆ with continuous trading (rebalancing) an investor's horizon is very short; that is, instantaneous;
  - ⋆ diffusion processes can be thought of as being instantaneously (locally) Normally distributed.

12.7 Example of Portfolio Choice with Stochastic Risk-free Rate and Only One Risky Asset

12.7.1 Utility Function

- An investor wishes to maximize the expected lifetime utility of the form

$$\mathbb{E}_0^{\mathbb{P}}\left[\int_0^T e^{-\delta t}u(C_t)\,dt\right],$$

where

- ⋆ δ is the subjective rate of time preference and
- ⋆ $C(t)$ is the consumption at t.

12.7.2 Securities Market

- Assume that the investor can trade in a market with two securities.
- The first security is a stock, with price process

$$\frac{d\mathcal{P}^S(t)}{\mathcal{P}^S(t)} = \mu_S\, dt + \sigma_S\, dB_S^{\mathbb{P}}(t), \quad \text{where } \sigma_S \text{ and } \mu_S \text{ are constants.}$$

- The second security is a bond, with price process

$$\frac{d\mathcal{P}^D(t)}{\mathcal{P}^D(t)} = r(t)\, dt,$$

with $r(t)$ being stochastic and following the Vasicek (1977) process:

$$dr(t) = a\Big(b - r(t)\Big)dt + \sigma_r\, dB_r^{\mathbb{P}}(t),$$

where a, b, and σ_r are constants, and $dB_S^{\mathbb{P}}(t) \times dB_r^{\mathbb{P}}(t) = \rho_{Sr}$.

12.7.3 Wealth Process of the Investors

- The dynamic budget constraint is:

$$dW = W\Big[w\,\mu_S + (1-w)r(t)\Big]dt - C\,dt + W\,w\,\sigma_S dB_S^{\mathbb{P}}.$$

12.7.4 State Variables and Bellman Equation

- The state variables for this problem are t, $W(t)$, and $r(t)$.
- Thus, the value function is: $J\Big(t, W(t), r(t)\Big)$.
- Therefore, the Bellman equation is:

$$\begin{aligned} 0 = \max_{\{C,w\}} \Big\{ & e^{-\delta\, t}\, u(C) + J_t \\ & + J_W\, W \left(w\,\mu_S + (1-w)r(t) - \frac{C}{W} \right) + \frac{1}{2} J_{WW}\, W^2\, w^2\, \sigma_S^2 \\ & + J_r\, a\,(b - r(t)) + \frac{1}{2} J_{rr}\, \sigma_r^2 + J_{Wr}\, W\, w\, \sigma_r\, \sigma_S\, \rho_{Sr} \Big\}. \end{aligned}$$

12.7.5 First-Order Condition for Consumption and Portfolio

- We start by re-writing the Bellman equation:

$$0 = \max_{\{C,w\}} \Big\{ e^{-\delta t}\, u(C) + J_t + J_W\, W \left(w\, \mu_S + (1-w) r(t) - \frac{C}{W} \right) + \frac{1}{2} J_{WW}\, W^2\, w^2\, \sigma_S^2 + J_r\, a\, (b - r(t)) + \frac{1}{2} J_{rr}\, \sigma_r^2 + J_{Wr}\, W\, w\, \sigma_r\, \sigma_S\, \rho_{Sr} \Big\}.$$

- Differentiating the Bellman equation with respect to consumption, we get:

$$0 = e^{-\delta t}\, u'(C) - J_W.$$

- Differentiating the Bellman equation above with respect to the proportion of wealth invested in the risky asset, we get:

$$0 = J_W\, W \Big(\mu_S - r(t) \Big) + J_{WW}\, W^2\, w\, \sigma_S^2 + J_{Wr}\, W\, \sigma_r\, \sigma_S\, \rho_{Sr}.$$

12.7.6 Interpretation of the Optimal Portfolio Weight

- We start by re-writing the first-order condition with respect to w:

$$0 = J_W\, W \Big(\mu_S - r(t) \Big) + J_{WW}\, W^2\, w\, \sigma_S^2 + J_{Wr}\, W\, \sigma_r\, \sigma_S\, \rho_{Sr}.$$

- Expressing the above expression with w on the left-hand side, we get:

$$w(t) = -\frac{J_W}{W\, J_{WW}} \frac{\Big(\mu_S - r(t) \Big)}{\sigma_S^2} - \frac{J_{Wr}}{W\, J_{WW}} \frac{\sigma_r\, \sigma_S\, \rho_{Sr}}{\sigma_S^2}, \quad \text{where}$$

  - ⋆ the first term is the (conditional) "myopic" mean-variance portfolio, and
  - ⋆ the second term is the "intertemporal hedging portfolio."

12.8 General Case with Stochastic Opportunity Set and Multiple Assets

- We now study the consumption-portfolio problem when there are $n > 1$ risky assets and investment opportunity set is not constant.

- Define the indirect utility-of-wealth function as

$$J(W, x, t) = \max_{C_s, \{w_{is}\}} \mathbb{E}_t^{\mathbb{P}} \left[\int_t^T U(C_s, s) ds + \text{Beq}(W_T, T) \right]. \tag{12.39}$$

- Then, using the result we have derived in (12.11) on Page 506, we get:

$$0 = \max_{C_s, \{w_{is}\}} \left[U(C_t, t) + \frac{1}{dt} \mathbb{E}_t^{\mathbb{P}} \, dJ(W, x, t) \right], \tag{12.40}$$

where the expression for $\frac{1}{dt} \mathbb{E}_t^{\mathbb{P}} \, dJ(W, x, t)$ is given on the next page.

- Using Ito's formula and equations (12.5) and (12.2) that describe the processes for wealth and the state variables, and assuming that $k = 1$ so there is only a single state variable x, we have that:

$$\begin{aligned} &\mathbb{E}_t^{\mathbb{P}} \, dJ(W, x, t) \\ &= \mathbb{E}_t^{\mathbb{P}} \left[J_t \, dt + J_W (dW) + J_x (dx) + \frac{1}{2} J_{WW} (dW)^2 + \frac{1}{2} J_{xx} (dx)^2 + J_{Wx} (dW)(dx) \right]. \end{aligned}$$

- After substituting the processes for wealth and the state variable, and keeping in mind that the expected value of the change in a Brownian is zero:

$$\begin{aligned} \frac{1}{dt} \mathbb{E}_t^{\mathbb{P}} \, dJ(W, x, t) \;=\; & J_t + J_W \Big(W \sum_{i=1}^{n} w_i (\mu_i - r) + (r\, W - C) \Big) + J_x \big(a(x, t) \big) \\ & + \frac{1}{2} J_{WW} \Big(W^2 \sum_{i=1}^{n} \sum_{j=1}^{n} w_i \, w_j \, \sigma_{ij} \Big) + \frac{1}{2} J_{xx} \big(b(x, t) \big)^2 + J_{Wx} \Big(W \sum_{i=1}^{n} w_i \, \sigma_{ix} \Big). \end{aligned} \tag{12.41}$$

where σ_{ij} is the covariance between the returns on stocks i and j,
and σ_{ix} is the covariance between the returns on stock i and state variable x.

12.8.1 First-order condition for consumption

- The first-order condition is obtained by differentiating (12.40) with respect to consumption (and remembering to account for (12.41))

$$0 = U_c - J_W \qquad \text{or equivalently} \qquad U_C = J_W. \tag{12.42}$$

- This equation is the same as Equation (12.31) on Page 516 for the case of a constant investment opportunity set.

- This equation says that the investor consumes up to the point that
  - ⋆ the marginal utility from consumption is equal to
  - ⋆ the marginal (indirect) utility from saving, that is, from having an additional unit of wealth.

- It is the exact analog of the equation we had in the two-date setting.

12.8.2 First-order condition for portfolio choice

- The first-order condition for portfolio weight w_i is obtained by differentiating (12.40) with respect to w_i (and remembering to account for (12.41)):

$$0 = W\, J_W(\mu_i - r) + W^2\, J_{WW} \sum_{j=1}^{n} w_j\, \sigma_{ij} + \underbrace{W\, J_{Wx}\, \sigma_{ix}}_{\text{new term}}. \tag{12.43}$$

- There will be n such linear equations, one for each risky asset.

- These n equations need to be solved simultaneously.

- We start by re-writing Equation (12.43), the first-order condition for w_i:

$$0 = W\, J_W(\mu_i - r) + W^2\, J_{WW} \sum_{j=1}^{n} w_j\, \sigma_{ij} + \underbrace{W\, J_{Wx}\, \sigma_{ix}}_{\text{new term}}. \qquad (12.43)$$

- Denote by $\mathbf{V} \equiv [\sigma_{ij}]$ the $n \times n$ instantaneous covariance matrix whose i,j-th element is σ_{ij}; and v_{ij} is the i,j-the element of $\mathbf{V}^{-1} \equiv [v_{ij}]$.

- Then, (12.43) can be written as:

$$w_i = \overbrace{\left(\frac{-J_W}{W\, J_{WW}}\right)}^{1/[\text{relative risk aversion}]} \overbrace{\sum_{j=1}^{n} v_{ij}(\mu_i - r)}^{\text{mean-var portfolio}} - \overbrace{\left(\frac{J_{Wx}}{W\, J_{WW}}\right) \sum_{j=1}^{n} v_{ij}\, \overbrace{\sigma_{jx}}^{\substack{\text{covariance of} \\ x \text{ with asset } j}}}^{\substack{\text{intertemporal hedging portfolio} \\ \text{Effect of } x \text{ on } J_W}}. \qquad (12.44)$$

12.8.3 Understanding the Intertemporal Hedging Portfolio

$$w_i = \overbrace{\left(\frac{-J_W}{W\, J_{WW}}\right)}^{1/[\text{relative risk aversion}]} \overbrace{\sum_{j=1}^{n} v_{ij}(\mu_i - r)}^{\text{mean-var portfolio}} - \overbrace{\left(\frac{J_{Wx}}{W\, J_{WW}}\right) \sum_{j=1}^{n} v_{ij}\, \overbrace{\sigma_{jx}}^{\substack{\text{covariance of} \\ x \text{ with asset } j}}}^{\substack{\text{intertemporal hedging portfolio} \\ \text{Effect of } x \text{ on } J_W}}.$$

- Observe that, in general, $-J_W/(J_{WW})$ and $-J_{Wx}/(J_{WW})$ will differ across investors.

- Thus, in contrast to the case with constant-investment-opportunities (where $J_{Wx} = 0$), the ratio of risky weights w_i/w_j will not be the same across investors.

- Thus, we will not get two-fund separation.

- However, if there is only one state variable, then a three-fund separation result will hold; that is, investors will be satisfied investing in three funds:
  - ⋆ a fund that has the risk-free asset;
  - ⋆ a second fund that is the optimal mean-variance myopic portfolio with weights, $\sum_{j=1}^{n} v_{ij}(\mu_j - r)$; this is the "tactical portfolio".
  - ⋆ a third fund that consists of risky assets that have the maximum absolute covariance with the state variable x; that is, $\sum_{j=1}^{n} v_{ij}\sigma_{jx}$; this is the strategic portfolio.
- While the composition of the three funds is described above,
  - ⋆ the magnitude of the holdings of the third fund is $-J_{Wx}/J_{WW}$;
  - ⋆ the magnitude of the holdings of second fund is $-J_W/J_{WW}$;
  - ⋆ with the remaining proportion of wealth invested in the first fund, which consists of the risk-free asset.

- Differentiating the first-order condition for consumption, $J_W = U_C$, with respect to W gives: $J_{WW} = U_{CC} \times (\partial C/\partial W)$.
- Using this result, we can write:

$$-\frac{J_W}{J_{WW}} = -\frac{U_C}{U_{CC}(\partial C/\partial W)} > 0 \quad \text{... because of concavity of } U(C). \tag{12.45}$$

- Similarly, $J_{Wx} = U_{CC} \times (\partial C/\partial x)$

$$-\frac{J_{Wx}}{J_{WW}} = -\frac{U_{CC}(\partial C/\partial x)}{U_{CC}(\partial C/\partial W)} = -\frac{\partial C/\partial x}{\partial C/\partial W} \lessgtr 0. \tag{12.46}$$

- Aim of "intertemporal hedging component" of the portfolio is to hedge against unfavorable shifts in investment opportunities.
- An unfavorable shift is defined as a change in x such that consumption falls for a given level of current wealth; that is,
  - ⋆ if $(\partial C/\partial x) < 0$ then an increase in x is unfavorable;
  - ⋆ if $(\partial C/\partial x) > 0$, then it is a decrease in x that is unfavorable.

- For example, suppose that $\mathbf{V}$ is a diagonal matrix, so that $v_{ii} = 1/\sigma_{ii}$ and $v_{ij} = 0$ for $i \neq j$.

- Then, the intertemporal-hedging component of the portfolio is:

$$-\frac{J_{Wx}}{J_{WW}} = -\frac{\partial C/\partial x}{\partial C/\partial W}\, v_{ii}\, \sigma_{ix} > 0 \text{ if and only if } \left(\frac{\partial C}{\partial x}\right) \sigma_{ix} < 0.$$

- The above expression says that
  - ⋆ if $(\partial C/\partial x) < 0$ and if $\sigma_{ix} > 0$,
  - ⋆ then there is a positive hedging demand for Asset i, and
  - ⋆ hence this asset will be held in greater amounts than suggested by the single-period mean-variance model.

12.8.4 Solving for optimal consumption and portfolio weights

- Observe that optimal consumption in Equation (12.42) and the portfolio weights in (12.44) are expressed in terms of J and its derivatives.

- Thus, to solve for the optimal consumption and portfolio policies, we need to find J.

- The solution for $J(W, x, t)$ can be obtained by substituting the expressions in (12.42) and (12.44) for optimal consumption and portfolio weights into (12.40) and solving the resulting nonlinear partial differential equation.

- For some special cases, one can solve this partial differential equation analytically; but, in general, one needs to solve this using numerical methods.

- In the appendix, we look at one such special case, where the investor's utility is given by a log function.

12.9 Summary

1. The objective of the investor is to

- to maximize the expected lifetime utility from consumption and bequest,

$$\max_{C_s,\{w_{is}\}} \mathbb{E}_t^{\mathbb{P}}\left[\int_t^T U(C_s,s)ds + \text{Beq}(W_T,T)\right]; \tag{12.47}$$

- by choosing the optimal consumption and portfolio policy,
- subject to the intertemporal budget constraint:

$$dW = W\sum_{i=1}^n w_i\,(\mu_i - r)\,dt + (r\,W - C)\,dt + W\sum_{i=1}^n w_i\sigma_i\,dB_{s_i}^{\mathbb{P}}.$$

- The first-order condition for consumption is

$$0 = U_C - J_W \qquad \text{or equivalently} \qquad U_C = J_W.$$

which says that the investor consumes up to the point that the marginal utility from consumption is equal to the marginal (indirect) utility from saving, that is, from having an additional unit of wealth.

- The first-order condition for portfolio weight w_i is

$$w_i = \overbrace{\left(\frac{-J_W}{W\,J_{WW}}\right)}^{1/[\text{relative risk aversion}]} \overbrace{\sum_{j=1}^n v_{ij}(\mu_j - r)}^{\text{mean-var portfolio}} - \overbrace{\left(\frac{\overbrace{J_{Wx}}^{\text{Effect of } x \text{ on } J_W}}{W\,J_{WW}}\right)\sum_{j=1}^n v_{ij}\,\overbrace{\sigma_{jx}}^{\substack{\text{covariance of}\\ x \text{ with asset } j}}}^{\text{intertemporal hedging portfolio}}.$$

2. Special cases with zero intertemporal hedging:

- If the investment opportunity set is constant, then
  - ⋆ the investor holds only the myopic portfolio,
  - ⋆ because there is nothing to hedge.

- If the investor has an elasticity of intertemporal substitution equal to 1 (for example, if investor has log utility), then
  - ⋆ the investor holds only the myopic portfolio,
  - ⋆ because the substitution effect cancels out the income effect, leading to a net zero demand for hedging.

12.10 Appendix: Special Case with Stochastic Investment Opportunities and Log Utility

- We now consider the case where the investor's utility function is logarithmic:

$$U(C_t, t) = e^{-\delta t} \log C_t, \quad \text{and} \quad \text{Beq}(W_T, T) = e^{-\delta T} \log W_T.$$

- The log utility function is a special case of the $C^{1-\gamma}/(1-\gamma)$, when $\gamma = 1$.

- As we saw before in the two-date problem, $\gamma = 1$ implies that the substitution and income effects are equal and opposite, and so they net out.

- That is, the elasticity of intertemporal substitution (EIS) equals one.

- Because the income and substitution effects cancel out, the investor with logarithmic preferences behaves **myopically**; that is, the investor optimizes only one period ahead, ignoring what might happen beyond the immediate next date (instant).

- For this utility function, one can solve the consumption-portfolio choice problem analytically:

$$J(W,x,t) = \alpha(t)e^{-\delta t}\log W_t + F(x,t) \quad \ldots \text{where } F(x,t) \text{ is some function to be identified}$$

$$C_t = \frac{1}{\alpha(t)}W_t$$

$$w_{it} = \sum_{j=1}^{n} v_{ij,t}(\mu_{j,t} - r_t) \quad \ldots \text{Note: no intertemporal hedging term!}$$

$$\alpha(t) = \frac{1}{\delta}\left[1 - (1-\delta)e^{-\delta(T-t)}\right]$$

- In the limit, as $T \to \infty$: $\alpha(t) = \frac{1}{\delta}$.

- There are a number of things to note about this solution.
- In the solution to the value function

$$J(W,x,t) = \alpha(t)e^{-\delta t}\log W_t + F(x,t)$$

  - ⋆ Note that the state variable x enters only additively through $F(x,t)$.
  - ⋆ As a result, x has no effect on consumption and portfolio choices (in particular, $J_{Wx} = 0$).
  - ⋆ The reason is, as we saw in the two-date problem, that the substitution and income effects are equal and opposite for $\gamma = 1$, and so net out.
- Consumption is a proportion of wealth; in the limit, as $T \to \infty$, consumption is a constant proportion δ of wealth.
- The optimal portfolio is the conditional mean-variance portfolio; it changes as moments of asset returns change, but, it does **not** have an intertemporal hedging part (because EIS $= 1$ and so the hedging demand is zero).

12.11 Suggestions for Further Readings

12.11.1 Source material for the lecture notes

- Portfolio choice using the dynamic programming approach is discussed in Back (2017, Chapter 14) and Dumas and Luciano (2017, Chapter 9).

- The consumption and portfolio choice problem in continuous time was first studied in two papers by Merton (1969, 1971) that form the foundation of the material for this chapter. The careful step-by-step derivation of the dynamic budget equation is given in Merton (1990, Chapter 3).

- This work then provided the foundation for the Intertemporal CAPM (Merton 1973a), which extended the single-period CAPM to a continuous-time setting, and we will study this in the next chapter.

Solving the Consumption-Portfolio Problem

- In the previous chapter, we studied the martingale approach to the optimal consumption and portfolio problem in complete markets, developed by Karatzas, Lehoczky, and Shreve (1987) and Cox and Huang (1989), with the solution characterized by a linear PDE.

- The dynamic programming approach adopted in Merton (1969, 1971) leads to a solution characterized by a nonlinear PDE but does not require markets to be complete.

- When Merton (1990) published his book that consists mostly of his published papers, he added a chapter (Chapter 6) to show how the dynamic programming approach he had proposed is equivalent to the Cox-Huang martingale approach if markets are complete.

- When the first-order conditions for optimal consumption and portfolio are substituted into the Hamilton-Jacobi-Bellman equation, one obtains a nonlinear partial differential equation, which is most inconvenient.

- Sometimes, nonlinear differential equations can be simplified by a change of dependent variable (change of unknown function).
  - ⋆ This can be done when the dependent variable (here J) is part of a nonlinear term whereas the independent variables (here W, x, t) appear in a linear way.
  - ⋆ Engineering mathematicians call this trick the "hodograph transformation" because it was first used when solving a problem in mechanics bearing that name.

- Consider the differential of the Hamilton-Jacobi-Bellman equation with respect to W and operate on it a change of unknown function, known in Finance as the "Cox and Huang (1989) transformation", which gives a linear differential equation for wealth, which we identified under the martingale approach:

$$\eta = \frac{\partial J}{\partial W}(W, x, t) \longrightarrow W = F(\eta, x, t), \quad \text{where } F(\cdot) \text{ is the inverse function for marginal utility of wealth.}$$

- Note from the first-order condition for consumption that $\partial J/\partial W = \partial U/\partial C$, which of course is the stochastic discount factor, M; that is, $W = F(M, x, t)$.

- Typically, it is difficult to solve nonlinear PDEs analytically, and one has to resort to numerical methods.

- Campbell (1993), in an important paper, shows how one can approximate the nonlinear term in order to get a PDE that can be solved analytically (with one computation done numerically).
  - ⋆ This approximation underlies the solution for the models studied in Campbell and Viceira (1999, 2001) and the book Campbell and Viceira (2002).
  - ⋆ This material is explained also in the excellent book Campbell (2018).

- Liu (2007) showed, for a class of "quadratic models" of consumption and portfolio choice, how to get closed-form solutions for the nonlinear PDEs that characterize the solution when using dynamic programming.
  - ⋆ Essentially, he "reverse-engineered" the entire set of models for which one could solve the resulting PDEs analytically.

Fund-Separation Results

- When the investment opportunity set is nonrandom, then we get two-fund separation, just like we do in the two-date model of Markowitz and Sharpe.
- Merton (1971) showed that when the investment opportunity set is stochastic, then an investor would like to hold $2 + k$ funds, if there are k additional state variables that affect the investment opportunity set.
- Nielsen and Vassalou (2006) show that
  - ⋆ rather than the additional k intertemporal hedging funds needed according to Merton to hedge k sources of risk,
  - ⋆ one needs only two additional funds (in addition to the risk-free asset and market portfolio),
    - ∗ one to hedge changes in the intercept of the capital market line (given by the riskfree rate), and
    - ∗ the second to hedge the changes in the slope of the capital market line (given by the market price of risk).

- Breeden (1979) shows that even if there are k additional sources of risk,
  - ⋆ if instead of measuring the risk of a security with respect to the aggregate market and the k risk factors,
  - ⋆ one measured the risk of the security with respect to its covariation with aggregate consumption,
  - ⋆ then one would get the Consumption CAPM that has only a single beta,
  - ⋆ with the risk of the security now measured by its beta with respect to aggregate consumption.

12.11.2 Potential applications and extensions of lecture notes

- Kim and Omberg (1996) characterize the optimal portfolio when the risk premium is stochastic and given by a mean-reverting Ornstein-Uhlenbeck process.
- Brennan, Schwartz, and Lagnado (1997) apply dynamic programming to the problem of asset allocation.
- The important role played by the intertemporal hedging component of the optimal portfolio is described also in the book by Campbell and Viceira (2002) and the more recent book by Campbell (2018).
- The distinct roles played by risk aversion and elasticity of intertemporal substitution parameters are explained in Svensson (1989) for the case of a constant investment opportunity set, and by Bhamra and Uppal (2006) for the case of a stochastic investment opportunity set.

- **Jump processes** are discussed in Back (2017, Pages 375–376).
- Merton (1971, Section 8) allows for jumps, modeled via Poisson processes, in the context of portfolio choice.
- Das and Uppal (2004) study international portfolio choice with jumps used to model the *systemic risk* of international markets dropping concurrently.
- There are several models of option pricing with jumps. For instance,
  - ⋆ Merton (1976) allows for jumps in the stock price when valuing options; Naik (1993) allows for jumps in volatility; Das and Sundaram (1999) allow for jumps in the level and volatility of the underlying asset price.
  - ⋆ Bakshi, Cao, and Chen (1997) and Bates (1996, 2008) provide empirical tests of option pricing models with jumps.
  - ⋆ Naik and Lee (1990) develop a general equilibrium model for pricing of options on the market when its return may have jumps.
- Duffie, Pan, and Singleton (2000) study asset pricing allowing for jumps.

- **Dynamic programming** is a very important concept that forms the foundation of a large number of models in Finance and Macroeconomics.
  - ⋆ Several worked out examples of dynamic programming applied to the portfolio choice problem are provided in the book by Ingersoll (1987)—in discrete time (Chapter 11) and in continuous time (Chapter 13).
  - ⋆ Ljungqvist and Sargent (2018) is a good starting point to study dynamic programming applied to problems in Macroeconomics.
  - ⋆ For a deep understanding of dynamic programming, see Stokey and Lucas (1989).
- **Review papers:**
  - ⋆ Campbell (2000) – general review of asset pricing and portfolio choice
  - ⋆ Sundaresan (2000) – general review of continuous time methods
  - ⋆ Brandt (2010) – focus is on portfolio choice problems
  - ⋆ Wachter (2010) – focus is on asset allocation.

12.11.3 Ph.D. elective courses offered on related topics

- Lionel Martellini, a professor at EDHEC, has taught:
  - ⋆ "*Dynamic Asset Allocation*" in 2009
  - ⋆ "*Dynamic Asset Allocation-Long-term Investing with Short-term Constraints*" in 2012
  - ⋆ "*Dynamic Risk and Asset Allocation Decisions: From Investment Problems to Investment Solutions*" in 2014.
- Pietro Veronesi taught the course "*Recent Developments in Portfolio Allocation and Macro-Finance Model*" in 2009.
- Michael Brandt has taught courses on "*Portfolio Choice*" in 2011, 2013.
- Jerome Detemple taught "*Strategic Asset Allocation*" in 2013 and 2016.
- Rama Cont taught "*Monte Carlo Methods*" in 2010 and 2013.
- Marcel Rindisbacher taught "*Monte Carlo Methods*" in 2017.

Lecture Notes for 2025

Continuous Time Financial Economics

Chapter 13

Partial- and General-Equilibrium Asset Pricing

Raman Uppal

EDHEC Business School

Chapter contents

13.1 Plan for the Day . . . 557
13.2 Main Points for Today . . . 558
13.3 Partial-Equilibrium Asset Returns in Continuous-Time . . . 559
13.3.1 Assumptions of the Merton Model (repeated) . . . 560
13.3.2 Stochastic Discount Factor (repeated) . . . 562
13.3.3 First-Order Condition for Optimal Portfolio Weight . . . 565
13.3.4 Envelope Condition . . . 566
13.3.5 First-Order Condition for Optimal Consumption . . . 567
13.3.6 Definition of Absolute Risk Aversion . . . 569
13.3.7 The Consumption CAPM (CCAPM) . . . 570
13.3.8 The Intertemporal CAPM (ICAPM) . . . 577
13.3.9 The Conditional CAPM . . . 582
13.4 General-Equilibrium Asset Pricing in an Exchange Economy . . . 584
13.4.1 Assumptions of the Lucas Model . . . 586
13.4.2 Asset Prices in the Lucas Model . . . 590
13.4.3 Interpreting the Interest Rate and Market Price of Risk in the Lucas Model . . . 593
13.5 General-Equilibrium Asset Pricing in a Production Economy . . . 595
13.5.1 The Production Process . . . 596
13.5.2 Investor Preferences and Investment Opportunities . . . 599
13.5.3 Optimization Problem and Dynamic Budget Constraint . . . 600
13.5.4 State Variables, Value Function, and Bellman Equation . . . 601
13.5.5 First-order Conditions for Consumption and Portfolio Policy . . . 602
13.5.6 Equilibrium Risk-Free Interest Rate and Its Interpretation . . . 603
13.6 Summary . . . 605

13.7 Suggestions for Further Readings . . . 608
13.7.1 Source material for the lecture notes . . . 608
13.7.2 Potential applications and extensions of lecture notes . . . 609
13.7.3 Ph.D. elective courses offered on related topics . . . 612

13.1 Plan for the Day

Last Time

- Continuous-time model of portfolio choice using dynamic programming

Today

- **Equilibrium asset pricing in continuous time**
  - ⋆ Partial equilibrium models of asset pricing in continuous time
  - ⋆ General equilibrium models of asset pricing in continuous time

For Next Time

- Review the material covered in class.
- Read Back, Chapters 10 and 14 and Dumas & Luciano, Chapter 11 and 12.
- Do the practice problems.

13.2 Main Points for Today

1. Asset prices in a dynamic partial-equilibrium model with continuous trading.
   - Consumption CAPM
   - Intertemporal CAPM
   - Conditional CAPM
2. Asset prices in a dynamic general-equilibrium model with continuous trading.
   - Risk-free rate (r_t) in general equilibrium
   - Market price of risk (λ_t) in general equilibrium

   Recall that these two quantities, the risk-free rate and the market price of risk, fully characterize the dynamics of the stochastic discount factor:

   $$\frac{dM_t}{M_t} = -r_t\, dt - \lambda_t\, dB_t^{\mathbb{P}}, \quad \text{where } \lambda_t = \frac{\mu_t - r_t}{\sigma_t}.$$

13.3 Partial-Equilibrium Asset Returns in Continuous-Time

- We now derive the relation between return and risk for a continuous-time economy where investors have time-separable utility, as in Merton (1973a).
- We find that if investment opportunities are constant over time then the single-period CAPM holds in this economy.
- If investment opportunities are changing, the single-beta CAPM needs to be replaced by a multi-beta Intertemporal CAPM (ICAPM).
  - ⋆ The additional betas reflect prices sources of risk from additional state variables that affect investment opportunities.
- Breeden (1979) shows that the multi-beta CAPM can be collapsed to a single consumption-beta model called the Consumption CAPM (CCAPM).

13.3.1 Assumptions of the Merton Model (repeated)

Assumptions of the Model: Financial securities

- The assumptions below are the same as the ones described in Section 12.4.2 of Chapter 12; we reproduce them below only for convenience.
- The investor can allocate her wealth across n risky and one risk-free asset.
- Define $\mathcal{P}_t^{S_i}$ to be the price of the i-th risky asset at date t, which follows

$$\frac{d\mathcal{P}_t^{S_i}}{\mathcal{P}_t^{S_i}} = \mu_i(x,t)\,dt + \sigma_i(x,t)\,dB_{s_i}^{\mathbb{P}}, \quad i = \{1,2,\ldots,n\}, \tag{13.1}$$

  and the correlation between Brownians is $(dB_{s_i}^{\mathbb{P}})(dB_{s_j}^{\mathbb{P}}) = \rho_{s_i s_j} dt$.
- We will denote the instantaneous risk-free rate by $r(x,t)$.

Assumptions of the Model: State variables driving returns

- Note that μ_i, σ_i and r are allowed to be functions of time t and a $k \times 1$ vector of state variables, which we denote by $x(t) = (x_1, x_2, \ldots, x_k)$.

- If μ_i, σ_i or r are time-varying then the investor is said to face a changing investment opportunities.

- The state variables themselves can follow diffusion processes. Let the i-th state variable follow the process

$$dx_i = a_i(x,t)\,dt + b_i(x,t)\,dB^{\mathbb{P}}_{x_i}, \quad i = \{1, 2, \ldots, k\}. \tag{13.2}$$

- The correlations between the Brownians are: $(dB^{\mathbb{P}}_{x_i})(dB^{\mathbb{P}}_{x_j}) = \rho_{x_i x_j} dt$ and between $(dB^{\mathbb{P}}_{s_i})(dB^{\mathbb{P}}_{x_j}) = \rho_{s_i x_j} dt$.

- So, up to $(n + k)$ Brownian motions affect distribution of n stock returns (implying that the market is incomplete).

13.3.2 Stochastic Discount Factor (repeated)

Definition 13.1 (SDF)
We define M to be an SDF process if

1. $M_0 = 1$;

2. $M_t > 0$ for all t with probability 1; and,

3. $M\,\mathcal{P}^D$ and $M\,\mathcal{P}^{S_n}$ are local martingales for $n = 1, 2, \ldots, N$.

- We have shown previously that a stochastic process M is an SDF process if and only if $M_0 = 1$ and

$$\frac{dM_t}{M_t} = -r_t\, dt - \lambda_t^\top\, dB_t^{\mathbb{P}}, \tag{13.3}$$

where the stochastic process λ_t satisfies

$$\sigma_t\, \lambda_t = \mu_t - r_t \mathbf{1}. \tag{13.4}$$

- The vector λ_t is called the vector of market prices of risk.
- If there are as many risky assets as Brownian motions, then markets are complete, and given our assumption that there are no redundant assets, the volatility matrix σ_t is non-singular so from equation (13.4)

$$\lambda_t = \sigma_t^{-1}\Big(\mu_t - r_t \mathbf{1}\Big), \tag{13.5}$$

and there is at most one SDF process.

- If there is only a single risky asset, then this simplifies to

$$\lambda_t = \frac{\mu_t - r_t}{\sigma_t}, \tag{13.6}$$

which is the Sharpe ratio of the risky asset.

- Thus, in the case of a single risky asset, the SDF is:

$$\frac{dM_t}{M_t} = -r_t\, dt - \left(\frac{\mu_t - r_t}{\sigma_t}\right) dB_t^{\mathbb{P}}, \tag{13.7}$$

and the risk-premium on Asset n is:

$$(\mu_n - r)\, dt = -\left(\frac{dM}{M}\right)\left(\frac{d\mathcal{P}_n}{\mathcal{P}_n}\right) = -\text{Cov}\left[\frac{dM}{M}, \frac{d\mathcal{P}_n}{\mathcal{P}_n}\right]. \tag{13.8}$$

- We will use these results several times in today's class.

13.3.3 First-Order Condition for Optimal Portfolio Weight

- The optimal portfolio weight in (12.44) on Page 531 of Chapter 12 is:

$$w_i = \overbrace{\left(\frac{-J_W}{W\,J_{WW}}\right)}^{1/[\text{relative risk aversion}]} \overbrace{\sum_{j=1}^{n} v_{ij}(\mu_j - r)}^{\text{mean-var portfolio}} - \overbrace{\left(\frac{J_{Wx}}{W\,J_{WW}}\right)}^{\text{Effect of } x \text{ on } J_W} \sum_{j=1}^{n} v_{ij} \overbrace{\sigma_{jx}}^{\substack{\text{covariance of} \\ x \text{ with asset } j}} . \qquad (12.44)$$

$\overbrace{\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad}^{\text{intertemporal hedging portfolio}}$

- We can re-write the above equation as:

$$w_i = \overbrace{\left(\frac{-J_W}{W\,J_{WW}}\right)}^{1/[\text{relative risk aversion}]} \left[\overbrace{\sum_{j=1}^{n} v_{ij}(\mu_j - r)}^{\text{mean-var portfolio}} + \overbrace{\left(\frac{J_{Wx}}{J_W}\right)}^{\text{Effect of } x \text{ on } J_W} \sum_{j=1}^{n} v_{ij} \overbrace{\sigma_{jx}}^{\substack{\text{covariance of} \\ x \text{ with asset } j}} \right]. \qquad (12.44)$$

$\overbrace{\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad}^{\text{intertemporal hedging portfolio}}$

where

$$\frac{J_{Wx}}{J_W} = \frac{\partial \log J_W}{\partial x} = \text{"sensitivity of log of marginal utility of wealth" to state variable } x.$$

13.3.4 Envelope Condition

- Recall also the envelope condition in Equation (12.42) on Page 529 of Chapter 12:

$$U_C = J_W. \qquad (12.42)$$

- Recall also that for power utility

$$U(C_t) = \frac{1}{1-\gamma} e^{-\delta t} C_t^{1-\gamma}, \qquad (13.9)$$

δ is the subjective rate of time preference, and

γ is the investor's relative risk aversion (RRA), which is defined as:

$$\text{RRA} \equiv -C\frac{U''(C)}{U'(C)} = -C\left(\frac{-\gamma\, e^{-\delta t} C_t^{-\gamma-1}}{e^{-\delta t} C_t^{-\gamma}}\right) = \gamma\left(\frac{e^{-\delta t} C_t^{-\gamma}}{e^{-\delta t} C_t^{-\gamma}}\right) = \gamma. \qquad (13.10)$$

13.3.5 First-Order Condition for Optimal Consumption

- Finally, recall from the Martingale approach, that the first-order condition for consumption, which is given in Equation (11.31) on Page 451 of Chapter 11:

$$\frac{\partial U(C_s)}{\partial C_s} = U'(C_s) = e^{-\delta s} C_s^{-\gamma} = \lambda_{BC} M_s, \tag{13.11}$$

so that

$$\begin{aligned}
\text{MRS}_{0,t} &= \frac{\frac{\partial U(C_t)}{\partial C_t}}{\frac{\partial U(C_0)}{\partial C_0}} = \frac{\lambda_{BC}\, M_t}{\lambda_{BC}\, M_0} \\
&= \frac{e^{-\delta t}\, u'(C_t)}{e^{-\delta 0}\, u'(C_0)} = \frac{M_t}{M_0} && \text{. . . canceling out } \lambda_{BC} \\
&= \frac{e^{-\delta t}\, u'(C_t)}{u'(C_0)} = M_t && \text{. . . } e^{-\delta \times 0} = 1 \text{ and } M_0 = 1 \qquad (13.12) \\
&= \frac{e^{-\delta t}\, C_t^{-\gamma}}{C_0^{-\gamma}} = M_t && \text{. . . substituting power function for } u(c) \text{ from (13.9).} \qquad (13.13)
\end{aligned}$$

- So far, we did not have to distinguish between one investor and another.
- But now we will want to add up the demands of risky assets over all investors.
- Let us use h to refer to a particular individual, with $h = \{1, 2, \ldots, H\}$.
- Then, we can write Equation (13.12) on the previous page for investor h as:

$$\frac{e^{-\delta^h t}\, (u^h)'(C_t^h)}{(u^h)'(C_0^h)} = M_t^h. \tag{13.14}$$

- This condition should hold
  - ⋆ regardless of whether market is complete or incomplete;
  - ⋆ regardless of whether there is nonportfolio (labor) income;
  - ⋆ regardless of whether the horizon is finite or infinite.
- The only requirement is that utility is additive.

13.3.6 Definition of Absolute Risk Aversion

- Re-writing the last result on the previous page, for Investor h:

$$\frac{e^{-\delta^h t}\,(u^h)'(C_t^h)}{(u^h)'(C_0^h)} = M_t^h \equiv g^h(t, C_t^h) \qquad \ldots \text{defining } g^h(t, C_t^h) \tag{13.14}$$

- Let us now define Absolute Risk Aversion (ARA):

$$\alpha_t = \frac{-u''(C_t)}{u'(C_t)}, \tag{13.15}$$

which for Investor h can be written as:

$$\alpha_t^h = \frac{-(u^h)''(C_t^h)}{(u^h)'(C_t^h)} = -\frac{\partial \log g^h(t, C_t^h)}{\partial C_t^h}. \tag{13.16}$$

13.3.7 The Consumption CAPM (CCAPM)

- We start by re-writing Equation (13.8) on Page 564:

$$(\mu_n - r)dt = -\left(\frac{d\mathcal{P}^{S_n}}{\mathcal{P}^{S_n}}\right)\left(\frac{dM}{M}\right) \tag{13.17}$$

- We wish to express the stochastic part of dM/M in terms of consumption.

- Using Itô's formula on $M_t^h \equiv g^h(t, C_t^h)$, gives:

$$dg^h(t, C_t^h) = \underbrace{dt \text{ terms involving } g_t^h, g_c^h, g_{cc}^h}_{\text{deterministic terms}} + \underbrace{g_c^h(t, C_t^h)\, dC_t^h}_{\text{stochastic term}} \tag{13.18}$$

- So, the next step is to find a nice expression for $g_c^h(t, C_t^h) = \frac{\partial g^h(t, C_t^h)}{\partial C_t^h}$.

- We start by re-writing Equation (13.14):

$$g^h(t, C_t^h) \equiv M_t^h = \frac{e^{-\delta^h t}\, u'(C_t^h)}{u'(C_0^h)}. \tag{13.14}$$

- Thus,

$$\begin{aligned}
\frac{\partial g^h(t, C_t^h)}{\partial C_t^h} &= \frac{e^{-\delta^h t}}{u'(C_0^h)} \frac{\partial}{\partial C_t^h} u'(C_t^h) \\
&= \frac{e^{-\delta^h t}}{u'(C_0^h)} u''(C_t^h) && \ldots \text{differentiating } u'(C_t^h) \\
&= \frac{e^{-\delta^h t}}{u'(C_0^h)} \frac{u'(C_t^h)}{u'(C_t^h)} u''(C_t^h) && \ldots \text{multiplying and dividing by the same term } u'(C_t^h) \\
&= \left(\frac{e^{-\delta^h t} u'(C_t^h)}{u'(C_0^h)}\right) \left(\frac{u''(C_t^h)}{u'(C_t^h)}\right) && \ldots \text{rearranging the terms in red} \\
&= M_t^h \left(-\alpha_t^h\right) && \ldots \text{using definition of } M_t^h \text{ and } \alpha_t^h
\end{aligned} \tag{13.19}$$

- We start by re-writing the last expression on the previous page: $g_c^h(t, C_t^h) = -M_t^h \alpha_t^h$.

- This implies that

$$\frac{dM}{M} = \frac{dg^h(t, C_t^h)}{g^h(t, C_t^h)} = \text{some deterministic terms} - \alpha_t^h dC_t^h. \tag{13.20}$$

- Substituting this in Equation (13.8) on Page 564, which is reproduced below:

$$(\mu_n - r)dt = -\left(\frac{d\mathcal{P}^{S_n}}{\mathcal{P}^{S_n}}\right)\left(\frac{dM}{M}\right) \tag{13.8}$$

we get that:

$$\underbrace{(\mu_n - r)dt}_{\text{risk premium}} = \underbrace{\alpha^h}_{\text{risk aversion}} \times \underbrace{\left(\frac{d\mathcal{P}^{S_n}}{\mathcal{P}^{S_n}}\right)\left(dC^h\right)}_{\text{cov between return \& consumption}}. \tag{13.21}$$

- The result that we have on the previous page relates the risk premium on security n to the consumption of a single investor h.

- We now extend this result so it applies at the level of the aggregate market:

$$(\mu_n - r)dt = \alpha^h \left(\frac{d\mathcal{P}^{S_n}}{\mathcal{P}^{S_n}}\right)\left(dC^h\right) \quad \text{. . . last expression on previous page}$$

$$\frac{1}{\alpha^h}(\mu_n - r)dt = \left(\frac{d\mathcal{P}^{S_n}}{\mathcal{P}^{S_n}}\right)\left(dC^h\right) \quad \text{. . . divide both sides by } \alpha^h$$

$$\sum_{h=1}^{H}\frac{1}{\alpha^h}(\mu_n - r)dt = \left(\frac{d\mathcal{P}^{S_n}}{\mathcal{P}^{S_n}}\right)\left(d\sum_{h=1}^{H}C^h\right) \quad \text{. . . sum over all } H \text{ investors}$$

$$\frac{1}{\alpha}(\mu_n - r)dt = \left(\frac{d\mathcal{P}^{S_n}}{\mathcal{P}^{S_n}}\right)(dC) \quad \text{. . . use definitions } \frac{1}{\alpha} \equiv \sum_{h=1}^{H}\frac{1}{\alpha^h} \text{ and } C \equiv \sum_{h=1}^{H}C^h$$

$$(\mu_n - r)dt = \alpha\left(\frac{d\mathcal{P}^{S_n}}{\mathcal{P}^{S_n}}\right)(dC) \quad \text{. . . multiplying both sides by } \alpha.$$

- The result on the previous page is the Consumption CAPM (Breeden 1979):

$$(\mu_n - r)dt = \alpha\left(\frac{d\mathcal{P}^{S_n}}{\mathcal{P}^{S_n}}\right)(dC), \tag{13.22}$$

where α is the (usually stochastic) coefficient of absolute risk aversion, and C is aggregate consumption in the economy.

- This equation tells us that the risk premium on a security is proportional to the conditional covariance of its of return with aggregate consumption.

- This is a sensible result, but for a long time it was not confirmed in the data because of the difficulty in measuring aggregate consumption accurately.

- In recent years, researchers have found ways to measure aggregate consumption more accurately, and thus, there is stronger support for this empirically.

- To write the Consumption CAPM in its familiar form, we need to eliminate α from Equation (13.22).

- We assume there exists an asset whose dividend is aggregate consumption (in the standard CAPM, this is the "market" portfolio).

- Then, applying the Consumption CAPM to the portfolio that pays aggregate consumption, we get:

$$(\mu_{agg} - r)dt = \alpha \left(\frac{d\mathcal{P}^{agg}}{\mathcal{P}^{agg}}\right)(dC) \quad \ldots \text{ replacing asset } n \text{ by } agg \text{ in: } (\mu_n - r)dt = \alpha \left(\frac{d\mathcal{P}^{S_n}}{\mathcal{P}^{S_n}}\right)(dC)$$

$$\alpha = \frac{(\mu_{agg} - r)dt}{\left(\frac{d\mathcal{P}^{agg}}{\mathcal{P}^{agg}}\right)(dC)} \quad \ldots \text{ isolating } \alpha \text{ on the left-hand side}$$

$$= \frac{(\mu_{agg} - r)dt}{\text{Var}\left(\frac{d\mathcal{P}^{agg}}{\mathcal{P}^{agg}}\right)}. \quad \ldots \text{ because } \left(\frac{d\mathcal{P}^{agg}}{\mathcal{P}^{agg}}\right)(dC) = \text{Var}\left(\frac{d\mathcal{P}^{agg}}{\mathcal{P}^{agg}}\right)$$

- Finally, substitute this result for α into the Consumption CAPM:

$$(\mu_n - r)dt = \alpha \left(\frac{d\mathcal{P}^{S_n}}{\mathcal{P}^{S_n}}\right)(dC) \quad \ldots \text{ from (13.22) on page 574}$$

$$= \frac{(\mu_{agg} - r)dt}{\text{Var}\left(\frac{d\mathcal{P}^{agg}}{\mathcal{P}^{agg}}\right)} \overbrace{\left[\left(\frac{d\mathcal{P}^{S_n}}{\mathcal{P}^{S_n}}\right)(dC)\right]}^{\text{covariance}} \quad \ldots \text{ substituting our result for } \alpha \text{ from previous page}$$

$$= \beta_{agg,n}(\mu_{agg} - r)dt, \qquad (13.23)$$

where

$$\beta_{agg,n} = \frac{\left(\frac{d\mathcal{P}^{S_n}}{\mathcal{P}^{S_n}}\right)(dC)}{\text{Var}\left(\frac{d\mathcal{P}^{agg}}{\mathcal{P}^{agg}}\right)}. \quad \ldots \text{ rearranging terms} \qquad (13.24)$$

13.3.8 The Intertemporal CAPM (ICAPM)

- Assume now that the market is Markovian, and for simplicity, assume that there is only a single state variable X.

- From the first-order condition for consumption, we have that:

$$M_t^h = \frac{e^{-\delta^h t}\,(u^h)'(C_t^h)}{(u^h)'(C_0^h)}$$

- Substitute into the above the envelope condition in Equation (12.42) on Page 12.42 of Chapter 12:

$$U_C = J_W, \tag{12.42}$$

which then leads to:

$$M_t^h = \frac{e^{-\delta\, t}\,u'(C_t^h)}{u'(C_0^h)} = \frac{e^{-\delta\, t}\,J_W^h(t, W_t^h, X_t)}{J_W^h(0, W_0^h, X_0)} = g^h(t, W_t^h, X_t).$$

- From this point on, the derivation proceeds exactly as it did for the Consumption CAPM.

- We start by re-writing Equation (13.8) on Page 564:

$$(\mu_n - r)dt = -\left(\frac{d\mathcal{P}^{S_n}}{\mathcal{P}^{S_n}}\right)\left(\frac{dM}{M}\right). \tag{13.25}$$

- We wish to identify the stochastic part of dM/M.

- Using Itô's formula on $M_t^h \equiv g^h(t, W_t^h, X_t)$ gives:

$$\begin{aligned} dg^h(t, W_t^h, X_t) = &\underbrace{dt \text{ terms involving } g_t^h, g_W^h, g_{WW}^h, g_X^h, g_{XX}^h, g_{WX}^h}_{\text{deterministic terms}} \\ &+ \underbrace{g_W(t, W_t^h, X_t)\,dW_t^h + g_X(t, W_t^h, X_t)\,dX_t}_{\text{stochastic terms}} \end{aligned} \tag{13.26}$$

- So, the next step is to find a nice expressions for $g_W^h(t, W_t^h, X_t)$ and $g_X^h(t, W_t^h, X_t)$.

- We start by re-writing Equation (13.14):

$$g^h(t, W_t^h, X_t) \equiv M_t^h = \frac{e^{-\delta^h t} J_W(t, W_t^h, X_t)}{J_W(t, W_0^h, X_t)}. \tag{13.14}$$

- Thus, proceeding as before, we get:

$$\frac{\partial g^h(t, W_t^h, X_t)}{\partial W_t^h} = M_t^h \left[\frac{J_{WW}(t, W_t^h, X_t)}{J_W(t, W_t^h, X_t)}\right] = M_t^h \left(-\alpha_t^h\right). \tag{13.27}$$

- Similarly:

$$\frac{\partial g^h(t, W_t^h, X_t)}{\partial X_t} = M_t^h \underbrace{\left[\frac{J_{WX}(t, W_t^h, X_t)}{J_W(t, W_t^h, X_t)}\right]}_{\equiv -\eta_t^h} = M_t^h(-\eta_t^h). \tag{13.28}$$

- This implies that

$$\frac{dM}{M} = \frac{dg^h(t, W_t^h, X_t)}{g^h(t, W_t^h, X_t)} = \text{some deterministic terms} - \alpha_t^h dW_t^h - \eta_t^h dX_t. \tag{13.29}$$

- Substituting the result in (13.29) on the previous page into Equation (13.8) on Page 564, which is reproduced below:

$$(\mu_n - r)dt = -\left(\frac{d\mathcal{P}^{S_n}}{\mathcal{P}^{S_n}}\right)\left(\frac{dM}{M}\right), \tag{13.8}$$

  we get that:

$$(\mu_n - r)dt = \alpha^h \left(\frac{d\mathcal{P}^{S_n}}{\mathcal{P}^{S_n}}\right)(dW^h) + \eta^h \left(\frac{d\mathcal{P}^{S_n}}{\mathcal{P}^{S_n}}\right)(dX). \tag{13.30}$$

- Dividing this by α^h, summing over h, and rearranging gives:

$$(\mu_n - r)dt = \alpha \left(\frac{d\mathcal{P}^{S_n}}{\mathcal{P}^{S_n}}\right)(dW) + \eta \left(\frac{d\mathcal{P}^{S_n}}{\mathcal{P}^{S_n}}\right)(dX), \tag{13.31}$$

  where

  - $W_t = \sum_{h=1}^H W_t^h$ is market wealth;
  - $\alpha_t = \frac{1}{\sum_{h=1}^H \alpha_t^h}$;
  - $\eta_t = \alpha_t \sum_{h=1}^H \frac{\eta_t^h}{\alpha_t^h}$.

- The equation below is the Intertemporal CAPM developed by Merton (1973a):

$$(\mu_n - r)dt = \alpha \left(\frac{d\mathcal{P}^{S_n}}{\mathcal{P}^{S_n}} \right) (dW) + \eta \left(\frac{d\mathcal{P}^{S_n}}{\mathcal{P}^{S_n}} \right) (dX) . \tag{13.31}$$

- It says that the risk premium on a security depends on its conditional covariance with aggregate (market) wealth and the state variable X.

13.3.9 The Conditional CAPM

- The Intertemporal CAPM simplifies to a conditional CAPM if $J^h_{WX} = 0$ for each investor h and each state variable X.

- Then, $\eta = 0$, and we get the conditional CAPM:

$$(\mu_n - r)dt = \alpha \left(\frac{d\mathcal{P}^{S_n}}{\mathcal{P}^{S_n}} \right) (dW) . \tag{13.32}$$

  or, written in its more traditional form:

$$(\mu_n - r)dt = \beta_{mkt,n} (\mu_{mkt} - r) . \tag{13.33}$$

- This looks like the one-period CAPM, but the difference is that the relation is conditional on information at date t; that is, the model actually is:

$$(\mu_{n,t} - r_t)dt = \beta_{mkt,n,t} (\mu_{mkt,t} - r_t) . \tag{13.34}$$

- The conditional CAPM would hold if
  - ⋆ either each investor has log utility;
  - ⋆ or, the capital market line were constant.
- It is unlikely that all investors have log utility. And, the empirical evidence suggests that both the risk-free rate and the Sharpe ratio change over time, so the capital market line is unlikely to be constant.
- Thus, the Intertemporal CAPM is a much better model of asset premia than the conditional CAPM.

13.4 General-Equilibrium Asset Pricing in an Exchange Economy

- The results derived so far, are all partial equilibrium. That is, the process of returns on the risky assets is given exogenously, rather than being determined in the model.
- This means that the Merton model relates only returns on Asset n to returns on aggregate wealth, rather than to the fundamental parameters of the investor's utility and the output process.
- For example, the Merton model does not tell us the relation between the deep (basic) parameters of the model and
  - ⋆ the risk-free rate, and
  - ⋆ the market price of risk.

- In this section, we study a general equilibrium model of asset pricing, and we identity the relation between the deep parameters of the model to
  - ⋆ the risk-free rate of interest; and,
  - ⋆ the market price of risk.
- The stochastic discount factor is fully described by the risk-free rate and the market price of risk, so once we know these quantities, we can solve for any other asset price (such as the price of a derivative contract).

13.4.1 Assumptions of the Lucas Model

- The model we study is a continuous-time version of Lucas (1978).
- We assume that there is a single agent in the economy,
  - ⋆ or that there are many agents but they are all identical,
  - ⋆ or that the many agents can be "aggregated" (Rubinstein 1974).
- One can relax the assumption that there is only a "single" agent:
  - ⋆ Bhamra and Uppal (2014) show how to characterize asset prices in general equilibrium with agents who can differ in both their preferences and beliefs if markets are complete.
  - ⋆ Dumas and Lyasoff (2012), Buss, Uppal, and Vilkov (2015), and Buss, Dumas, Uppal, and Vilkov (2016) show how to characterize asset prices with heterogeneous agents even when markets are incomplete.

Endowment Process

- A single non-storable consumption good, Y_t, with the evolution of aggregate output:

$$\frac{dY_t}{Y_t} = \mu_Y \, dt + \sigma_Y \, dB_t^{\mathbb{P}}, \qquad \text{where } \mu_Y \text{ and } \sigma_Y \text{ are constants.}$$

Utility Function

- The representative agent is assumed to have time-separable power utility:

$$U(C,t) = e^{-\delta t} \frac{C_t^{1-\gamma}}{1-\gamma};$$

Financial assets

- A risky stock with the following return process.

$$\frac{d\mathcal{P}_t^S + Y_t dt}{\mathcal{P}_t^S} = dR_t = \mu_{R,t} \, dt + \sigma_{R,t} \, dB_t^{\mathbb{P}}.$$

- A locally riskfree bond

$$\frac{d\mathcal{P}_t^D}{\mathcal{P}_t^D} = r_t \, dt.$$

- $\mu_{R,t}, \sigma_{R,t}, r_t$ to be determined endogenously in equilibrium.

Optimization Problem

- Maximize lifetime utility,

$$\mathbb{E}_t^{\mathbb{P}}\left[\int_t^{\infty} e^{-\delta(u-t)}\frac{C_u^{1-\gamma}}{1-\gamma}du\right],$$

subject to the static budget constraint:

$$\mathbb{E}_0^{\mathbb{P}}\left[\int_0^{\infty}\frac{M_u}{M_0}C_u\,du\right] \leq \mathcal{P}_0^S,$$

in which

- ⋆ M_u is the marginal utility of the investor at date u (state-price density, stochastic discount factor), and
- ⋆ the investor is endowed with a single share of the stock, whose price at $t = 0$ is $\mathcal{P}_0^S$.

13.4.2 Asset Prices in the Lucas Model

- We start with the first-order condition for consumption, which we derived already in Equation (13.12) on Page 567:

$$M_t = \frac{e^{-\delta t}\,u'(C_t)}{u'(C_0)} \qquad (13.12)$$

$$= \frac{e^{-\delta t}\,C_t^{-\gamma}}{C_0^{-\gamma}} \qquad \ldots \text{substituting power utility for } u(C_t)$$

$$= e^{-\delta t}\,C_0^{\gamma}\,C_t^{-\gamma} \qquad \ldots \text{rearranging terms}$$

- Define

$$g(t, C_t) = M_t \equiv e^{-\delta t}\,C_0^{\gamma}\,C_t^{-\gamma}$$

so that that partial derivatives, which we will need for Itô's formula, are:

$$g_t(t, C_t) = -\delta\,g(t, C_t);$$

$$g_c(t, C_t) = -\gamma\,\frac{g(t, C_t)}{C_t};$$

$$g_{cc}(t, C_t) = \gamma(1+\gamma)\,\frac{g(t, C_t)}{C_t^2}.$$

- Note that in equilibrium, aggregate consumption has to equal aggregate endowment; that is, $C_t = Y_t$, which implies that

$$\frac{dC_t}{C_t} = \frac{dY_t}{Y_t} = \mu_Y\, dt + \sigma_Y\, dB_t^{\mathbb{P}}.$$

- Applying Itô's formula to $g(t, C_t) = M_t \equiv e^{-\delta\, t}\, C_0^{\gamma}\, C_t^{-\gamma}$:

$$\begin{aligned} dM &= g_t\, dt + g_c\, dC + \frac{1}{2} g_{cc} (dC)^2 && \text{. . . applying Itô's formula to } g(t, C_t) \\ &= -\delta\, g\, dt - \gamma \frac{g}{C_t}\, C_t \left(\mu_Y\, dt + \sigma_Y\, dB_t^{\mathbb{P}}\right) \\ &\quad + \frac{1}{2}\gamma(1+\gamma)\frac{g}{C_t^2}\, C_t^2 \left(\sigma_Y\, dB_t^{\mathbb{P}}\right)^2 && \text{. . . substituting } dC \\ \frac{dM}{M} &= -\left[\delta + \gamma\, \mu_Y - \frac{1}{2}\gamma(1+\gamma)\sigma_Y^2\right] dt - \gamma\sigma_Y dB_t^{\mathbb{P}}. && \text{. . . collecting } dt \text{ and } dB_t^{\mathbb{P}} \text{ terms} \end{aligned}$$

- We start by re-writing the last result on the previous page:

$$\frac{dM}{M} = -\left[\delta + \gamma\, \mu_Y - \frac{1}{2}\gamma(1+\gamma)\sigma_Y^2\right] dt - \gamma\sigma_Y dB_t^{\mathbb{P}}. \tag{13.35}$$

- We also know, from Equation (13.7) on Page 564 that

$$\frac{dM}{M} = -r\, dt - \left(\frac{\mu_R - r}{\sigma_R}\right) dB_t^{\mathbb{P}}. \tag{13.7}$$

- Comparing the drift and diffusion terms in Equations (13.35) and (13.7), we conclude that:

$$r = \delta + \gamma\, \mu_Y - \frac{1}{2}\gamma(1+\gamma)\sigma_Y^2, \tag{13.36}$$

$$\frac{\mu_R - r}{\sigma_R} = \gamma\, \sigma_Y. \tag{13.37}$$

13.4.3 Interpreting the Interest Rate and Market Price of Risk in the Lucas Model

- We see from the expression for the interest rate:

$$r = \delta + \gamma\,\mu_Y - \frac{1}{2}\gamma(1+\gamma)\sigma_Y^2, \tag{13.38}$$

  - ⋆ The interest rate increases with δ, because as δ increases the investor values future consumption less, and so decreases savings (or equivalently, increases borrowing), which leads to an increase in the interest rate.
  - ⋆ The interest rate increases with μ_Y because an increase in the growth rate of consumption increases the desire to borrow.
  - ⋆ The interest rate decreases with σ_Y^2 because an increase in the risk of the growth rate of consumption reduces the desire to borrow or equivalently, increases savings, referred to as the *precautionary saving demand* (Kimball 1990).

- Similarly, we see from the expression for the market price of risk:

$$\frac{\mu_R - r}{\sigma_R} = \gamma\,\sigma_Y. \tag{13.39}$$

  - ⋆ The market price of risk increases with γ, because the compensation for bearing risk needs to increase with an increase in risk aversion.
  - ⋆ The market price of risk increases with σ_Y, because the compensation for bearing risk needs to increase with the risk of the growth rate of aggregate consumption.

- Note that because there is only one state variable, aggregate endowment, the volatility of stock returns is equal to the volatility of aggregate endowments:

$$\sigma_R = \sigma_Y.$$

- Substituting this back into Equation (13.39), we get the discount rate:

$$\mu_R = r + \gamma\,\sigma_Y\,\sigma_R = r + \gamma\,\sigma_Y^2 \qquad \ldots \text{ expected return} = \text{risk-free rate} + \text{risk premium}$$

13.5 General-Equilibrium Asset Pricing in a Production Economy

- In the previous section, the model we studied was of an exchange economy, where output was given exogenously (by the process for dY_t).
- In this section, we study a general equilibrium production economy, based on the paper by Cox, Ingersoll, and Ross (1985b).
- We assume that there is a single good (such as "corn") that can either be consumed or invested in the production process.
- There is a representative investor who chooses optimally how much to consume and how much to invest at each instant; the investment this instant then determines the output next instant.
- Thus, in this economy output is determined endogenously.

13.5.1 The Production Process

- We assume that the production sector has constant returns to scale, which simplifies solving the model.
  - ⋆ This assumption can be (and has been) relaxed.
- The output each instant, $Y(t)$, depends on the capital stock, $K(t)$:

$$Y(t) = \mu K(t) \tag{13.40}$$

  for some constant technology level, μ.
- In macroeconomics, this is often written as $Y(t) = A K(t)$, and therefore, the model is called the "$A K$" model.

- The level of the firm's capital stock can be increased by investment at the rate $I(t)$. Thus, the capital accumulation equation is:

$$dK(t) = I(t)\,dt + \sigma\,K(t)\,dB_t^{\mathbb{P}}, \tag{13.41}$$

where σ is the exogenous shock to the firm's capital stock and $dB_t^{\mathbb{P}}$ is the increment to a Brownian motion under the physical probability measure $\mathbb{P}$.

- A firm's output flow is divided between its investment and dividend flows:

$$Y(t) = I(t) + D(t), \tag{13.42}$$

where the household uses the dividend for consumption:

$$C(t) = D(t). \qquad \text{. . . this is the market clearing condition in the real sector}$$

- We can therefore write the capital accumulation equation as follows:

$$\begin{aligned} dK(t) &= I(t)\,dt + \sigma\,K(t)\,dB_t^{\mathbb{P}} && \text{. . . from capital accumulation equation (13.41)} \\ &= \Big[Y(t) - D(t)\Big]dt + \sigma\,K(t)\,dB_t^{\mathbb{P}} && \text{. . . because } I(t) = Y(t) - D(t) \text{ from (13.42)} \\ &= \Big[\mu\,K(t) - D(t)\Big]dt + \sigma\,K(t)\,dB_t^{\mathbb{P}} && \text{. . . because } Y(t) = \mu\,K(t) \text{ in (13.40)} \end{aligned} \tag{13.43}$$

- In this model, physical capital K can be converted, one-for-one, into financial capital (the stock), implying that $\mathcal{P}^S(t) = K(t)$, and hence,

$$\begin{aligned} d\mathcal{P}^S(t) &= \Big[\mu\,\mathcal{P}^S(t) - D(t)\Big]dt + \sigma\,\mathcal{P}^S(t)\,dB_t^{\mathbb{P}} && \text{. . . setting } K(t) = \mathcal{P}^S(t) \text{ in (13.43} \\ d\mathcal{P}^S(t) + D(t)\,dt &= \mu\,\mathcal{P}^S(t)dt + \sigma\,\mathcal{P}^S(t)\,dB_t^{\mathbb{P}} && \text{. . . adding } D(t)\,dt \text{ to both sides} \\ \frac{d\mathcal{P}^S(t) + D(t)\,dt}{\mathcal{P}^S(t)} &= \mu\,dt + \sigma\,dB_t^{\mathbb{P}}, && \text{. . . dividing both sides by } \mathcal{P}^S(t) \end{aligned}$$

so that the expected stock return is μ and its volatility is σ.

13.5.2 Investor Preferences and Investment Opportunities

- We assume an infinite-horizon representative-agent economy (that is, all agents are identical) with time-separable power utility function:

$$U(C,t) = e^{-\delta t}\frac{C_t^{1-\gamma}}{1-\gamma}.$$

- Each investor in this economy can choose to invest in this production process or to borrow/lend at the risk-free rate $r(t)$, which will be determined endogenously in equilibrium.
  - ⋆ This is the only quantity to be determined in equilibrium.
  - ⋆ The stock-return process has mean μ and volatility σ, exactly as specified exogenously for the production process.
- Each investor also needs to decide how much to consume each period (with the remainder of the aggregate output reinvested).

13.5.3 Optimization Problem and Dynamic Budget Constraint

- Denote by w the share of the investor's wealth that is invested in the risky production process and $1-w$ the share invested in the risk-free asset.
- Then, the representative investor needs to solve the following problem:

$$\max_{C,w}\ \mathbb{E}_0^{\mathbb{P}}\int_0^{\infty} U[C(t)]\,dt,$$

subject to the following dynamic budget constraint:

$$dW = W\left[w\,\mu + (1-w)\,r - \frac{C}{W}\right]dt + W\,w\,\sigma\,dB^{\mathbb{P}}.$$

13.5.4 State Variables, Value Function, and Bellman Equation

- In this model, the investment opportunity set is constant over time.
- So the only state variable is $W(t)$, the wealth of the representative investor.
  - ⋆ This wealth is also equal to the aggregate capital stock in the economy, $W(t) = K(t) = \mathcal{P}^S(t)$.
- Thus, the value function is $V(W,t)$ and the Bellman equation is:

$$0 = \max_{C,w} \left\{ U[C(t)] + \frac{1}{dt}\mathbb{E}_t^{\mathbb{P}}\left[dV(W,t)\right] \right\} \quad \text{... from Bellman's principle of optimality}$$

$$= \max_{C,w} \left\{ U[C(t)] + \frac{1}{dt}\mathbb{E}_t\left[V_t + V_W(dW) + \frac{1}{2}V_{WW}(dW)^2\right] \right\} \quad \text{... applying Itô's formula to } V(W,t)$$

$$= \max_{C,w} \left\{ U[C(t)] + V_t + W\,V_W\left[w\,\mu + (1-w)\,r - \frac{C}{W}\right] + \frac{1}{2}W^2\,V_{WW}\,w^2\sigma^2 \right\} \quad \text{... substituting } dW$$

13.5.5 First-order Conditions for Consumption and Portfolio Policy

- The first-order condition for consumption, C, is:

$$0 = U_C - V_W.$$

- The first-order condition for the portfolio weight, w, is:

$$0 = W\,V_W(\mu - r) + W^2\,V_{WW}\,w\,\sigma^2,$$

$$w = \left(\frac{-V_W}{W\,V_{WW}}\right)\left(\frac{\mu - r}{\sigma^2}\right) \quad \text{... isolating } w \text{ on the left-hand side}$$

which, one can show, is

$$= \frac{1}{\gamma}\left(\frac{\mu - r}{\sigma^2}\right). \quad \text{... because one can show that } \frac{-V_W}{W\,V_{WW}} = \frac{1}{\gamma}$$

13.5.6 Equilibrium Risk-Free Interest Rate and Its Interpretation

- We use the market-clearing condition in the financial sector to obtain the risk-free rate in equilibrium.

- The market-clearing condition in the financial sector is that
  - ⋆ the entire portfolio weight be invested in the risky asset (because in equilibrium the representative investor needs to hold the entire share of the risky asset),
  - ⋆ or equivalently, that there is nothing invested in the risk-free asset (because there cannot be any borrowing or lending in equilibrium when there is only a single representative investor).

- Using the first-order condition for the portfolio weight, we get:

$$w = 1$$
$$\frac{1}{\gamma}\frac{\mu - r}{\sigma^2} = 1$$
$$\mu - r = \gamma\,\sigma^2 \quad \text{implying that:} \quad r = \mu - \gamma\,\sigma^2 \quad \text{or, equivalently:}$$

$$\boxed{\mu = r + \gamma\,\sigma^2.} \tag{13.44}$$

- We see that the risk premium, $\mu - r$, is the product of two terms: σ^2, the amount of risk in the economy and γ, the investor's risk aversion.

- Alternatively, μ, the expected return on capital (the discount rate) consists of the risk-free interest rate, r, *plus* an adjustment for risk, $\gamma\,\sigma^2$.

13.6 Summary

1. The Consumption CAPM states that:

$$(\mu_n - r)dt = \alpha \left(\frac{d\mathcal{P}^{S_n}}{\mathcal{P}^{S_n}} \right) (dC),$$

where α is the (usually stochastic) coefficient of absolute risk aversion, and C is aggregate consumption in the economy.

2. The Intertemporal CAPM developed by Merton (1973a) states that:

$$(\mu_n - r)dt = \alpha \left(\frac{d\mathcal{P}^{S_n}}{\mathcal{P}^{S_n}} \right) (dW) + \eta \left(\frac{d\mathcal{P}^{S_n}}{\mathcal{P}^{S_n}} \right) (dX)\,; \tag{13.31}$$

that is, the risk premium on a security is proportional to the conditional covariance with aggregate (market) wealth and the state variable X.

3. The Intertemporal CAPM simplifies to the conditional CAPM when $J^h_{WX} = 0$ for each investor h and each state variable X:

$$(\mu_n - r)dt = \alpha \left(\frac{d\mathcal{P}^{S_n}}{\mathcal{P}^{S_n}} \right) (dW)\,.$$

or, written in its more traditional form:

$$(\mu_{n,t} - r_t)dt = \beta_{mkt,n,t}\,(\mu_{mkt,t} - r_t)\,,$$

which looks like the traditional single-period CAPM,
but is conditional on information at date t.

4. In the continuous-time version of the general equilibrium endowment economy Lucas (1978) model:

$$r = \delta + \gamma\,\mu_Y - \frac{1}{2}\gamma(1+\gamma)\sigma_Y^2,$$

$$\frac{\mu_R - r}{\sigma_R} = \gamma\,\sigma_Y.$$

5. In the general equilibrium production economy of Cox, Ingersoll, and Ross (1985b):

$$r = \mu - \gamma\,\sigma^2$$

$$\frac{\mu - r}{\sigma} = \gamma\,\sigma.$$

13.7 Suggestions for Further Readings

13.7.1 Source material for the lecture notes

- The material on the Intertemporal CAPM is based on Back (2017, Chapter 14). See also Dumas and Luciano (2017, Chapters 11 & 12).
- The three models of equilibrium we have discussed are:
  - ⋆ Intertemporal CAPM (Merton 1973a), a special case of which is the Consumption CAPM of Breeden (1979);
  - ⋆ Exchange/endowment economy (Lucas 1978) ;
  - ⋆ Production economy (Cox, Ingersoll, and Ross 1985b).
- There are thousands of papers that build on these three models
  (you can go to Google Scholar and search for the papers that cite these ones).
- There are also a large number of papers that test empirically the implications of these models.

13.7.2 Potential applications and extensions of lecture notes

- Different preferences; for example:
  - ⋆ internal/external habit
    - ∗ Detemple and Zapatero (1991), Schroder and Skiadas (2002), Abel (1990), Campbell and Cochrane (1999), Chan and Kogan (2002), Dumas and Luciano (2017, Chapter 20).
  - ⋆ Epstein-Zin-Weil utility
    - ∗ Epstein and Zin (1989), Weil (1989, 1990), Duffie and Epstein (1992), Dumas, Uppal, and Wang (2000), Dumas and Uppal (2001), Dumas and Luciano (2017, Chapter 21).
  - ⋆ ambiguity aversion or a desire for "robustness"
    - ∗ Epstein and Wang (1994) in discrete time, Chen and Epstein (2002) in continuous time, Epstein and Schneider (2008) with a nice application, and the review paper Epstein and Schneider (2010)
    - ∗ Maenhout (1999), Hansen and Sargent (2007) Uppal and Wang (2003), Bhamra and Uppal (2019), Bhamra, Uppal, and Walden (2020).
- Different production processes; for example, with non-constant returns to scale or costs for adjusting capital.
  - ⋆ Kogan (2001, 2004), Jermann (1998), Ai, Croce, and Li (2013).

- Different endowment process; for example:
  - ⋆ with long-run risk
    - ∗ Bansal and Yaron (2004).
  - ⋆ with disaster risk
    - ∗ Barro (2006, 2009), Backus, Chernov, and Martin (2011), Wachter (2013), Tsai and Wachter (2015), Ghosh, Julliard, and Taylor (2017).
- Different belief processes; for example, with "sentiment risk:"
  - ⋆ Harrison and Kreps (1978), Scheinkman and Xiong (2003), Dumas, Kurshev, and Uppal (2009), Dumas and Luciano (2017, Chapter 18).
- With parameter uncertainty:
  - ⋆ Collin-Dufresne, Johannes, and Lochstoer (2016).

- Modeling life cycle features; for example, age-dependent labor income.
  - ⋆ Cocco, Gomes, and Maenhout (2005), Gomes and Michaelides (2003, 2005, 2008).
- Multiple agents who are heterogeneous with respect to their preferences and/or beliefs.
  - ⋆ Dumas (1989), Wang (1996), Guvenen (2009), Bhamra and Uppal (2014), Borovička (2020).
- Modeling explicitly the role of financial intermediaries:
  - ⋆ He and Krishnamurthy (2013, 2018), Brunnermeier and Sannikov (2014, 2016).
- General-equilibrium models with incomplete markets
  - ⋆ Basak and Cuoco (1998), Bhamra and Uppal (2008), Dumas and Lyasoff (2012), Gabaix, Lasry, Lions, and Moll (2016), Nuño and Moll (2018), and Ahn, Kaplan, Moll, Winberry, and Wolf (2018).

13.7.3 Ph.D. elective courses offered on related topics

- Pietro Veronesi taught a course on "*Recent Developments in Portfolio Allocation and Macro-Finance Model*" in 2009 and 2012.
- Ravi Bansal taught a course on "*Long-Run Risks in Asset Prices*" in 2011.
- Federico Bandi taught a course on "*The Econometrics of Continuous-Time Models*" in 2012.
- Harjoat Bhamra taught a course on "*Monetary Policy and Asset Prices*" in 2016.
- Michael Chernov taught a course on "*Macro-Based Valuation Across Assets and Horizons*" in 2018.
- José Luis Peydró taught a course on "*Systemic Risk and Macroprudential Policy*" in 2019.
- Jesus Fernandez-Villaverde taught a course on "*Numerical Solutions and Bayesian Estimation Methods for General Equilibrium Asset Pricing Models*" in 2019.
- Jaume Ventura taught a course on "*The Macroeconomics of Credit and Asset Bubbles*" in September 2020.
- Harjoat Bhamra taught a course on *Asset Pricing and Monetary Policy* in 2023.
- Lukas Schmid taught a course on Models and Methods in Macro-Finance in 2024.
- Robert Kosowski taught a course on *Alternative Risk Premia and Replication of Financial Anomalies* in 2025.

Lecture Notes for 2025

Continuous Time Financial Economics

Chapter 14

Course Summary: Key Ideas Covered in Part 2

Raman Uppal

EDHEC Business School

Chapter contents

14.8 Valuation of Fixed Income Securities 617
14.9 Models of the Term Structure of Interest Rates 621
14.10 Portfolio Choice: Mean-Variance and Two-Date Models 631
14.11 Portfolio Choice Using the Martingale Approach 635
14.12 Portfolio Choice Using Dynamic Programming 638
14.13 Partial- and General-Equilibrium Asset Pricing 641

* Note that each section number above matches the number of the corresponding chapter.

14.8 Valuation of Fixed Income Securities

1. The yield of a zero-coupon bond with \$1 face value having $\tau = T - t$ years to maturity and price $\mathcal{P}^Z(t,T)$ is defined to be the rate $y(\tau)$ such at:

$$\mathcal{P}^Z(t,T) = \frac{\text{amount received}}{\text{discount factor}} = \frac{1_\tau}{e^{y\,\tau}} = e^{-y\,\tau} \quad \Leftrightarrow \quad y = -\frac{1}{\tau}\log \mathcal{P}^Z(t,T);$$

2. The yield curve is the collection of yields for different maturities.

3. For a bond with cashflows $C_1, C_2, \ldots, C_N$ at dates $u_1 < u_2 < \ldots < u_N$ and price P at date $t < u_1$, the continuously compounded yield to maturity of a bond at date t is defined to be the rate $\mathbf{y}$ such that

$$\mathcal{P}^B = \sum_{j=1}^{N} e^{-\mathbf{y}\tau_j} C_j,$$

where $\tau_j = u_j - t$ denotes the time remaining until j-th cash flow is paid.

4. "Macaulay duration" of the bond, or simply duration is the term

$$-\frac{\frac{d\mathcal{P}^B}{\mathcal{P}^B}}{d\mathbf{y}} = \sum_{j=1}^{N} \tau_j \times \frac{e^{-\mathbf{y}\tau_j} C_j}{P},$$

which is a weighted average of the times-to-maturity τ_j of the cashflows, with the weights being the fraction of bond value that the cashflow constitutes.

5. A standard (plain vanilla) interest rate swap is the exchange of fixed for floating interest rate payments on a given notional principal.

$$[\text{Value to fixed-rate payer}]_t = \underbrace{[\mathcal{P}^Z(t,t_0) - \mathcal{P}^Z(t,t_N)]}_{\text{value of floating leg}} - \underbrace{[\bar{\mathcal{R}}\Delta t \sum_{i=1}^{N} \mathcal{P}^Z(t,t_i)]}_{\text{value of fixed leg}}.$$

6. The value at $0 < t_i$ of a caplet with reset date t_i and payment date t_{i+1} is

$$\mathcal{P}^Z(0,t_{i+1})\mathcal{R}_i^{fwd}(0)\,\Delta t\,N(d_1) - \mathcal{P}^Z(0,t_{i+1})\bar{\mathcal{R}}\,\Delta t\,N(d_2);$$

where σ is the constant volatility of $\mathcal{R}_i^{fwd}(t)$, and

$$d_1 = \frac{\log\left(\frac{\mathcal{R}_i^{fwd}(0)}{\bar{\mathcal{R}}}\right) + \frac{1}{2}\sigma^2 t_i}{\sigma\sqrt{t_i}}; \qquad d_2 = d_1 - \sigma\sqrt{t_i}.$$

7. A payer swaption gives the owner of the option the right to enter into the swap as the payer of the fixed rate, for a given swap rate;

- Value of swap to fixed-rate payer: $\underbrace{[\mathcal{P}^Z(t,t_0) - \mathcal{P}^Z(t,t_N)]}_{V^{\text{floating}}(t)} - \underbrace{[\bar{\mathcal{R}}\Delta t \sum_{i=1}^{N} \mathcal{P}^Z(t,t_i)]}_{V^{\text{fixed}}(t)}$.

- Thus, value of European payer swaption at date 0 is

$$= \underbrace{\left[\mathcal{P}^Z(0,t_0) - \mathcal{P}^Z(0,t_N)\right]}_{V^{\text{floating}}(0)} N(d_1) - \underbrace{\left[\bar{\mathcal{R}}\Delta t \sum_{i=1}^{N} \mathcal{P}^Z(0,t_i)\right]}_{V^{\text{fixed}}(0)} N(d_2)$$

$$d_1 = \frac{\log\left(\frac{V^{\text{floating}}(0)}{V^{\text{fixed}}(0)}\right) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}; \qquad d_2 = d_1 - \sigma\sqrt{T},$$

where σ is the constant volatility of the forward swap rate $\mathcal{R}^{swap}(t)$.

14.9 Models of the Term Structure of Interest Rates

1. Under the risk neutral measure, the price $\mathcal{P}^Z(t,u)$ at date t of a discount bond maturing at u must be:

$$\mathcal{P}^Z(t,u) = E_t^{\mathbb{Q}}\left[\exp\left(-\int_t^u r(s)\,ds\right)\right].$$

2. The forward rate at t for an instantaneous loan is given by

$$r_t^{fwd} = -\frac{\partial \log \mathcal{P}^Z(0,t)}{\partial t}.$$

3. We study a number of term structure models. For each model we:

3.1. Specify the process for the interest rate, $dr(t)$;

3.2. Solve for the short-rate, $r(t)$;

3.3. Use the short rate to find discount-bond prices, $\mathcal{P}^Z(t,u)$;

3.4. Find the process (return) for discount-bond prices, $d\mathcal{P}^Z(t,u)$;

3.5. Use these results to value options on discount bonds and options on discount rates (caps, and floors).

3.6. Understand how to hedge in these models.

3.7. Discuss how to estimate the parameters of these models and how to hedge under these models.

4. In the basic Vasicek (1977) model, with $B^{\mathbb{Q}}$ denoting a Brownian motion under the risk-neutral measure

- It is assumed that the short rate follows the process:
$$dr(t) = \kappa\left[\theta - r(t)\right]dt + \sigma dB^{\mathbb{Q}}(t),$$
- The level of the interest rate is:
$$r_t = \theta - e^{-\kappa t}\left[\theta - r_0\right] + \sigma\int_0^t e^{\kappa(s-t)}dB_s^{\mathbb{Q}}$$
- The mean of the interest rate is
$$\mathbb{E}^{\mathbb{Q}}[r_t] = r_0\, e^{-\kappa t} + \theta\left(1 - e^{-\kappa t}\right).$$
- The variance of the interest rate is
$$\text{var}(r_t) = \sigma^2 E\left\{\int_0^t e^{2\kappa(s-t)}\,ds\right\} = \frac{\sigma^2}{2\kappa}\left(1 - e^{-2\kappa t}\right).$$
- The price of discount bonds is:
$$\begin{aligned}\mathcal{P}^Z(t,u) &= \exp\left[-a(\tau) - b(\tau)r(t)\right],\\ a(\tau) &= \theta\tau - \frac{\theta}{\kappa}\left(1 - e^{-\kappa\tau}\right) - \frac{\sigma^2}{4\kappa^3}\left(2\kappa\tau - e^{-2\kappa\tau} + 4e^{-\kappa\tau} - 3\right)\\ b(\tau) &= \frac{1}{\kappa}\left(1 - e^{-\kappa\tau}\right).\end{aligned}$$

5. Hedging in the Vasicek Model

- The Vasicek model has a single Brownian driving the short interest rate:
  - ⋆ Thus, there is only a single Brownian motion driving all bond prices
  - ⋆ So the bond prices are perfectly correlated in the Vasicek model.
  - ⋆ Therefore, the hedge ratio for hedging one bond with another will depend only on the relative volatilities of the bond returns.

6. The general Hull and White model is:

$$dr(t) = \theta(t)\,dt - \kappa(t)\,r(t)\,dt + \sigma(t)\,dB^{\mathbb{Q}}(t).$$

- In the first special case, which we refer to as the Hull and White model, κ and σ are constants and only $\theta(t)$ is a (non-random) function of time:

$$\begin{aligned} dr(t) =& \theta(t)\,dt - \kappa\,r(t)\,dt + \sigma\,dB^{\mathbb{Q}}(t), \\ \text{or:}\quad dr(t) =& \kappa\left(\frac{\theta(t)}{\kappa} - r(t)\right)dt + \sigma\,dB^{\mathbb{Q}}(t), \end{aligned}$$

  where $\theta(t)/\kappa$ is the time-varying long run mean of the short-rate process.

- In the second special case we set $\kappa = 0$ to get:
  $dr(t) = \theta(t)\,dt + \sigma\,dB^{\mathbb{Q}}(t)$,
  which is the continuous time Ho and Lee (1986) model, which has no mean reversion.

- The interest rate in the Hull and White model is:

$$\begin{aligned} r(s) &= \phi(s) + \widehat{r}(s), \\ \phi(s) &= \int_0^s e^{-\kappa(s-y)}\theta(y)\,dy, \\ \widehat{r}(s) &= e^{-\kappa s}r(0) + \sigma\int_0^s e^{-\kappa(s-y)}dB^{\mathbb{Q}}(y). \end{aligned}$$

- The interest rate in the Ho and Lee model, where $\kappa = 0$, is:

$$\begin{aligned} r(s) &= \phi(s) + \widehat{r}(s), \quad \text{where} \\ \phi(s) &= \int_0^s \theta(y)\,dy, \\ \widehat{r}(s) &= r(0) + \sigma\,B(s). \end{aligned}$$

- The price at t of a discount bond in the extended Vasicek model is

$$\mathcal{P}^Z(t,u) = \exp\Big(\underbrace{-\int_t^u \phi(s)ds}_{\text{new term}}\ \underbrace{-a(\tau) - b(\tau)\widehat{r}(t)}_{\text{from Vasicek model}}\Big).$$

7. Because

- the volatilities of discount bond returns in the Hull and White model are the same as in the Vasicek model with $\kappa > 0$; and,
- the volatilities of discount bond returns in the continuous time Ho and Lee model are the same as in the Vasicek model with $\kappa = 0$;

Thus, the computation of the hedge ratios in these models is analogous to the computations for the hedge ratio in the Vasicek model.

8. In the extended Vasicek model, the value at date 0 of a call option with exercise price K maturing at date T on a discount bond maturing at date $T' > T$ is:

$$\text{Call Price}_0 = [\mathcal{P}^Z]^{mkt}(0,T')N(d_1) - [\mathcal{P}^Z]^{mkt}(0,T)\,KN(d_2),$$

where

$$d_1 = \frac{\log\left(\frac{[\mathcal{P}^Z]^{mkt}(0,T')}{[\mathcal{P}^Z]^{mkt}(0,T)K}\right) + \frac{1}{2}\sigma_{avg}^2\,T}{\sigma_{avg}\sqrt{T}}$$

$$d_2 = d_1 - \sigma_{avg}\sqrt{T},$$

with

$$\sigma_{avg} = \sigma\,\frac{\left(e^{-\kappa T} - e^{-\kappa T'}\right)}{\kappa}\sqrt{\frac{e^{2\kappa T}-1}{2\kappa\,T}}.$$

9. Consider a floorlet with reset date t_i and payment date t_{i+1}. Define $\Delta t = t_{i+1} - t_i$ and let $\bar{R}$ denote the floor rate.

Then, in the extended Vasicek model, at date 0 the values of the floorlet is:

$$\text{Floorlet Price}_0 = (1 + \bar{\mathcal{R}}\Delta t)P^{mkt}(0, t_{i+1})N(d_1) - P^{mkt}(0, t_i)N(d_2)$$

where

$$d_1 = \frac{\log\left(\frac{(1+\bar{\mathcal{R}}\Delta t)P^{mkt}(0,t_i+1)}{P^{mkt}(0,t_i)}\right) + \frac{1}{2}\sigma_{avg}^2 t_i}{\sigma_{avg}\sqrt{t_i}}$$

$$d_2 = d_1 - \sigma_{avg}\sqrt{t_i}\,,$$

where

- for the continuous-time Ho and Lee model $\sigma_{avg} = \sigma\Delta t$ and,
- for the Hull and White model

$$\sigma_{avg} = \frac{\sigma\, e^{-\kappa t_i}\left(1 - e^{-\kappa\Delta t}\right)}{\kappa}\sqrt{\frac{e^{2\kappa t_i} - 1}{2\kappa\, t_i}}.$$

10. We also discussed briefly the following continuous-time term structure models:

- General Hull and White;
- Cox, Ingersoll and Ross;
- Longstaff and Schwartz;
- Affine Models;
- Heath, Jarrow and Morton.

14.10 Portfolio Choice: Mean-Variance and Two-Date Models

1. Insights of modern portfolio theory

- Investors hold frontier portfolios; large asset base improves the frontier.
- An individual asset's risk is its contribution to portfolio risk.
  - ⋆ Variance of each asset contributes little to portfolio risk; covariances between assets determine portfolio risk.
- When there is risk-free asset, frontier portfolios are linear combinations of the risk-free asset and the tangent portfolio.
- The optimal mean-variance portfolio is given by
  $$w = \frac{1}{\gamma} \mathbf{V}^{-1} (\mu - r_F \mathbf{1}).$$
- For any asset i:
  - ⋆ Its risk is given by its beta with respect to a frontier portfolio.
  - ⋆ Its return per unit of risk is the premium on the frontier portfolio.

2. Implications of the Capital Asset Pricing Model (CAPM):

- The market portfolio is an efficient frontier portfolio.
- The portfolio frontier consists of portfolios of the market portfolio and the risk-free asset.
- Systematic risk is that part of total risk that is correlated with the market portfolio and cannot be diversified away, and so needs to be rewarded; unsystematic risk is not rewarded:
  $$\mathbb{E}^{\mathbb{P}}[r_i] - r_F = \beta_{im} \left(\mathbb{E}^{\mathbb{P}}[r_m] - r_F \right), \quad \text{where} \quad \beta_{im} = \sigma_{im} / \sigma_m^2.$$
  - ⋆ Two securities with the same total risk can have very different exposure to systematic risk.
- The Capital Market Line gives tradeoff between **portfolio** risk and return; the price of one unit risk, i.e. market price of risk for an efficient portfolio is the risk premium on market portfolio per unit of its volatility: $\frac{\mathbb{E}[r_m] - r_F}{\sigma_m}$.

3. Consumption-portfolio choice in a two-date model

- First-order condition for assets i and j in an optimal portfolio:

$$\mathbb{E}^{\mathbb{P}} U'[C_T] R_i = \mathbb{E}^{\mathbb{P}} U'[C_T] R_j, \quad \text{for any two assets } i \text{ and } j.$$

- First-order condition for optimal intertemporal consumption:

$$P_i \, U'(C_0) = \mathbb{E}^{\mathbb{P}} \left[\delta \, U'(C_T) X_i \right]$$

- The equation above implies that the price of any asset must be

$$P_i = \mathbb{E}^{\mathbb{P}} \left[\left(\frac{\delta \, U'(C_T)}{U'(C_0)} \right) X_i \right] = \mathbb{E}^{\mathbb{P}} \left[m_{0,T} \, X_i \right],$$

where $m_{0,T}$ is the marginal rate of substitution between initial and end-of-period consumption.

- We also have the result that:

$$\mathbb{E}^{\mathbb{P}} [R_i] - R_f = (-) \frac{\text{Cov} \left(U'(C_T), R_i \right)}{\mathbb{E}^{\mathbb{P}} \left[U'(C_T) \right]},$$

which says that:

⋆ the risk-premium decreases as the covariance between marginal utility and return increases;

⋆ that is, investors are willing to accept a lower risk premium from those assets whose returns come in states with high marginal utility (that is, low consumption).

14.11 Portfolio Choice Using the Martingale Approach

1. Under the martingale approach to consumption and portfolio choice:

- We view the optimally invested wealth as an asset that provides a dividend equal to the consumption of the investor.

- Thus, current wealth must equal the expected discounted value of the stream of future consumption (dividends) plus discounted terminal wealth:

$$W_t = \mathbb{E}_t^{\mathbb{P}}\left[\int_t^T \frac{M_s}{M_t} C_s ds + \frac{M_T}{M_t} W_T\right],$$

  which is the static budget constraint.

- The objective of the investor, subject to the static budget constraint, is:

$$\max_{C_s, W_T} \mathbb{E}_t^{\mathbb{P}}\left[\int_t^T U(C_s, s) ds + \text{Beq}(W_T, T)\right] + \lambda_{BC}\left(M_t W_t - \mathbb{E}_t^{\mathbb{P}}\left[\int_t^T M_s C_s ds + M_T W_T\right]\right).$$

- The first-order conditions for C_s and W_T are:

$$\begin{aligned} \partial U(C_s, s)/\partial C_s &= \lambda_{BC} M_s \quad \Longleftrightarrow \quad C_s = g(\lambda_{BC} M_s, s) \\ \partial \text{Beq}(W_T, T)/\partial W_T &= \lambda_{BC}\, M_T \quad \Longleftrightarrow \quad W_T = G(\lambda_{BC}\, M_T, T). \end{aligned}$$

- In the expressions above, λ_{BC} can be identified by substituting the expressions for optimal consumption and optimal terminal wealth into the static budget constraint.

- The first-order condition for the optimal portfolio weight vector is:

$$\mathbf{w} = -\frac{W_M\, M}{W}\mathbf{V}^{-1}(\boldsymbol{\mu} - r\mathbf{1}) + \frac{W_x}{W}(\boldsymbol{\sigma}^\top)^{-1}\boldsymbol{\sigma}_x,$$

  which is the same as the one we will obtain using dynamic programming when markets are complete.

- In the expression above, wealth$W(M, x, t)$ (and hence also its derivatives, W_M and W_x) can be identified by solving a linear partial differential equation.

14.12 Portfolio Choice Using Dynamic Programming

1. The objective of the investor is to

   - to maximize the expected lifetime utility from consumption and bequest,

$$\max_{C_s,\{w_{is}\}} \mathbb{E}_t^{\mathbb{P}}\left[\int_t^T U(C_s, s)ds + \text{Beq}(W_T, T)\right]; \tag{14.1}$$

   - by choosing the optimal consumption and portfolio policy,
   - subject to the intertemporal budget constraint:

$$dW = W\sum_{i=1}^{n} w_i\,(\mu_i - r)\,dt + (r\,W - C)\,dt + W\sum_{i=1}^{n} w_i\sigma_i\,dB_{s_i}^{\mathbb{P}}.$$

- The first-order condition for consumption is

$$0 = U_C - J_W \qquad \text{or equivalently} \qquad U_C = J_W.$$

which says that the investor consumes up to the point that the marginal utility from consumption is equal to the marginal (indirect) utility from saving, that is, from having an additional unit of wealth.

- The first-order condition for portfolio weight w_i is

$$w_i = \overbrace{\left(\frac{-J_W}{W\,J_{WW}}\right)}^{1/[\text{relative risk aversion}]} \overbrace{\sum_{j=1}^{n} v_{ij}(\mu_j - r)}^{\text{mean-var portfolio}} - \overbrace{\left(\frac{\overbrace{J_{Wx}}^{\text{Effect of } x \text{ on } J_W}}{W\,J_{WW}}\right) \sum_{j=1}^{n} v_{ij}\,\overbrace{\sigma_{jx}}^{\substack{\text{covariance of} \\ x \text{ with asset } j}}}^{\text{intertemporal hedging portfolio}}.$$

2. Special cases with zero intertemporal hedging:

- If the investment opportunity set is constant, then
  - ⋆ the investor holds only the myopic portfolio,
  - ⋆ because there is nothing to hedge.

- If the investor has an elasticity of intertemporal substitution equal to 1 (for example, if investor has log utility), then
  - ⋆ the investor holds only the myopic portfolio,
  - ⋆ because the substitution effect cancels out the income effect, leading to a net zero demand for hedging.

14.13 Partial- and General-Equilibrium Asset Pricing

1. The Consumption CAPM states that:

$$(\mu_n - r)dt = \alpha \left(\frac{d\mathcal{P}^{S_n}}{\mathcal{P}^{S_n}} \right) (dC),$$

where α is the (usually stochastic) coefficient of absolute risk aversion, and C is aggregate consumption in the economy.

2. The Intertemporal CAPM developed by Merton (1973a) states that:

$$(\mu_n - r)dt = \alpha \left(\frac{d\mathcal{P}^{S_n}}{\mathcal{P}^{S_n}} \right) (dW) + \eta \left(\frac{d\mathcal{P}^{S_n}}{\mathcal{P}^{S_n}} \right) (dX)\,; \tag{13.31}$$

that is, the risk premium on a security is proportional to the conditional covariance with aggregate (market) wealth and the state variable X.

3. The Intertemporal CAPM simplifies to the conditional CAPM when $J^h_{WX} = 0$ for each investor h and each state variable X:

$$(\mu_n - r)dt = \alpha \left(\frac{d\mathcal{P}^{S_n}}{\mathcal{P}^{S_n}} \right) (dW)\,.$$

or, written in its more traditional form:

$$(\mu_{n,t} - r_t)dt = \beta_{mkt,n,t}\,(\mu_{mkt,t} - r_t)\,,$$

which looks like the traditional single-period CAPM,
but is conditional on information at date t.

4. In the continuous-time version of the general equilibrium endowment economy Lucas model:

$$r = \delta + \gamma\,\mu_Y - \frac{1}{2}\gamma(1+\gamma)\sigma_Y^2,$$

$$\frac{\mu_R - r}{\sigma_R} = \gamma\,\sigma_Y.$$

5. In the general equilibrium model of a production economy studied in Cox, Ingersoll, and Ross (1985b):

$$r = \mu - \gamma\,\sigma^2,$$

$$\frac{\mu - r}{\sigma} = \gamma\,\sigma.$$

Lecture Notes for 2025

Continuous Time Financial Economics

Appendix A

Practice Questions for Part 2

Raman Uppal

EDHEC Business School

Enclosed below are some practice questions that will help you improve your understanding of the material covered in class; your answers to these questions do **not** have to be submitted to me (and the solutions are included below).

1. Suppose that you are given the following prices of discount bonds:

$$\begin{aligned}\mathcal{P}^Z(0,0.5) &= 0.995;\\ \mathcal{P}^Z(0,1.0) &= 0.988;\\ \mathcal{P}^Z(0,1.5) &= 0.978;\\ \mathcal{P}^Z(0,2.0) &= 0.966.\end{aligned}$$

Find the continuously-compounded yields for the discount bonds with maturities 0.5 years and 1.5 years.

From the result we derived in class, we know that

$$y = \frac{-1}{\tau}\log \mathcal{P}^Z(t,T).$$

- Thus, the yield for a discount bond with maturity 0.5 years is

$$y(0.5) = \frac{-1}{0.5}\log(0.995) = 0.010025 \text{ p.a.}$$

- Similarly, the yield for a discount bond with maturity 1.5 years is

$$y(1.5) = \frac{-1}{1.5}\log(0.978) = 0.014830 \text{ p.a.}$$

2. Suppose that you are given the following discount-bond prices:

$$\begin{aligned}\mathcal{P}^Z(0,0.5) &= 0.995;\\ \mathcal{P}^Z(0,1.0) &= 0.988.\end{aligned}$$

Compute the six-month and one-year LIBOR rates.

- We know that

$$\begin{aligned}\mathcal{P}^Z(0,u) &= \frac{1}{1+\mathcal{R}_i u}\\ \mathcal{R}_i &= \frac{\frac{1}{\mathcal{P}^Z(0,u)}-1}{u}.\end{aligned}$$

- So, using the result in the expression above

$$\text{6-month LIBOR} = \frac{\frac{1}{\mathcal{P}^Z(0,0.5)}-1}{0.5} = \frac{\frac{1}{0.995}-1}{0.5} = 0.010050 \text{ p.a.}$$

- Similarly,

$$\text{1-year LIBOR} = \frac{\frac{1}{\mathcal{P}^Z(0,1.0)}-1}{1.0} = \frac{\frac{1}{0.988}-1}{1.0} = 0.012145 \text{ p.a.}$$

3. Suppose that you are given the following discount-bond prices:

$$\mathcal{P}^Z(0,0.5) = 0.995;$$
$$\mathcal{P}^Z(0,1.0) = 0.988;$$
$$\mathcal{P}^Z(0,1.5) = 0.978;$$
$$\mathcal{P}^Z(0,2.0) = 0.966.$$

Compute the spot swap rate for a two-year swap with semi-annual payments.

- We know from the results derived in class that

$$\mathcal{P}^Z(t,t_0) - \mathcal{P}^Z(t,t_N) = \mathcal{R}^{swap}(t)\Delta t \sum_{i=1}^{N} \mathcal{P}^Z(t,t_i),$$

$$\begin{aligned}
\mathcal{R}^{swap}(t) &= \frac{\mathcal{P}^Z(t,t_0) - \mathcal{P}^Z(t,t_N)}{\Delta t \sum_{i=1}^{N} \mathcal{P}^Z(t,t_i)} \\
&= \frac{\mathcal{P}^Z(t_0,t_0) - \mathcal{P}^Z(t_0,t_N)}{\Delta t \sum_{i=1}^{N} \mathcal{P}^Z(t_0,t_i)} \\
&= \frac{\mathcal{P}^Z(0,0) - \mathcal{P}^Z(0,2.0)}{0.5 \sum_{i=1}^{4} \mathcal{P}^Z(0,i/2)} \\
&= \frac{1 - 0.9666}{0.5\,(0.995 + 0.988 + 0.978 + 0.966)} \\
&= 0.017316 \text{ p.a.}
\end{aligned}$$

4. Suppose that you are given the following discount-bond prices:

$$\begin{aligned}
\mathcal{P}^Z(0,0.5) &= 0.995;\\
\mathcal{P}^Z(0,1.0) &= 0.988;\\
\mathcal{P}^Z(0,1.5) &= 0.978;\\
\mathcal{P}^Z(0,2.0) &= 0.966;\\
\mathcal{P}^Z(0,2.5) &= 0.951;\\
\mathcal{P}^Z(0,3.0) &= 0.935.
\end{aligned}$$

Compute the *forward swap rate* for a two-year swap with semi-annual payments beginning one year from now.

- We know from our work in class that:

$$\mathcal{P}^Z(t,t_0) = \mathcal{P}^Z(t,t_N) + \mathcal{R}^{swap}(t)\Delta t \sum_{i=1}^{N} \mathcal{P}^Z(t,t_i),$$

$$\begin{aligned}
\mathcal{R}^{swap}(t) &= \frac{\mathcal{P}^Z(t,t_0) - \mathcal{P}^Z(t,t_N)}{\Delta t \sum_{i=1}^{N} \mathcal{P}^Z(t,t_i)}\\
&= \frac{\mathcal{P}^Z(0,1.0) - \mathcal{P}^Z(0,3.0)}{\Delta t \sum_{i=1}^{N} \mathcal{P}^Z(0,1+i/2)}\\
&= \frac{\mathcal{P}^Z(0,1.0) - \mathcal{P}^Z(0,3.0)}{0.5 \sum_{i=1}^{4} \mathcal{P}^Z(0,1+i/2)}\\
&= \frac{0.988 - 0.935}{0.5\,(0.978 + 0.966 + 0.951 + 0.935)}\\
&= 0.027676 \text{ p.a.}
\end{aligned}$$

5. Consider a two-year coupon bond with \$1 face value and a semi-annual coupon of 0.03, with the first coupon being six months away. Compute

5.1. the price of the bond;

5.2. the yield to maturity of the bond (using Solver in Excel); and,

5.3. the duration of the bond.

You are given the following following discount-bond prices:

$$\begin{aligned}\mathcal{P}^Z(0,0.5) &= 0.995;\\ \mathcal{P}^Z(0,1.0) &= 0.988;\\ \mathcal{P}^Z(0,1.5) &= 0.978;\\ \mathcal{P}^Z(0,2.0) &= 0.966.\end{aligned}$$

5.1. The price of a coupon bond is the present value of all the cash flows to the bond, which consist of the coupons and the terminating principal:

$$\begin{aligned}\mathcal{P}^B &= (0.995 \times 0.03) + (0.988 \times 0.03) + (0.978 \times 0.03) + (0.966 \times 1.03)\\ &= \$1.08381.\end{aligned}$$

5.2. The expression for the yield to maturity is:

$$\mathcal{P}^B = \sum_{j=1}^{N} e^{-\mathbf{y}\tau_j} C_j.$$

which we can write explicitly as

$$1.08381 = \sum_{j=1}^{4} e^{-\mathbf{y}(j/2)} C_j, \quad \text{with } C_1 = C_2 = C_3 = 0.03; C_4 = 1.03.$$

We now need to solve for $\mathbf{y}$ (in Excel, you can use Solver to do this) and the solution we get is:

$$\mathbf{y} = 0.017117.$$

5.3. Recall the definition of duration:

$$\text{Duration} = \sum_{j=1}^{N} \tau_j \times \frac{e^{-\mathbf{y}\tau_j} C_j}{\mathcal{P}^B}$$

We compute each term in the definition of duration, noting that we have already computed $\mathcal{P}^B$ and $\mathbf{y}$:

Date	Cashflow	Discount Factor	Weight	Contribution to Duration
τ_j	C_j	$e^{-\mathbf{y}\tau_j}$	$\frac{e^{-\mathbf{y}\tau_j}C_j}{\mathcal{P}^B}$	$\tau_j \times \frac{e^{-\mathbf{y}\tau_j}C_j}{\mathcal{P}^B}$
0.5	0.03	0.991477911	0.027444236	0.013722118
1.0	0.03	0.983028449	0.027210354	0.027210354
1.5	0.03	0.974650993	0.026978465	0.040467697
2.0	1.03	0.966344931	0.918366946	1.836733891
				1.918134

6. Calculate the value of a 3×10 European *receiver* swaption (an option maturing in 3 years to enter into a 10-year swap as the receiver):

- with the underlying swap having semiannual cash flows,
- the fixed rate being equal to 0.045
- the forward swap rate having a constant volatility equal to 0.10, and
- the prices of discount bonds of different maturities being as given below.

Note that the data below has been typed in such a way that you can cut it and paste it into an Excel spreadsheet. Then, using the command "Data" $\rightarrow$ "Text to Columns" and choosing the option "Fixed width" will give you two columns with the data below. If you have trouble cutting-and-pasting, please send me an email asking for an Excel spreadsheet with the data.

Maturity	Discount Bond Price
0.5	0.993734709
1.0	0.985033134
1.5	0.974047703
2.0	0.960943178
2.5	0.945893780
3.0	0.929080399
3.5	0.910687928
4.0	0.890902754
4.5	0.869910453
5.0	0.847893704
5.5	0.825030437
6.0	0.801492231
6.5	0.777442961
7.0	0.753037689
7.5	0.728421802
8.0	0.703730365
8.5	0.679087699
9.0	0.654607146
9.5	0.630391011
10.0	0.606530660
10.5	0.583106753
11.0	0.560189589
11.5	0.537839549
12.0	0.516107604
12.5	0.495035897
13.0	0.474658349

- The value of European receiver swaption at date 0 is

$$= \left[\bar{\mathcal{R}}\Delta t \sum_{i=1}^{N} \mathcal{P}^Z(0,t_i)\right] N(-d_2) - \left[\mathcal{P}^Z(0,t_0) - \mathcal{P}^Z(0,t_N)\right] N(-d_1)$$

$$d_1 = \frac{\log\left(\frac{\mathcal{P}^Z(0,t_0)-\mathcal{P}^Z(0,t_N)}{\bar{\mathcal{R}}\Delta t \sum_{i=1}^{N} \mathcal{P}^Z(0,t_i)}\right) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}; \qquad d_2 = d_1 - \sigma\sqrt{T}.$$

- We first calculate d_1 and d_2.

$$d1 = \frac{\log\left[\frac{\mathcal{P}^Z(0,3)-\mathcal{P}^Z(0,13)}{0.045\times 0.5\times \Sigma_{i=1}^{20} \mathcal{P}^Z(0,3+i/2)}\right] + \frac{1}{2}\times 0.1^2 \times 3}{0.1\sqrt{3}} = 2.26614$$

$$d_2 = d_1 - 0.1\sqrt{3} = 2.09294.$$

- Plugging these values back into the formula, we get that the value of the European receiver swaption at date $t = 0$ is 0.000336499.

7. In Lecture 7, we derived the price of a discount bond in the Vasicek model for the special case of $\kappa = 0$. Please derive the price of a discount bond for the case where $\kappa \neq 0$. The expressions you need to derive are (17.6b) and (17.6c) on page 336 of the textbook by K. Back, "Asset Pricing and Portfolio Choice Theory."

8. In Lecture 8, in the section titled "Martingale Approach to Consumption and Portfolio Choice in Discrete Time" we derived the optimal consumption and portfolio of an investor whose lifetime utility is given by

$$U(C_0) + \mathsf{E}\,U(C_2).$$

That is, lifetime utility depends only on consumption at dates 0 and 2. Please derive the optimal consumption and portfolio rules for the same numerical example, assuming now that lifetime utility depends on consumption at dates 0, 1, and 2; that is,

$$U(C_0) + \mathsf{E}\,U(C_1) + \mathsf{E}\,U(C_2).$$

9. Question 13.3 from Chapter 13 of Kerry Back's book "Asset Pricing and Portfolio Choice Theory" (first edition).

Lecture Notes for 2025

Continuous Time Financial Economics

Appendix B

Homework Questions for Part 2 . . . To be handed in

Raman Uppal

EDHEC Business School

Instructions for Homework Questions:

To get credit for your homework, you need to follow these instructions.

- You need to submit your solutions to only the **five** questions given below. Most of these questions are from previous exams, so working on these questions will give you an idea of what to expect on the exam. (Of course, the questions that appear on your exam will *appear* to you to be much more difficult!)
- Each question has several parts that are designed to help you get to the final answer.
- Each question is worth 10 points, and the total homework is worth 50 points.
- If you think that there is any error in the questions, please send me an email.
- You should work on these questions independently; this is *not* group work. You are welcome to consult the class notes (lecture slides) and any text book, but please do not search for solutions to these questions on the internet.
- When submitting your work, please submit only a PDF file (please do not send .doc files, because they may appear differently on my computer compared to what they appear on your computer; in any case, it is best to use LaTeX).
- Please name your file using the following convention: `LASTNAME-Firstname-HW2.pdf`. For example, if I were submitting a file it would be named: `UPPAL-Raman-HW2.pdf`.

- Please type your answers legibly and neatly – if I cannot read your work, I will not be able to give you credit for it. This is a good opportunity to learn LaTeX.
- Please use a font size of 12 points and margins of 1.5 inches.
- Please start each question on a new page.
- Your homework should be emailed to `brigitte.bogaerts-chevillotte@edhec.edu`. Please do NOT copy me on your email.
- The deadline for submitting your answers to these questions is 2 June 2025 (the deadline for the homework for the course on *Financial Econometrics* may be different).
- Assignments will not be accepted (or graded) after the deadline and a grade of 0 will be assigned in that case.
- The LaTeX file with these questions is available from Brigitte; please email her if you do not find it in the course folder on Blackboard. You can use this file to start your homework; that way, you will not need to retype the questions.

1. Derivative pricing and hedging

In this question, we study the prices of bonds of different maturities in the Vasicek (1977) model. We will assume that there is a single source of uncertainty. Let this "single factor" be the yield $r(t)$ on the instant-maturity bond.

- Denote by t the date today, by T_k the maturity date of the kth bond, and by $\tau_k = T_k - t$ the time-to-maturity of this bond.
- Denote by $P(t, \tau_k)$ the price of a discount bond that makes a single payment of $\$1$ in τ_k periods from today.
- Thus, the price of a discount bond with 0 time to maturity is: $P(t, 0) = 1$, and the instantaneous rate of return on a bond is given by: $\frac{dP(t,\tau_k)}{P(t,\tau_k)}$. The instantaneous yield, $r(t)$, then is defined as
$$\lim_{\tau \to 0} \frac{dP(t,\tau)}{P(t,\tau)} = r(t)\, dt.$$
- The Vasicek model assumes that $r(t)$ follows an Ornstein-Uhlenbeck process:
$$dr(t) = \kappa\big[r_\infty - r(t)\big]\, dt + \sigma_r\, dB_r,$$
where κ, r_∞ and σ_r are positive constants, and B_r is a Brownian motion under the physical measure.

Please answer the following questions.

1.1. Assume that prices for bonds of all maturities depend on only a single factor, $r(t)$. Denote the price of a τ-maturity bond by $P[r(t), \tau_k]$. Then, write down the process for the return on this bond, that is, $\frac{dP[r(t),\tau_k]}{P[r(t),\tau_k]}$; that is, find the mean and volatility of the bond's rate of return, denoted by $\mu_p(r, \tau_k)$ and $\sigma_p(\tau_k)$, respectively. *Note that this question does not require you to solve for $P[r(t), \tau_k]$ in terms of the interest rate; your answer should be in terms of $P[r(t), \tau_k]$ and its derivatives.*

1.2. The Sharpe ratio (or, equivalently, the market price of risk) is the expected return in excess of the risk-free rate divided by the volatility of the return. Assume that the Sharpe ratio of a bond is constant over time; denote this constant by SR. Use the expressions for the mean and volatility of the bond's rate of return to derive the differential equation in terms of SR that bonds of all maturities must satisfy in this model.

1.3. Consider a portfolio $H(t)$ that consists of two assets: 1 unit long of a bond of maturity τ_1 and x units short of a bond with maturity τ_2. Write down the rate of return on this portfolio. Find the x that makes the rate of the return on portfolio $H(t)$ is risk-free.

1.4. Use the expression for the rate of return on portfolio $H(t)$ to derive an expression relating the Sharpe ratio of the first bond with maturity τ_1 to the Sharpe ratio of the second bond with maturity τ_2. Provide an economic interpretation for this condition.

2. Portfolio choice using dynamic programming

We consider a securities market model with two securities.

- The first security is a bond, with price process
$$\frac{dR(t)}{R(t)} = r\,dt,$$
where r is a constant.

- The second security is a stock, with price process
$$\frac{dS(t)}{S(t)} = \mu(t)dt + \sigma\,dB(t),$$
where σ is a constant, and $\mu(t)$ is given by
$$d\mu(t) = \kappa\big(\theta - \mu(t)\big)dt - s\,dB(t),$$
for some constants κ, θ, and s; that is, the expected return on the stock is stochastic.

- An investor with initial wealth W, maximizes utility over terminal consumption, where utility has constant relative risk aversion:
$$U\big(C(T)\big) = \frac{\big[C(T)\big]^{1-\gamma}}{1-\gamma}.$$

Based on this information, please answer the following questions.

2.1. Define $\eta(t)$ as
$$\eta(t) \equiv \frac{\mu(t) - r}{\sigma},$$
and $\theta_\eta \equiv (\theta - r)/\sigma$ and $s_\eta \equiv s/\sigma$. Determine the process followed by $\eta(t)$.

2.2. Formulate the investor's problem as a dynamic programming problem, denoting by w the fraction of wealth invested in the risky asset at time t. That is, write the objective function and the dynamic budget constraint.

2.3. Explain why the investor's value function can be written in the form $J\big(W(t), \eta(t), t\big)$; that is, why is J a function of $W(t)$, $\eta(t)$, and t. Then, write down the Bellman equation that characterizes $J\big(W(t), \eta(t), t\big)$.

2.4. From the Bellman equation, find the expression for the optimal proportion of wealth invested in the risky asset, $w(t)$, and provide an interpretation for it.

2.5. Explain whether or not this problem can be solved using the martingale pricing approach, where the budget constraint is written in a static rather than dynamic form.

3. Intertemporal consumption and portfolio choice with two risky assets

- We consider the intertemporal consumption-portfolio problem of an investor "a".
- The objective of this investor is to choose her portfolio policy and consumption policy to maximize a time-separable expected utility function that depends on the rate of consumption at all future dates:

$$\max E_0 \left[\int_0^\infty e^{-\delta s} \frac{C^{1-\gamma_a}}{1-\gamma_a} ds \right].$$

- Suppose that the investor can allocate wealth across 2 risky assets and a bank account. The price S_n of each of the risky assets is given by:

$$\frac{dS_n}{S_n} = \mu_n dt + \sigma_n dB_n, \qquad n = \{1, 2\},$$

where μ_n and σ_n are constants, B_n is a Brownian motion (under the physical measure), and the correlation between B_1 and B_2 is zero.

- Let the risk-free interest rate earned on money invested in the bank account be denoted by r, which is constant.
- Denote the value function of the investor by J and the proportion of wealth allocated to the two risky assets by w_1 and w_2.

Based on the above, please answer the following three questions.

3.1. Identify the state variables of the investor's problem and write the dynamic optimization problem of the investor along with his budget constraint.

3.2. What is the Bellman equation that characterizes the solution to the investor's problem?

3.3. Derive the first-order conditions for the investor's portfolio problem.

3.4. Show that the investors's optimal portfolio policy can be written as an investment in two funds: (i) a fund consisting of *only* the risk-free asset; (ii) a fund consisting of *only* the two risky assets (please identify the weights assigned to each of the two risky assets in this fund; denote these weights by x_1 and x_2).

3.5. Now consider the intertemporal consumption-portfolio problem of investor "b" who wishes to maximize:

$$\max E_0 \left[\int_0^\infty e^{-\delta s} \frac{C^{1-\gamma_b}}{1-\gamma_b} ds \right].$$

Will this investor choose to invest in the same two funds as investor "a" or will she want to invest in funds that are different? Explain why.

4. Intertemporal consumption, portfolio, and labor choice

- Consider an individual's intertemporal consumption, *labor*, and asset-allocation problem for the case of a risk-free asset and a single risky asset.
- The individual maximizes her expected lifetime utility, which is of the form:
$$E_0\left[\int_0^T e^{-\delta t}u(C_t, L_t)dt\right],$$
where C_t is the individual's consumption at date t and L_t is the amount of effort that the individual exerts at date t. The function $u(C_t, L_t)$ is an increasing concave function of C_t but a decreasing concave function of L_t.
- The risk-free asset pays a constant rate of return equal to r per unit of time.
- The price of the risk asset $S(t)$ satisfies the process:
$$\frac{dS(t)}{S(t)} = \mu_s\,dt + \sigma_s\,dB_s, \quad \text{where } \mu_s \text{ and } \sigma_s \text{ are constants.}$$
- For each unit of labor effort exerted at date t, the individual receives an instantaneous flow of labor income of $L(t)\,y(t)\,dt$. The return to effort, or the wage rate, $y(t)$, is stochastic and follows the process:
$$dy(t) = \mu_y\,dt + \sigma_y\,dB_y, \quad \text{where } \mu_y \text{ and } \sigma_y \text{ are constants and } (dB_s)\times(dB_y) = \rho\,dt.$$

Based on the above, please answer the following questions.

4.1. Letting W denote the investor's wealth and w the proportion of wealth invested in the risky asset, write down the dynamic (intertemporal) budget constraint for this problem.

4.2. Denote the value function by J. What are the state variables for the optimization problem of the investor? Write down the Bellman equation?

4.3. Write the first-order conditions that the optimal consumption, labor-effort, and portfolio policies must satisfy.

4.4. Show how the optimal demand for the risky asset, w, can be written as the sum of two terms: one term that would be present even if y was constant and another term that is present only because y is stochastic. Provide an economic interpretation for the two terms.

5. Portfolio choice with inflation

- Consider the following consumption and portfolio choice problem. An individual must choose between two different assets, a stock and a short (instantaneous) maturity, default-free bond. *In addition, the individual faces a stochastic rate of inflation*, that is, uncertain changes in the price level (e.g., the Consumer Price Index) denoted by $P(t)$.
- The *price level* (currency price of the consumption good) follows the process
$$\frac{dP(t)}{P(t)} = \mu_\pi\, dt + \sigma_\pi\, dB_\pi(t).$$
- The nominal (currency value) of the stock is $S(t)$ and it satisfies the process
$$\frac{dS(t)}{S(t)} = \mu_S\, dt + \sigma_S\, dB_S(t).$$
- The nominal (currency value) of the bond (bank deposit) is given by $D(t)$. It pays an instantaneous nominal rate of return equal to r. Hence, its nominal price satisfies the process
$$\frac{dD(t)}{D(t)} = r\, dt.$$
Assume that μ_π, σ_π, μ_S, σ_S, and r are constant and that the correlation ρ between the two Brownian motions is also constant: $dB_\pi(t) \times dB_S(t) = \rho\, dt$.

Based on the above, please answer the following questions.

5.1. Denote the real *price* of the stock by $R^S(t) = S(t)/P(t)$, and denote the real *price* of the bank deposit by $R^D(t) = D(t)/P(t)$. What processes do the real (consumption-good value) rates of return on the stock and the bond satisfy; that is, find $dR^S(t)/R^S(t)$ and $dR^D(t)/R^D(t)$?

5.2. Let $c(t)$ be the individual's date t *real* rate of consumption and let $w(t)$ be the proportion of real wealth, $W(t)$, that is invested in the stock. Derive the process followed by real wealth. That is, find the dynamic budget constraint.

5.3. Assume that an infinitely-lived individual chooses $c(t)$ and $w(t)$ to solve the following:
$$\max_{c(t),w(t)} E_0 \int_0^\infty U\big[c(t), t\big]\, dt,$$
subject to the dynamic budget constraint derived in Q 5.2 above. Assuming that $U\big[c(t), t\big]$ is a concave utility function, solve for the individual's optimal choice of $w(t)$ in terms of the indirect utility-of-wealth function, J.

5.4. How does the optimal portfolio weight $w(t)$ vary with the correlation ρ? What is the economic intuition for this comparative-static result?

Blank page.

Lecture Notes for 2025

Continuous Time Financial Economics

Appendix C

Information about the Final Exam

Raman Uppal

EDHEC Business School

C.1 Information about the exam

- The exam tests your understanding of the material covered in class
  - ⋆ The exam does this by asking you to apply the material covered in class to *new problems* that were not covered in class.
- There are **two questions** on the exam.
  - ⋆ The first question is on the first part of the course
    - ∗ Material on valuation and hedging of options (options on equities, exchange rates, interest rates, swaps, etc.).
  - ⋆ The second question is on the second part of the course
    - ∗ Material on portfolio selection and equilibrium asset pricing.
- Each question consists of several parts (usually three or four); the parts are designed to guide you to the final answer.
- The two questions have equal weight on the exam.

C.2 Rules for the exam

- You have two hours to work on the exam.
- The exam is closed book (no notes allowed and no AI).
- During the exam, you are not allowed to have the following with you:
  - ⋆ Any sheet with notes on the course material;
  - ⋆ The slides (lecture notes) for the course;
  - ⋆ Any textbook;
  - ⋆ A mobile phone.
  - ⋆ Access to the internet.

C.3 Suggestions on how to prepare for the exam

- The exam is designed to test your understanding of the material that we covered in class;
  - ⋆ The exam does this by asking you to apply the material covered in class to two new problems;
- The above implies that
  1. You must understand very well the material covered in class
  2. You must be well rested so that you can think clearly on how to apply the material covered in class to the new problems.
- So, in my view, the best way to prepare for the exam is to:
  - ⋆ Review carefully the material covered in the course slides and homeworks;
  - ⋆ Sleep well the night before the exam and stay relaxed during the exam so that you can think.

Blank page.

Lecture Notes for 2025

Continuous Time Financial Economics

Appendix D

Sample Final Exam

Raman Uppal

EDHEC Business School

Instructions for the final exam

1. Start by putting your name on the first page of both handouts that you have received—without that, it will not be possible to give you credit for the exam.

2. You must return **all** the material you have received (questions + answers) to get credit.

3. Credit will be given for providing the logic underlying your answer, so please explain clearly the reasoning behind your answer.

4. I can only give points for material that I can read, understand, and interpret. So, it is to your advantage to express your ideas clearly, and to highlight key arguments and quantities that you would like me to focus on when reading the exam.

5. Please put your answer in the boxed space provided on the answer sheet—this is where I will be looking for your response. There is space below the boxes to show your work.

6. There is no reward for long answers, and incorrect/illogical statements will be penalized. So, please keep your answers brief and to the point.

7. If something is not clear, please make a reasonable assumption and continue. Please highlight your assumptions so that I can identify them when grading the exam.

Rules for the exam

1. You have **two hours** to work on the exam.
2. You are **not allowed** to have the following with you during the exam:
   - Any sheet with handwritten or typed notes;
   - Any slides (lecture notes) for the course;
   - Any books;
   - A mobile phone.
   - Access to the internet.

Q1. Options and hedging

- In class, we studied the Black-Scholes model of option pricing that assumes that the interest rate is constant. We also studied Merton's model of option pricing that does not require the assumption of a constant interest rate; the Merton model also allows for volatility that is changing over time, as long as the changes are deterministic (non-stochastic).
- This question is about *implementing* Merton's formula when the discount bond price is given by Vasicek's model (that is, we analyze just the expression for volatility for implementing this option-pricing model).
- Suppose the price $P^S(t)$ of a non-dividend paying stock has a constant volatility σ_S.
- Let the price of a discount bond be denoted by $P^D(t,T)$.
- Let the constant $\kappa > 0$ denote the rate of mean reversion of the short rate process and the constant $\phi > 0$ the (absolute) volatility of the short rate process.
- Assume that that volatility at date t of a discount bond maturing at $T > t$ is
$$\sigma_D(t) = \frac{\phi}{\kappa}\left(1 - e^{-\kappa(T-t)}\right).$$
- Assume that the return on the stock and bond have constant correlation denoted by ρ.

Please answer the following questions, each of which is worth 2.5 points.

Q1.1 Find the volatility at t of $\frac{P^S(t)}{P^D(t,T)}$; that is, the volatility term of the process
$$\frac{d\left(\frac{P^S(t)}{P^D(t,T)}\right)}{\frac{P^S(t)}{P^D(t,T)}}.$$
Denote this volatility of $\frac{P^S(t)}{P^D(t,T)}$ by $\boldsymbol{\sigma}(t)$.

Q1.2 Define
$$\boldsymbol{\sigma}_{avg} = \sqrt{\frac{1}{T}\int_0^T [\boldsymbol{\sigma}(t)]^2\, dt}.$$
Show that
$$\boldsymbol{\sigma}_{avg}^2 = \sigma_S^2 + \frac{1}{\kappa^2}\left[\phi^2 - 2\kappa\rho\sigma_S\phi - \left(2\phi^2 - 2\kappa\rho\sigma_S\phi\right)\left(\frac{1-e^{-\kappa T}}{\kappa T}\right) + \phi^2\left(\frac{1-e^{-2\kappa T}}{2\,\kappa T}\right)\right].$$

Q1.3 For small T, use l'Hôpital's rule to show that $\boldsymbol{\sigma}_{avg} \approx \sigma_S$.

Q1.4 For large T, show that $\boldsymbol{\sigma}_{avg} > \sigma_S$ if ρ is sufficiently small.

END of first question ♠

Q2. Portfolio choice and general equilibrium asset pricing

- Consider an infinite-horizon representative-agent economy (that is, all agents are identical) with time-separable power utility function:

$$U(C,t) = e^{-\delta t}\frac{C_t^{1-\gamma}}{1-\gamma}.$$

- In our course, we studied an "endowment" economy, where the amount of output available for consumption was given *exogenously*.
- In this question, we study a "production" economy, where there is a single good (such as "corn") that can either be consumed or invested in the production process. The representative investor chooses how much to consume and how much to invest each instant; the investment this instant then determines the output next instant. Thus, in this economy output is determined *endogenously*.
- We assume that the production sector has constant returns to scale. The output each instant, denoted by $Y(t)$, depends on the capital stock, $K(t)$, as described by $\frac{dY(t)}{Y(t)} = \mu\, K(t)\, dt + \sigma\, K(t)\, dB_t$, where μ, σ are constants; $B(t)$ is a Brownian motion.
- The change in the capital stock is the change in the aggregate output, less consumption:

$$\frac{dK(t)}{K(t)} = \frac{dY_t}{Y_t} - C(t)\, dt.$$

- Each investor in this economy can choose to invest in this production process or to borrow/lend at the risk-free rate $r(t)$, which will be determined endogenously in equilibrium.
- Each investor also needs to decide how much to consume each period (with the remainder of the aggregate output reinvested).

Based on the above, please answer the following questions, each of which is worth 2.5 points.

Q2.1 Write the optimization problem of the representative agent along with the dynamic budget constraint that this optimization problem must satisfy.

Q2.2 Denote the value function by V. What are the state variables for the optimization problem of the representative investor? What is the Bellman equation?

Q2.3 Write the first-order conditions that the optimal consumption policy and portfolio policy of the representative investor must satisfy.

Q2.4 Use the market-clearing condition (that in equilibrium aggregate borrowing must equal zero) to obtain the risk-free rate in equilibrium. Use this to obtain the expression for the risk premium. Then, provide an economic interpretation for the risk-premium.

END of second question ♠

Lecture Notes for 2025

Continuous Time Financial Economics

Appendix E

Guide To Notation

Raman Uppal

EDHEC Business School

E.1 Roman letters

Symbol	Definition	Appears for first time in Chapter
1_A	random variable whose value is 1 if the event (set of states) A occurs, 0 otherwise	1
$B(t)$	Brownian motion's level at t	2
c or C	Consumption	1
Cov$[x, y]$	Covariance between x and y	3
E	Expectation operator (unconditional, at $t = 0$)	1
E_t	Conditional expectation operator at t	1
$\mathcal{E}$	Elasticity of an option	1
f	Exercise boundary for an American option	5
K	Strike price of an option	1
m	Stochastic discount factor (SDF) or state price density in a single-period (two-date) model	1
M_t	Stochastic discount factor (SDF) process	3
$\mathcal{P}$	Price of an asset	1
$\mathcal{P}^{A_k}$	Price of Arrow-Debreu state security paying one unit in state k	1
$\mathcal{P}^B$	Price of a coupon bond	6
$\mathcal{P}^C$	Price of a call option	1
$\mathcal{P}^D$	Price of bank deposit or money-market account; $\mathcal{P}_0^D = 1$	1
$\mathcal{P}^{Fwd}$	Price of forward contract	6
$\mathcal{P}^S$	Price of stock	1
$\mathcal{P}^X$	Price of foreign currency; that is, the spot exchange rate	6
$\mathcal{P}^Z$	Price of zero-coupon discount bond	6
$\mathcal{P}^{Z^*}$	Price of zero-coupon bond in foreign currency	6
$\mathbb{P}$	Physical probability measure (could be objective or subjective)	1

prob$^{\mathbb{P}}$	Physical probability under the measure $\mathbb{P}$, with the measure $\mathbb{P}$ written explicitly	1
prob$^{\mathbb{Q}}$	Risk-neutral probability (RNP) under the measure $\mathbb{Q}$; prob$^{\mathbb{Q}}(A) = E^{\mathbb{P}}[\xi_T \, 1_A]$	1
prob$^{\mathbb{S}}$	Probability when the numeraire is the stock, so under the measure $\mathbb{S}$	1
q	Dividend yield on an asset	2
$\mathbb{Q}$	Risk-neutral probability measure	1
r	Risk-free interest rate	1
r^*	Risk-free interest rate in a foreign currency	6
R_f	Gross risk-free interest rate ($R_f = 1 + r$) or ($R_f = e^r$)	1
R_i	Gross rate of return on asset i	1
R_{mkt}	Gross rate of return on market portfolio	1
R_z	Gross rate of return on zero-beta portfolio	1
RNP	Abbreviation for "risk-neutral probability"	1
$\mathbb{S}$	Probability measure when stock is the numeraire	1
SDF	Abbreviation for "stochastic discount factor"	1
t	Time, usually today (current date)	1
T	Terminal date	1
T'	Used to refer to dates after the terminal date; that is, $T' > T$	2
$\top$	Transpose operator (for example, to change row vectors into column vectors)	3
$\mathcal{U}$	Utility, which is a function of lifetime consumption	3
U	Utility, which is a function of one period consumption	3
u	Utility, which is a function of one period consumption, with $U(C_t) = e^{-\delta t} C_t$	3
$V(t, \mathcal{P}_t^S)$	Value of an option at date t	5
Var$[x]$	Variance of x	3
w_n	Portfolio weight; proportion of wealth invested in asset n; also denoted by π_n	1
W	Wealth	2
y	Yield on a bond	6
y^*	Yield on a bond in foreign currency	6

E.2 Greek letters

Symbol	Definition	Appears for first time in Chapter
Δ	Delta of an option, measures change in option price as price of underlying asset changes	5
Δt	Change in t	5
$\Delta Y(t)$	Change in $Y(t)$	5
Γ	Gamma of an option, measures change in option delta as stock price changes	5
λ	Factor risk premium; or, market price(s) of risk; also known as Sharpe ratio	3
λ	Factor risk premium; or, market price(s) of risk; also known as Sharpe ratio	3
μ	Drift of an Itô process	2
π_n	Portfolio weight; proportion of wealth invested in asset n; also denoted by w_n	5
ρ	Correlation coefficient, correlation between two Brownian motions	2
ρ	Rho of an option, measures change in option value as interest rate changes	5
σ	Volatility (scalar or matrix); diffusion coefficient of an Itô process	2
Σ	Variance-covariance matrix, also denoted by V	3
τ	Stopping time (the first time a particular condition is satisfied)	2
Θ	Theta, measures change in option value as t increases (time-to-maturity, T, decreases)	2
$\mathcal{V}$	Vega of an option, measures change in option value as volatility changes	5
ψ	Factor risk premium divided by variance of the factor, $\psi = \lambda/\text{Var}[f]$ or $\psi = \Sigma_F^{-1}/\lambda$	3
ξ_T	Radon-Nikodym derivative, $\xi_T = \frac{d\mathbb{Q}}{d\mathbb{P}} = \frac{d\text{prob}^{\mathbb{Q}}}{d\text{prob}^{\mathbb{P}}} = M_T \mathcal{P}_T^D$	4
ξ_t	Conditional expectation of ξ_T; that is, $\xi_t = E_t[\xi_T]$	4

Blank page.

Lecture Notes for 2025

Continuous Time Financial Economics

Appendix F

List of Ph.D. Elective Courses Offered by EDHEC

Raman Uppal

EDHEC Business School

F.1 List of Ph.D. Courses Offered in the Past

Year	Instructor	Affiliation	Course title
2009	Lionel Martellini	EDHEC	Dynamic Asset Allocation Decision
2009	Pietro Veronesi	U. of Chicago	Recent Developments in Portfolio Allocation and Macro-Finance Model
2010	Rama Cont	Columbia	Monte Carlo Methods in Finance
2010	Tim Bollerslev	Duke	Volatility Modelling
2010	Harrison Hong	Princeton	Behavioural Finance
2010	Ekkehart Boehmer	EDHEC	Microstructure
2010	Francis Diebold	U. of Penn.	The Yield Curve
2011	Peter Christoffersen	Toronto	Risk Management and Extreme Risks
2011	Yacine Aït-Sahalia	Princeton	Financial Econometrics of Jumps
2011	Antonio Mello	UW-Madison	Advanced Corporate Finance
2011	Jerome Detemple	Boston	Derivaties Securities
2011	Nicholas Polson	Chicago	Bayesian Methods in Financial Econometrics
2011	Allan Timmermann	UC San Diego	Predictive Modeling and Forecast Evaluation in Financial Markets
2011	Michael Brandt	Duke	Portfolio Choice
2011	Ravi Bansal	Duke	Long-Run Risks in Asset Prices
2011	Tarun Ramadorai	Oxford	Risk and Return in Hedge Funds
2012	Mikhail Chernov	LSE	Empirical Option Pricing
2012	Rama Cont	Columbia	Credit Risk Modelling
2012	Ekkehart Boehmer	EDHEC	Microstructure
2012	Federico Bandi	Johns Hopkins	The Econometrics of Continuous-Time Models
2012	Harrison Hong	Princeton	Behavioural Finance
2012	Tim Bollerslev	Duke	Volatility Modelling
2012	Lionel Martellini	EDHEC	Dynamic Asset Allocation-Long-term Investing with Short-term Constraints
2012	Pietro Veronesi	U. of Chicago	Macroeconomic and Political Uncertainty and the Dynamics of Asset Prices

Year	Instructor	Affiliation	Course title
2012	Jianqing Fan	Princeton	Portfolio Problems with High-Dimension Covariance Matrices
2012	Sanjiv Das	Santa Clara	Credit Risk Modelling
2013	Michael Brandt	Duke	Portfolio Allocation
2013	Tarun Ramadorai	Oxford	International Finance
2013	Jerome Detemple	Boston	Strategic Asset Allocation
2013	Torben Andersen	Northwestern	High-Frequency Asset Pricing
2013	Vikas Agarwal	Georgia State	Hedge Funds
2013	Allan Timmermann	UC San Diego	Predictive Modeling in Financial Markets
2013	Federico Bandi	Johns Hopkins	Estimation of Continuous-time Models
2013	Harrison Hong	Princeton	Behavioural Finance
2013	Peter Christoffersen	Toronto	Risk Management and Extreme Risks
2013	Rama Cont	Imperial	Monte-Carlo Methods in Finance
2014	Allan Timmermann	UC San Diego	Markov Switching and Time-varying Parameter Models in Finance
2014	Tim Bollerslev	Duke	Volatility Modelling
2014	Ekkehart Boehmer	EDHEC	Microstructure
2014	Yacine Aït-Sahalia	Princeton	An Introduction to High Frequency Financial Econometrics
2014	Mikhail Chernov	LSE	Empirical Option Pricing
2014	Sanjiv Das	Santa Clara	Data Science, Analytics, Algorithms
2014	Lionel Martellini	EDHEC	Dynamic Risk and Asset Allocation Decisions
2014	Ludovic Phalippou	Oxford	Private Equity
2014	Pietro Veronesi	U. of Chicago	Macroeconomic and Political Uncertainty and Dynamics of Asset Prices
2015	Federico Bandi	Johns Hopkins	The Econometrics of Continuous-Time Models
2015	Torben Andersen	Northwestern	High-Frequency Asset Pricing
2015	Peter Christoffersen	Toronto	Risk Management and Extreme Risks
2015	Harrison Hong	Princeton	Behavioral Finance and Asset Management
2016	Jerome Detemple	Boston	Dynamic Asset Allocation
2016	Vikas Agarwal	Georgia State	Asset Management (Mutual Funds & Hedge Funds)

Year	Instructor	Affiliation	Course title
2016	Francis Diebold	U. of Penn.	Aspects of Real-Time Measurement and Monitoring in Financial Markets, Macro Fundamentals and the Interface
2016	Tim Bollerslev	Duke	Volatility Modeling
2016	Ludovic Phalippou	Oxford	Private Equity
2016	Pietro Veronesi	U. of Chicago	Macroeconomic and Political Uncertainty and the Dynamics of Asset Prices
2017	Rama Cont	Imperial	Liquidity Risk
2017	Riccardo Rebonato	EDHEC	Term Structure Modelling in the P and Q measures
2017	Sanjiv Das	Santa Clara	Data Science, Analytics, Algorithms
2017	Harjoat Bhamra	Imperial College	Monetary Policy and Asset Prices
2017	Marcel Rindisbacher	Boston Univ.	Monte Carlo Methods
2017	Harrison Hong	Columbia Univ.	Behavioral Finance and Asset Management
2017	Federico Bandi	Johns Hopkins	Continuous-time Financial Econometrics
2018	Michael Chernov	LSE, now UCLA	Macro-Based Valuation Across Assets and Horizons
2018	Jeffrey M. Wooldridge	Michigan State Univ.	A Course in Microeconometrics
2018	Yacine Aït-Sahalia	Princeton	Topics in Financial Econometrics
2018	Tarun Ramadorai	Imperial College	International Finance
2018	Allan Timmermann	UC San Diego	Predictive Modeling and Forecast Evaluation in Financial Markets
2019	Bryan Kelly	Yale	Empirical Asset Pricing
2019	José Luis Peydró	Pompeu Fabra Univ.	Systemic Risk and Macroprudential Policy
2019	Toni Whited	Michigan	Dynamic Corporate Finance
2019	Michael Brandt	Duke	Macroeconomic Nowcasting
2019	Jesus Fernandez-Villaverde	Wharton	Numerical Solutions and Bayesian Estimation Methods for General Equilibrium Asset Pricing Models
2020	Jeffrey M. Wooldridge	Michigan State Univ.	Panel Data Analysis
2020	Albert Menkveld	Vrije U. of Amsterdam	Market Microstructure
2020	Jaume Ventura	Barcelona	The Macroeconomics of Credit and Asset Bubbles
2020	Federico Bandi	Johns Hopkins	Nonparametric Methods in Finance

Year	Instructor	Affiliation	Course title
2021	Yacine Aït-Sahalia	Princeton	Topics in Financial Econometrics
2021	CEPR Conference	EDHEC	Household Finance
2021	Pietro Veronesi	U. of of Chicago	Fixed Income Securities
2021	Sanjiv Das	Santa Clara	Data Science, Analytics, Algorithms
2022	Stefano Giglio	Yale University	Empirical Asset Pricing
2022	Andrew Patton	Duke University	Financial Econometrics
2022	Vikas Agarwal	Georgia State University	Asset Management: Mutual Funds and Hedge Funds
2022	Harrison Hong	Columbia University	Behavioral Finance and Asset Management
2022	Marcin Kacperczyk	Imperial College	Climate/Sustainable Finance
2023	Gordon Phillips	Dartmouth University	Finance and Industrial Organization
2023	Jeffrey Wooldridge	Michigan State	Topics in Microeconometrics
2023	Yacine Aït-Sahalia	Princeton University	Machine Learning with Applications in Finance
2023	Ludovic Phalippou	Oxford University	Private Equity
2023	Harjoat Bhamra	Imperial College	Asset Pricing and Monetary Policy
2024	Frederico Belo	INSEAD	Production-based Asset Pricing: Theory and Empirics
2024	Federico Bandi	Johns Hopkins	Spectral Asset Pricing
2024	Lukas Schmid	Univ. of Southern California	Models and Methods in Macro-Finance
2024	Pietro Veronesi	Chicago Booth	Fixed Income
2024	Tarun Ramadorai	Imperial College	Household Finance
2025	Robert Kosowski	Imperial College	Alternative Risk Premia and Replication of Financial Anomalies
2025	Albert Menkveld	Univ. of Amsterdam	Market Microstructure

F.2 List of Ph.D. Courses Scheduled (subject to change)

Year	Instructor	Affiliation	Course title
2025	Ayako Yasuda	Berkeley	Sustainable Finance and Impact Investing

Bibliography

Abel, A. B. 1990. Asset prices under habit formation and catching up with the joneses. *American Economic Review* 80 (2): 38. (Cited on page 609).

Agarwal, V., K. A. Mullally, and N. Y. Naik. 2015. The economics and finance of hedge funds: A review of the academic literature. *Foundations and Trends in Finance* 10 (1): 1–111. (Cited on page 36).

Ahn, D.-H., R. F. Dittmar, and A. R. Gallant. 2002. Quadratic term structure models: Theory and evidence. *Review of Financial Studies* 15 (1): 243–288. (Cited on pages 155, 254, 286).

Ahn, S., G. Kaplan, B. Moll, T. Winberry, and C. Wolf. 2018. When inequality matters for macro and macro matters for inequality. *NBER Macroeconomics Annual* 32 (1): 1–75. (Cited on page 611).

Ai, H., M. M. Croce, and K. Li. 2013. Toward a quantitative general equilibrium asset pricing model with intangible capital. *Review of Financial Studies* 26 (2): 491–530. (Cited on page 609).

Aït-Sahalia, Y. 2009. Estimating and testing continuous-time models in finance: The role of transition densities. *Annual Review of Financial Economics* 1 (1): 341–359. (Cited on page 37).

Alessie, R., S. Hochguertel, and A. Van Soest. 2002. Household portfolios in the Netherlands. In *Household portfolios,* edited by L. Guiso, M. Haliassos, and T. Jappelli. Cambridge: MIT Press. (Cited on page 387).

Allen, F., J. Qian, and M. Qian. 2019. A review of China's institutions. *Annual Review of Financial Economics* 11 (1): 39–64. (Cited on page 39).

Ang, A., W. N. Goetzmann, and S. M. Schaefer. 2011. The efficient market theory and evidence: Implications for active investment management. *Foundations and Trends in Finance* 5 (3): 157–242. (Cited on page 36).

Ao, M., Y. Li, and X. Zheng. 2019. Approaching mean-variance efficiency for large portfolios. *Review of Financial Studies* 32 (7): 2890–2919. (Cited on page 390).

Aragon, G. O., and W. E. Ferson. 2007. Portfolio performance evaluation. *Foundations and Trends in Finance* 2 (2): 83–190. (Cited on page 36).

Arrondel, L., and B. Lefebvre. 2001. Behavior of household portfolios in france: The role of housing. *Review of Income and Wealth* 47 (4): 489–514. (Cited on page 387).

Back, K. 2005. *A course in derivative securities: Introduction to theory and computation.* NY: Springer. (Cited on pages 34, 143, 286).

———. 2010. Martingale pricing. *Annual Review of Financial Economics* 2 (1): 235–250. (Cited on page 478).

———. 2017. *Asset pricing and portfolio choice theory.* Second edition. NY: Oxford University Press. (Cited on pages 24, 31, 33, 42, 143, 286, 383, 474, 543, 550, 608).

Backus, D., M. Chernov, and I. Martin. 2011. Disasters implied by equity index options. *Jounral of Finance* 66 (6): 1969–2012. (Cited on page 610).

Bakshi, G. S., C. Cao, and Z. Chen. 1997. Empirical performance of alternative option pricing models. *Journal of Finance* 52 (5): 2003–2049. (Cited on page 550).

Banks, J., and S. Smith. 2002. UK household portfolios. In *Household portfolios,* edited by L. Guiso, M. Haliassos, and T. Jappelli. Cambridge, MA: MIT Press. (Cited on page 387).

Bansal, R., and A. Yaron. 2004. Risks for the long run: A potential resolution of asset pricing puzzles. *Journal of Finance* 59 (4): 1481–1509. (Cited on page 610).

Barasinska, N., D. Schäfer, and A. Stephan. 2008. Financial risk aversion and household asset diversification. SOEP Papers on Multidisciplinary Panel Data Research. (Cited on page 387).

Barber, B. M., and T. Odean. 2000. Trading is hazardous to your wealth: The common stock investment performance of individual investors. *Journal of Finance* 55:773–806. (Cited on page 388).

———. 2001. Boys will be boys: Gender, overconfidence, and common stock investment. *Quarterly Journal of Economics* 116 (1): 261–292. (Cited on page 388).

———. 2003. Are individual investors tax savvy? evidence from retail and discount brokerage accounts. *Journal of Public Economics* 88 (1-2): 419–442. (Cited on page 388).

———. 2013. The behavior of individual investors. In *Handbook of economics and finance,* edited by G. M. Constantinides, M. Harris, and R. M. Stulz, vol. 2. Elsevier, Amsterdam. (Cited on page 388).

Barberis, N., and R. H. Thaler. 2003. A survey of behavioral finance. In *Handbook of the economics of finance, 1(b),* edited by G. M. Constantinides, M. Harris, and R. M. Stulz, 1053–1125. Elsevier. (Cited on page 388).

Barberis, N. C. 2013. Thirty years of prospect theory in economics: a review and assessment. *Journal of Economic Perspectives* 27 (1): 173–96. (Cited on page 388).

Barro, R. J. 2006. Rare disasters and asset markets in the twentieth century. *Quarterly Journal of Economics* 121 (3): 823–866. (Cited on page 610).

———. 2009. Rare disasters, asset prices, and welfare costs. *American Economic Review* 99 (1): 243–264. (Cited on page 610).

Basak, S., and G. Chabakauri. 2010. Dynamic mean-variance asset allocation. *Review of Financial Studies* 23 (8): 2970–3016. (Cited on page 391).

———. 2012. Dynamic hedging in incomplete markets: A simple solution. *Review of Financial Studies* 25 (6): 1845–1896. (Cited on page 391).

Basak, S., and D. Cuoco. 1998. An equilibrium model with restricted stock market participation. *Review of Financial Studies* 11 (2): 309–341. (Cited on page 611).

Bates, D. S. 1996. Testing option pricing models. Chap. 20 in *Handbook of statistics (statistical methods in finance)*, edited by G. Maddala and C. Rao, 14:567–611. Amsterdam: Elsevier. (Cited on page 550).

———. 2008. The market for crash risk. *Journal of Economic Dynamics and Control* 32, no. 7 (July): 2291–2321. (Cited on page 550).

Baxter, M., and A. Rennie. 1996. *Financial calculus: An introduction to derivative pricing*. Cambridge university press. (Cited on page 256).

Bekaert, G., C. R. Harvey, A. Kiguel, and X. Wang. 2016. Globalization and asset returns. *Annual Review of Financial Economics* 8 (1): 221–288. (Cited on page 39).

Ben-David, I., F. Franzoni, and R. Moussawi. 2017. Exchange-traded funds. *Annual Review of Financial Economics* 9 (1): 169–189. (Cited on page 39).

Benartzi, S., and R. Thaler. 2001. Naive diversification strategies in defined contribution saving plans. *American Economic Review* 91 (1): 7998. (Cited on page 388).

Benartzi, S., R. H. Thaler, S. P. Utkus, and C. R. Sunstein. 2007. The law and economics of company stock in 401(k) plans. *Journal of Law and Economics* 50:45–79. (Cited on page 387).

Berk, J. B., and J. H. van Binsbergen. 2017. Mutual funds in equilibrium. *Annual Review of Financial Economics* 9 (1): 147–167. (Cited on page 39).

Bhamra, H. S., and R. Uppal. 2006. The role of risk aversion and intertemporal substitution in dynamic consumption-portfolio choice with recursive utility. *Journal of Economic Dynamics and Control* 30:967–991. (Cited on page 549).

———. 2008. The effect of introducing a non-redundant derivative on the volatility of stock-market returns when agents differ in risk aversion. *Review of Financial Studies*, forthcoming. (Cited on page 611).

———. 2014. Asset prices with heterogeneity in preferences and beliefs. *Review of Financial Studies* 27 (2): 519–580. (Cited on pages 586, 611).

———. 2019. Does household finance matter? Small financial errors with large social costs. *American Economic Review* 109 (3): 1116–54. (Cited on pages 387, 609).

Bhamra, H. S., R. Uppal, and J. Walden. 2020. The geography of beliefs. SSRN Working Paper. (Cited on pages 387, 609).

Bielecki, T. R., and M. Rutkowski. 2013. *Credit risk: modeling, valuation and hedging*. Springer Science & Business Media. (Cited on pages 147, 288).

Black, F. 1976. The pricing of commodity contracts. *Journal of Financial Economics* 3 (1-2): 167–179. (Cited on pages 29, 94, 96, 98, 99, 114).

Black, F., E. Derman, and W. Toy. 1990. A one-factor model of interest rates and its application to treasury bond options. *Financial Analysts Journal* 46 (1): 33–39. (Cited on pages 69, 268–273, 286).

Black, F., and P. Karasinski. 1991. Bond and option pricing when short rates are lognormal. *Financial Analysts Journal* 47 (4): 52–59. (Cited on pages 69, 268, 286).

Black, F., and M. S. Scholes. 1973. The pricing of options and corporate liabilities. *Journal of Political Economy* 81 (3): 637–654. (Cited on pages 29, 49, 157).

Bodie, Z., J. Detemple, and M. Rindisbacher. 2009. Life-cycle finance and the design of pension plans. *Annual Review of Financial Economics* 1 (1): 249–286. (Cited on page 37).

Borovička, J. 2020. Survival and long-run dynamics with heterogeneous beliefs under recursive preferences. *Journal of Political Economy* 128 (1): 000–000. (Cited on page 611).

Börsch-Supan, A., and A. Eymann. 2002. Household portfolios in Germany. In *Household portfolios,* edited by L. Guiso, M. Haliassos, and T. Jappelli. Cambridge: MIT Press. (Cited on page 387).

Boyle, P., L. Garlappi, R. Uppal, and T. Wang. 2012. Keynes meets Markowitz: The trade-off between familiarity and diversification. *Management Science* 58 (2): 253–272. (Cited on page 388).

Brace, A., D. Gątarek, and M. Musiela. 1997. The market model of interest rate dynamics. *Mathematical Finance* 7 (2): 127–155. (Cited on pages 30, 145).

Brandt, M. W. 2010. Portfolio choice problems. In *Handbook of financial econometrics,* edited by Y. Ait-Sahalia and L. P. Hansen, 1:269–336. Forthcoming. North Holland. (Cited on pages 478, 551).

Brav, A., W. Jiang, and H. Kim. 2010. Hedge fund activism: A review. *Foundations and Trends in Finance* 4 (3): 185–246. (Cited on page 36).

———. 2015. Recent advances in research on hedge fund activism: Value creation and identification. *Annual Review of Financial Economics* 7 (1): 579–595. (Cited on page 39).

Breeden, D. T. 1979. An intertemporal asset pricing model with stochastic consumption and investment opportunities. *Journal of Financial Economics* 7 (3): 265–296. (Cited on pages 559, 574, 608).

Brennan, M. J., E. S. Schwartz, and R. Lagnado. 1997. Strategic asset allocation. *Journal of Economic Dynamics and Control* 21 (8-9): 1377–1403. (Cited on page 549).

Brigo, D., and F. Mercurio. 2007. *Interest rate models – theory and practice: With smile, inflation and credit.* Springer Science & Business Media. (Cited on pages 146, 147, 287, 288).

Britten-Jones, M. 1998. *Fixed income and interest rate derivative analysis.* Elsevier. (Cited on page 146).

Brunnermeier, M. K., and Y. Sannikov. 2014. A macroeconomic model with a financial sector. *American Economic Review* 104 (2): 379–421. (Cited on page 611).

———. 2016. Macro, money, and finance: A continuous-time approach. In *Handbook of Macroeconomics,* vol. 2B. 1497-1545: Elsevier. (Cited on pages 35, 611).

Burnside, C., M. Eichenbaum, and S. Rebelo. 2011. Carry trade and momentum in currency markets. *Annual Review of Financial Economics* 3 (1): 511–535. (Cited on page 38).

Buss, A., B. Dumas, R. Uppal, and G. Vilkov. 2016. The intended and unintended consequences of financial-market regulations: A general-equilibrium analysis. *Journal of Monetary Economics* 81:25–43. (Cited on page 586).

Buss, A., R. Uppal, and G. Vilkov. 2015. Where experience matters: Asset allocation and asset pricing with opaque and illiquid assets. Working Paper, EDHEC, February. (Cited on page 586).

Cairns, A. J. G. 2018. *Interest rate models: An introduction.* Princeton University Press. (Cited on pages 146, 147, 287, 288).

Calvet, L. E., J. Y. Campbell, and P. Sodini. 2007. Down or out: Assessing the welfare costs of household investment mistakes. *Journal of Political Economy* 115 (5): 707–747. (Cited on page 387).

Campbell, J. Y. 1993. Intertemporal asset pricing without consumption data. *American Economic Review* 83:487–512. (Cited on page 546).

———. 2000. Asset pricing at the millennium. *Journal of Finance* 55 (4): 1515–67. (Cited on pages 35, 478, 551).

———. 2006. Household finance. *Journal of Finance* 61 (4): 1553–1604. (Cited on page 387).

———. 2018. *Finance decisions and markets: A course in asset pricing.* Princeton University Press. (Cited on pages 24, 34, 384, 546, 549).

Campbell, J. Y., and J. H. Cochrane. 1999. By force of habit: A consumption-based explanation of aggregate stock market behavior. *Journal of Political Economy* 107 (2): 205–251. (Cited on page 609).

Campbell, J. Y., A. W. Lo, and A. C. MacKinlay. 1997. *The econometrics of financial markets.* Princeton, New Jersey: Princeton University Press. (Cited on page 34).

Campbell, J. Y., T. Ramadorai, and B. Ranish. 2012. Do stock traders learn from experience? evidence from an emerging market. Working Paper, Harvard University and Oxford University. (Cited on page 387).

Campbell, J. Y., and L. M. Viceira. 1999. Consumption and portfolio decisions when expected returns are time varying. *Quarterly Journal of Economics* 114:433–495. (Cited on pages 475, 546).

———. 2001. Who should buy long-term bonds? *American Economic Review* 91:99–127. (Cited on pages 475, 546).

———. 2002. *Strategic asset allocation: Portfolio choice for long-term investors.* New York: Oxford University Press. (Cited on pages 475, 546, 549).

Cass, D., and J. E. Stiglitz. 1970. The structure of investor preferences and asset returns, and separability in portfolio allocation: A contribution to the pure theory of mutual funds. *Journal of Economic Theory* 2 (2): 122–160. (Cited on page 384).

Chan, Y. L., and L. Kogan. 2002. Catching up with the joneses: Heterogeneous preferences and the dynamics of asset prices. *Journal of Political Economy* 110 (6): 1255–1285. (Cited on page 609).

Chen, Z., and L. G. Epstein. 2002. Ambiguity, risk, and asset returns in continuous time. *Econometrica* 70 (4): 1403–1443. (Cited on page 609).

Choi, J. J. 2015. Contributions to defined contribution pension plans. *Annual Review of Financial Economics* 7 (1): 161–178. (Cited on page 39).

Christensen, J. H. E., F. X. Diebold, and G. D. Rudebusch. 2011. The affine arbitrage-free class of Nelson-Siegel term structure models. *Journal of Econometrics* 164 (1): 4–20. (Cited on page 146).

Cocco, J. F., F. J. Gomes, and P. J. Maenhout. 2005. Consumption and portfolio choice over the life cycle. Forthcoming, *Review of Financial Studies* 18 (2): 491–533. (Cited on page 611).

Cochrane, J. H. 2005. *Asset pricing.* Revised. Princeton, NJ: Princeton University Press. (Cited on page 34).

———. 2012. Continuous-time linear models. *Foundations and Trends in Finance* 6 (3): 165–219. (Cited on page 36).

Collin-Dufresne, P., M. Johannes, and L. A. Lochstoer. 2016. Parameter learning in general equilibrium: The asset pricing implications. *American Economic Review* 106, no. 3 (March): 664–98. (Cited on page 610).

Cox, J. C., and C.-f. Huang. 1989. Optimal consumption and portfolio policies when asset prices follow a diffusion process. *Journal of Economic Theory* 49 (1): 33–83. (Cited on pages 30, 474, 544).

Cox, J. C., J. E. Ingersoll Jr., and S. A. Ross. 1985a. A theory of the term structure of interest rates. *Econometrica* 53 (2): 385–407. (Cited on pages 69, 155, 247–249, 251, 266, 286).

———. 1985b. An intertemporal general equilibrium model of asset prices. *Econometrica* 53 (2): 363–84. (Cited on pages 30, 595, 607, 608, 643).

Cox, J. C., and S. A. Ross. 1976. The valuation of options for alternative stochastic processes. *Journal of Financial Economics* 3 (1-2): 145–166. (Cited on page 29).

Cox, J. C., S. A. Ross, and M. Rubinstein. 1979. Option pricing: A simplified approach. *Journal of Financial Economics* 7:229–263. (Cited on pages 29, 271).

Cvitanic, J., and F. Zapatero. 2004. *Introduction to the economics and mathematics of financial markets*. First. MIT Press. (Cited on page 34).

Cvitanić, J., and I. Karatzas. 1992. Convex duality in constrained portfolio optimization. *Annals of Applied Probability* 2:767–818. (Cited on page 477).

Dai, Q., and K. J. Singleton. 2000. Specification analysis of affine term structure models. *Journal of Finance* 55 (5): 1943–1978. (Cited on pages 155, 252, 286).

Das, S. R., and R. K. Sundaram. 1999. Of smiles and smirks: A term structure perspective. *Journal of Financial and Quantitative Analysis* 34 (2): 211–239. (Cited on page 550).

Das, S. R., and R. Uppal. 2004. Systemic risk and international portfolio choice. *The Journal of Finance* 59 (6): 2809–2834. (Cited on page 550).

De Finetti, B. 1940. *Il problema dei pieni*. Istituto italiano degli attuari. (Cited on page 384).

DeMiguel, V., L. Garlappi, F. J. Nogales, and R. Uppal. 2009. A generalized approach to portfolio optimization: Improving performance by constraining portfolio norms. *Management Science* 55 (5): 798–812. (Cited on page 390).

DeMiguel, V., L. Garlappi, and R. Uppal. 2009. Optimal versus naive diversification: How inefficient is the $1/N$ portfolio strategy? *Review of Financial Studies* 22 (5): 1915–1953. (Cited on page 389).

DeMiguel, V., A. Martin-Utrera, F. J. Nogales, and R. Uppal. 2019. A transaction-cost perspective on the multitude of firm characteristics. Forthcoming in *Review of Financial Studies*. (Cited on page 386).

Demirgüç-Kunt, A., and R. Levine. 2009. Finance and inequality: Theory and evidence. *Annual Review of Financial Economics* 1 (1): 287–318. (Cited on page 37).

Detemple, J. B., R. Garcia, and M. Rindisbacher. 2003. A Monte Carlo method for optimal portfolios. *Journal of Finance* 58:410–446. (Cited on page 478).

Detemple, J. B., and F. Zapatero. 1991. Asset prices in an exchange economy with habit formation. *Econometrica* 59 (6): 1633–57. (Cited on page 609).

Diebold, F. X., and C. Li. 2006. Forecasting the term structure of government bond yields. *Journal of Econometrics* 130 (2): 337–364. (Cited on page 146).

Diebold, F. X., M. Piazzesi, and G. D. Rudebusch. 2005. Modeling bond yields in finance and macroeconomics. *American Economic Review* 95 (2): 415–420. (Cited on page 146).

Duffee, G. R. 2002. Term premia and interest rate forecasts in affine models. *Journal of Finance* 57 (1): 405–443. (Cited on page 253).

Duffie, D. 2001. *Dynamic asset pricing theory.* Princeton, NJ: Princeton University Press. (Cited on page 34).

Duffie, D., and L. G. Epstein. 1992. Stochastic Differential Utility. *Econometrica* 60:353–394. (Cited on pages 476, 609).

Duffie, D., and R. Kan. 1996. A yield-factor model of interest rates. *Mathematical Finance* 6 (4): 379–406. (Cited on pages 155, 249, 252, 266, 286).

Duffie, D., J. Pan, and K. Singleton. 2000. Transform analysis and asset pricing for affine jump-diffusions. *Econometrica* 68:1343–1376. (Cited on page 550).

Duffie, D., and K. J. Singleton. 2012. *Credit risk: pricing, measurement, and management.* Princeton university press. (Cited on pages 147, 288).

Dumas, B. 1989. Two-person dynamic equilibrium in the capital market. *Review of Financial Studies* 2 (2): 157–188. (Cited on page 611).

Dumas, B., A. Kurshev, and R. Uppal. 2009. Equilibrium portfolio strategies in the presence of sentiment risk and excess volatility. *The Journal of Finance* 64 (2): 579–629. (Cited on page 610).

Dumas, B., and E. Luciano. 2017. *The economics of continuous-time finance.* Cambridge, MA: MIT Press. (Cited on pages 24, 31, 33, 42, 143, 286, 383, 474, 477, 543, 608–610).

Dumas, B., and A. Lyasoff. 2012. Incomplete-market equilibria solved recursively on an event tree. *Journal of Finance* 67 (5): 1897–1941. (Cited on pages 586, 611).

Dumas, B., and R. Uppal. 2001. Global diversification, growth and welfare with imperfectly integrated markets for goods. *Review of Financial Studies* 14:227–305. (Cited on pages 476, 609).

Dumas, B., R. Uppal, and T. Wang. 2000. Efficient intertemporal allocations with recursive utility. *Journal of Economic Theory* 93 (2): 240–59. (Cited on pages 476, 609).

Epstein, L., and S. Zin. 1989. Substitution, risk aversion and the temporal behavior of consumption and asset returns: A theoretical framework. *Econometrica* 57 (4): 937–969. (Cited on pages 476, 609).

Epstein, L. G., and M. Schneider. 2008. Ambiguity, information quality, and asset pricing. *The Journal of Finance* 63 (1): 197–228. (Cited on page 609).

———. 2010. Ambiguity and asset markets. *Annual Review of Financial Economics* 2 (1): 315–346. (Cited on page 609).

Epstein, L. G., and T. Wang. 1994. Intertemporal asset pricing under knightian uncertainty. *Econometrica* 62 (2): 283–322. (Cited on page 609).

Fagereng, A., C. Gottlieb, and L. Guiso. 2017. Asset market participation and portfolio choice over the life-cycle. *Journal of Finance* 72 (2): 705–750. (Cited on page 387).

Fama, E. F., and K. R. French. 2015. A five-factor asset pricing model. *Journal of Financial Economics* 116 (1): 1–22. (Cited on pages 354, 386).

Ferson, W. E. 2010. Investment performance evaluation. *Annual Review of Financial Economics* 2 (1): 207–234. (Cited on page 37).

Filipovic, D. 2009. *Term-structure models. A graduate course*. Springer. (Cited on pages 146, 147, 287, 288).

Frydman, C., and D. Jenter. 2010. CEO compensation. *Annual Review of Financial Economics* 2 (1): 75–102. (Cited on page 37).

Gabaix, X., J.-M. Lasry, P.-L. Lions, and B. Moll. 2016. The dynamics of inequality. *Econometrica* 84 (6): 2071–2111. (Cited on page 611).

Garlappi, L., R. Uppal, and T. Wang. 2007. Portfolio selection with parameter and model uncertainty: A multi-prior approach. *The Review of Financial Studies* 20:41–81. (Cited on page 388).

Garman, M. B., and S. W. Kohlhagen. 1983. Foreign currency option values. *Journal of International Money and Finance* 2 (3): 231–237. (Cited on page 29).

Geman, H., N. El Karoui, and J.-C. Rochet. 1995. Changes of numeraire, changes of probability measure and option pricing. *Journal of Applied Probability* 32 (2): 443–458. (Cited on page 29).

Ghosh, A., C. Julliard, and A. P. Taylor. 2017. What is the Consumption-CAPM missing? An information-theoretic framework for the analysis of asset pricing models. *The Review of Financial Studies* 30 (2): 442–504. (Cited on page 610).

———. 2019. An information-theoretic asset pricing model. London School of Economics working paper. (Cited on page 390).

Gibson, R., F.-S. Lhabitant, and D. Talay. 2010. Modeling the term structure of interest rates: A review of the literature. *Foundations and Trends in Finance* 5 (1–2): 1–156. (Cited on pages 36, 144).

Glasserman, P. 2004. *Monte Carlo methods in financial engineering*. Springer-Verlag. (Cited on page 478).

Gollier, C. 2001. *The economics of risk and time.* Cambridge, MA: MIT Press. (Cited on page 34).

Gomes, F., and A. Michaelides. 2003. Portfolio choice with internal habit formation: A life-cycle model with uninsurable labor income risk. *Review of Economic Dynamics* 6 (4): 729–766. (Cited on page 611).

———. 2005. Optimal life-cycle asset allocation: Understanding the empirical evidence. *Journal of Finance* 60 (2): 869–904. (Cited on page 611).

———. 2008. Asset pricing with limited risk sharing and heterogeneous agents. *Review of Financial Economics* 21 (1): 415–448. (Cited on page 611).

Grinblatt, M., and M. Keloharju. 2001. How distance, language, and culture influence stockholdings and trades. *Journal of Finance* 56 (3): 1053–1073. (Cited on page 387).

Guiso, L., M. Haliassos, and T. Jappelli. 2002. *Household portfolios.* Cambridge, MA: MIT Press. (Cited on page 387).

Guiso, L., and T. Jappelli. 2002. Household portfolios in Italy. In *Household portfolios,* edited by L. Guiso, M. Haliassos, and T. Jappelli. Cambridge: MIT Press. (Cited on page 387).

Guiso, L., and P. Sodini. 2013. Household finance: An emerging field. In *Handbook of the economics of finance,* edited by G. M. Constantinides, M. Harris, and R. M. Stulz. Amsterdam: Elsevier. (Cited on page 387).

Guvenen, F. 2009. A parsimonious macroeconomic model for asset pricing. *Econometrica* 77 (6): 1711–1750. (Cited on page 611).

Haliassos, M. 2002. Stockholding: Recent lessons from theory and computations. Working Papers in Economics, University of Cyprus. (Cited on page 387).

Hansen, L. P., and T. J. Sargent. 2007. *Robustness.* Princeton, NJ: Princeton University Press. (Cited on pages 388, 609).

Harrison, J. M., and D. M. Kreps. 1978. Speculative investor behavior in a stock market with heterogeneous expectations. *Quarterly Journal of Economics* 92:323–336. (Cited on page 610).

Harrison, M., and D. Kreps. 1979. Martingales and arbitrage in multiperiod securities markets. *Journal of Economic Theory* 20:381–408. (Cited on page 29).

He, H., and N. D. Pearson. 1991. Consumption and portfolio policies with incomplete markets and short-sale constraints: The infinite-dimensional case. *Journal of Economic Theory* 54:259–304. (Cited on page 477).

He, Z., and A. Krishnamurthy. 2013. Intermediary asset pricing. *American Economic Review* 103 (2): 732–70. (Cited on page 611).

———. 2018. Intermediary asset pricing and the financial crisis. *Annual Review of Financial Economics* 10 (1): 173–197. (Cited on page 611).

Heath, D., R. Jarrow, and A. Morton. 1992. Bond pricing and the term structure of interest rates: A new methodology for contingent claims valuation. *Econometrica* 60 (1): 77–105. (Cited on pages 145, 155, 255, 256, 266, 286).

Hirshleifer, D. 2015. Behavioral finance. *Annual Review of Financial Economics* 7 (1): 133–159. (Cited on page 39).

Ho, T. S. Y., and S.-B. Lee. 1986. Term structure movements and pricing interest rate contingent claims. *Journal of Finance* 41 (5): 1011–1029. (Cited on pages 69, 155, 217, 261, 267, 268, 286, 625).

Hou, K., H. Mo, C. Xue, and L. Zhang. 2019. Which factors? *Review of Finance* 23 (1): 1–35. (Cited on pages 354, 386).

Hou, K., C. Xue, and L. Zhang. 2015. Digesting anomalies: An investment approach. *Review of Financial Studies* 28 (3): 650–705. (Cited on pages 354, 386).

———. 2017a. A comparison of new factor models. *Fisher College of Business Working Paper,* nos. 2015-03: 05. (Cited on pages 354, 386).

———. 2017b. Replicating anomalies. *The Review of Financial Studies.* (Cited on pages 354, 386).

Huang, C.-F., and R. Litzenberger. 1988. *Foundations for financial economics.* New Jersey: Prentice Hall. (Cited on page 384).

Huberman, G. 2001. Familiarity breeds investment. *Review of Financial Studies* 14 (3): 659–680. (Cited on page 387).

Hull, J., and A. White. 1990. Pricing interest-rate-derivative securities. *Review of Financial Studies* 3 (4): 573–592. (Cited on pages 155, 214, 215, 246, 266, 286).

Hull, J. C. 2017. *Options, futures, and other derivatives.* Upper Saddle River, NJ: Prentice Hall. (Cited on page 34).

Ilmanen, A. 2011. *Expected returns: An investor's guide to harvesting market rewards.* West Sussex, UK: Wiley Finance. (Cited on page 390).

Ingersoll, J. E., Jr. 1987. *Theory of financial decision making.* Maryland: Rowman & Littlefield. (Cited on pages 34, 551).

Jagannathan, R., E. Schaumburg, and G. Zhou. 2010. Cross-sectional asset pricing tests. *Annual Review of Financial Economics* 2 (1): 49–74. (Cited on page 37).

Jamshidian, F. 1997. LIBOR and swap market models and measures. *Finance and Stochastics* 1 (4): 293–330. (Cited on page 145).

Jarrow, R. A. 2002. *Modeling fixed-income securities and interest rate options.* Stanford University Press. (Cited on page 287).

———. 2009. Credit risk models. *Annual Review of Financial Economics* 1 (1): 37–68. (Cited on pages 37, 147, 288).

———. 2019. *Modeling fixed-income securities and interest rate options.* 3rd. Chapman & Hall/CRC Financial Mathematics Series. (Cited on page 146).

Jegadeesh, N., and S. Titman. 2011. Momentum. *Annual Review of Financial Economics* 3 (1): 493–509. (Cited on page 38).

Jermann, U. J. 1998. Asset pricing in production economies. *Journal of Monetary Economics* 41 (2): 257–275. (Cited on page 609).

Kan, R., and G. Zhou. 2007. Optimal portfolio choice with parameter uncertainty. *Journal of Financial and Quantitative Analysis* 42 (3): 621–656. (Cited on page 390).

Karatzas, I., J. P. Lehoczky, and S. E. Shreve. 1987. Optimal portfolio and consumption decisions for a small investor on a finite horizon. *SIAM Journal of Control and Optimization* 25 (6): 1557–1586. (Cited on pages 30, 474, 544).

Karatzas, I., J. P. Lehoczky, S. E. Shreve, and G. L. Xu. 1991. Martingale and duality methods for utility maximization in an incomplete market. *SIAM Journal of Control and Optimization* 29:702–730. (Cited on page 477).

Karatzas, I., and S. E. Shreve. 1998. *Methods of mathematica finance.* New York: Springer-Verlag. (Cited on pages 474, 477).

Kim, T. S., and E. Omberg. 1996. Dynamic nonmyopic portfolio behavior. *Review of Financial Studies* 9 (1): 141–61. (Cited on page 549).

Kimball, M. S. 1990. Precautionary saving in the small and in the large. *Econometrica* 58 (1): 53–73. (Cited on page 593).

Kogan, L. 2001. An equilibrium model of irreversible investment. *Journal of Financial Economics* 62 (2): 201–245. (Cited on page 609).

———. 2004. Asset prices and real investment. *Journal of Financial Economics* 73:411–431. (Cited on page 609).

Korteweg, A. 2019. Risk adjustment in private equity returns. *Annual Review of Financial Economics* 11 (1): 131–152. (Cited on page 39).

Laeven, L. 2011. Banking crises: A review. *Annual Review of Financial Economics* 3 (1): 17–40. (Cited on page 38).

Lando, D. 2009. Credit risk modeling. In *Handbook of financial time series,* 787–798. Springer. (Cited on pages 147, 288).

Ledoit, O., and M. Wolf. 2003. Improved estimation of the covariance matrix of stock returns with an application to portfolio selection. *Journal of Empirical Finance* 10 (5): 603–621. (Cited on page 390).

———. 2004a. A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis* 88 (January): 365–411. (Cited on page 390).

———. 2004b. Honey, I shrunk the sample covariance matrix. *Journal of Portfolio Management* 30, no. 4 (December): 110–119. (Cited on page 390).

———. 2017. Nonlinear shrinkage of the covariance matrix for portfolio selection: Markowitz meets goldilocks. *The Review of Financial Studies* 30 (12): 4349–4388. (Cited on page 390).

Leippold, M., and L. Wu. 2002. Asset pricing under the quadratic class. *Journal of Financial and Quantitative Analysis* 37 (2): 271–295. (Cited on page 254).

LeRoy, S. F., and J. Werner. 2001. *Principles of financial economics.* Cambridge, UK: Cambridge University Press. (Cited on page 34).

Lintner, J. 1969. The aggregation of investor's diverse judgments and preferences in purely competitive security markets. *Journal of Financial and Quantitative Analysis* 4 (4): 347–400. (Cited on page 384).

Litzenberger, R., J. Castura, and R. Gorelick. 2012. The impacts of automation and high frequency trading on market quality. *Annual Review of Financial Economics* 4 (1): 59–98. (Cited on page 38).

Liu, J. 2007. Portfolio selection in stochastic environments. *Review of Financial Studies* 20 (1): 1–39. (Cited on page 546).

Ljungqvist, L., and T. J. Sargent. 2018. *Recursive macroeconomic theory.* MIT press. (Cited on page 551).

Longstaff, F., and E. Schwartz. 1992. Interest rate volatility and the term structure: A two factor general equilibrium model. *Journal of Finance* 47:1259–1282. (Cited on pages 155, 248, 249, 252, 266, 286).

Lototsky, S. V., and B. L. Rozovsky. 2017. *Stochastic partial differential equations.* Springer. (Cited on page 478).

Lucas, R. J. 1978. Asset prices in an exchange economy. *Econometrica* 46 (6): 1429–1445. (Cited on pages 30, 586, 607, 608).

Madhavan, A. 2014. Exchange-traded funds: An overview of institutions, trading, and impacts. *Annual Review of Financial Economics* 6 (1): 311–341. (Cited on page 38).

Maenhout, P. 1999. Robust portfolio rules and asset pricing. Working paper, Harvard University. (Cited on page 609).

Margrabe, W. 1978. The value of an option to exchange one asset for another. *Journal of Finance* 33 (1): 177–186. (Cited on pages 29, 114).

Markowitz, H. M. 1952. Portfolio selection. *Journal of Finance* 7 (1): 77–91. (Cited on pages 30, 384).

———. 1959. *Portfolio selection: Efficient diversification of investments.* New York: Wiley. (Cited on page 384).

———. 2006. de Finetti scoops Markowitz. *Journal of Investment Management* 4 (3). (Cited on page 384).

McCulloch, J. H. 1971. Measuring the term structure of interest rates. *Journal of Business* 44 (1): 19–31. (Cited on page 144).

———. 1975. The tax-adjusted yield curve. *Journal of Finance* 30 (3): 811–830. (Cited on page 144).

Mehra, R. 2007. The equity premium puzzle: A review. *Foundations and Trends in Finance* 2 (1): 1–81. (Cited on page 36).

Merton, R. C. 1969. Lifetime portfolio selection under uncertainty: The continuous-time case. *Review of Economics and Statistics:* 247–257. (Cited on pages 30, 385, 474, 543, 544).

———. 1971. Optimum consumption and portfolio rules in a continuous time model. *Journal of Economic Theory* 3:373–413. (Cited on pages 30, 385, 474, 543, 544, 547, 550).

———. 1972. An analytic derivation of the efficient portfolio frontier. *Journal of Financial and Quantitative Analysis* 7 (4): 1851–72. (Cited on page 384).

———. 1973a. An intertemporal capital asset pricing model. *Econometrica* 41 (5): 867–888. (Cited on pages 30, 385, 543, 559, 581, 605, 608, 641).

———. 1973b. The theory of rational option pricing. *Bell Journal of Economics and Management Science* 4 (1): 141–183. (Cited on page 29).

Merton, R. C. 1976. Option pricing when underlying stock returns are discontinuous. *Journal of Financial Economics* 3 (1/2): 125–44. (Cited on page 550).

———. 1990. *Continuous time finance.* Cambridge MA: Basil Blackwell. (Cited on pages 29, 34, 385, 474, 543, 544).

Meucci, A. 2009. *Risk and asset allocation.* Springer. (Cited on page 390).

Miltersen, K. R., K. Sandmann, and D. Sondermann. 1997. Closed form solutions for term structure derivatives with log-normal interest rates. *Journal of Finance* 52 (1): 409–430. (Cited on page 145).

Mitchell, S. O., and S. Utkus. 2004. The role of company stock in defined contribution plans. In *The pension challenge: risk transfers and retirement income security,* edited by S. O. Mitchell and K. Smetters. Oxford University Press. (Cited on page 387).

Mossin, J. 1966. Equilibrium in a capital asset market. *Econometrica:* 768–783. (Cited on page 384).

Munk, C. 2013. *Financial asset pricing theory.* Oxford. (Cited on page 34).

Musiela, M., and M. Rutkowski. 1997. Continuous-time term structure models: Forward measure approach. *Finance and Stochastics* 1 (4): 261–291. (Cited on page 145).

———. 2005. *Martingale methods in financial modelling.* Springer, Heidelberg Berlin New York. (Cited on page 145).

Musto, D. K. 2011. The economics of mutual funds. *Annual Review of Financial Economics* 3 (1): 159–172. (Cited on page 38).

Naik, V., and M. Lee. 1990. General equilibrium pricing of options on the market portfolio with discontinuous returns. *Review of Financial Studies* 3:493–521. (Cited on page 550).

Naik, V. 1993. Option valuation and hedging strategies with jumps in the volatility of asset returns. *Journal of Finance* 48 (5): 1969–1984. (Cited on page 550).

Nawalkha, S., and G. Soto. 2017. A review of term structure estimation methiods. *Alternative Investment Analyst Review* 6 (1): 67–75. (Cited on page 144).

Nelson, C. R., and A. F. Siegel. 1987. Parsimonious modeling of yield curves. *Journal of Business* 60 (4): 473–489. (Cited on pages 139, 144).

Nielsen, L. T., and M. Vassalou. 2006. The instantaneous capital market line. *Economic Theory* 28 (3): 651–664. (Cited on page 547).

Nuño, G., and B. Moll. 2018. Social optima in economies with heterogeneous agents. *Review of Economic Dynamics* 28:150–180. (Cited on page 611).

Nymand-Andersen, P. 2018. *Yield curve modelling and a conceptual framework for estimating yield curves: Evidence from the european central bank's yield curves.* Technical report. ECB Statistics Paper. (Cited on page 144).

Odean, T. 1998. Are investors reluctant to realize their losses? *Journal of Finance* 53 (5): 1775–1798. (Cited on page 388).

Odean, T. 1999. Do investors trade too much? *American Economic Review* 89:1279–1298. (Cited on page 388).

Pascucci, A. 2011. *PDE and martingale methods in option pricing.* Springer Science & Business Media. (Cited on page 478).

Pennacchi, G. 2008. *Theory of asset pricing.* MA: Addison-Wesley. (Cited on page 34).

Piazzesi, M. 2010. Affine term structure models. In *Handbook of financial econometrics: tools and techniques,* 691–766. Elsevier. (Cited on page 287).

Pliska, S. 1997. *Introduction to mathematical finance: Discrete time models.* Blackwell. (Cited on pages 424, 477).

Polkovnichenko, V. 2005. Household portfolio diversification: A case for rank dependent preferences. *Review of Financial Studies* 18:1467–1502. (Cited on page 387).

Pressacco, F., and P. Serafini. 2007. The origins of the mean-variance approach in finance: Revisiting de Finetti 65 years later. *Decisions in Economics and Finance* 30 (1): 19–49. (Cited on page 384).

Raponi, V., R. Uppal, and P. Zaffaroni. 2020. Robust portfolio choice. Working Paper, SSRN eLibrary. (Cited on page 390).

Rebonato, R. 1996. *Interest-rate option models: Understanding, analysing and using models for exotic interest-rate options.* John Wiley & Sons. (Cited on pages 146, 287).

———. 2012. *Modern pricing of interest-rate derivatives: The LIBOR market model and beyond.* Princeton University Press. (Cited on page 145).

Roll, R. 1980. Orthogonal portfolios. *Journal of Financial and Quantitative Analysis* 15 (5): 1005–1023. (Cited on page 384).

Ross, S. 1976. The arbitrage theory of capital asset pricing. *Journal of Economic Theory* 13 (3): 341–360. (Cited on page 390).

Rubinstein, M. 1974. An aggregation theorem for securities markets. *Journal of Financial Economics* 1:225–244. (Cited on page 586).

———. 2002. Markowitz's "portfolio selection": A fifty-year retrospective. *The Journal of finance* 57 (3): 1041–1045. (Cited on page 385).

———. 2006a. *A history of the theory of investments: My annotated bibliography.* Hoboken, NJ: John Wiley & Sons. (Cited on pages 384, 385).

———. 2006b. Bruno de Finetti and mean-variance portfolio selection. *Journal of Investment Management* 4 (3). (Cited on page 384).

Samuelson, P. 1969. Lifetime portfolio selection by dynamic stochastic programming. *Review of Economics and Statistics* 51:239–246. (Cited on page 385).

Scheinkman, J. A., and W. Xiong. 2003. Overconfidence and speculative bubbles. *Journal of Political Economy* 111 (6): 1183–1219. (Cited on page 610).

Schönbucher, P. J. 2003. *Credit derivatives pricing models: models, pricing and implementation.* John Wiley & Sons. (Cited on pages 147, 288).

Schroder, M., and C. Skiadas. 1999. Optimal consumption and portfolio selection with stochastic differential utility. *Journal of Economic Theory* 89:68–126. (Cited on page 476).

———. 2002. An isomorphism between asset pricing models with and without linear habit formation. *Review of Financial Studies* 15 (4): 1189–1221. (Cited on page 609).

Sharpe, W. 1964. Capital asset prices: A theory of market equilibrium under conditions of risk. *Journal of Finance* 1919 (3): 425–442. (Cited on pages 30, 384, 385).

Shefrin, H. 2010. Behavioralizing finance. *Foundations and Trends in Finance* 4 (1-2): 1–184. (Cited on pages 36, 388).

Shefrin, H., and M. Statman. 1985. The disposition to sell winners too early and ride losers too long: Theory and evidence. *Journal of Finance* 40 (3): 777–790. (Cited on page 388).

———. 2000. Behavioral portfolio theory. *Journal of Financial and Quantitative Analysis* 35 (2): 127–151. (Cited on page 388).

Spatt, C. S. 2010. An informal perspective on the economics and regulation of securities markets. *Annual Review of Financial Economics* 2 (1): 127–143. (Cited on page 37).

Stokey, N. L., and J. Lucas Robert E. 1989. *Recursive methods in economic dynamics (with prescott, e.c.)* Cambridge, MA: Harvard University Press. (Cited on page 551).

Strahan, P. E. 2013. Too big to fail: causes, consequences, and policy responses. *Annual Review of Financial Economics* 5 (1): 43–61. (Cited on page 38).

Sundaram, R. K., and S. R. Das. 2015. *Derivatives: Principles and practice.* McGraw-Hill Irwin New York, NY. (Cited on pages 143, 287).

Sundaresan, S. M. 2000. Continuous-time methods in finance: A review and an assessment. *Journal of Finance* 55 (4): 1569–1622. (Cited on pages 35, 478, 551).

Svensson, L. E. O. 1989. Portfolio choice with non-expected utility in continuous time. *Economic Letters* 30:313–317. (Cited on page 549).

Svensson, L. E. 1994. *Estimating and interpreting forward interest rates: Sweden 1992-1994.* Technical report. National bureau of economic research. (Cited on pages 139, 144).

Tobin, J. 1958. Liquidity preference as behavior towards risk. *Review of Economic Studies* 25 (2): 68–85. (Cited on page 384).

Treynor, J. L. 1961a. Market value, time, and risk. Unpublished manuscript. (Cited on page 384).

———. 1961b. Toward a theory of market value of risky assets. Unpublished manuscript. (Cited on page 384).

Tsai, J., and J. A. Wachter. 2015. Disaster risk and its implications for asset pricing. *Annual Review of Financial Economics* 7 (1): 219–252. (Cited on page 610).

Tufano, P. 2009. Consumer finance. *Annual Review of Financial Economics* 1 (1): 227–247. (Cited on page 37).

Uppal, R. 1993. A general equilibrium model of international portfolio choice. *The Journal of Finance* 48 (2): 529–553. (Cited on page 475).

Uppal, R., and T. Wang. 2003. Model misspecification and underdiversification. *Journal of Finance* 58 (6): 2465–2486. (Cited on page 609).

Varian, H. 1993. A portfolio of Nobel laureates: Markowitz, Miller and Sharpe. *Journal of Economic Perspectives* 7 (1): 159–169. (Cited on page 384).

Vasicek, O. 1977. An equilibrium characterization of the term structure. *Journal of Financial Economics* 5 (2): 177–188. (Cited on pages 30, 69, 155, 173, 214, 247, 249, 251, 259, 268, 286, 623).

Vasicek, O. A., and H. G. Fong. 1982. Term structure modeling using exponential splines. *Journal of Finance* 37 (2): 339–348. (Cited on page 144).

Veronesi, P. 2010. *Fixed income securities: Valuation, risk, and risk management.* John Wiley & Sons. (Cited on pages 146, 287).

Vissing-Jorgensen, A. 2003. Perspectives on behavioral finance: Does " irrationality" disappear with wealth? evidence from expectations and actions. *NBER Macroeconomics Annual* 18:139–194. (Cited on page 388).

Wachter, J. A. 2002. Portfolio and consumption decisions under mean-reverting returns: An exact solution for complete markets. *Journal of Financial and Quantitative Analysis* 37:63–91. (Cited on page 475).

———. 2010. Asset allocation. *Annual Review of Financial Economics* 2 (1): 175–206. (Cited on pages 38, 478, 551).

———. 2013. Can time-varying risk of rare disasters explain aggregate stock market volatility? *The Journal of Finance* 68 (3): 987–1035. (Cited on page 610).

Wang, J. 1996. The term structure of interest rates in a pure exchange economy with heterogeneous investors. *Journal of Financial Economics* 41 (1): 75–110. (Cited on page 611).

Weil, P. 1989. The equity-premium puzzle and the risk-free rate puzzle. *Journal of Monetary Economics* 24:401–421. (Cited on page 609).

———. 1990. Nonexpected utility in macroeconomics. *Quarterly Journal of Economics* 105 (1): 29–42. (Cited on pages 476, 609).

Wermers, R. 2011. Performance measurement of mutual funds, hedge funds, and institutional accounts. *Annual Review of Financial Economics* 3 (1): 537–574. (Cited on page 38).

Worthington, A. C. 2009. Household asset portfolio diversification: Evidence from the household, income and labor dynamics in Australia (HILDA) survey. Working Paper, Griffith Business School, Griffith University. (Cited on page 387).

Zhang, Y., X. Li, and S. Guo. 2018. Portfolio selection problems with Markowitz's mean-variance framework: A review of literature. *Fuzzy Optimization and Decision Making* 17 (2): 125–158. (Cited on page 385).

Zhou, G. 2018. Measuring investor sentiment. *Annual Review of Financial Economics* 10 (1): 239–259. (Cited on page 39).

Some blank pages for miscellaneous notes.

Blank page.

End of slides for Part 2 of

Continuous Time Financial Economics